AOS-CX 10.18.xxxx Fundamentals Guide (8400 Switch Series)

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About this document

This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing HPE Aruba Networking switches on a network.

Subtopics

Applicable products

This document applies to the following products:

- HPE Aruba Networking 8400 Switch Series (JL366A, JL363A, JL687A)

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in Support and Other Resources.

Command syntax notation conventions

Convention	Usage
`example - text`	Identifies commands and their options and operands , code examples, filenames, pathnames, and output d isplayed in a command window. Items that appear li ke the example text in the previous column are to be entered exactly as shown and are required unless en closed in brackets (**[ ]**).

Convention	Usage
example - text	In code and screen examples, indicates text entered by a user.
Any of the following: • <example - text> • <example - text> • example - text • example - text	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: • For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (**< >**). Substitute the text—including the enclosing angle brackets—with an actual value. • For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
|	Vertical bar. A logical **OR** that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.
{ }	Braces. Indicates that at least one of the enclosed items is required.
[ ]	Brackets. Indicates that the enclosed item or items are optional.
… or ...	Ellipsis: • In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. • In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment.

The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term **switch**, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

```
switch>
```

Indicates the operator command context.

```
switch#
```

Indicates the manager command context.

```
switch(CONTEXT-NAME)#
```

Indicates the configuration context for a feature. For example:

```
switch(config-if)#
```

Identifies the **interface** context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch(config-vlan-100)#
```

When referring to this context, this document uses the syntax:

```
switch(config-vlan-<VLAN-ID>)#
```

Where <VLAN-ID> is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format: **member/slot/port.**

On the HPE Aruba Networking 8400 Switch Series

- member: Always 1. VSF is not supported on this switch.
- slot: Specifies physical location of a module in the switch chassis.
    - Management modules are on the front of the switch in slots 1/5 and 1/6.
    - Line modules are on the front of the switch in slots 1/1 through 1/4, and 1/7 through 1/10.
- port: Physical number of a port on a line module

For example, the logical interface **1/1/4** in software is associated with physical port 4 in slot 1 on member 1.

Identifying modular switch components

- Power supplies are on the front of the switch behind the bezel above the management modules. Power supplies are labeled in software in the format: member/power supply:
    - member: 1.
    - power supply: 1 to 4.
- Fans are on the rear of the switch and are labeled in software as: member/tray/fan:
    - member: 1.
    - tray: 1 to 4.
    - fan: 1 to 4.
- Fabric modules are not labeled on the switch but are labeled in software in the format: member/module:
    - member: 1.
    - member: 1 or 2.
- The display module on the rear of the switch is not labeled with a member or slot number.

About AOS-CX

AOS-CX is a new, modern, fully programmable operating system built using a database-centric design that ensures higher availability and dynamic software process changes for reduced downtime. In addition to robust hardware reliability, the AOS-CX operating system includes additional software elements not available with traditional systems, including:

- Automated visibility to help IT organizations scale: The HPE Aruba Networking Network Analytics Engine allows IT to monitor and troubleshoot network, system, application, and security-related issues easily through simple scripts. This engine comes with a built-in time series database that enables customers and developers to create software modules that allow historical troubleshooting, as well as analysis of historical trends to predict and avoid future problems due to scale, security, and performance bottlenecks.
- Programmability simplified: A switch that is running the AOS-CX operating system is fully programmable with a built-in Python interpreter as well as REST-based APIs, allowing easy integration with

other devices both on premise and in the cloud. This programmability accelerates IT organization understanding of and response to network issues. The database holds all aspects of the configuration, statistics, and status information in a highly structured and fully defined form.

- Faster resolution with network insights: With legacy switches, IT organizations must troubleshoot problems after the fact, using traditional tools like CLI and SNMP, augmented by separate, expensive monitoring, analytics, and troubleshooting solutions. These capabilities are built in to the AOS-CX operating system and are extensible.
- High availability: For switches that support active and standby management modules, the AOS-CX database can synchronize data between active and standby modules and maintain current configuration and state information during a failover to the standby management module.
- Ease of roll-back to previous configurations: The built-in database acts as a network record, enabling support for multiple configuration checkpoints and the ability to roll back to a previous configuration checkpoint.

Subtopics

AOS-CX system databases
Network Analytics Engine introduction
AOS-CX CLI
Aruba CX mobile app
HPE Aruba Networking NetEdit
Ansible modules
AOS-CX Web UI
AOS-CX REST API
In-band and out-of-band management
SNMP-based management support
User accounts

AOS-CX system databases

The AOS-CX operating system is a modular, database-centric operating system. Every aspect of the switch configuration and state information is modeled in the AOS-CX switch configuration and state database, including the following:

- Configuration information
- Status of all features
- Statistics

The AOS-CX operating system also includes a time series database, which acts as a built-in network record. The time series database makes the data seamlessly available to HPE Aruba Networking Network Analytics

Engine agents that use rules that evaluate network conditions over time. Time-series data about the resources monitored by agents are automatically collected and presented in graphs in the switch Web UI.

Network Analytics Engine introduction

The HPE Aruba Networking Network Analytics Engine is a first-of-its-kind built-in framework for network assurance and remediation. Combining the full automation and deep visibility capabilities of the AOS-CX operating system, this unique framework enables monitoring, collecting network data, evaluating conditions, and taking corrective actions through simple scripting agents.

This engine is integrated with the AOS-CX system configuration and time series databases, enabling you to examine historical trends and predict future problems due to scale, security, and performance bottlenecks. With that information, you can create software modules that automatically detect such issues and take appropriate actions.

With the faster network insights and automation provided by the HPE Aruba Networking Network Analytics Engine, you can reduce the time spent on manual tasks and address current and future demands driven by Mobility and IoT.

AOS-CX CLI

The AOS-CX CLI is an industry standard text-based command-line interface with hierarchical structure designed to reduce training time and increase productivity in multivendor installations.

The CLI gives you access to the full set of commands for the switch while providing the same password protection that is used in the Web UI. You can use the CLI to configure, manage, and monitor devices running the AOS-CX operating system.

Aruba CX mobile app

The Aruba CX mobile app enables you to use a mobile device to configure or access a supported AOS-CX switch. You can connect to the switch through Bluetooth or Wi-Fi.

You can use this application to do the following:

- Connect to the switch for the first time and configure basic operational settings—all without requiring you to connect a terminal emulator to the console port.
- View and change the configuration of individual switch features or settings.
- Manage the running configuration and startup configuration of the switch, including the following:
    - Transferring files between the switch and your mobile device

    - Sharing configuration files from your mobile device
    - Copying the running configuration to the startup configuration
- Access the switch CLI.

Refer to the Aruba CX mobile app **datasheet** for more information.

HPE Aruba Networking NetEdit

HPE Aruba Networking NetEdit enables the automation of multidevice configuration change workflows without the overhead of programming.

The key capabilities of NetEdit include the following:

- Intelligent configuration with validation for consistency and compliance
- Time savings by simultaneously viewing and editing multiple configurations
- Customized validation tests for corporate compliance and network design
- Automated large-scale configuration deployment without programming
- Ability to track changes to hardware, software, and configurations (whether made through NetEdit or directly on the switch) with automated versioning

For more information about HPE Aruba Networking NetEdit, search for NetEdit at the following website:

www.hpe.com/support/hpesc

Ansible modules

Ansible is an open-source IT automation platform.

HPE Aruba Networking publishes a set of Ansible configuration management modules designed for switches running AOS-CX software. The modules are available from the following places:

- The **arubanetworks.aoscx_role** role in the Ansible Galaxy at: **https://galaxy.ansible.com/arubanetworks/aoscx_role**
- The **aoscx-ansible-role** at the following GitHub repository: **https://github.com/aruba/aoscx-ansible-role**

AOS-CX Web UI

The Web UI gives you quick and easy visibility into what is happening on your switch, providing faster problem detection, diagnosis, and resolution. The Web UI provides dashboards and views to monitor the status of the switch, including easy to read indicators for: power supply, temperature, fans, CPU use, memory use, log entries, system information, firmware, interfaces, VLANs, and LAGs. In addition, you use the Web UI to access the Network Analytics Engine, run certain diagnostics, and modify some aspects of the switch configuration.

AOS-CX REST API

Switches running the AOS-CX software are fully programmable with a REST (REpresentational State Transfer) API, allowing easy integration with other devices both on premises and in the cloud. This programmability—combined with the HPE Aruba Networking Network Analytics Engine—accelerates network administrator understanding of and response to network issues.

The AOS-CX REST API enables programmatic access to the AOS-CX configuration and state database at the heart of the switch. By using a structured model, changes to the content and formatting of the CLI output do not affect the programs you write. And because the configuration is stored in a structured database instead of a text file, rolling back changes is easier than ever, thus dramatically reducing a risk of downtime and performance issues.

The AOS-CX REST API is a web service that performs operations on switch resources using HTTPS **POST**, **GET**, **PUT**, **DELETE**, and **PATCH** methods.

A switch resource is indicated by its Uniform Resource Identifier (URI). A URI can be made up of several components, including the host name or IP address, port number, the path, and an optional query string. The AOS-CX operating system includes the AOS-CX REST API Reference, which is a web interface based on the Swagger UI. The AOS-CX REST API Reference provides the reference documentation for the REST API, including resources URIs, models, methods, and errors. The AOS-CX REST API Reference shows most of the supported read and write methods for all switch resources.

In-band and out-of-band management

Management communications with a managed switch can be either of the following:

In band

In-band management communications occur through ports on the line modules of the switch, using common communications protocols such as SSH and SNMP.

When you use an in-band management connection, management traffic from that connection uses the same network infrastructure as user data. User data uses the `data plane`, which is responsible for moving data from source to destination. Management traffic that uses the data plane is more likely to be affected by traffic congestion and other issues affecting the user network.

Out of band

OOBM (out-of-band management) communications occur through a dedicated serial or USB console port or though a dedicated networked management port.

OOBM operates on a `management plane` that is separate from the `data plane` used by data traffic on the switch and by in-band management traffic. That separation means that OOBM can continue to function even during periods of traffic congestion, equipment malfunction, or attacks on the network. In addition, it can provide improved switch security: a properly configured switch can limit management access to the management port only, preventing malicious attempts to gain access through the data ports.

Networked OOBM typically occurs on a management network that connects multiple switches. It has the added advantage that it can be done from a central location and does not require an individual physical cable from the management station to the console port of each switch.

SNMP-based management support

The AOS-CX operating system provides SNMP read access to the switch. SNMP support includes support of industry-standard MIB (Management Information Base) plus private extensions, including SNMP events, alarms, history, statistics groups, and a private alarm extension group. SNMP access is disabled by default.

User accounts

To view or change configuration settings on the switch, users must log in with a valid account. Authentication of user accounts can be performed locally on the switch, or by using the services of an external TACACS+ or RADIUS server.

Two types of user accounts are supported:

- Operators: Operators can view configuration settings, but cannot change them. No operator accounts are created by default.
- Administrators: Administrators can view and change configuration settings. A default locally stored administrator account is created with username set to **admin** and no password. You set the administrator account password as part of the initial configuration procedure for the switch.

Initial Configuration

Perform the initial configuration of a factory default switch using one of the following methods:

- Load a switch configuration using zero-touch provisioning (ZTP). When ZTP is used, the configuration is loaded from a server automatically when the switch booted from the factory default configuration.

- Connect to the switch wirelessly with a mobile device through Bluetooth, and use the Aruba CX Mobile App to deploy an initial configuration from a provided template. The template you choose during the deployment process determines how the management interface is configured. Optionally, as the final deployment step, you can select to import the switch into NetEdit through a WiFI connection to the NetEdit server.
  Alternatively, you can use the Aruba CX Mobile App to manually configure switch settings and features for a subset of the features you can configure using the CLI. You can also access the CLI through the mobile application.
- Connect the management port on the switch to your network, and then use SSH client software to reach the switch from a computer connected to the same network. This requires that a DHCP server is installed on the network. Configure switch settings and features by executing CLI commands.
- Connect a computer running terminal emulation software to the console port on the switch. Configure switch settings and features by executing CLI commands.

NOTE

Reserved ports or ports used by other applications/services with in the system are not recommended to be used for other services. When two services use the same port there is chance of unexpected behaviors from these services. Best practices is to use unique port for each service across system.

Subtopics

Initial configuration using ZTP

Zero Touch Provisioning (ZTP) configures a switch automatically from a remote server.

Prerequisites

- The switch must be in the factory default configuration.
  Do not change the configuration of the switch from its factory default configuration in any way, including by setting the administrator password.

- Your network administrator or installation site coordinator must provide a Category 6 (Cat6) cable connected to the network that provides access to the servers used for Zero Touch Provisioning (ZTP) operations.

Procedure

1. Connect the network cable to the out-of-band management port on the switch.
   See the Installation Guide for switch to determine the location of the switch ports.

2. If the switch is powered on, power off the switch.

3. Power on the switch. During the ZTP operation, the switch might reboot if a new firmware image is being installed. ZTP goes to "Failed" state if the switch receives DHCP IP for vlan1 and does not receive any ZTP options within 60 seconds.

Initial configuration using the Aruba CX mobile app

This procedure describes how to use your mobile device to connect to the Bluetooth interface of the switch to connect to the switch for the first time so that you can configure basic operational settings using the Aruba CX mobile app.

Prerequisites

- You have obtained the USB Bluetooth adapter that was shipped with the switch. Information about the make and model of the supported adapter is included in the information about the HPE Aruba Networking Installer Mobile App in the Apple Store or Google Play.

- The HPE Aruba Networking Installer Mobile App must be installed on your mobile device.

- Bluetooth must be enabled on your mobile device.

- Your mobile device must be within the communication range of the Bluetooth adapter.

- If you are planning to import the switch into NetEdit, your mobile device must be able to use a Wi-Fi connection—not Bluetooth—to access the NetEdit server.

  If your mobile device does not support simultaneous Bluetooth and Wi-Fi connections, you must use the NetEdit interface to import the switch at a later time. You can use the **Devices** tab to display the IP address of the switches you configured using your mobile device.

- The switch must be installed and powered on, with the network operating system boot sequence complete.

  For information about installing and powering on the switch, see the Installation Guide for the switch.

  Because you are using this mobile application to configure the switch through the Bluetooth interface, it is not necessary to connect a console to the switch.

- Bluetooth and USB must be enabled on the switch. On switches shipped from the factory, Bluetooth and USB are enabled by default.

Procedure

1. Install the USB Bluetooth adapter in the USB port of the switch.
   For switches that have multiple management modules, you must install the USB Bluetooth adapter in the USB port of the active management module. Typically, the active management module is the module in slot 5.

   Switches shipped from the factory have both USB and Bluetooth enabled by default.

   For information about the location of the USB port on the switch, see the Installation Guide for the switch.

2. Use the Bluetooth settings on your mobile device to pair and connect the switch to your mobile device.
   If you are in range of multiple Bluetooth devices, more than one device is displayed on the list of available devices. Switches running the AOS-CX operating system are displayed in the following format:

   ```
   Switch_model - Serial_number
   ```

   For example: `8325-987654X1234567` or `8320-AB12CDE123`

   A switch supports one active Bluetooth connection at a time.

   On some Android devices, you might need to change the settings of the paired device to specify that it be used for Internet access.

3. Open the Aruba CX mobile app on your mobile device.
   The application attempts to connect to the switch using the switch Bluetooth IP address and the default switch login credentials. The **Home** screen of the application shows the status of the connection to the switch:

   - If the login attempt was successful, the Bluetooth icon is displayed and the status message shows the Bluetooth IP address of the switch. In addition, the connection graphic is green. You can continue to the next step.

- If the login attempt was not successful, but a response was received, the Bluetooth icon is displayed, but the status message is: **Login Required**. You can continue to the next step. When you tap one of the tiles, you will be prompted for login credentials.
- If the login attempt did not receive a response, the Bluetooth icon is not displayed, and the status message is: **No Connection**.

4. Create the initial switch configuration:

    - You can deploy an initial configuration to the switch. Through this process, you supply the information required by a configuration template that you choose from a list of templates provided by the application. Then you deploy the configuration to the switch and, optionally, import the switch into NetEdit.

      CAUTION

      When you deploy a switch configuration, it becomes the running configuration, replacing the entire existing configuration of the switch. All changes previously made to the factory default configuration are overwritten.

      If you plan to both deploy a switch configuration and customize the configuration of switch features, deploy the initial configuration first.

      To deploy an initial switch configuration, tap: **Initial Config** and follow the instructions in the application.

    - Alternatively, you can complete the initial configuration of the switch by tapping **Modify Config** and then selecting the features and settings to configure.
    - You can also use the **Modify Config** feature to configure some switch features after the initial configuration is complete. For more information about what you can configure using the Aruba CX mobile app, see the online help for the application.

Subtopics

Troubleshooting Bluetooth connections

Subtopics

Bluetooth connection IP addresses

The Bluetooth connection uses IP addresses in the 192.168.99.0/24 subnet.

Switch

192.168.99.1

Mobile device

192.168.99.10

Bluetooth is connected but the switch is not reachable

Symptom

The mobile device settings indicate that the device is connected to the switch through Bluetooth. However, the mobile application indicates that the switch is not reachable.

Solution 1

Cause

The mobile device is paired with a different nearby switch.

Action

1. Verify the model number and serial number of the switch to which you are attempting to connect.
2. Use the Bluetooth settings on your mobile device to pair and connect the switch to your mobile device. If you are in range of multiple Bluetooth devices, more than one device is displayed on the list of available devices. Switches running the AOS-CX operating system are displayed in the following format:

   `Switch_model-Serial_number`

   For example: `8325-987654X1234567` or `8320-AB12CDE123`

   A switch supports one active Bluetooth connection at a time.

   On some Android devices, you might need to change the settings of the paired device to specify that it be used for Internet access.

Solution 2

Cause

The mobile device is connected to a different network—such as through a Wi-Fi connection—that conflicts with the subnet used for the switch Bluetooth connection.

Action

Disconnect the mobile device from the network that is using the conflicting subnet.

For example, use the mobile device settings to turn off or disable Wi-Fi. If you choose to disable Wi-Fi on the mobile device, and you are not able to access cellular service, you will not be able to connect to the NetEdit server to import the switch, but you can still deploy a switch configuration.

Bluetooth is not connected

Symptom

Your mobile device cannot establish a Bluetooth connection to the switch.

Solution 1

Cause

Bluetooth is not enabled on your mobile device.

Action

- Use your mobile device settings application to enable Bluetooth.
- Use the Bluetooth settings on your mobile device to pair and connect the switch to your mobile device. If you are in range of multiple Bluetooth devices, more than one device is displayed on the list of available devices. Switches running the AOS-CX operating system are displayed in the following format:

  Switch_model-Serial_number

  For example: **8325-987654X1234567** or **8320-AB12CDE123**

  A switch supports one active Bluetooth connection at a time.

  On some Android devices, you might need to change the settings of the paired device to specify that it be used for Internet access.

Solution 2

Cause

Your mobile device is not within the broadcast range of the Bluetooth adapter.

Action

Move closer to the switch.

Devices can communicate through Bluetooth when they are close, typically within a few feet of each other.

Solution 3

Cause

Your mobile device is not paired with the switch.

Action

1. Use your mobile device settings application to enable Bluetooth.
2. Use the Bluetooth settings on your mobile device to pair and connect the switch to your mobile device. If you are in range of multiple Bluetooth devices, more than one device is displayed on the list of available devices. Switches running the AOS-CX operating system are displayed in the following format:

   Switch_model-Serial_number

   For example: **8325-987654X1234567** or **8320-AB12CDE123**

   A switch supports one active Bluetooth connection at a time.
3. On some Android devices, you might need to change the settings of the paired device to specify that it be used for Internet access.

Solution 4

Cause

Bluetooth is not enabled on the switch.

New switches are shipped from the factory with the USB port and Bluetooth enabled. However, an installed switch might have been configured to disable Bluetooth or disable the USB port, which the USB Bluetooth adapter uses.

Action

Use a different CLI connection to enable Bluetooth on the switch.

- Use the **show bluetooth** CLI command to show the Bluetooth configuration and the status of the Bluetooth adapter.
- To enable the USB port, enter the CLI command: **usb**
- An inserted USB drive must be mounted each time the switch boots or fails over to a different management module. To mount the drive, enter the CLI command: **usb mount**
- To enable Bluetooth, enter the CLI command: **bluetooth enable**

Solution 5

Cause

Another mobile device has already connected to the switch through Bluetooth. This cause is likely if your device is repeatedly disconnected within 1-2 seconds of establishing a connection.

Action

1. Use a different CLI connection to see if there is another device connected:
   Use the **show bluetooth** CLI command to show the Bluetooth configuration and the status of the Bluetooth adapter.

2. Either disconnect the other device or use that device to communicate with the switch.
   A switch can use Bluetooth to connect to one mobile device at a time.

Solution 6

Cause

The switch has been restarted since the mobile device was last paired with the switch, and the device is having difficulty establishing the Bluetooth connection.

Action

1. Use the Bluetooth mobile device settings to forget the switch device.

2. Use your mobile device settings application to disable Bluetooth. Use your mobile device settings application to enable Bluetooth.
   If you are in range of multiple Bluetooth devices, more than one device is displayed on the list of available devices. Switches running the AOS-CX operating system are displayed in the following format:

   Switch_model-Serial_number

   For example: **8325-987654X1234567** or **8320-AB12CDE123**

   A switch supports one active Bluetooth connection at a time.

   On some Android devices, you might need to change the settings of the paired device to specify that it be used for Internet access.

Solution 7

Cause

The USB Bluetooth adapter is not installed in the switch.

If the switch has multiple management modules, the USB Bluetooth adapter might be installed in the management module that is not the active management module.

Action

Install the USB Bluetooth adapter in the USB port of the switch.

For switches that have multiple management modules, you must install the USB Bluetooth adapter in the USB port of the active management module. Typically, for new switches, the active management module is the module in slot 5 (HPE Aruba Networking 8400 switches) or slot 1 (HPE Aruba Networking 6400 switches).

For information about the location of the USB port on the switch, see the Installation Guide for the switch.

Solution 8

Cause

A problem occurred with the Bluetooth feature on the switch. For example, the software daemon was stopped and then restarted.

Action

1. Use a different connection to the switch CLI to disable and then enable Bluetooth.
```
switch(config)# bluetooth disable
switch(config)# bluetooth enable
```
2. Use the Bluetooth mobile device settings to forget the switch device.
3. Use your mobile device settings application to disable Bluetooth.
4. Use your mobile device settings application to enable Bluetooth.
5. Use your mobile device settings application to enable Bluetooth.
If you are in range of multiple Bluetooth devices, more than one device is displayed on the list of available devices. Switches running the AOS-CX operating system are displayed in the following format:

   Switch_model-Serial_number

   For example: **8325-987654X1234567** or **8320-AB12CDE123**

   A switch supports one active Bluetooth connection at a time.

   On some Android devices, you might need to change the settings of the paired device to specify that it be used for Internet access.

Solution 9

Cause

A switch that is member of a stack (but is not the conductor switch), has a USB Bluetooth adapter installed, but mobile application has lost contact with that switch.

Action

Remove and then reinstall the USB Bluetooth adapter.

Do not remove the USB Bluetooth adapter from the conductor switch.

Initial configuration using the CLI

This procedure describes how to connect to the switch for the first time and configure basic operational settings using the CLI. In this procedure, you use a computer to connect to the switch using the either the console port or management port.

Procedure

1. Connect to the console port or the management port.
2. Log into the switch for the first time.
3. Configure switch time using the NTP client.

Subtopics

Connecting to the console port

Prerequisites

- A switch installed as described in its hardware installation guide.
- A computer with terminal emulation software.
- A JL448A HPE Aruba Networking X2 C2 RJ45 to DB9 console cable.

Procedure

1. Connect the console port on the switch to the serial port on the computer using a console cable.
2. Start the terminal emulation software on the computer and configure a new serial session with the following settings:
    - Speed: 115200 bps
    - Data bits: 8
    - Stop bits: 1
    - Parity: None
    - Flow control: None
3. Start the terminal emulation session.
4. Press **Enter** once. If the connection is successful, you are prompted to log in.

Optional console port speed setting

If desired, the console port speed can be set with the **console baud-rate** command. For example, setting the console port speed to 9600 bps:

```
switch(config)# console baud-rate 9600
This command will configure the baud rate immediately for the active serial
console session. After the command is executed the user will be prompted to
re-login. The serial console will be inaccessible until the terminal client
settings are updated to match the baud rate of the switch.
Continue (y/n)? y
```

Showing the console port current speed:

```
switch# show console
Baud Rate: 9600
```

For details on the **console baud-rate** and **show console** commands, see Switch system and hardware commands.

Connecting to the management port

Prerequisites

- Two Ethernet cables
- SSH client software

Procedure

1. By default, the management interface is set to automatically obtain an IP address from a DHCP server, and SSH support is enabled. If there is no DHCP server on your network, you must configure a static address on the management interface:
   a. Connect to the console port
   b. Configure the management interface.
2. Use an Ethernet cable to connect the management port to your network.
3. Use an Ethernet cable to connect your computer to the same network.
4. Start your SSH client software and configure a new session using the address assigned to the management interface. (If the management interface is set to operate as a DHCP client, retrieve the IP address assigned to the management interface from your DHCP server.)
5. Start the session. If the connection is successful, you are prompted to log in.

Logging into the switch for the first time

The first time you log in to the switch you must use the default administrator account. This account has no password, so you will be prompted on login to define one to safeguard the switch.

Procedure

1. When prompted to log in, specify **admin**. When prompted for the password, press **ENTER**. (By default, no password is defined.)
   For example:

   ```
   switch login: admin
   password:
   ```

2. Define a password for the **admin** account. The password can contain up to 32 alphanumeric characters in the range ASCII 32 to 127, which includes special characters such as asterisk (*), ampersand (&), exclamation point (!), dash (-), underscore (_), and question mark (?).
   For example:

   ```
   Please configure the 'admin' user account password.
   Enter new password: *******
   Confirm new password: *******
   switch#
   ```

3. You are placed into the manager command context, which is identified by the prompt: `switch#`, where `switch` is the model number of the switch. Enter the command `config` to change to the global configuration context `config`.
   For example:

   ```
   switch# config
   switch(config)#
   ```

Setting switch time using the NTP client

Prerequisites

- The IP address or domain name of an NTP server.
- If the NTP server uses authentication, obtain the password required to communicate with the NTP server.

Procedure

1. If the NTP server requires authentication, define the authentication key for the NTP client with the command **ntp authentication**.
2. Configure an NTP server with the command **ntp server**. When configuring a time backward more than five minutes on the HPE Aruba Networking 8320, 8325 , or 9300 Switch Series, a reboot is recommended to avoid unusual switch behavior.
3. By default, NTP traffic is sent on the default VRF. If you want to send NTP traffic on the management VRF, use the command **ntp vrf**.
4. Review your NTP configuration settings with the commands **show ntp servers** and `show ntp status`.
5. See the current switch time, date, and time zone with the command **show clock**.

Example

This example creates the following configuration:

- Defines the authentication key **1** with the password **myPassword**.
- Defines the NTP server **my-ntp.mydomain.com** and makes it the preferred server.
- Sets the switch to use the management VRF (**mgmt**) for all NTP traffic.

```
switch(config)# ntp authentication-key 1 md5 myPassword
switch(config)# ntp server my-ntp.mydomain.com key 10 prefer
```

```
switch(config)# ntp vrf mgmt
```

Configuring banners

Procedure

1. Configure the banner that is displayed when a user connects to a management interface. Use the command `banner motd`. For example:

```
switch(config)# banner motd ^

Enter a new banner. Terminate the banner with the delimiter you have
chosen.
 >  >  This is an example of a banner text which a connecting user
 >  >  will see before they are prompted for their password.
 >  >
```

```
> > As you can see it may span multiple lines and the input
> > will be terminated when the delimiter character is
> > encountered.^
Banner updated successfully!
```

2. Configure the banner that is displayed after a user is authenticated. Use the command `banner exec`. For example:

```
switch(config)# banner exec &

Enter a new banner. Terminate the banner with the delimiter you have
chosen.
> > This is an example of a different banner text. This time
> > the banner entered will be displayed after a user has
> > authenticated.
> >
> > & This text will not be included because it comes after the '&'
Banner updated successfully!
```

Configuring in-band management on a data port

Prerequisites

- A connection to the CLI via either the console port or the management port
- Ethernet cable

Procedure

1. Use an Ethernet cable to connect a data port to your network.
2. Configure a layer 3 interface on the data port.
3. Enable SSH support on the interface (on the default VRF) with the command `ssh server vrf default`.
   For example:

   switch# **config** switch(config)# **ssh server vrf default**
4. Enable the Web UI on the interface (on the default VRF) with the command **https-server vrf default**.
   For example:

   switch(config)# **https-server vrf default**

Using the Web UI

The Web UI is disabled by default. Follow these steps to enable it on the management port and log in.

The Web UI is enabled by default on the default VRF.

Prerequisites

- A connection to the switch CLI.

Procedure

1. Log in to the CLI.
2. Switch to `config` context and enable the Web UI on the management port VRF with the command **https-server vrf mgmt** .
   For example:

   ```
   switch# config
   switch(config)# https-server vrf mgmt
   switch# config
   switch(config)# https-server vrf default
   ```

3. Start your web browser and enter the IP address of the management port in the address bar,
   For example: **https://192.168.1.1**
4. The Web UI starts and you are prompted to log in.

Configuring the management interface

Prerequisites

A connection to the console port.

Procedure

1. Switch to the management interface context with the command `interface mgmt` .
2. By default, the management interface on the management port is enabled. If it was disabled, re-enable it with the command `no shutdown`.

3. Use the command `ip dhcp` to configure the management interface to automatically obtain an address from a DHCP server on the network (factory default setting). Or, assign a static IPv4 or IPv6 address, default gateway, and DNS server with the commands `ip address`, `ipv6 address`, `ip static`, `default-gateway`, and `nameserver`.

4. SSH is enabled by default on the management VRF. If disabled, enable SSH with the command `ssh server vrf mgmt`.

Examples

This example enables the management interface with dynamic addressing using DHCP :

```
switch(config)# interface mgmt
switch(config-if-mgmt)# no shutdown
switch(config-if-mgmt)# ip dhcp
```

This example enables the management interface with static addressing creating the following configuration:

- Sets a static IPv4 address of **198.168.100.10** with a mask of **24** bits.
- Sets the default gateway to **198.168.100.200**.
- Sets the DNS server to **198.168.100.201**.

```
switch(config)# interface mgmt
switch(config-if-mgmt)# no shutdown
switch(config-if-mgmt)# ip static 198.168.100.10/24
switch(config-if-mgmt)# default-gateway 198.168.100.200
switch(config-if-mgmt)# nameserver 198.168.100.201
```

Restoring the switch to factory default settings

Prerequisites

You are connected to the switch through its Console port.

> **NOTE**
> This procedure erases all user information and configuration settings Consider backing up your running configuration first.

1. Optionally, back up the running configuration with either **copy running-config** <REMOTE-URL> or **copy running-config <STORAGE-URL>**. The **json** storage format is required for later configuration restoration.

2. Switch to the configuration context with the command **config**.

3. Erase all user information and configuration, restoring the switch to its factory default state with the command **erase all zeroize**. Enter **Y** when prompted to continue. The switch automatically restarts.

4. Optionally restore your saved configuration (it must be in **json** format) with either **copy** <REMOTE-URL> **running-config** or **copy <STORAGE-URL> running-config** followed by **copy running-config startup-config**.

Example

Backing up the running configuration to a file on a remote server (using TFTP), resetting the switch to its factory default state, and then restoring the saved configuration.

```
switch# copy running-config tftp://192.168.1.10/backup_cfg json vrf mgmt

  % Total    % Received % Xferd  Average Speed   Time    Time     Time
Current
                                 Dload  Upload   Total   Spent    Left
Speed
100 10340    0     0  100 10340      0  1329k --:--:-- --:--:-- --:--:--
1329k
100 10340    0     0  100 10340      0  1313k --:--:-- --:--:-- --:--:--
1313k
switch#
switch#
switch# erase all zeroize
This will securely erase all customer data and reset the switch
to factory defaults. This will initiate a reboot and render the
switch unavailable until the zeroization is complete.
This should take several minutes to one hour to complete.
Continue (y/n)? y

The system is going down for zeroization.
[  OK  ] Stopped PSPO Module Daemon.
[  OK  ] Stopped AOS-CX Switch Daemon for BCM.
...
[  OK  ] Stopped Remount Root and Kernel File Systems.
[  OK  ] Reached target Shutdown.
reboot: Restarting system
Press Esc for boot options
ServiceOS Information:
    Version:           GT.01.03.0006
    Build Date:        2018-10-30 14:20:44 PDT
    Build ID:          ServiceOS:GT.01.03.0006:8ee0faaa52da:201810301420
```

```
    SHA:              xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
...
################ Preparing for zeroization #################
################ Storage zeroization #######################
################ WARNING: DO NOT POWER OFF UNTIL  ##########
################          ZEROIZATION IS COMPLETE ##########
################ This should take several minutes ##########
################ to one hour to complete          ##########
################ Restoring files ###########################
Boot Profiles:
0. Service OS Console
1. Primary Software Image [XL.10.02.0010]
2. Secondary Software Image [XL.10.02.0010]
Select profile(primary):
Booting primary software image...
Verifying Image...
Image Info:
        Name: AOS-CX
     Version: XL.10.02.0010
    Build Id: AOS-CX:XL.10.02.0010:feaf5b9b7f09:201901292014
  Build Date: 2019-01-29 12:43:50 PST
Extracting Image...
Loading Image...
Done.
kexec_core: Starting new kernel
System is initializing
fips_post_check[5473]: FIPS_POST: Cryptographic selftest started...SUCCESS
[  OK  ] Started Login banner readiness check.
...
8400X login: admin
Password:
switch#
switch#
switch# copy tftp://192.168.1.10/backup_cfg running-config json vrf mgmt

  % Total    % Received % Xferd  Average Speed   Time    Time     Time
Current
                                 Dload  Upload   Total   Spent    Left
Speed
100 10340  100 10340    0     0  2858k      0 --:--:-- --:--:-- --:--:--
2858k
100 10340  100 10340    0     0  2804k      0 --:--:-- --:--:-- --:--:--
2804k
```

```
Large configuration changes will take time to process, please be patient.
switch#
switch#
switch# copy running-config startup-config
Large configuration changes will take time to process, please be patient.
switch#
```

Limitations and considerations

Profile switching between VSX and VSF

- VSF and VSX are mutually exclusive features. Before switching between profiles, ensure the device is in standalone mode.
- Configuration data is not compatible between VSF and VSX profiles. Back up your configurations before changing profiles (switching will erase startup configurations and reboot the device).
- Never apply the VSX profile on an active VSF setup. Applying VSX profile on an active VSF commander may cause undesired behavior and potentially require a factory reset.
- Checkpoints created in one profile mode should not be applied to a switch in a different profile mode. For example, do not apply VSX checkpoints on a switch configured for VSF profile, or vice versa.
- Configuration downloads should match the current profile of the switch. Do not download VSX configurations to a switch in VSF profile mode, or vice versa.

Software downgrade considerations

- When performing a software downgrade on a device configured with either VSX or VSF profiles, ensure that the startup configuration is erased prior to the downgrade process. This prevents configuration incompatibilities that could arise between different software versions.
- Failure to erase the startup configuration before downgrade may result in unexpected behavior, including potential boot failures or feature inconsistencies when the device initializes with the older software version which does not support VSX.

REST API profile switching procedure

About this task

Complete the following steps when changing profiles via REST API:

Procedure

1. Ensure the device is in standalone mode.
2. Execute **erase startup-config**.
3. Set the **configured_profile** in System table to the desired profile (VSX or Basic).
4. Save the configuration.
5. Reboot the switch to activate the desired profile.

Results

It is best practice to always save the running configuration after changing profiles and before rebooting. If the running configuration is not saved after a profile change, a subsequent reboot may return the device to the Basic (default) profile.

Management interface commands

Select a command from the list in the left navigation menu.

Subtopics

default-gateway
ip static
nameserver
show interface mgmt

`default-gateway`

Syntax

```
default-gateway <IP-ADDR>
no default-gateway <IP-ADDR>
```

Description

Assigns an IPv4 or IPv6 default gateway to the management interface. An IPv4 default gateway can only be configured if a static IPv4 address was assigned to the management interface. An IPv6 default gateway can only be configured if a static IPv6 address was assigned to the management interface. The default gateway should be on the same network segment.

The **no** form of this command removes the default gateway from the management interface.

Parameter	Description
<IP-ADDR>	Specifies an IP address in IPv4 format (**x.x.x.x**), where **x** is a de cimal number from 0 to 255, or IPv6 format (**xxxx:xxxx:xxxx:x xxx:xxxx:xxxx:xxxx:xxxx**), where **x** is a hexadecimal number f rom 0 to F.

Examples

Setting a default gateway with the IPv4 address of **198.168.5.1**:

```
switch(config)# interface mgmt
switch(config-if-mgmt)# default-gateway 198.168.5.1
```

Setting an IPv6 address of **2001:DB8::1**:

```
switch(config)# interface mgmt
switch(config-if-mgmt)# default-gateway 2001:DB8::1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	`config-if-mgmt`	Administrators or local user group members with execution righ ts for this command.

`ip static`

Syntax

```
ip static <IP-ADDR>/<MASK>
no ip static <IP-ADDR>/<MASK>
```

Description

Assigns an IPv4 or IPv6 address to the management interface.

The **no** form of this command removes the IP address from the management interface and sets the interface to operate as a DHCP client.

Parameter	Description
<IP-ADDR>	Specifies an IP address in IPv4 format (**x.x.x.x**), where **x** is a de cimal number from 0 to 255, or IPv6 format (**xxxx:xxxx:xxxx:x xxx:xxxx:xxxx:xxxx:xxxx**), where **x** is a hexadecimal number f rom 0 to F.
<MASK>	Specifies the number of bits in an IPv4 or IPv6 address mask in CIDR format (**x**), where **x** is a decimal number from 0 to 32 for I Pv4, and 0 to 128 for IPv6.

Examples

Setting an IPv4 address of **198.51.100.1** with a mask of **24** bits:

```
switch(config)# interface mgmt
switch(config-if-mgmt)# ip static 198.51.100.1/24
```

Setting an IPv6 address of **2001:DB8::1** with a mask of **32** bits:

```
switch(config)# interface mgmt
switch(config-if-mgmt)# ip static 2001:DB8::1/32
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	`config-if-mgmt`	Administrators or local user group members with execution righ ts for this command.

`nameserver`

Syntax

```
nameserver <PRIMARY-IP-ADDR> [ <SECONDARY-IP-ADDR> ]
no nameserver <PRIMARY-IP-ADDR> [ <SECONDARY-IP-ADDR> ]
```

Description

Assigns a primary or secondary IPv4 or IPv6 DNS server to the management interface. IPv4 DNS servers can only be configured if a static IPv4 address was assigned to the management interface. IPv6 DNS servers can only be configured if a static IPv6 address was assigned to the management interface. The default gateway should be on the same network segment.

The **no** form of this command removes the DNS servers from the management interface.

Parameter	Description
<PRIMARY-IP-ADDR>	Specifies the IP address of the primary DNS server. Specify the address in IPv4 format (**x.x.x.x**), where **x** is a decimal number f rom 0 to 255, or IPv6 format (**xxxx:xxxx:xxxx:xxxx:xxxx:xxx x:xxxx:xxxx**), where **x** is a hexadecimal number from 0 to F.
<SECONDARY-IP-ADDR>	Specifies the IP address of the secondary DNS server. Specify th e address in IPv4 format (**x.x.x.x**), where **x** is a decimal number from 0 to 255, or IPv6 format (**xxxx:xxxx:xxxx:xxxx:xxxx:xxx x:xxxx:xxxx**), where **x** is a hexadecimal number from 0 to F.

Examples

Setting primary and secondary DNS servers with the IPv4 addresses of **198.168.5.1** and **198.168.5.2** :

```
switch(config)# interface mgmt
switch(config-if-mgmt)# nameserver 198.168.5.1 198.168.5.2
```

Setting primary and secondary DNS servers with the IPv6 addresses of **2001:DB8::1** and **2001:DB8::2**:

```
switch(config)# interface mgmt
switch(config-if-mgmt)# nameserver 2001:DB8::1 2001:DB8::2
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	`config-if-mgmt`	Administrators or local user group members with execution righ ts for this command.

`show interface mgmt`

Syntax

```
show interface mgmt [vsx-peer]
```

Description

Shows status and configuration information for the management interface.

Parameter	Description
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output f rom the VSX peer switch is not displayed. This parameter is avai lable on switches that support VSX.

Example

```
switch# show interface mgmt
  Address Mode                    : static
  Admin State                     : up
```

```
Mac Address                      : 02:42:ac:11:00:02
IPv4 address/subnet-mask         : 192.168.1.10/16
Default gateway IPv4             : 192.168.1.1
IPv6 address/prefix              : 2001:db8:0:1::129/64
IPv6 link local address/prefix: fe80::7272:cfff:fefd:e485/64
Default gateway IPv6             : 2001:db8:0:1::1
Primary Nameserver               : 2001::1
Secondary Nameserver             : 2001::2
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

NTP commands

Select a command from the list in the left navigation menu.

Subtopics

- **ntp authentication**
- **ntp authentication-key**
- **ntp disable**
- **ntp enable**
- **ntp conductor**
- **ntp server**
- **ntp trusted-key**
- **ntp vrf**
- **show ntp associations**
- **show ntp authentication-keys**
- **show ntp servers**

[show ntp statistics]
[show ntp status]

`ntp authentication`

Syntax

```
ntp authentication
no ntp authentication
```

Description

Enables support for authentication when communicating with an NTP server.

The **no** form of this command disables authentication support.

Examples

Enabling authentication support:

```
switch(config)# ntp authentication
```

Disabling authentication support:

```
switch(config)# no ntp authentication
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`ntp authentication-key`

Syntax

```
ntp authentication-key <KEY-ID> {md5 | sha1}
    [{ <PLAINTXT-KEY> [trusted] | ciphertext <ENCRYPTED-KEY> }]
no ntp authentication-key <KEY-ID>  {md5 | sha1}
    [{ <PLAINTXT-KEY> [trusted] | ciphertext <ENCRYPTED-KEY> }]
```

Description

Defines an authentication key that is used to secure the exchange with an NTP time server. This command provides protection against accidentally synchronizing to a time source that is not trusted.

The **no** form of this command removes the authentication key.

> **NOTE**
> Using "#" in the NTP password will prevent the NTP to syncronize when authentication is enabled

Parameter	Description
`<KEY-ID>`	Specifies the authentication key ID. Range: 1 to 65534.
`md5`	Selects MD5 key encryption.
`sha1`	Specifies SHA1 key encryption.
`<PLAINTXT-KEY>`	Specifies the plaintext authentication key. Range: 8 to 40 chara cters. The key may contain printable ASCII characters excluding "#" or be entered in hex. Keys longer than 20 characters are ass umed to be hex. To use an ASCII key longer than 20 characters, convert it to hex.
`trusted`	Specifies that this is a trusted key. When NTP authentication is enabled, the switch only synchronizes with time servers that tra nsmit packets containing a trusted key.
`ciphertext <ENCRYPTED-KEY>`	Specifies the ciphertext authentication key in Base64 format. T his is used to restore the NTP authentication key when copying configuration files between switches or when uploading a previ ously saved configuration. **NOTE**

Parameter	Description
	When the key is not provided on the command line, plaintext key prompting occurs upon pressing Enter, followed by prompting as to whether the key is to be trusted. The entered key characters are masked with asterisks.

Examples

Defining key 10 with MD5 encryption and a provided plaintext trusted key:

```
switch(config)# ntp authentication-key 10 md5 F8245a0b trusted
```

Defining key 5 with SHA1 encryption and a prompted plaintext trusted key:

```
switch(config)# ntp authentication-key 5 sha1
Enter the NTP authentication key: *********
Re-Enter the NTP authentication key: *********
Configure the key as trusted (y/n)? y
```

Removing key 10:

```
switch(config)# no ntp authentication-key 10
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

`ntp disable`

Syntax

```
ntp disable
```

Description

Disables the NTP client on the switch. The NTP client is disabled by default.

Examples

Disabling the NTP client.

```
switch(config)# ntp disable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`ntp enable`

Syntax

```
ntp enable
no ntp enable
```

Description

Enables the NTP client on the switch to automatically adjust the local time and date on the switch. The NTP client is disabled by default.

The **no** form of this command disables the NTP client.

Examples

Enabling the NTP client.

```
switch(config)# ntp enable
```

Disabling the NTP client.

```
switch(config)# no ntp enable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`ntp conductor`

Syntax

```
ntp conductor vrf <VRF-NAME> [stratum <NUMBER>]
no ntp conductor vrf <VRF-NAME> [stratum <NUMBER>]
```

Description

Sets the switch as the conductor time source for NTP clients on the specified VRF. By default, the switch operates at stratum level 8. The switch cannot function as both NTP conductor and client on the same VRF.

The **no** form of this command stops the switch from operating as the conductor time source on the specified VRF.

Parameter	Description
`vrf <VRF-NAME>`	Specifies the VRF on which to act as conductor time source.
`stratum <NUMBER>`	Specifies the stratum level at which the switch operates. Range: 1 - 15. Default: 8.

Examples

Setting the switch to act as conductor time source on VRF **primary-vrf** with a stratum level of **9**.

```
switch(config)# ntp conductor vrf primary-vry statum 9
```

Stops the switch from acting as conductor time source on VRF **primary-vrf**.

```
switch(config)# no ntp conductor vrf primary-vry
```

Command History

Release	Modification
10.08	Inclusive language.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	`config`	Administrators or local user group members with execution righ ts for this command.

`ntp server`

Syntax

```
ntp server <IP-ADDR> [key <KEY-NUM>] [minpoll <MIN-NUM>] [maxpoll <MAX-NUM>]
[burst | iburst][prefer] [version <VER-NUM>]
no ntp server <IP-ADDR>
            <IP-ADDR> [key <KEY-NUM>] [minpoll <MIN-NUM>] [maxpoll <MAX-
NUM>][burst | iburst] [prefer] [version <VER-NUM>]
```

Description

Defines an NTP server to use for time synchronization, or updates the settings of an existing server with new values. Up to eight servers can be defined.

The **no** form of this command removes a configured NTP server.

> **NOTE**
> The default NTP version is 4; it is backwards compatible with version 3.

Parameter	Description
server *<IP-ADDR>*	Specifies the address of an NTP server as a DNS name, an IPv4 address (**x.x.x.x**), where **x** is a decimal number from 0 to 255, or an IPv6 address (**xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx**), where **x** is a hexadecimal number from 0 to F. When specifying an IPv4 address, you can remove leading zeros. For example, the address **192.169.005.100** becomes **192.168.5.100**. When specifying an IPv6 address, you can use two colons (::) to represent consecutive zeros (but only once), remove leading zeros, and collapse a hextet of four zeros to a single 0. For example, this address **2222:0000:3333:0000:0000:0000:4444:0055** becomes **2222:0:3333::4444:55** .
key *<KEY-NUM>*	Specifies the key to use when communicating with the server. A trusted key must be defined with the command **ntp authentication - key** and authentication must be enabled with the command **ntp authentication**. Range: 1 to 65534.
minpoll *<MIN-NUM>*	Specifies the minimum polling interval in seconds, as a power of 2. Range: 4 to 17. Default: 6 (64 seconds).
maxpoll *<MAX-NUM>*	Specifies the maximum polling interval in seconds, as a power of 2. Range: 4 to 17. Default: 10 (1024 seconds).
burst	Send a burst of packets instead of just one when connected to the server. Useful for reducing phase noise when the polling interval is long.
iburst	Send a burst of six packets when not connected to the server. Useful for reducing synchronization time at startup.
prefer	Make this the preferred server.
version *<VER-NUM>*	Specifies the version number to use for all outgoing NTP packets. Range: 3 or 4. Default: 4. **NOTE**

Parameter	Description
	NTP is backwards compatible.

Usage

For features such as Activate and ZTP, a switch that has a factory default configuration will automatically be configured with **pool.ntp.org**. NTP server configurations via DHCP options are supported. The DHCP server can be configured with maximum of two NTP server addresses which will be supported on the switch. Only IPV4 addresses are supported.

> **NOTE**
>
> When configuring a time backward more than five minutes , a reboot is recommended to avoid unusual switch behavior.

NTP uses a stratum to describe the distance between a network device and an authoritative time source:

- A stratum 1 time server is directly attached to an authoritative time source (such as a radio or atomic clock or a GPS time source).
- A stratum 2 NTP server receives its time through NTP from a stratum 1 time server.

When using multiple servers with same stratum setting, the best practice to configure a preferred server, so NTP will attempt to use the preferred server as the primary NTP connection. If a preferred server is not manually set when NTP is enabled, the configured server with the lowest stratum will automatically be set as the preferred server. If there are servers with the same stratum, this auto prefer status will prevent AOS-CX from toggling between different servers as the primary server. Auto prefer selection of servers with same stratum (if not manually selected) may change after reconfiguring the switch, or after executing the **reboot** command.

Examples

Defining the ntp server pool.ntp.org, using iburst, and NTP version 4.

```
switch(config)# ntp server pool.ntp.org iburst version 4
```

Removing the ntp server pool.ntp.org.

```
switch(config)# no ntp server pool.ntp.org
```

Defining the ntp server my-ntp.mydomain.com and makes it the preferred server.

```
switch(config)# ntp server my-ntp.mydomain.com prefer
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`ntp trusted-key`

Syntax

```
ntp trusted-key <KEY-ID>
no ntp trusted-key <KEY-ID>
```

Description

Sets a key as trusted. When NTP authentication is enabled, the switch only synchronizes with time servers that transmit packets containing a trusted key.

The **no** form of this command removes the trusted designation from a key.

Parameter	Description
<KEY-ID>	Specifies the identification number of the key to set as trusted. Range: 1 to 65534.

Examples

Defining key 10 as a trusted key.

```
switch(config)# ntp trusted-key 10
```

Removing trusted designation from key 10:

```
switch(config)# no ntp trusted-key 10
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`ntp vrf`

Syntax

```
ntp vrf <VRF-NAME>
no ntp vrf <VRF-NAME>
```

Description

Specifies the VRF on which the NTP client communicates with an NTP server. The switch cannot function as both NTP conductor and client on the same VRF.

The **no** form of the command returns to default VRF.

Parameter	Description
<VRF-NAME>	Specifies the name of a VRF.

Example

Setting the switch to use the default VRF for NTP client traffic.

```
switch(config)# ntp vrf default
```

Setting the switch to use the default management VRF for NTP client traffic.

```
switch(config)# ntp vrf mgmt
```

Returning the switch to use the default VRF for NTP client traffic.

```
switch(config)# no ntp vrf
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`show ntp associations`

Syntax

```
show ntp associations [vsx-peer]
```

Description

Shows the status of the connection to each NTP server. The following information is displayed for each server:

- Tally code : The first character is the Tally code:
  - (blank): No state information available (e.g. non-responding server)
  - x : Out of tolerance (discarded by intersection algorithm)
  - . : Discarded by table overflow (not used)

- - : Out of tolerance (discarded by the cluster algorithm)
- + : Good and a preferred remote peer or server (included by the combine algorithm)
- # : Good remote peer or server, but not utilized (ready as a backup source)
- * : Remote peer or server presently used as a primary reference
- o : PPS peer (when the prefer peer is valid)

- ID: Server number.
- NAME: NTP server FQDN/IP address (Only the first 24 characters of the name are displayed).
- REMOTE: Remote server IP address.
- REF_ID: Reference ID for the remote server (Can be an IP address).
- ST: (Stratum) Number of hops between the NTP client and the reference clock.
- LAST: Time since the last packet was received in seconds unless another unit is indicated.
- POLL: Interval (in seconds) between NTP poll packets. Maximum (1024) reached as server and client sync.
- REACH: 8-bit octal number that displays status of the last eight NTP messages (377 = all messages received).

Parameter	Description
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output f rom the VSX peer switch is not displayed. This parameter is avai lable on switches that support VSX.

Example

```
switch# show ntp associations
----------------------------------------------------------------------
 ID             NAME           REMOTE          REF-ID ST LAST POLL REACH
----------------------------------------------------------------------
  1       192.0.1.1       192.0.1.1          .INIT. 16    -   64     0
* 2   time.apple.com    17.253.2.253         .GPSs.  2   70  128   377
----------------------------------------------------------------------
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show ntp authentication-keys`

Syntax

```
show ntp authentication-keys [vsx-peer]
```

Description

Shows the currently defined authentication keys.

Parameter	Description
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output f rom the VSX peer switch is not displayed. This parameter is avai lable on switches that support VSX.

Examples

```
switch# show ntp authentication-keys
--------------------------------
Auth key  Trusted   MD5 password
--------------------------------
 10           No     **********
 20          Yes     **********
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution righ ts for this command.

show ntp servers

Syntax

```
show ntp servers[vsx-peer]
```

Description

Shows all configured NTP servers, including any DHCP servers, default pool servers or any server with the status **auto prefer**.

Parameter	Description
vsx-peer	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output f rom the VSX peer switch is not displayed. This parameter is avai lable on switches that support VSX.

Example

```
switch# show ntp servers
------------------------------------------------
     NTP SERVER KEYID MINPOLL MAXPOLL OPTION VER
------------------------------------------------
     192.0.1.18     -       5       10 iburst   3
     192.0.1.19     -       6       10   none   4
```

```
     192.0.1.20      -        6        8  burst    3 prefer
------------------------------------------------
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show ntp statistics`

Syntax

```
show ntp statistics [vsx-peer]
```

Description

Shows global NTP statistics. The following information is displayed:

- Rx-pkts: Total NTP packets received.
- Current Version Rx-pkts: Number of NTP packets that match the current NTP version.
- Old Version Rx-pkts: Number of NTP packets that match the previous NTP version.
- Error pkts: Packets dropped due to all other error reasons.
- Auth-failed pkts: Packets dropped due to authentication failure.
- Declined pkts: Packets denied access for any reason.
- Restricted pkts: Packets dropped due to NTP access control.
- Rate-limited pkts: Number of packets discarded due to rate limitation.

- KOD pkts: Number of Kiss of Death packets sent.

Parameter	Description
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output f rom the VSX peer switch is not displayed. This parameter is avai lable on switches that support VSX.

Examples

```
switch(config)# show ntp statistics
                Rx-pkts 100
Current Version Rx-pkts 80
    Old Version Rx-pkts 20
               Err-pkts 2
       Auth-failed-pkts 1
          Declined-pkts 0
        Restricted-pkts 0
      Rate-limited-pkts 0
               KoD-pkts 0
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show ntp status`

Syntax

```
show ntp status [vsx-peer]
```

Description

Shows the status of NTP on the switch.

Parameter	Description
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output f rom the VSX peer switch is not displayed. This parameter is avai lable on switches that support VSX.

Examples

Displaying the status information when the switch is not synced to an NTP server:

```
switch# show ntp status
NTP is enabled.
NTP authentication is enabled.
NTP is using the default VRF for NTP server connections.
Wed Nov 23 23:29:10 PDT 2016
NTP uptime: 187 days, 1 hours, 37 minutes, 48 seconds
Not synchronized with an NTP server.
```

Displaying the status information when the switch is synced to an NTP server:

```
switch# show ntp status
NTP is enabled.
NTP authentication is enabled.
NTP is using the default VRF for NTP server connections.
Wed Nov 23 23:29:10 PDT 2016
NTP uptime: 187 days, 1 hours, 37 minutes, 48 seconds
Synchronized to NTP Server 17.253.2.253 at stratum 2.
Poll interval = 1024 seconds.
Time accuracy is within 0.994 seconds
Reference time: Thu Jan 28 2016 0:57:06.647 (UTC)
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

Telnet access

Telnet server enables switches to accept Telnet connections from clients to manage the switch. The user authentication is password based authentication (RADIUS, TACACS+ or locally stored password). The Telnet server can be enabled on a desired VRF using the `telnet server` command. The maximum number of Telnet sessions per VRF is five.

In the default configuration, Telnet access is disabled.

Subtopics

Telnet commands

Telnet commands

Select a command from the list in the left navigation menu.

Subtopics

show telnet server sessions
show telnet server
telnet server

`show telnet server sessions`

Syntax

```
show telnet server sessions [vrf <VRF-NAME> | all-vrfs]
```

Description

Shows all active Telnet sessions for the specified VRF or all VRFs. If no VRF is provided, the Telnet sessions on the **default** VRF are shown.

Parameter	Description
`vrf <VRF-NAME>`	Specifies the Telnet sessions for a specific VRF.
`all-vrfs`	Specifies the Telnet sessions for all VRFs

Examples

Showing the Telnet session on the **default** VRF:

```
switch(config)# show telnet server sessions
TELNET sessions on VRF default:
    IPv4 TELNET Sessions:
        Server IP         : 10.1.1.1
        Client IP         : 10.1.1.2
        Client Port       : 58835
```

The following example shows the Telnet sessions on all VRFs. If only the **default** VRF is supported by the switch, only the **default** VRF will display in the output of this command.

```
switch(config)# show telnet server sessions all-vrfs

TELNET sessions on VRF mgmt:
    IPv4 TELNET Sessions:
        Server IP         : 10.1.1.1
        Client IP         : 10.1.1.2
        Client Port       : 58835
TELNET sessions on VRF default:
    IPv4 TELNET Sessions:
        Server IP         : 20.1.1.1
        Client IP         : 20.1.1.2
        Client Port       : 58837
```

Command History

Release	Modification
10.11	Command introduced on all other platforms.

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`show telnet server`

Syntax

```
show telnet server
```

Description

Shows the Telnet server configuration.

Examples

Showing the Telnet server configuration:

```
switch(config)# show telnet server
TELNET Server Configuration:
    IP Version          : IPv4
    TCP Port            : 23
    Enabled VRFs        : default, vrf1, vrf2,
                          red, green
```

Command History

Release	Modification
10.11	Command introduced on all other platforms.

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`telnet server`

Syntax

```
telnet server vrf <VRF-NAME>
no telnet server vrf <VRF-NAME>
```

Description

Enables the Telnet server on the desired VRF. Telnet server is disabled by default.

The **no** form of this command disables the Telnet server.

Parameter	Description
`vrf <VRF-NAME>`	Specifies the VRF on which the Telnet server will be enabled or disabled.

Examples

Configuring the Telnet server on the **default** VRF:

```
switch(config)# telnet server vrf default
```

Configuring the Telnet server on the **mgmt** VRF:

```
switch(config)# telnet server vrf mgmt
```

Command History

Release	Modification
10.11	Command introduced on 9300, 1000, 4100i and 8xxx platforms.

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

Interface configuration

Subtopics

- Configuring a layer 2 interface
- Configuring a layer 3 interface
- Single source IP address
- Priority-based flow control (PFC)
- Flow control and lossless buffering
- Forward error correction
- Unsupported transceiver support
- Configuring an interface persona
- Monitor mode
- Interface commands

Configuring a layer 2 interface

Procedure

1. Change to the interface configuration context for the interface with the command `interface`.
2. By default, interfaces are layer 3. To create a layer 2 interface, disable routing with the command `no routing`.
3. Set the interface MTU (maximum transmission unit) with the command `mtu`.
4. Review interface configuration settings with the command `show interface`.

Configuring a layer 3 interface

Procedure

1. Change to the interface configuration context for the interface with the command **interface**.
2. Interfaces are layer 3 by default. If you previously set the interface to layer 2, then enable routing support with the command **routing**.
3. Assign an IPv4 address with the command **ip address**, or an IPv6 address with the command **ipv6 address**.

4. If required, enable support for layer 3 counters with the command **l3-counters**.

5. If required, set the IP MTU with the command **ip mtu**.

6. Review interface configuration settings with the command **show interface**.

Examples

This example creates the following configuration:

- Configures interface **1/1/1** as a layer 3 interface.
- Defines an IPv4 address of **10.10.20.209** with a 24-bit mask.

```
switch# config
switch(config)# interface 1/1/1
switch(config-if)# ip address 10.10.20.209/24
```

This example creates the following configuration:

- Configures interface **1/1/2** as a layer 3 interface.
- Defines an IPv6 address of **2001:0db8:85a3::8a2e:0370:7334** with a 24-bit mask.
- Enables layer 3 transmit and receive counters.

```
switch# config
switch(config)# interface 1/1/2
switch(config-if)# ipv6 address 2001:0db8:85a3::8a2e:0370:7334/24
switch(config-if)# l3-counters tx
switch(config-if)# l3-counters rx
```

- Configures interface **1/1/1** as a layer 3 interface.
- Defines an IPv4 address of **10.10.20.209** with a 24-bit mask.

```
switch# config
switch(config)# interface 1/1/1
switch(config-if)# routing
switch(config-if)# ip address 10.10.20.209/24
```

- Configures interface **1/3/1** as a layer 3 interface.
- Defines an IPv6 address of **2001:0db8:85a3::8a2e:0370:7334** with a 24-bit mask.
- Enables layer 3 transmit and receive counters.

```
switch# config
switch(config)# interface 1/3/1
switch(config-if)# routing
switch(config-if)# ipv6 address 2001:0db8:85a3::8a2e:0370:7334/24
switch(config-if)# l3-counters tx
switch(config-if)# l3-counters rx
```

Single source IP address

Certain IP-based protocols used by the switch (such as RADIUS, sFlow, TACACS, and TFTP), use a client-server model in which the client's source IP address uniquely identifies the client in packets sent to the server. By default, the source IP address is defined as the IP address of the outgoing switch interface on which the client is communicating with the server. Since the switch can have multiple routing interfaces, outgoing packets can potentially be sent on different paths at different times. This can result in different source IP addresses being used for a client, which can create a client identification problem on the server. For example, it can be difficult to interpret system logs and accounting data on the server when the same client is associated with multiple IP addresses.

To resolve this issue, you can use the commands `ip source-interface` and `ipv6 source-interface` to define a single source IP address that applies to all supported protocols (RADIUS, sFlow, TACACS, and TFTP), or an individual address for each protocol. This ensures that all traffic sent by a client to a server uses the same IP address.

Priority-based flow control (PFC)

About this task

Priority-based flow control (PFC) is defined in the IEEE 802.1Qbb standard. It is used to eliminate packet loss due to congestion on a network link.

For priority-based flow control, care must be taken when configuring quality of service.

Procedure

1. Configure the global CoS map (`qos cos-map` command) and queue profile (`qos queue-profile` command) before configuring PFC on any interface.
    - Each code point in the CoS Map to be used for PFC must be assigned a unique local-priority value.
    - Each local-priority used for PFC must be the sole local-priority mapped to a queue in the queue profile.

2. CoS map or queue profile changes for non-PFC priorities can be modified without any impact on PFC functionality.
3. All interfaces with PFC enabled must also be configured with `qos trust cos` or `qos trust dscp.`
4. For PFC to function properly with VSX and MCLAG, PFC must be configured on the inter-switch-link.

Results

See also commands flow-control and show interface flow-control.

Refer to the Quality of Service Guide for more information about buffering and queuing for flow-controlled traffic.

Flow control and lossless buffering

Requirements for proper lossless buffering

The switch uses a virtual output queue (VOQ) architecture where most packet buffering for all destination ports occurs on the ingress line module. Each line module has between 1.5 and 4 GB of buffer memory with 256 MB used for traffic destined for multiple ports (Multidest) and the remainder for traffic destined for one port (Unidest). By default, all unicast packets arriving on ports of the same line module share the Unidest pool. Any packet can be dropped if the destination port VOQ has exceeded its limit. The VOQ limit is either 1 MB for JL365 modules, or 1.7 MB for JL366 and JL687 modules. The TX Drops columns of show interface <IF-NAME> command will display the number of drops that have occurred.

Priority-based Flow Control (PFC) reconfigures buffering. After the first interface is configured for PFC the switch will move some Unidest buffers to a new Headroom pool. Unicast traffic arriving at PFC-configured ports with the specified PCP will still use the Unidest pool with other traffic but will be marked as lossless. Lossless-marked packets will never be dropped when a destination port VOQ has exceeded its limit. All other unicast traffic arriving on non-PFC ports, or on PFC-configured ports with different PCPs also shares the Unidest pool. However, such traffic is still subject to dropping when a destination port VOQ has exceeded its limit.

Each PFC-configured-port priority is given a limit on the amount of lossless buffering it can store in the Unidest pool. The limit is either 1 MB for JL365A modules, or 1.7 Mbytes for JL366A and JL687A modules. Once the limit is exceeded, the switch will send a PFC pause frame from the port to stop further packets from arriving. Due to link propagation time and processing time on the receiving switch, additional packets in flight are allowed to be received and will be stored in the Headroom pool. The Headroom pool size is sufficient to store all packets in flight arriving after the priority pause frame is sent. The switch will periodically transmit priority pause frames until all the port’s lossless packets are gone from the Headroom pool and its lossless packets in the Unidest pool go below half of the pool's capacity.

When the switch receives a PFC pause frame for the configured PCP, the port will cease transmitting further packets from that queue until the timeout specified in the Quanta field has expired or a resume frame is received.

Forward error correction

Forward error correction (FEC) is used to control errors in transmissions where the source sends redundant data and the destination only recognizes the data portion that contains no apparent errors. FEC does not require a handshake between the source and destination at the cost of a higher forward channel bandwidth. It is therefore best used in scenarios where re-transmissions are costly or impossible, such as using multicast one-way communication.

Unsupported transceiver support

Transceiver products (optical, DAC, AOCs) that are listed as supported by a switch model are detailed in the Transceiver Guide. Transceiver products that are not listed, are considered unsupported; this would include transceivers that are:

- Non-HPE Aruba Networking branded products
- HPE branded products that were designed for non-AOS-CX switch models (e.g. Comware)
- HPE branded products designated for use in HPE Compute Servers or Storage
- Transceivers originally designated for use in HPE Aruba Networking WLAN controllers or former Mobility Access Switch (MAS) products
- End-of-life HPE Aruba Networking Transceivers

The unsupported transceiver mode (UT-mode) is designed to allow the possible use of these unsupported products. Not all unsupported products can be recognized and enabled; they may be unable to be identified (do not follow the proper MSA standards for identification). These unsupported transceiver products are enabled only on a best-effort basis and there are no guarantees implied for their continued operation.

This feature is enabled by default. A periodic system log will be generated by default at an interval of 24 hours listing the ports on which unsupported transceivers are present. The log interval is configurable and can be disabled by setting the log-interval to **none**.

Configuring an interface persona

A persona is a template interface. It is a mechanism intended to ease the process of configuring one or multiple interfaces that behave in the same manner. For example, a persona could be an uplink interface with a well-known collection of VLANs to which it belongs. Instead of manually configuring each interface one-by-one, the persona collects the common configuration settings and allows them to be applied on the interface or a collection of interfaces.

Persona configuration is similar to configuring other interfaces. Most of the commands in the interface context are available. Because of the nature of the commands, several of them do not apply to the persona (for example, applying the same IP address configuration to several ports) and have been removed from the available list.

Configuration AAA and port-access commands is not supported within the interface context when configuring a persona:

```
switch(config)# interface persona access
switch(config-if)#port-access <command>
switch(config-if)#aaa <command>
```

In addition, the following commands are not supported:

```
aaa authentication port-access dot1x authenticator macsec
aaa authentication port-access dot1x supplicant
aaa authentication port-access dot1x authenticator mka cak-length
arp ipv4 <IPV4_ADDR> mac <MAC_ADDR>
downshift-enable
energy-efficient-ethernet
error-control { auto | none | base-r-fec | rs-fec }
ip bootp-gateway <IPV4-ADDR>
ip forward-protocol udp <IPV4-ADDR> {<PORT-NUM> | <PROTOCOL>}
ip helper-address <IPV4-ADDR> [vrf <VRF-NAME>]
ip igmp router-alert-check [enable | disable]
ip igmp snooping [auto vlan <VLAN-LIST>]
ip igmp snooping [blocked vlan <VLAN-LIST>]
ip igmp snooping [fastleave vlan <VLAN-LIST>]
ip igmp snooping [forced-fastleave vlan <VLAN-LIST>]
ip igmp snooping [forward vlan <VLAN-LIST>]
ip ospf <PROCESS-ID> area <AREA-ID>
ip ospf passive
ip rip <PROCESS-ID> {all-ip | ip-address}
ip rip all-ip disable
ip rip all-ip enable
ip rip all-ip receive disable
ip rip all-ip send disable
ipv6 helper-address multicast {all-dhcp-servers | <MULTICAST-IPV6-ADDR>}
egress <PORT-NUM>
ipv6 mld snooping [auto vlan <VLAN-LIST>]
ipv6 mld snooping [blocked vlan <VLAN-LIST>]
ipv6 mld snooping [fastleave vlan <VLAN-LIST>]
ipv6 mld snooping [forced-fastleave vlan <VLAN-LIST>]
ipv6 mld snooping [forward vlan <VLAN-LIST>]
ipv6 neighbor <IPV6-ADDR> mac <MAC-ADDR>
ipv6 ospfv3 <PROCESS-ID> area <AREA-ID>
ipv6 ospfv3 passive
ipv6 ripng <PROCESS-ID>
```

```
   lacp port-id <PORT-ID>
   lacp port-priority <PORT-PRIORITY>
   lag <ID>
   link-poe
   lldp dot3 poe
   lldp med poe
   lldp med poe [priority-override]
   persona {access | uplink | custom <PERSONA-NAME>} [copy | attach]
   port-access port-security sticky-learn mac-address
   power-over-ethernet
   power-over-ethernet allocate-by {usage | class}
   power-over-ethernet assigned-class {3 | 4 | 6 | 8}
   power-over-ethernet pd-class-override
   power-over-ethernet power-pairs {alt-a|alt-a-and-alt-b}
   power-over-ethernet pre-std-detect
   power-over-ethernet priority {critical|high|low}
   ptp lag-role {primary | secondary}
   spanning-tree cost <PORT-COST>
   spanning-tree instance <INSTANCE-ID> cost <PORT-COST>
   spanning-tree instance <INSTANCE-ID> port-priority <PRIORITY-MULTIPLIER>
   spanning-tree port-priority <PRIORITY-MULTIPLIER>
   spanning-tree vlan <A:1-4094> cost <0-200000000>
   spanning-tree vlan <A:1-4094> port-priority <0-15>
   split [confirm]
   track by <OBJECT-ID>
   vsx-sync {access-lists | qos | rate-limits | vlans | policies | irdp}
```

Modes

There are three supported modes:

1. Copy—The **copy** mode is a one-step configuration that copies the persona configuration to an interface. Further changes to the persona will not be applied to the interfaces using that mode.

2. Attach—Unlike the **copy** mode, besides applying the configuration to the interface immediately, the **attach** mode also follows the persona configuration. It means that the subsequent changes to the persona will be applied to the interfaces attached to it.

3. Label—The **label** mode follows the interface persona configuration. The two exceptions are **copy** and **attach** modes. The **label** mode classifies the purpose of an interface and maintains functionality without requiring a persona to exist.

Predefined and custom persona names

There are two predefined interface persona names:

- uplink

- access

These names have no predefined configuration. To use them, they must be manually configured as needed. You can also create personas with a custom name. These custom personas can be created and configured in the same manner as the predefined ones. The only difference is the command used to apply them to the interface. The procedure below provides the details.

Creating and configuring an interface persona

About this task

Follow these steps to create and configure an interface persona:

Procedure

1. Create the interface **persona** with the command **interface persona** **<** PERSONA-NAME **>**.
2. Set the interface configuration as any other physical interface.
3. Review the configuration with the command **show running-configuration current-context**.
4. Switch to an interface context with the command **interface** **<** PORT **>**.
5. Apply the persona configuration to the interface and set the mode with the command **persona custom <PERSONA-NAME> <mode>**. Note that **<custom>** is listed among the three arguments required for configuration.

Results

For information on this feature, see the related video on the **AirHeads Broadcasting Channel**.

> **NOTE**
>
> All interface configurations related to Device-Profile are supported persona interface.

Examples

To copy a predefined persona name configuration to an interface:

1. Configure the interface persona:

```
switch(config)# interface persona uplink
switch(config-if)# no shutdown
switch(config-if)# no routing
switch(config-if)# vlan access 100
switch(config-if)# exit
```

2. Apply the configuration with **copy** mode:

```
switch(config)# interface 1/1/1
switch(config-if)# persona uplink copy
switch(config-if)# exit
```

To attach a custom persona name to several interfaces simultaneously:

1. Configure the interface persona:

```
switch(config)# interface persona mytemplate
switch(config-if)# no shutdown
switch(config-if)# vrf attach upstream
switch(config-if)# exit
```

2. Apply the configuration with **attach** mode:

```
switch(config)# interface 1/1/1-1/1/24
switch(config-if)# persona custom mytemplate attach
switch(config-if)# exit
```

To label a custom persona name to an interface:

1. Apply the configuration with **label** mode:

```
switch(config)# interface 1/1/1
switch(config-if)# persona custom mypersona
switch(config-if)# exit
link-poe
lldp dot3 poe
lldp med poe
lldp med poe [priority-override]
persona {access | uplink | custom <PERSONA-NAME>} [copy | attach]
port-access port-security sticky-learn mac-address
power-over-ethernet
power-over-ethernet allocate-by {usage | class}
power-over-ethernet assigned-class {3 | 4 | 6 | 8}
power-over-ethernet pd-class-override
power-over-ethernet power-pairs {alt-a|alt-a-and-alt-b}
power-over-ethernet pre-std-detect
power-over-ethernet priority {critical|high|low}
ptp lag-role {primary | secondary}
spanning-tree cost <PORT-COST>
spanning-tree instance <INSTANCE-ID> cost <PORT-COST>
spanning-tree instance <INSTANCE-ID> port-priority <PRIORITY-MULTIPLIER>
spanning-tree port-priority <PRIORITY-MULTIPLIER>
spanning-tree vlan <A:1-4094> cost <0-200000000>
spanning-tree vlan <A:1-4094> port-priority <0-15>
split [confirm]
track by <OBJECT-ID>
vsx-sync {access-lists | qos | rate-limits | vlans | policies | irdp}
```

Modes

There are three supported modes:

1. Copy—The **copy** mode is a one-step configuration that copies the persona configuration to an interface. Further changes to the persona will not be applied to the interfaces using that mode.

2. Attach—Unlike the **copy** mode, besides applying the configuration to the interface immediately, the **attach** mode also follows the persona configuration. It means that the subsequent changes to the persona will be applied to the interfaces attached to it.

3. Label—The **label** mode follows the interface persona configuration. The two exceptions are **copy** and **attach** modes. The **label** mode classifies the purpose of an interface and maintains functionality without requiring a persona to exist.

Predefined and custom persona names

There are two predefined interface persona names:

- uplink
- access

These names have no predefined configuration. To use them, they must be manually configured as needed. You can also create personas with a custom name. These custom personas can be created and configured in the same manner as the predefined ones. The only difference is the command used to apply them to the interface. The procedure below provides the details.

Creating and configuring an interface persona

About this task

Follow these steps to create and configure an interface persona:

Procedure

1. Create the interface **persona** with the command **interface persona** **<** PERSONA-NAME **>**.
2. Set the interface configuration as any other physical interface.
3. Review the configuration with the command **show running-configuration current-context**.
4. Switch to an interface context with the command **interface** **<** PORT **>**.
5. Apply the persona configuration to the interface and set the mode with the command **persona custom <PERSONA-NAME> <mode>**. Note that **<custom>** is listed among the three arguments required for configuration.

Results

For information on this feature, see the related video on the **AirHeads Broadcasting Channel**.

> **NOTE**
>
> All interface configurations related to Device-Profile are supported persona interface.

Examples

To copy a predefined persona name configuration to an interface:

1. Configure the interface persona:

```
switch(config)# interface persona uplink
switch(config-if)# no shutdown
switch(config-if)# no routing
switch(config-if)# vlan access 100
switch(config-if)# exit
```

2. Apply the configuration with **copy** mode:

```
switch(config)# interface 1/1/1
switch(config-if)# persona uplink copy
switch(config-if)# exit
```

To attach a custom persona name to several interfaces simultaneously:

1. Configure the interface persona:

```
switch(config)# interface persona mytemplate
switch(config-if)# no shutdown
switch(config-if)# vrf attach upstream
switch(config-if)# exit
```

2. Apply the configuration with **attach** mode:

```
switch(config)# interface 1/1/1-1/1/24
switch(config-if)# persona custom mytemplate attach
switch(config-if)# exit
```

To label a custom persona name to an interface:

1. Apply the configuration with **label** mode:

```
switch(config)# interface 1/1/1
switch(config-if)# persona custom mypersona
switch(config-if)# exit
```

Monitor mode

Monitor mode displays interface statistics in real time, updating frequently. The update interval varies by product, number of visible ports, and display options in use. View the current update interval in the help menu.

Formatting options can be supplied either at command launch or while monitor mode is active. Press `?` to see the available formatting options available in the current command context. Exit the help menu with `q`.

Monitor mode detects terminal size to adjust the number of interfaces and statistics viewable on screen. Additional interfaces or statistics columns can be viewed with the navigation keys: Arrow keys, PageUp, PageDown, Home, and End. Serial console does not automatically detect terminal setting changes, resulting in unexpected output. Recommended recovery steps are to exit and restart.

Monitor mode refreshes data automatically until exited with `q`.

Interface commands

Select a command from the list in the left navigation menu

Subtopics

- **allow-unsupported-transceiver**
- **default interface**
- **description**
- **error-control**
- **flow-control**
- **interface**
- **interface loopback**
- **interface vlan**
- **ip address**
- **ip mtu**
- **ip tcp mss**
- **ip unnumbered**
- **ipv6 address**
- **ipv6 source-interface**
- **ipv6 tcp mss**
- **l3-counters**
- **mtu**
- **persona**
- **rate-interval**
- **routing**
- **show allow-unsupported-transceiver**
- **show interface**
- **show interface dom**

show interface flow-control
show interface link-diagnostics
show interface statistics
show interface transceiver
show interface supported-transceivers
show interface utilization
show ip interface
show ip source-interface
show ipv6 interface
show ipv6 source-interface
show system interface-group
shutdown
split
system interface-group

allow-unsupported-transceiver

Syntax

```
allow-unsupported-transceiver [confirm | log-interval {none | <INTERVAL>}]
no allow-unsupported-transceiver
```

Description

Allows unsupported transceivers to be enabled or establish connections. Transceivers with speeds up to 100G are enabled by this command.

> **NOTE**
> The 8400 Series Switches will enable unsupported transceivers for speeds up to 100G when running AOS-CX 10.10 or later.

The **no** form of this command disallows using unsupported transceivers.

Parameter	Description
`confirm`	Specifies that unsupported transceiver warnings are to be auto matically confirmed.
`log-interval none`	Disables unsupported transceiver logging.
`log-interval <INTERVAL>`	Sets the unsupported transceiver logging interval in minutes. Default: 1440 minutes. Range: 1440 to 10080 minutes.

Usage

When none of the parameters are specified it will display a warning message to accept the warranty terms. With **confirm** option the warning message is displayed but the user is not prompted to **(y/n)** answering. Warranty terms must be agreed to as part of enablement and the support is on best effort basis.

Examples

Allowing unsupported transceivers with follow-up confirmation:

```
switch(config)# allow-unsupported-transceiver
Warning: The use of unsupported transceivers, DACs, and AOCs is at your
own risk and may void support and warranty. Please see HPE Warranty terms
and conditions.
Do you agree and do you want to continue (y/n)? y
```

Allowing unsupported transceivers with confirmation in command syntax:

```
switch(config)# allow-unsupported-transceiver confirm
Warning: The use of unsupported transceivers, DACs, and AOCs is at your
own risk and may void support and warranty. Please see HPE Warranty terms
and conditions.
```

Configuring unsupported transceiver logging with an interval of every 48 hours:

```
switch(config)# allow-unsupported-transceiver log-interval 2880
```

Disabling unsupported transceiver logging:

```
switch(config)# allow-unsupported-transceiver log-interval none
```

Disallowing unsupported transceivers with follow-up confirmation:

```
switch(config)# no allow-unsupported-transceivers
Warning: Unsupported transceivers, DACs, and AOCs will be disabled,
which could impact network connectivity. Use 'show allow-unsupported-
transceiver'
to identify unsupported transceivers, DACs, and AOCs.
Ccontinue (y/n)? y
```

Disallowing unsupported transceivers with confirmation in command syntax:

```
switch(config)# no allow-unsupported-transceiver confirm
Warning: Unsupported transceivers, DACs, and AOCs will be disabled,
which could impact network connectivity. Use 'show allow unsupported-
transceiver'
to identify unsupported transceivers, DACs, and AOCs.
switch(config)#
```

Command History

Release	Modification
10.10	Up to 100G support enabled for unsupported transceivers on 8400 (up to 100 G) series switches in UT mode.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	`config`	Administrators or local user group members with execution righ ts for this command.

`default interface`

Syntax

```
default interface <INTERFACE-ID>
```

Description

Sets an interface (or a range of interfaces) to factory default values.

Parameter	Description
<INTERFACE-ID>	Specifies the ID of a single interface or range of interfaces. Format: **member/slot/port** or **member/slot/port - member/slot/po rt** to specify a range.

Examples

Resetting an interface:

```
switch(config)# default default interface 1/1/1
```

Resetting an range of interfaces:

```
switch(config)# default default interface 1/1/1-1/1/10
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`description`

Syntax

```
description <DESCRIPTION>
no description
```

Description

Associates descriptive information with an interface to help administrators and operators identify the purpose or role of an interface.

The **no** form of this command removes a description from an interface.

Parameter	Description
<DESCRIPTION>	Specify a description for the interface. Range: 1 to 64 ASCII cha racters (including space, excluding question mark).

Examples

Setting the description for an interface to **DataLink 01**:

```
switch(config-if)# description DataLink 01
```

Removing the description for an interface.

```
switch(config-if)# no description
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config-if`	Administrators or local user group members with execution righ ts for this command.

`error-control`

Syntax

```
error-control {auto | none | base-r-fec | rs-fec}
no error-control {auto | none | base-r-fec | rs-fec}
```

Description

Configures the forward error correction (FEC) mode to use for an interface. When not configured, the system will automatically select the FEC mode based on the installed transceiver. In most cases, the standard FEC mode will work best, but certain link partners may require a non-standard mode.

The **no** and **auto** forms of this command configure the interface to automatically use the standard FEC mode of the currently installed transceiver.

NOTE

FEC configuration only applies to transceivers, DACs, or AOCs running at 25G or 100G. 100G DACs are a special case. They can only set FEC to none when auto-negotiation is disabled through the **speed override** command. The default for the installed transceiver is used in all other cases.

NOTE

Transceivers for which FEC is auto-negotiated will request the mode configured by this command, but may resolve to a different mode. The applied FEC mode is displayed as a commented line in the configuration shown with the **show run** command. It is also displayed with **show interface** command.

Parameter	Description
`auto`	Use the transceiver default.
`none`	Do not use any FEC.
`base-r-fec`	Use IEEE BASE - R (Firecode) FEC.
`rs-fec`	Use IEEE RS (Reed - Solomon) FEC.

Command History

Release	Modification
10.11	Command enabled on 6400 and 8400 Switch Series.
10.08.1021	Command introduced.

Command Information

Platforms	Command context	Authority
8400	`config-if`	Administrators or local user group members with execution righ ts for this command.

`flow-control`

Syntax

```
flow-control rx
no flow-control rx
flow-control priority rxtx <PRIORITY>
no flow-control priority rxtx <PRIORITY>
```

Description

Command **flow-control** enables negotiation of receive flow control on the current interface. The switch advertises RX support to the link partner. The final configuration is determined based on the capabilities of both partners.

Command **flow-control priority** enables priority-based flow control (PFC). A single priority is supported..

Each invocation of this command replaces the previous configuration.

The **no** form of these commands disables any configured flow control on the selected interface.

Parameter	Description
`rx`	Enables the ability to cease and resume transmission when rece iving IEEE 802.3x LLFC pause frames on this interface.
`<PRIORITY>`	Specifies the packet priority for which PFC will operate. Range: 0 to 7. One priority can be specified.

Usage (flow control)

- For interfaces that auto-negotiate, link-level flow control is subject to negotiation, plus speed and other parameters. Both ends of the link must negotiate the same flow control mode for it to be applied.
- For interfaces that do not auto-negotiate, the configured link-level flow control mode is always applied and the user is responsible for ensuring that both ends of the link are configured for the same mode.
- All members of a LAG must have the same flow control configuration.
- Lossless flow control is only supported for single destination unicast traffic. Replicated traffic (for example, broadcast, multicast, mirroring) cannot be guaranteed to be lossless.
- Lossless behavior is not supported when operating in a VSF stack configuration.

Usage (priority-based flow control (PFC))

- PFC will only operate correctly between interfaces with the same priority configuration. Traffic flow will not be lossless between interfaces with different priorities or between interfaces where one has PFC enabled and the other does not.
- Any queue used by protocol or control traffic must not be configured for lossless behavior. Routing protocols and VSX-synchronization messages use local-priority 7, therefore the CoS priority mapped to local-priority 7 should not be used in any lossless configuration.
  For example, in a default configuration, the CoS map assigns local-priority 7 to packets arriving with VLAN priority 7. This means that lossless pools should not be configured to use priority 7, and that any `flow-control priority` mode should not be configured on priority 7, since that VLAN priority maps to local-priority 7.
- PFC is enabled for the given Priority Code Point. Only one priority may be enabled for PFC per interface. PFC is only supported on JL365A 8400X 8-port 40GbE QSFP+ Advanced Module, JL366A HPE Aruba Networking 8400X 6-port 40GbE/100GbE QSFP28 Adv Mod, and JL687A 8400X-32Y 32p 1/10/25G SFP/SFP+/SFP28 Module line modules.
- PFC will only operate correctly between interfaces with the same priority configuration. Traffic flow will not be lossless between interfaces with different priorities or between interfaces where one has PFC enabled and the other does not.

Examples

Enabling support for RX flow control:

```
switch(config)# interface 1/1/1
switch(config-if)# flow-control rx
```

Disabling support for RX flow control:

```
switch(config)# interface 1/1/1
switch(config-if)# no flow-control rx
```

```
switch(config)# interface 1/1/1
switch(config-if)# flow-control rxtx pool 2
```

```
switch(config)# interface 1/1/1
switch(config-if)# no flow-control rxtx pool 2
```

```
switch(config)# interface 1/1/1
switch(config-if)# flow-control priority rx 3,4
```

```
switch(config)# interface 1/1/1
switch(config-if)# no flow-control priority rx 3,4
```

Enabling support for priority RXTX flow control:

```
switch(config)# interface 1/1/1
switch(config-if)# flow-control priority rxtx 2
```

Disabling support for priority RXTX flow control:

```
switch(config)# interface 1/1/1
switch(config-if)#
```

Command History

Release	Modification
10.10	Adjusted the priority syntax.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config-if`	Administrators or local user group members with execution righ ts for this command.

`interface`

Syntax

```
interface <PORT-NUM>
```

Description

Switches to the **config-if** context for a physical port. This is where you define the configuration settings for the logical interface associated with the physical port.

Parameter	Description
`<PORT-NUM>`	Specifies a physical port number. Format: **member/slot/port**.

Examples

Configuring an interface:

```
switch(config)# interface 1/1/1
switch(config-if)#
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`interface loopback`

Syntax

```
interface loopback <ID>
no interface loopback <ID>
```

Description

Creates a loopback interface and changes to the **config-loopback-if** context. Loopback interfaces are layer 3.

The **no** form of this command deletes a loopback interface.

Parameter	Description
`<INSTANCE>`	Specifies the loopback interface ID. Range: 1 to 256

Examples

```
switch# config
switch(config)# interface loopback 1
switch(config-loopback-if)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	`config`	Administrators or local user group members with execution righ ts for this command.

`interface vlan`

Syntax

```
interface vlan <VLAN-ID>
no interface vlan <VLAN-ID>
```

Description

Creates an interface VLAN also know as an SVI (switched virtual interface) and changes to the **config-if-vlan** context. The specified VLAN must already be defined on the switch.

The **no** form of this command deletes an interface VLAN.

Parameter	Description
`<VLAN-ID>`	Specifies the loopback interface ID.
none	Do not reserve any internal VLANs.

Examples

```
switch# config
switch(config)# vlan 10
switch(config-vlan-10)# exit
switch(config)# interface vlan 10
switch(config-if-vlan)#
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`ip address`

Syntax

```
ip address <IP-ADDR>/<MASK>
             [secondary]
no ip address <IP-ADDR>/<MASK>
             [secondary]
```

Description

Sets an IPv4 address for the current layer 3 interface.

The **no** form of this command removes the IPv4 address from the interface.

Parameter	Description
<IP-ADDR>	Specifies an IP address in IPv4 format (**x.x.x.x**), where **x** is a de cimal number from 0 to 255. You can remove leading zeros. For example, the address **192.169.005.100** becomes **192.168.5.1 00**.
<MASK>	Specifies the number of bits in the address mask in CIDR format (**x**), where **x** is a decimal number from 0 to 128.
`secondary`	Specifies a secondary IP address. (Supported on 6300, except for S3L75A, S3L76A, S3L77A.)

Examples

Setting the IP address on interface **1/1/1** to **192.168.100.1** with a mask of **24** bits:

```
switch(config)# interface 1/1/1
switch(config-if)# ip address 192.168.100.1/24
```

Removing the IP address **192.168.100.1** with a mask of **24** bits from interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# No ip address 192.168.100.1/24
```

Assigning the IP address **192.168.20.1** with a mask of **24** bits to loopback interface **1**:

```
switch(config)# interface loopback 1
switch(config-loopback-if)# ip address 192.168.20.1/24
```

```
switch(config)# interface 1/1/1
switch(config-if)# routing
switch(config-if)# ip address 192.168.100.1/24
```

```
switch(config)# interface loopback 1
switch(config-loopback-if)# routing
switch(config-loopback-if)# ip address 192.168.20.1/24
```

Assigning the IP address **192.168.199.1** with a mask of **24** bits to interface VLAN **10**:

```
switch(config)# interface vlan 10
switch(config-if-vlan)# ip address 192.168.199.1/24
```

Removing the IP address **192.168.199.1** with a mask of **24** bits from interface VLAN **10**:

```
switch(config)# interface vlan 10
switch(config-if-vlan)# no ip address 192.168.199.1/24
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority

All platforms	`config-if-vlan`
	`config-if`
	`config-loopback-if`
		Administrators or local user group members with execution righ ts for this command.

`ip mtu`

Syntax

```
ip mtu <VALUE>
no ip mtu
```

Description

Sets the IP MTU (maximum transmission unit) for an interface. This defines the largest IP packet that can be sent or received by the interface. This value should be less than or equal to the overall MTU for the interface.

The **no** form of this command sets the IP MTU to the default value 1500.

Parameter	Description
<VALUE>	Specifies the IP MTU in bytes. Range: 68 to 9198. Default: 150 0.

Usage

The IP MTU value for subinterface must be less than or equal to the parent MTU for the subinterface. The subinterface uses its IP MTU value and not the parent IP MTU value.

Examples

Setting the IP MTU to 576 bytes:

```
switch(config-if)# ip mtu 576
```

Setting the IP MTU to the default value:

```
switch(config-if)# no ip mtu
```

```
switch(config)# interface 1/1/1.10
switch(config-subif)# ip mtu 6000
```

Command History

Release	Modification
10.08	Subinterface support added.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority

All platforms	`config-if-vlan`
	`config-if`
	`config-subif`
		Administrators or local user group members with execution righ ts for this command.

`ip tcp mss`

Syntax

```
ip tcp mss <VALUE>
no ip tcp mss <VALUE>
```

Description

Sets the TCP (Transmission Control Protocol) MSS (Maximum Segment Size) for an interface. When configured, the MSS option in the TCP header is set in the TCP handshake for negotiation during connection establishment. Once the TCP session is established, changing the TCP MSS value on the interface will not affect the existing session.

The **no** form of this command removes the TCP MSS configuration from the interface.

> **NOTE**
> TCP MSS is supported on ROP, SVI, subinterface, and L3 GRE tunnel interface.

Parameter	Description
<VALUE>	Specifies the IPv4 TCP MSS in bytes. Range: 68 to 65495.

Examples

Configuring the IPv4 TCP MSS for the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip tcp mss 1460
```

Removing the IPv4 TCP MSS configuration from the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip tcp mss 1460
```

Command History

Release	Modification
10.14.1000	Command introduced.

Command Information

Platforms	Command context	Authority

All platforms	`config-if-vlan`
	`config-if`
	`config-gre-if`
		Administrators or local user group members with execution righ ts for this command.

`ip unnumbered`

Syntax

```
ip unnumbered <ifname>
no ip unnumbered <ifname>
```

Description

The IP unnumbered feature allows the switch to process IP packets without configuring a unique IP address on an interface by telling the router to use the IP address of another selected interface as the address for that link. Once set, this port will borrow the user-configured primary IPv4 address from an lender interface and use that IP address in its L3 control plane exchange.

The **no** version of this command removes IP unnumbered support for this port. Once unset, this port will remove borrowed IPv4 address.

Parameter	Description
<Ifname >	Name of the interface

Usage

The port where IP unnumbered is configured is known as the borrower interface. Only route-only ports can borrow an address.

The port the address is borrowed from is known as the lender interface. Only a loopback interface can be used as a lender interface. The unnumbered interface and lender interface must belong to the same VRF. Additionally, the lender interface for an IP unnumbered interface cannot itself configured as unnumbered.

The following caveats apply to this feature:

- Only the primary IPv4 address can be borrowed on an IP unnumbered interface. You cannot configure a secondary IP address for an interface that is already configured as an IP unnumbered interface.
- If an IP address is configured on an interface that is already configured as an IP unnumbered interface, it will function as a normal IP interface.
- An IPv6 unnumbered configuration on the interface is not supported. If **ip unnumbered** is enabled on the interface, only the IPv4 address is borrowed. However, Any IPv6 configurations can co-exist with an ip unnumbered interface.
- An IP unnumbered interface cannot have multiple lender interfaces for the same address family.
- The **ip unnumbered** command requires that the interface borrowing the address is a routed port. The **IP unnumbered** feature is only supported on devices connected with point-to-point interfaces.
- An unnumbered IP address can not be used as tunnel source.
- The unnumbered interface and lender interface must belong to the same VRF.
- Deleting a lender interface removes all the associated IP unnumbered references.

Examples

Configuring IP unnumbered for a route-only port:

```
switch(config)# interface 1/1/1
switch(config-if)# ip unnumbered loopback1
```

The following examples removes IP unnumbered support for this port. Once unset, this port will remove the borrowed IPv4 address.

```
switch(config)# interface 1/1/1
switch(config-if)# no ip unnumbered loopback1
```

If an IP address is configured on an interface that is already configured as an IP unnumbered interface (or vice versa), both configurations can coexist on the interface. The switch's arbitration logic ensures that static IP configuration is given priority over unnumbered.

```
switch(config)# interface 1/1/1
switch(config-if)# ip unnumbered interface loopback 1
switch(config-if)# ip address 10.16.1.1/24
switch(config-if)# ip address 192.168.1.2/24 secondary
or
switch(config)# interface 1/1/1
switch(config-if)# ip address 10.16.1.1/24
switch(config-if)# ip unnumbered interface loopback 1
switch(config-if)# ip address 192.168.1.2/24 secondary
switch(config)# sh running-config interface
switch(config-if)# do sh running-config interface 1/1/1
interface 1/1/1
   no shutdown
```

```
    ip address 10.16.1.1/24
    ip address 192.168.1.2/24 secondary
    ip unnumbered interface loopback 1
      ! ip unnumbered is ignored when static ip is configured
```

If loopback lender interface does not exist during IP unnumbered configuration, a prompt will be displayed in CLI to allow the user to create that interface.

```
switch(config)# interface 1/1/1
switch(config-if)# ip unnumbered interface loopback 1
Lender port not found.
Do you want to create (y/n)? y
switch(config-if)#
Failure case:
switch(config)# ip unnumbered interface loopback 1
Lender port not found.
Do you want to create (y/n)? y
Lender port: loopback1 creation failed
switch(config-if)
```

Command History

Release	Modification
10.14.1000	Command introduced

Command Information

Platforms	Command context	Authority
8400	`config-if`	Administrators or local user group members with execution righ ts for this command.

`ipv6 address`

Syntax

```
ipv6 address <IPV6-ADDR>/<MASK>{eui64 | [tag <ID>]}
no ipv6 address <IPV6-ADDR>/<MASK>
```

Description

Sets an IPv6 address on the interface.

The **no** form of this command removes the IPv6 address on the interface.

> **NOTE**
> This command automatically creates an IPv6 link-local address on the interface. However, it does not add the **ipv6 address link-local** command to the running configuration. If you remove the IPv6 address, the link-local address is also removed. To maintain the link-local address, you must manually execute the **ipv6 address link-local** command.

Parameter	Description
<IPV6-ADDR>	Specifies the IP address in IPv6 format (**xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx**), where **x** is a hexadecimal number from 0 to F. You can use two colons (::) to represent consecutive zeros (but only once), remove leading zeros, and collapse a hextet of four zeros to a single 0. For example, this address **2222:0000:3333:0000:0000:0000:4444:0055** becomes **2222:0:3333::4444:55**.
<MASK>	Specifies the number of bits in the address mask in CIDR format (**x**), where **x** is a decimal number from 0 to 128.
`eui64`	Configure the IPv6 address in the EUI - 64 bit format.
`tag` *<ID>*	Configure route tag for connected routes. Range: 0 to 4294967295. Default: 0.

Examples

Setting the IPv6 address **2001:0db8:85a3::8a2e:0370:7334** with a mask of 24 bits:

```
switch(config-if)# ipv6 address 2001:0db8:85a3::8a2e:0370:7334/24
```

Removing the IP address **2001:0db8:85a3::8a2e:0370:7334** with mask of 24 bits:

```
switch(config-if)# no ipv6 address 2001:0db8:85a3::8a2e:0370:7334/24
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config-if`	Administrators or local user group members with execution righ ts for this command.

`ipv6 source-interface`

Syntax

```
ipv6 source-interface {sflow |  tftp |  radius | tacacs | ntp | syslog |
ubt |


dhcp-relay|
simplivity | http | sftp-scp | ssh-client | dns | all} {interface <IFNAME>
| <IPV6-ADDR>} [vrf <VRF-NAME>]
no ipv6 source-interface {sflow |  tftp |  radius | tacacs | ntp | syslog |
ubt |


dhcp-relay|
simplivity | http | sftp-scp | ssh-client | dns | all} [interface <IFNAME>
| <IPV6-ADDR>] [vrf <VRF-NAME>]
```

Description

Sets a single source IPv6 address for a feature on the switch. This ensures that all traffic sent the feature has the same source IPv6 address regardless of how it egresses the switch. You can define a single global address that applies to all supported features, or an individual address for each feature.

This command provides two ways to set the source IPv6 addresses: either by specifying a static IPv6 address, or by using the address assigned to a switch interface. If you define both options, then the static IPv6 address takes precedence.

The `no` form of this command deletes the single source IPv6 address for all supported protocols, or a specific protocol.

Parameter	Description
`sflow | tftp | radius | tacacs | ntp | syslog | ubt | dhcp-relay | simplivity | http | sftp-scp | ssh-client | dns | all`	Sets a single source IPv6 address for a specific protocol. The `all` option sets a global address that applies to all protocols that do not have an address set.
`interface <IFNAME>`	Specifies the name of the interface from which the specified protocol obtains its source IPv6 address.
`<IPV6-ADDR>`	Specifies the source IPv6 address to use for the specified protocol. The IP address must be defined on the switch, and it must exist on the specified VRF (which is the `default` VRF, if the `vrf` option is not used). Specify the IP address in IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where `x` is a hexadecimal number from 0 to F.
`vrf <VRF-NAME>`	Specifies the name of the VRF from which the specified protocol sets its source IPv6 address.

Examples

Configuring the IPv6 address `2001:DB8::1` as the global single source address:

```
switch# config
switch(config)# ipv6 source-interface all 2001:DB8::1/32
```

Configuring the IPv6 address `2001:DB8::1` on VRF `sflow-vrf` on interface `1/1/2` as the single source address for sFlow:

```
switch(config)# vrf sflow-vrf
switch(config-vrf)# exit
switch(config)# interface 1/1/2
switch(config-if)# no shutdown
switch(config-if)# vrf attach sflow-vrf
switch(config-if)# ipv6 address 2001:DB8::1/32
switch(config-if)# exit
switch(config)# ipv6 source-interface sflow interface 1/1/2 vrf sflow-vrf
```

Stop the source IPv6 address from using the IP address on interface **1/1/1** on VRF `one` .

```
switch(config)# no ipv6 source-interface all interface 1/1/1 vrf one
```

Clear the source IPv6 address `2001:DB8::1`.

```
switch(config)# no ipv6 source-interface all 2001:DB8::1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`ipv6 tcp mss`

Syntax

```
ipv6 tcp mss <VALUE>
no ipv6 tcp mss <VALUE>
```

Description

Sets the TCP (Transmission Control Protocol) MSS (Maximum Segment Size) for an interface. When configured, the MSS option in the TCP header is set in the TCP handshake for negotiation during connection establishment. Once the TCP session is established, changing the TCP MSS value on the interface will not affect the existing session.

The **no** form of this command removes the TCP MSS configuration from the interface.

> **NOTE**
> TCP MSS is supported on ROP, SVI, subinterface, and L3 GRE tunnel interface.

Parameter	Description
<VALUE>	Specifies the IPv6 TCP MSS in bytes. Range: 68 to 65495.

Examples

Configuring the IPv6 TCP MSS for the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 tcp mss 1440
```

Removing the IPv6 TCP MSS configuration from the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ipv6 tcp mss 1440
```

Command History

Release	Modification
10.14.1000	Command introduced.

Command Information

Platforms	Command context	Authority

All platforms	`config-if-vlan`
	`config-if`
	`config-gre-if`
		Administrators or local user group members with execution righ ts for this command.

`l3-counters`

Syntax

```
l3-counters [rx | tx]
no l3-counters [rx | tx]
```

Description

Enables counters on a layer 3 interface. By default, all interfaces are layer 3. To change a layer 2 interface to layer 3, use the **routing** command.

The **no** form of this command, with no specification, disables both transmit and receive counters on a layer 3 interface. To disable transmit (**tx**) or receive (**rx**) counters only, specify the counter type you want to disable.

Parameter	Description
`rx`	Specifies receive counters.
`tx`	Specifies transmit counters.

Examples

Enabling layer 3 transmit counters on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# l3-counters tx
```

Disabling layer 3 transmit and receive counters on interface **1/1/2**:

```
switch(config)# interface 1/1/2
switch(config-if)# no l3-counters
```

```
switch(config)# interface 1/1/1.10
switch(config-subif)# l3-counters
```

```
switch(config)# interface 1/1/1.10
switch(config-subif)# no l3-counters
```

```
switch(config)# interface 1/1/1.10
switch(config-subif)# l3-counters rx
```

```
switch(config)# interface 1/1/1.10
switch(config-subif)# no l3-counters tx
```

```
switch(config)# interface 1/1/1
switch(config-if)# routing
switch(config-if)# l3-counters
```

```
switch(config)# interface 1/3/1
switch(config-if)# routing
switch(config-if)# l3-counters
```

```
switch(config)# interface 1/1/1.10
switch(config-if)# routing
switch(config-if)# l3-counters tx
```

```
switch(config)# interface 1/3/1.10
switch(config-if)# routing
switch(config-if)# l3-counters tx
```

Command History

Release	Modification
10.08	Added support for 13 counters on subinterfaces
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	`config-if`	Administrators or local user group members with execution righ ts for this command.

`mtu`

Syntax

```
mtu <VALUE>
no mtu
```

Description

Sets the MTU (maximum transmission unit) for an interface. This defines the maximum size of a layer 2 (Ethernet) frame. Frames larger than the MTU (1500 bytes by default) are dropped.

To support jumbo frames (frames larger than 1522 bytes), increase the MTU as required by your network. A frame size of up to 9198 bytes is supported.

The largest possible layer 1 frame will be 18 bytes larger than the MTU value to allow for link layer headers and trailers.

The **no** form of this command sets the MTU to the default value 1500.

Parameter	Description
<VALUE>	Specifies the MTU in bytes. Range: 46 to 9198. Default: 1500.

Examples

Setting the MTU on interface **1/1/1** to 1000 bytes:

```
switch(config)# interface 1/1/1
switch(config-if)# no routing
switch(config-if)# mtu 1000
```

Setting the MTU on interface **1/1/1** to the default value:

```
switch(config)# interface 1/1/1
switch(config-if)# no routing
switch(config-if)# no mtu
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config-if`	Administrators or local user group members with execution righ ts for this command.

`persona`

Syntax

```
persona {access | uplink | custom <PERSONA-NAME>} [copy | attach]
no persona {access | uplink | custom <PERSONA-NAME>} [copy | attach]
```

Description

Associates one of three persona types with an interface to classify the purpose or role of an interface. On the 10000 Switch Series, “access” persona ports are typically connected to workloads / VMs, and the “uplink” (fabric) persona ports are connected to the core / spine.

The **no** form of this command removes the interface persona.

Parameter	Description
`access`	Selects the **access** persona type.
`uplink`	Selects the **uplink** persona type.
`custom <PERSONA-NAME>`	Selects the **custom** persona type with a user - provided name. Range: 1 to 64 printable ASCII characters including space.
`copy`	Specifies the mode: copies settings from the persona interface of the same name.
`attach`	Specifies the mode: attaches the specified interface to the pers ona interface of the same name.

Usage

- If the mode is specified, either copy or attach, the interface configuration is dependent on the interface template whose name is "access", "uplink", or "<PERSONA-NAME>". On the other hand, if the mode is not specified, then the persona is just a label in the interface, and its configuration is not modified even if the interface persona exists. When configuring the mode, one of the following options is possible:
  - The **copy** option performs a one-time copy of the template interface. Subsequent changes to the template are not copied and the 'persona' setting is just a label. If the mode is set to **copy** and the interface persona does not exist, then the CLI command fails with the message "Interface persona not found".
  - The **attach** option performs a copy of the template interface, and subsequent changes to the template interface configuration are immediately applied to all attached interfaces. The template interface does not need to exist before attaching other interfaces to it. After attaching a template,

the copied settings can be modified for an individual interface. However, any change in the attached template will overwrite the modified values with the new template values.

- When a mode is specified, it should match an interface created with the command interface persona <PERSONA-NAME>. The only exception to this rule is when the mode is set to `attach` and the persona does not already exist.
- The mode is only available to be configured for an interface that meets the following conditions:
  - IS a physical interface
  - IS NOT a LAG member
  - IS NOT a persona interface

Examples

Configuring an access persona:

```
switch(config)# interface 1/1/1
switch(config-if)# persona access
```

Configuring an uplink persona:

```
switch(config)# interface 1/1/1
switch(config-if)# persona uplink
```

Configuring a custom persona named "mypersona":

```
switch(config)# interface 1/1/1
switch(config-if)#persona custom mypersona
```

Removing the persona setting.

```
switch(config-if)# no persona
```

Copying a predefined persona name configuration to an interface:

1. Configuring the interface persona:

```
switch(config)# interface persona uplink
switch(config-if)# no shutdown
switch(config-if)# no routing
switch(config-if)# vlan access 100
switch(config-if)# exit
```

2. Applying the configuration from the persona named "mypersona" with **copy** mode:

```
switch(config)# interface 1/1/1
switch(config-if)# persona custom mypersona copy
```

```
switch(config-if)# exit
```

Attaching a custom persona name named "mypersona" to several interfaces simultaneously:

1. Configuring an interface persona named "mypersona":

```
switch(config)# interface persona mypersona
switch(config-if)# no shutdown
switch(config-if)# vrf attach upstream
switch(config-if)# exit
```

2. Applying the "mypersona" configuration with **attach** mode:

```
switch(config)# interface 1/1/1-1/1/24
switch(config-if)# persona custom mypersona attach
switch(config-if)# exit
```

Command History

Release	Modification
10.10	Added optional parameters: attach, copy.
10.09	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	`config-if`	Administrators or local user group members with execution righ ts for this command.

`rate-interval`

Syntax

```
rate-interval <VALUE>
no rate-interval
```

Description

This command sets the time interval to calculate interface rates. Lower intervals are more useful for detecting traffic bursts, but may increase computation load to the overall system. Intervals must be a multiple of five seconds. The command-line interface will not accept a rate interval value that is not a multiple of five.

The **no** form of this command sets the rate collection interval to the default value of 300 seconds.

Parameter	Description
<VALUE>	The statistics rate collection interval in seconds. The supported range is 5 - 300 seconds, where the number of seconds is a mu ltiple of five. **NOTE** The supported range for HPE Aruba Networkin g 6400 and 8400 Switch series is 30 - 300 se conds.

Examples

Setting the rate collection interval to 50 seconds

```
switch(config)# interface 1/1/1
switch(config-if)# rate-interval 50
```

Setting the rate collection interval to the default value:

```
switch(config-if)# no rate-interval
```

The following example shows the command-line interface warning that appears while configuring an invalid rate-interval.

```
switch(config)# interface 1/1/1
switch(config-if)# rate interval 6
            The interval must be a multiple of 5.
```

Usage

The rate collection interval must be configured in the multiples of 5. Any other value will be rejected and the CLI will display the error message, **The interval must be a multiple of 5.**

Command History

Release	Modification
10.12.1000	Command supported on all platforms.

Release	Modification
10.12	Command Introduced on 6300 and 8360 Switch series.

Command Information

Platforms	Command context	Authority
All platforms	`config-if`	Administrators or local user group members with execution righ ts for this command.

`routing`

Syntax

```
routing
no routing
```

Description

Enables routing support on an interface, creating a L3 (layer 3) interface on which the switch can route IPv4/IPv6 traffic to other devices.

By default, routing is enabled on all interfaces.

The **no** form of this command disables routing support on an interface, creating a L2 (layer 2) interface.

CAUTION

If you enable this configuration, collection of flow tracking statistics is disabled.

Examples

Enabling routing support on an interface:

```
switch(config-if)# routing
```

Disabling routing support on an interface:

```
switch(config-if)# no routing
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	`config-if`	Administrators or local user group members with execution righ ts for this command.

`show allow-unsupported-transceiver`

Syntax

```
show allow-unsupported-transceiver
```

Description

Displays configuration and status of unsupported transceivers.

Examples

Showing unallowed unsupported transceivers:

```
switch(config)# show allow-unsupported-transceiver
Allow unsupported transceivers : no
Logging interval               : 1440 minutes
----------------------------------------------
Port       Type           Status
----------------------------------------------
1/1/31     SFP-SX         unsupported
1/1/32     SFP-1G-BXD     unsupported
1/1/2      SFP28DAC3      unsupported
```

Showing allowed unsupported transceivers:

```
switch# show allow-unsupported-transceiver
Allow unsupported transceivers : yes
Logging interval               : 1440 minutes
----------------------------------------------
```

```
Port        Type            Status
---------------------------------------------
1/1/31      SFP-SX          unsupported-allowed
1/1/32      SFP-1G-BXD      unsupported-allowed
1/1/2       SFP28DAC3       unsupported
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config-if`	Administrators or local user group members with execution righ ts for this command.

`show interface`

Syntax

```
show interface [<IFNNAME>|<IFRANGE>] [brief | physical]
show interface [<IFNNAME>|<IFRANGE>] [extended [non-zero] | [human-
readable]]
show interface [<IFNNAME>] monitor [human-readable]
show interface [lag | loopback | tunnel | vlan ] [<ID>] [brief]
show interface lag [<LAG-ID>] [extended [non-zero] | [human-readable]]
show interface lag [<LAG-ID>] monitor [human-readable]
```

Description

Shows active configurations and operational status information for interfaces.

Parameter	Description
	Specifies a interface name.

Parameter	Description
<IFNAME>	
<IFRANGE>	Specifies the port identifier range.
`brief`	Shows brief info in tabular format.
`physical`	Shows the physical connection info in tabular format.
`extended`	Shows additional statistics, including the **tx filtered** and **rx filte red** counters. • Rx filter packets are protocol packets received when the p rotocol is disabled on the switch and there is only one port i n the VLAN. Protocols include OSPF, PIM, RIP, LACP, and LL DP. • An example of a Tx filtered packet would be a multicast pac ket being filtered from going out of the ingress port.
`human-readable`	Shows statistics rounded to the nearest power of 1000, for exa mple, 1K, 345M, 2G. This is available only in the CLI interface o utput.
`non-zero`	Shows only non zero statistics.
`LAG`	Shows LAG interface information.
`monitor`	Continuously monitor interface statistics.
`LOOPBACK`	Shows loopback interface information.
`TUNNEL`	Shows tunnel interface information.
`VLAN`	Shows VLAN interface information.
<LAG-ID>	Specifies the LAG number. Range: 1 - 256
<LOOPBACK-ID>	Specifies the LOOPBACK number. Range: 0 - 255

Parameter	Description
<TUNNEL-ID>	Specifies the tunnel ID. Range: 1 - 255
<VLAN-ID>	Specifies the VLAN ID. Range: 1 - 4094
VXLAN	Shows the VXLAN interface information.
<VXLAN-ID>	Specifies the VXLAN interface identifier. Default: 1

Examples

Showing interface information when it is configured as a route-only port:

```
switch# show interface 1/1/1
Interface 1/1/1 is up
Admin state is up
Link state: up for 2 days (since Sun Jun 21 05:30:22 UTC 2020)
Link transitions: 1
Description: backup data center link
Hardware: Ethernet, MAC Address: 70:72:cf:fd:e7:b4
MTU 1500
Type 1GbT
Full-duplex
qos trust none
Speed 1000 Mb/s
Auto-negotiation is on
Flow-control: off
Error-control: off


L3 Counters: Rx Enabled, Tx Enabled
Rate collection interval: 300 seconds
Rates                             RX                   TX         Total (RX+TX)
------------- -------------------- -------------------- --------------------
Mbits / sec                     0.00                 0.00                 0.00
KPkts / sec                     0.00                 0.00                 0.00
```

```
Unicast                        0.00                 0.00                 0.00
Multicast                      0.00                 0.00                 0.00
Broadcast                      0.00                 0.00                 0.00
Utilization %                  0.00                 0.00                 0.00
Statistics                       RX                   TX                Total
------------- -------------------- -------------------- --------------------
Packets                           0                    0                    0
Unicast                           0                    0                    0
Multicast                         0                    0                    0
Broadcast                         0                    0                    0
Bytes                             0                    0                    0
Jumbos                            0                    0                    0
Dropped                           0                    0                    0
Filtered                          0                    0                    0
Pause Frames                      0                    0                    0
L3 Packets                        0                    0                    0
L3 Bytes                          0                    0                    0
Errors                            0                    0                    0
CRC/FCS                           0                  n/a                    0
Collision                       n/a                    0                    0
Runts                             0                  n/a                    0
Giants                            0                  n/a                    0
Other                             0                    0                    0
MDI mode: MDIX
VLAN Mode: native-untagged
Native VLAN: 1
Allowed VLAN List: all
Rate collection interval: 300 seconds
Rates                            RX                   TX        Total (RX+TX)
------------- -------------------- -------------------- --------------------
Mbits / sec                    0.00                 0.00                 0.00
KPkts / sec                    0.00                 0.00                 0.00
Unicast                        0.00                 0.00                 0.00
Multicast                      0.00                 0.00                 0.00
Broadcast                      0.00                 0.00                 0.00
Utilization %                  0.00                 0.00                 0.00
Statistics                       RX                   TX                Total
------------- -------------------- -------------------- --------------------
Packets                           0                    0                    0
Unicast                           0                    0                    0
Multicast                         0                    0                    0
Broadcast                         0                    0                    0
Bytes                             0                    0                    0
```

```
Jumbos                              0                    0                    0
Dropped                             0                    0                    0
Filtered                            0                    0                    0
Pause Frames                        0                    0                    0
Errors                              0                    0                    0
CRC/FCS                             0                  n/a                    0
Collision                         n/a                    0                    0
Runts                               0                  n/a                    0
Giants                              0                  n/a                    0
```

Showing information when the interface is currently linked at a downshifted speed:

```
switch(config-if)# show interface 1/1/1
Interface 1/1/1 is up
...
Auto-negotiation is on with downshift active
```

```
switch(config-if)# show interface 1/1/1
Interface 1/1/1 is up
...
Energy-Efficient Ethernet is enabled and active
```

Showing information when the interface is shut down during a VSX split:

```
switch(config-if)# show interface 1/1/1
Interface 1/1/1 is down
Admin state is up
State information: Disabled by VSX
Link state: down for 3 days (since Tue Mar 16 05:20:47 UTC 2021)
Link transitions: 0
Description:
Hardware: Ethernet, MAC Address: 04:09:73:62:90:e7
MTU 1500
Type SFP+DAC3
Full-duplex
qos trust none
Speed 0 Mb/s
Auto-negotiation is off
Flow-control: off
Error-control: off
VLAN Mode: native-untagged
Native VLAN: 1
Allowed VLAN List: 1502-1505
Rate collection interval: 300 seconds
Rate                               RX                   TX         Total
(RX+TX)
---------------- -------------------- --------------------
--------------------
```

```
Mbits / sec                      0.00                 0.00
0.00
KPkts / sec                      0.00                 0.00
0.00
Unicast                          0.00                 0.00
0.00
Multicast                        0.00                 0.00
0.00
Broadcast                        0.00                 0.00
0.00
Utilization                      0.00                 0.00
0.00
Statistic                          RX                   TX
Total
---------------- -------------------- --------------------
--------------------
Packets                             0
0                    0
Unicast                             0
0                    0
Multicast                           0
0                    0
Broadcast                           0
0                    0
Bytes                               0
0                    0
Jumbos                              0
0                    0
Dropped                             0
0                    0
Pause Frames                        0
0                    0
Errors                              0
0                    0
CRC/FCS                             0
n/a                    0
Collision                         n/a
0                    0
Runts                               0
n/a                    0
Giants                              0
n/a                    0
```

```
switch(config-if)# show interface 1/1/1 physical
----------------------------------------------------------------------------
----------------------------------------------------------------
                            Link    Admin         Speed           Flow-
Control          EEE        PoE Power                     Port
Port         Type           Status    Config    Status | Config    Status |
Config    Status | Config  (Watts)     State Information  Description
----------------------------------------------------------------------------
----------------------------------------------------------------
1/1/1        1GbT           up        up        1G        auto       off
off       on        on      --         10M/100M/1G        --
```

Showing the monitor information:

> **NOTE**
>
> In monitor mode, the CLI refreshes data automatically until it is exited by entering **q**. Pressing **?** opens the help menu to display which options are available in this context.

```
Interface 1/1/1 is up
Rate                                   RX                   TX        Total
(RX+TX)
---------------- -------------------- --------------------
--------------------
MBits / sec                      30196.43             30196.43
60392.85
MPkts / sec                      58977.39             58977.40
117954.79
Unicast                              0.00                 0.00
0.00
Multicast                        58977.39             58977.40
117954.79
Broadcast                            0.00                 0.00
0.00
Utilization %                       75.49                75.49
150.98
Statistic                              RX                   TX        Total
(RX+TX)
---------------- -------------------- --------------------
--------------------
Packets                        4756527649           4756527865
9513055514
Unicast                                 0
0                    0
```

```
Multicast                    4756527649           4756527865
9513055514
Broadcast                             2
0                    2
Bytes                      304417778668         304417795428
608835574096
Jumbos                                0
0                    0
Dropped                               0           19028847730
19028847730
Pause Frames                          0
0                    0
Errors                                0
0                    0
CRC/FCS                               0
n/a                    0
help: ?, quit: q
Help for Interface Monitor
h  Toggle human-readable mode
c  Clear interface statistics
Does not apply to rates
Arrows, PgUp, PgDn, Home, End
Navigate interface statistics
Delay: 2
help: ?, quit: q
```

Showing the output for interface 1/1/1 in human-readable format:

> **NOTE**
>
> In human-readable format, the **< 1** symbol for **Utilization** indicates that the amount of packets is between zero and one. This is true in cases where the number of bytes increases but the number of packets and the **Utilization** value is not displayed even in the normal output, where the human-readable parameter is not included in the command.

```
switch(config-if)# show interface 1/1/1 human-readable
Interface 1/1/1 is up
Rate                                 RX                    TX         Total
(RX+TX)
---------------- -------------------- --------------------
--------------------
Bits / sec                           3M
3M                   6M
Pkts / sec                          316                   316
633
```

```
Unicast                          319                  319
638
Multicast                          0
0                    0
Broadcast                          0
0                    0
Utilization %                    < 1                  < 1
< 1
Statistic                         RX                   TX
Total
---------------- -------------------- --------------------
--------------------
Packets                         577K
577K                   1M
Unicast                         577K
577K                   1M
Multicast                          0
51                   51
Broadcast                          0
15                   15
Bytes                           744M
745M                   1G
Jumbos                             0
0                    0
Dropped                            0
0                    0
Filtered                           0
0                    0
Pause Frames                       0
0                    0
Errors                             0
0                    0
CRC/FCS                            0
n/a                    0
Collision                        n/a
0                    0
Runts                              0
n/a                    0
Giants                             0
n/a                    0
```

Showing information about extended counters:

> **NOTE**
>
> The output of the `show interface extended` command varies depending on the switch model and configuration.

```
switch(config-if)# show interface 1/1/17 extended
-------------------------------------------------------------------
Interface 1/1/17
-------------------------------------------------------------------
Statistics                                              Value
-------------------------------------------------------------------
Dot1d Tp Port In Frames                                 547
Dot1d Tp Port Out Frames                                608
Dot3 In Pause Frames                                    0
Dot3 Out Pause Frames                                   0
Ethernet Stats Broadcast Packets                        19
Ethernet Stats Bytes                                    40162
Ethernet Stats Packets                                  342
...
-------------------------------------------------------------------
Error-Statistics                                        Value
-------------------------------------------------------------------
Dot1d Base Port MTU Exceeded Discards                   0
Dot3 Control In Unknown Opcodes                         0
Dot3 Stats Alignment Errors                             0
Dot3 Stats FCS Errors                                   0
Dot3 Stats Frame Too Longs                              0
Dot3 Stats Internal Mac Transmit Errors                 0
Ethernet RX Oversize Packets                            0
...
```

Showing interface link-status:

```
switch# show interface link-status
-------------------------------------------------------------


Port           Type           Physical    Link         Last
                              Link State  Transitions  Change
-------------------------------------------------------------
1/1/1          1G-BT          down        0            --
1/1/2          1G-BT          up          1            1 minute ago (Fri
Mar 09 12:36:56 UTC 2018)
1/1/3          1G-BT          up          1            1 minute ago (Fri
Mar 09 12:36:56 UTC 2018)
```

```
1/1/4          --             down         0            --
1/1/5          --             down         0            --
```

Showing interface loopback 1 link-status:

```
-------------------------------------------------------------
                              Physical     Link         Last
Port           Type           Link State   Transitions  Change
-------------------------------------------------------------
loopback1      --             up           --           --
```

Showing interface 1/1/2-1/1/3 link-status:

```
-------------------------------------------------------------
                              Physical     Link         Last
Port           Type           Link State   Transitions  Change
-------------------------------------------------------------
1/1/2          1G-BT          up           1            1 minute ago (Fri
Mar 09 12:36:56 UTC 2018)
1/1/3          1G-BT          up           1            1 minute ago (Fri
Mar 09 12:36:56 UTC 2018)
```

Showing interface link-status:

```
switch# show interface link-status
-------------------------------------------------------------------------
Port           Type            Physical    Link         Link Flaps  Last
                               Link State  Transitions Ignored     Change
-------------------------------------------------------------------------
1/1/1          1G-BT           down        0           0           --
1/1/2          1G-BT           up          1           0           1 minute
ago (Fri Mar 09 12:36:56 UTC 2018)
1/1/3          1G-BT           up          1           0           1 minute
ago (Fri Mar 09 12:36:56 UTC 2018)
1/1/4          --              down        0           0           --
1/1/5          --              down        0           0           --
```

Showing state information when interface is blocked:

```
8360(config-if)# show interface 1/1/1
Interface 1/1/1 is up (Blocked)
Admin state is up
State information: Blocked by UDLD
Link state: up for 1 minute (since Mon Jun 10 09:25:27 UTC 2024)
Link transitions: 1
Description:
Persona:
Hardware: Ethernet, MAC Address: 00:fd:45:67:85:91
MTU 1500
Type 10G-LR / 10G SFP+ LR
```

```
Full-duplex
qos trust none
Speed 10000 Mb/s
Auto-negotiation is off
Flow-control: off
Error-control: off
VLAN Mode: access
Access VLAN: 1
Rate collection interval: 300 seconds
Rate                                   RX                   TX          Total
(RX+TX)
---------------- -------------------- --------------------
--------------------
Mbits / sec                          0.00                 0.00
0.00
KPkts / sec                          0.00                 0.00
0.00
Unicast                              0.00                 0.00
0.00
Multicast                            0.00                 0.00
0.00
Broadcast                            0.00                 0.00
0.00
Utilization %                        0.00                 0.00
0.00
Statistic                              RX                   TX
Total
---------------- -------------------- --------------------
--------------------
Packets                                15
15                     30
Unicast                                12
12                     24
Multicast                               3
3                      6
Broadcast                               0
0                      0
Bytes                                1350                 1350
2700
Jumbos                                  0
0                      0
Dropped                                 0
0                      0
```

```
Pause Frames                                    0
0                    0
Errors                                          0
0                    0
CRC/FCS                                         0
n/a                    0
Collision                                     n/a
0                    0
Runts                                           0
n/a                    0
Giants                                          0
n/a                    0
```

Command History

Release	Modification
10.15	Added state information when port goes into down state.
10.11	Added `monitor` parameter.
10.10	Added `human-readable` parameter.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show interface dom`

Syntax

```
show interface [<INTERFACE-ID>] dom [detail] [vsx-peer]
```

Description

Shows diagnostics information and alarm/warning flags for the optical transceivers (SFP, SFP+, QSFP+). This information is known as DOM (Digital Optical Monitoring). DOM information also consists of vendor determined thresholds which trigger high/low alarms and warning flags.

Parameter	Description
<INTERFACE-ID>	Specifies an interface. Format: **member/slot/port**.
`detail`	Show detailed information.
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output f rom the VSX peer switch is not displayed. This parameter is avai lable on switches that support VSX.

Example

```
switch# show interface dom
----------------------------------------------------------------------------
------
Port    Type      Channel  Temperature Voltage  Tx Bias  Rx Power      Tx
Power
                 (Celsius)   (Volts)   (mA)      (mW/dBm)       (mW/dBm)
----------------------------------------------------------------------------
------
1/1/1   SFP+SR             47.65       3.31     8.40     0.08, -10.96
0.63, -2.49
1/1/2   SFP+SR             n/a         n/a      n/a      n/a              n/a
1/1/3   SFP+DA3            42.10       3.24     n/a      n/a              n/a
1/1/4   QSFP+SR4  1        44.46       3.30     6.12     0.08, -10.96
0.63, -1.95
2                  44.46       3.30      6.04      0.08, -10.96  0.63, -2.00
3                  44.46       3.30      6.51      0.08, -10.96  0.60, -2.16
4                  44.46       3.30      6.19      0.08, -10.96  0.63, -1.94
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show interface flow-control`

Syntax

```
show interface [<IFNNAME>|<IFRANGE>] flow-control [detail]
```

Description

Shows the flow control configuration, status, and statistics of the specified interface for interfaces on which flow control is enabled.

> **NOTE**
>
> If **detail** is not specified, this command shows a summary of all flow controlled interfaces with one interface per line. If **detail** is specified, this command shows flow control detailed statistics.

> **NOTE**
>
> As of AOS-CX 10.10, the separate **show flow-control** command has been removed, with it being effectively replaced by this command.

Parameter	Description
`<IFNNAME>|<IFRANGE>`	Specifies the interface (port) name or range. When no interface range is specified, only interfaces with flow control enabled in t he configuration or status are shown.
`detail`	Shows detailed information.

Examples

Showing summary flow control information:

```
switch# show interface flow-control
----------- -------------------------------------
Port        Flow
            Control
----------- -------------------------------------
1/1/1       config: llfc rx
            status: llfc rx
1/1/2       config: llfc rx
            status: none
```

Showing summary flow control information with PFC:

```
switch# show interface flow-control
----------- -------------------------------------
Port        Flow
            Control
----------- -------------------------------------
1/1/1       config: pfc rxtx-1,2
            status: pfc rxtx-1,2
1/1/2       config: pfc rxtx-5
            status: none
```

Showing summary flow control information with PFC:

```
switch# show interface flow-control
  Flow Control Watchdog Settings
      Trigger Timeout:  100 milliseconds
      Resume Time:      100 milliseconds
----------- ------------------------------------- ------------- --------
Port        Flow                                  Watchdog      Watchdog
            Control                               Status        Timeouts
----------- ------------------------------------- ------------- --------
1/1/1       config: llfc rx
            status: llfc rx
1/1/2       config: llfc rx                       incompatible         0
            status: llfc rx
1/1/10      config: pfc rxtx-1,2                  enabled           1234
            status: pfc rxtx-1,2
1/1/12      config: pfc rxtx-1,2                  error                0
            status: pfc rxtx-1,2
1/1/32:4    config: pfc rxtx-5
            status: pfc rxtx-5
```

Showing summary flow control information where the configuration does not match status due to a reboot required to apply PFC configuration in hardware:

```
switch# show interface flow-control
 Flow Control Watchdog Settings
```

```
   Trigger Timeout:  100 milliseconds (actual: not applied)
   Resume Time:      100 milliseconds (actual: not applied)
 ----------- -------------------------------------- ------------- --------
 Port        Flow                                   Watchdog      Watchdog
             Control                                Status        Timeouts
 ----------- -------------------------------------- ------------- --------
 1/1/1       config: llfc rx
             status: llfc rx
 1/1/2       config: llfc rx                        incompatible         0
             status: llfc rx
 1/1/10      config: pfc rxtx-1,2                   pending           1234
             status: none
 1/1/12      config: pfc rxtx-1,2                   pending              0
             status: none
 1/1/32:4    config: pfc rxtx-5
             status: none
```

Showing detailed flow control information with RX flow control enabled:

```
switch# show interface 1/1/1 flow-control detail
Interface 1/1/1 is up
 Admin state is up
 Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
 Flow-control: llfc rx
 Statistics                             RX
 -------------------- --------------------
 Dot3 Pause Frames                       0
```

Showing detailed flow control information with RX flow control enabled:

```
switch# show interface 1/1/1 flow-control detail
Interface 1/1/1 is up
 Admin state is up
 Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
 Flow-control: llfc rx
 Flow-control watchdog: disabled
 Statistics                             RX
 -------------------- --------------------
 Dot3 Pause Frames                       0
```

Showing detailed flow control information with RXTX flow control enabled:

```
switch# show interface 1/1/1 flow-control detail
Interface 1/1/1 is up
 Admin state is up
 Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
 Flow-control: llfc rxtx
 Statistics                             RX                   TX
```

```
 -------------------- -------------------- --------------------
 Dot3 Pause Frames                       0                    0
```

Showing detailed flow control information with PFC enabled:

```
switch# show interface 1/1/1 flow-control detail
Interface 1/1/1 is up
 Admin state is up
 Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
 Flow-control: pfc rxtx-4,5
 Statistics                             RX                   TX
 -------------------- -------------------- --------------------
 Priority 0 Pauses                       0                    0
 Priority 1 Pauses                       0                    0
 Priority 2 Pauses                       0                    0
 Priority 3 Pauses                       0                    0
 Priority 4 Pauses                       0                    0
 Priority 5 Pauses                       0                    0
 Priority 6 Pauses                       0                    0
 Priority 7 Pauses                       0                    0
 Total Pause Frames                      0                    0
```

Showing detailed flow control information with PFC enabled and flow control watchdog disabled:

```
switch# show interface 1/1/1 flow-control detail
Interface 1/1/1 is up
 Admin state is up
 Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
 Flow-control: pfc rxtx-4,5
 Flow-control watchdog: disabled
 Statistics                             RX                   TX
 -------------------- -------------------- --------------------
 Priority 0 Pauses                       0                    0
 Priority 1 Pauses                       0                    0
 Priority 2 Pauses                       0                    0
 Priority 3 Pauses                       0                    0
 Priority 4 Pauses                       0                    0
 Priority 5 Pauses                       0                    0
 Priority 6 Pauses                       0                    0
 Priority 7 Pauses                       0                    0
 Total Pause Frames                      0                    0
Interface 1/1/1 is up
 Admin state is up
 Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
 Flow-control: pfc rxtx-4,5
 Flow-control watchdog: disabled
 Statistics                             RX                   TX
```

```
-------------------- -------------------- --------------------
Priority 0 Pauses                     0                    0
Priority 1 Pauses                     0                    0
Priority 2 Pauses                     0                    0
Priority 3 Pauses                     0                    0
Priority 4 Pauses                     0                    0
Priority 5 Pauses                     0                    0
Priority 6 Pauses                     0                    0
Priority 7 Pauses                     0                    0
Total Pause Frames                    0                    0
```

Showing detailed flow control information with both PFC and flow control watchdog enabled:

```
switch# show interface 1/1/1 flow-control detail
Interface 1/1/1 is up
 Admin state is up
 Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
 Flow-control: pfc rxtx-4,5
 Flow-control watchdog: enabled
 Statistics                          RX                   TX
 -------------------- -------------------- --------------------
 Priority 0 Pauses                     0                    0
 Priority 1 Pauses                     0                    0
 Priority 2 Pauses                     0                    0
 Priority 3 Pauses                     0                    0
 Priority 4 Pauses                     0                    0
 Priority 5 Pauses                     0                    0
 Priority 6 Pauses                     0                    0
 Priority 7 Pauses                     0                    0
 Total Pause Frames                    0                    0
 Queue        Watchdog Timeouts
 ------------ -----------------
 Queue 0                      0
 Queue 1                      0
 Queue 2                      0
 Queue 3                      0
 Queue 4                      0
 Queue 5                      0
 Queue 6                      0
 Queue 7                      0
```

Showing detailed flow control information when flow control watchdog is enabled in the configuration but it could not be applied because the configured flow control mode is not compatible with watchdog:

```
switch# show interface 1/1/1 flow-control detail
Interface 1/1/1 is up
 Admin state is up
```

```
 Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
 Flow-control: llfc rx
 Flow-control watchdog: incompatible
```

Showing detailed flow control information when flow control watchdog is enabled in the configuration but could not be applied because a compatible flow control mode first requires a reboot:

```
switch# show interface 1/1/1 flow-control detail
Interface 1/1/1 is up
 Admin state is up
 Link state: up for 3 minutes (since Thu Apr 07 16:38:02 UTC 2022)
 Flow-control: off
 Flow-control watchdog: pending
```

Command History

Release	Modification
10.10	Examples updated with new and changed output elements.
10.08	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manage r (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

show interface link-diagnostics

Syntax

```
show interface [<IFNNAME>|<IFRANGE>] link-diagnostics
```

Description

Shows information about select link diagnostics, including MAC, PHY, and Transceiver diagnostics.

Parameter	Description
`<IFNAME>`	Specifies a interface name.
`<IFRANGE>`	Specifies the port identifier range.
link - diagnostics	Shows link diagnostics information.

Examples

> **NOTE**
>
> AOS-CX 10.18 adds Link Delay Information:
>
> - **Link-Up Failure**: The number of times a link-up was detected, but was lost when the link-up delay time expired.
> - **Link-Down Recovery**: The number of times a link-down was detected, but was recovered when the link-down delay time expired.
> -

Showing interface information for 1/1/12 link diagnostics:

```
switch# show interface 1/1/12 link-diagnostics
----------------------------------------------------------------------------
------
Interface 1/1/12          Current State            Changes        Last
Change
----------------------------------------------------------------------------
------
Port Status               Waiting for link
Link Down Reason          Local fault
Transceiver
No transceiver           --                           0         --
TX Fault                 --                           0         --
Transceiver error        --                           0         --
System error             --                           0         --
DOM error                --                           0         --
```

```
Configuring transceiver   --                             0         --
Transceiver disabled      --                             0         --
Transceiver unsupported   --                             0         --
MAC and PHY
TX Disabled               --                             0         --
Autoneg Incomplete        --                             0         --
No Signal Detected        --                             0         --
No RX Lock                True                           1         22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)
Remote Fault              --                             0         --
Local Fault               True                           1         22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)
Link State                Down                           1         16
seconds ago (Tue Nov 14 14:30:54 UTC 2023)
Link Delay Information                                Count         Last
Change
  Link-Up Failure                                         2         22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)
  Link-Down Recovery                                      1         16
seconds ago (Tue Nov 14 14:30:54 UTC 2023)

switch# show interface 1/1/12 link-diagnostics
----------------------------------------------------------------------------
------
Interface 1/1/12           Current State           Changes          Last
Change
----------------------------------------------------------------------------
------
Port Status                Link up
Link Down Reason           --
Transceiver
No transceiver             --                             0         --
TX Fault                   --                             0         --
Transceiver error          --                             0         --
System error               --                             0         --
DOM error                  --                             0         --
Configuring transceiver    --                             0         --
Transceiver disabled       --                             0         --
Transceiver unsupported    --                             0         --
MAC and PHY
TX Disabled                --                             2         22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)
Autoneg Incomplete         --                             0         --
```

```
No Signal Detected          --                             2         22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)
No RX Lock                  --                             2         22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)
Remote Fault                --                             2         21
seconds ago (Tue Nov 14 14:30:49 UTC 2023)
Local Fault                 --                             2         22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)
  Link State                Up                             1         16
seconds ago (Tue Nov 14 14:30:54 UTC 2023)
Link Delay Information                                 Count         Last
Change
  Link-Up Failure                                          2         22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)
  Link-Down Recovery                                       1         16
seconds ago (Tue Nov 14 14:30:54 UTC 2023)
```

Showing interface information for 1/1/7-1/1/8 link-diagnostics:

```
switch# show interface 1/1/7-1/1/8 link-diagnostics
----------------------------------------------------------------------------
----
Interface 1/1/7             Current State            Changes         Last
Change

----------------------------------------------------------------------------
---
Port Status                 Link up
Link Down Reason            --
Transceiver
No transceiver              --                             0         --
TX Fault                    --                             0         --
Transceiver error           --                             0         --
 System error               --                             0         --
DOM error                   --                             0         --
Configuring transceiver     --                             0         --
Transceiver disabled        --                             0         --
Transceiver unsupported     --                             0         --
MAC and PHY
TX Disabled                 --                             2         22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)
Autoneg Incomplete          --                             0         --
No Signal Detected          --                             2         22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)
 No RX Lock                 --                             2         22
```

```
seconds ago (Tue Nov 14 14:30:48 UTC 2023)
Remote Fault                --                              2          21
seconds ago (Tue Nov 14 14:30:49 UTC 2023)
 Local Fault                --                              2          22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)
 Link State                Up                               1          16
seconds ago (Tue Nov 14 14:30:54 UTC 2023)
Link Delay Information                                  Count          Last
Change
  Link-Up Failure                                           2          22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)
  Link-Down Recovery                                        1          16
seconds ago (Tue Nov 14 14:30:54 UTC 2023)
----------------------------------------------------------------------------
----
Interface 1/1/8             Current State             Changes          Last
Change
----------------------------------------------------------------------------
----
Port Status                Link up
Link Down Reason           --
Transceiver
No transceiver             --                               0          --
 TX Fault                  --                               0          --
Transceiver error          --                               0          --
System error               --                               0          --
 DOM error                 --                               0          --
 Configuring transceiver   --                               0          --
Transceiver disabled      --                               0          --
Transceiver unsupported   --                               0          --
 MAC and PHY
TX Disabled               --                               2          22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)
Autoneg Incomplete        --                               0          --
No Signal Detected        --                               2          22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)
No RX Lock                --                               2          22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)
 Remote Fault             --                               2          21
seconds ago (Tue Nov 14 14:30:49 UTC 2023)
Local Fault               --                               2          22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)
Link State                Up                               1          16
```

```
seconds ago (Tue Nov 14 14:30:54 UTC 2023)
Link Delay Information                                  Count          Last
Change
  Link-Up Failure                                           2          22
seconds ago (Tue Nov 14 14:30:48 UTC 2023)
  Link-Down Recovery                                        1          16
seconds ago (Tue Nov 14 14:30:54 UTC 2023)
```

Command History

Release	Modification
10.18	Introduces Link Delay Information.
10.14	Command introduced.

Command Information

Platforms	Command context	Authority
9300P	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show interface statistics`

Syntax

```
show interface [<IFNAME>|<IFRANGE>] statistics [non-zero] [human-readable]
show interface [<IFNAME>|<IFRANGE>] statistics monitor [non-zero] [human-
readable]
show interface [<IFNAME>|<IFRANGE>] error-statistics [non-zero] [human-
readable]
show interface [<IFNAME>|<IFRANGE>] error-statistics monitor [non-zero]
[human-readable]
show interface lag [<LAG-ID>] statistics [non-zero] [human-readable]
show interface lag [<LAG-ID>] statistics monitor [non-zero] [human-readable]
show interface lag [<LAG-ID>] error-statistics [non-zero] [human-readable]
show interface lag [<LAG-ID>] error-statistics monitor [non-zero] [human-
readable]
```

```
show interface vxlan <VXLAN-ID> statistics [non-zero] [human-readable]
```

Description

Shows statistics for switch interfaces such as packets transmitted and received, bytes transmitted and received, broadcast and multicast packets.

Parameter	Description
`<IFNAME>`	Specifies a interface name.
`<IFRANGE>`	Specifies the port identifier range.
`LAG`	Shows LAG interface information.
`<LAG-ID>`	Specifies the LAG number. Range: 1 - 256
`VXLAN`	Shows the VXLAN interface information.
`<VXLAN-ID>`	Specifies the VXLAN interface identifier. Default: 1
`monitor`	Continuously monitor interface statistics.
`human-readable`	Shows statistics rounded to the nearest power of 1000, for exa mple, 1K, 345M, 2G.
`non-zero`	Shows only non zero statistics.

Examples

Showing statistics of all interfaces:

```
show interface statistics
----------------------------------------------------------------------------------------------------------------------------------------------------
Interface                 RX Bytes  RX Packets  RX Drops  TX Bytes  TX Packets  TX Drops  RX Broadcast  RX Multicast  TX Broadcast  TX Multicast  RX Pause  TX Pause
----------------------------------------------------------------------------------------------------------------------------------------------------
1/1/1                      2727136        1975         0     17796         195         0            82          1788            96            54         0         0
1/1/10                           0           0         0         0           0         0             0             0             0             0         0         0
1/1/11                           0           0         0         0           0         0             0             0             0             0         0         0
1/1/12                           0           0         0         0           0         0             0             0             0             0         0         0
...
1/1/30       - lag1              0           0         0     11271          92         0             0             0             0            51         0         0
1/1/31       - lag2           2360          25        50   2732119        2040         0             0             0           178          1839         0         0
1/1/32       - lag2              0           0         0     11373          93         0             0             0             0            51         0         0
vlan1                            0           0         0         0           0         0             0             0             0             0         0         0
```

Showing statistics of all interfaces with only non-zero statistics:

```
show interface statistics non-zero
----------------------------------------------------------------------------------------------------------------------------------------------------
Interface                       RX Bytes  RX Packets  RX Drops  TX Bytes  TX Packets  TX Drops RX Broadcast  RX Multicast  TX Broadcast TX Multicast  RX Pause  TX Pause
----------------------------------------------------------------------------------------------------------------------------------------------------
1/1/1                            2727136        1975         0     17796         195         0           82          1788            96           54         0         0
1/1/30       - lag1                    0           0         0     11271          92         0            0             0             0           51         0         0
1/1/31       - lag2                 2360          25        50   2732119        2040         0            0             0           178         1839         0         0
1/1/32       - lag2                    0           0         0     11373          93         0            0             0             0           51         0         0
```

Showing statistics of all interfaces in the human-readable format:

```
show interface statistics human-readable
----------------------------------------------------------------------------------------------------------
Interface            RX Bytes  RX Pkts  RX Drops  TX Bytes   TX Pkts  TX Drops  RX Bcast  RX Mcast  TX Bcast  TX Mcast  RX Pause  TX Pause
----------------------------------------------------------------------------------------------------------
1/1/1                    744M     577K         0      745M      578K         0         0         0        73       287         0         0
1/1/2                    474M     367K         0      475M      369K         0         0         0        73       288         0         0
1/1/3                       0        0         0         0         0         0         0         0         0         0         0         0
```

Showing statistics of a single interfaces:

```
show interface 1/1/2 statistics
----------------------------------------------------------------------------------------------------------------------------------------------------
Interface                       RX Bytes  RX Packets  RX Drops  TX Bytes  TX Packets  TX Drops  RX Broadcast  RX Multicast  TX Broadcast  TX Multicast  RX Pause  TX Pause
----------------------------------------------------------------------------------------------------------------------------------------------------
1/1/2                            2725080        1931         0     25877         253         0            21          1788            65            55         0         0
```

Showing statistics of all members of a LAG interface:

```
show interface lag1 statistics
----------------------------------------------------------------------------------------------------------------------------------------------------
Interface                      RX Bytes  RX Packets  RX Drops  TX Bytes  TX Packets  TX Drops  RX Broadcast  RX Multicast  TX Broadcast  TX Multicast  RX Pause  TX Pause
----------------------------------------------------------------------------------------------------------------------------------------------------
1/1/3        - lag1                2424          26         0   2734082        2062         0             0             0           191          1848         0         0
1/1/30       - lag1                   0           0         0     12383         100         0             0             0             0            59         0         0
lag1                               2424          26         0   2746465        2162         0             0             0           191          1907         0         0
```

Showing error statistics of all interfaces:

```
show interface error-statistics
------------------------------------------------------------------------------------
Interface                          RX Errors   TX Errors    Giants    Runts    CRC/FCS   Collisions
------------------------------------------------------------------------------------
1/1/1                                    190          20    100647        0          0            0
1/1/10                                     0           0       100      290       7165          949
1/1/11                                     0           0         0        0          0            0
1/1/12                                     0           0         0        0          0            0
...
1/1/30 - lag1                           1500         500     45800        0          0            0
1/1/31 - lag2                              0           0        11       27          0            0
1/1/32 - lag2                              0           0         0        0          6           18
```

Showing monitor statistics:

> **NOTE**
> The rows and columns of show interface monitor statistics depends on the length of width of the client terminal. The CLI can be navigated using the arrow keys as well as the PageUp, PageDown, Home, and End keys.

```
show interface statistics monitor
-----------------------------------------------------------------
Interface                         RX Bytes          RX Packets >>
-----------------------------------------------------------------
1/1/1                        3440525421984         53758209526
1/1/2                        3440526607008         53758228042
1/1/3                        3440527785312         53758246453
1/1/30                       3440559671264         53758744653
1/1/31                       3440560851680         53758763098
1/1/32                       3440562028704         53758781489

                                                 help: ?, quit: q
```

```
Help for Interface Monitor

f    Toggle full statistics
h    Toggle human-readable mode
n    Toggle non-zero mode
r    Toggle rate display

c    Clear interface statistics
        Does not apply to rates

Arrows, PgUp, PgDn, Home, End
        Navigate interface statistics

Delay:2
                                                                   help: ?, quit: q
```

Showing monitor error statistics in human-readable format:

```
show interface 1/1/1-1/1/3,1/1/30-1/1/32 error-statistics monitor human-readable
-------------------------------------------------------------------------------------------
Interface                 RX Errors   TX Errors   RX Giants    RX Runts     CRC/FCS  Collisions
-------------------------------------------------------------------------------------------
1/1/1                             0           0           0           0           0           0
1/1/2                             0           0           0           0           0           0
1/1/3                             0           0           0           0           0           0
1/1/30                            0           0           0           0           0           0
1/1/31                            0           0           0           0           0           0
1/1/32                            0           0           0           0           0           0


Human-readable                                                             help: ?, quit: q
```

```
Help for Interface Monitor

h    Toggle human-readable mode
n    Toggle non-zero mode

c    Clear interface statistics
        Does not apply to rates

Arrows, PgUp, PgDn, Home, End
        Navigate interface statistics

Delay:2
                                                              help: ?, quit: q
```

Command History

Release	Modification
10.11	Added `moitor` parameter.
10.10	Added `human-readable` parameter.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show interface transceiver`

Syntax

```
show interface [<INTERFACE-ID>] transceiver [detail | threshold-violations]
[vsx-peer]
```

Description

Displays information about transceivers present in the switch. The information shown varies for different transceiver types and manufacturers. Only basic information is shown for unsupported HPE and third-party transceivers installed in the switch and they are also identified with an asterisk in the output.

Parameter	Description
<INTERFACE-ID>	Specifies the name or range of an interface on the switch. Use t he format **member/slot/port** (for example, **1/3/1**).
`detail`	Show detailed information for the interfaces.
`threshold-violations`	Show threshold violations for transceivers.
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output f rom the VSX peer switch is not displayed. This parameter is avai lable on switches that support VSX.

Example

Showing summary transceiver information with identification of unsupported transceivers:

```
switch(config)# show interface transceiver
---------------------------------------------------------------------
Port      Type           Product      Serial          Part
Number        Number          Number
---------------------------------------------------------------------
1/1/1     SFP+SR         J9150A       MYxxxxxxxx      1990-3657
1/1/2     SFP+ER*        --           --              --
1/2/1     QSFP+SR4       JH233A       MYxxxxxxxx      2005-1234
1/2/2     QSFP+ER4*      --           --              --
1/3/1     SFP28DAC3      844477-B21   MYxxxxxxxx      77fc-7ce7
* unsupported transceiver
switch(config)# show interface transceiver
--------------------------------------------------------------------
Port      Type           Product      Serial          Part
                 Number        Number          Number
--------------------------------------------------------------------
1/1/15    SFP+SR         J9150D       xxxxxxxxxx      1990-4634
1/1/16    SFP+SR         J9150D       xxxxxxxxxx      1990-4634
```

Showing detailed transceiver information:

```
switch(conf#) show interface transceiver detailing
Transceiver in 1/1/1
   Interface Name     : 1/1/1
   Type               : SFP+SR
   Connector Type     : LC
```

```
   Wavelength          : 850nm
   Transfer Distance   : 0m (SMF), 30m (OM1), 80m (OM2), 300m (OM3)
   Diagnostic Support  : DOM
   Product Number      : J9150A
   Serial Number       : MYxxxxxxx
   Part Number         : 1990-3657
Status
   Temperature : 47.65C
   Voltage     : 3.31V
   Tx Bias     : 8.40mA
   Rx Power    : 0.08mW, -10.96dBm
   Tx Power    : 0.56mW, -2.49dBm
Recent Alarms :
   Rx power low alarm
   Rx power low warning
   Recent Errors :
   Rx loss of signal
Transceiver in 1/1/2
   Interface Name      : 1/1/2
   Type                : unknown
   Connector Type      : ??
   Wavelength          : ??
   Transfer Distance   : ??
   Diagnostic Support  : ??
   Product Number      : ??
   Serial Number       : ??
   Part Number         : ??
Transceiver in 1/2/1
   Interface Name      : 1/2/1
   Type                : QSFP+SR4
   Connector Type      : MPO
   Wavelength          : 850nm
   Transfer Distance   : 0m (SMF), 0m (OM1), 0m (OM2), 100m (OM3)
   Diagnostic Support  : DOM
   Product Number      : JH233A
   Serial Number       : MYxxxxxxx
   Part Number         : 2005-1234
Status
   Temperature : 44.46C
   Voltage     : 3.30V
----------------------------------------------
          Tx Bias  Rx Power     Tx Power
Channel#  (mA)     (mW/dBm)     (mW/dBm)
```

```
-----------------------------------------------
1          6.12     0.00, -inf   0.63, -1.95
2          6.04     0.00, -inf   0.63, -2.00
3          6.51     0.00, -inf   0.60, -2.16
4          6.19     0.00, -inf   0.63, -1.94
Recent Alarms :
 Channel 1 :
   Rx power low alarm
   Rx power low warning
 Channel 2 :
   Rx power low alarm
   Rx power low warning
 Channel 3 :
   Rx power low alarm
   Rx power low warning
 Channel 4 :
   Rx power low alarm
   Rx power low warning
Recent Errors :
 Channel 1 :
   Rx Loss of Signal
 Channel 2 :
   Rx Loss of Signal
 Channel 3 :
   Rx Loss of Signal
 Channel 4 :
   Rx Loss of Signal
Transceiver in 1/2/2
  Interface Name     : 1/2/2
  Type               : unknown
  Connector Type     : ??
  Wavelength         : ??
  Transfer Distance  : ??
  Diagnostic Support : ??
  Product Number     : ??
  Serial Number      : ??
  Part Number        : ??
Transceiver in 1/3/1
  Interface Name     : 1/3/1
  Type               : SFP28DAC3
  Connector Type     : Copper Pigtail
  Transfer Distance  : 0.00km (SMF), 0m (OM1), 0m (OM2), 0m (OM3)
  Diagnostic Support : None
```

```
  Product Number      : 844477-B21
  Serial Number       : MYxxxxxxx
  Part Number         : 77fc-7ce7
switch(config)# show interface transceiver detail
Transceiver in 1/1/15
 Interface Name      : 1/1/15
 Type                : SFP+SR
 Connector Type      : LC
 Wavelength          : 850nm
 Transfer Distance   : 0.00km (SMF), 20m (OM1), 80m (OM2), 300m (OM3)
 Diagnostic Support  : DOM
 Product Number      : J9150D
 Serial Number       : xxxxxxxxxx
 Part Number         : 1990-4634
Status
 Temperature : 30.38C
 Voltage     : 3.26V
 Tx Bias     : 5.54mA
 Rx Power    : 0.56mW, -2.52dBm
 Tx Power    : 0.62mW, -2.08dBm
 Recent Alarms:
 Recent Errors:
Transceiver in 1/1/16
 Interface Name      : 1/1/16
 Type                : SFP+SR
 Connector Type      : LC
 Wavelength          : 850nm
 Transfer Distance   : 0.00km (SMF), 20m (OM1), 80m (OM2), 300m (OM3)
 Diagnostic Support  : DOM
 Product Number      : J9150D
 Serial Number       : xxxxxxxxxx
 Part Number         : 1990-4634
Status
 Temperature : 30.62C
 Voltage     : 3.28V
 Tx Bias     : 5.64mA
 Rx Power    : 0.61mW, -2.15dBm
 Tx Power    : 0.59mW, -2.29dBm
Recent Alarms:
Recent Errors:
```

Showing detailed transceiver information with identification of unsupported transceivers:

```
Transceiver in 1/1/2
 Interface Name      : 1/1/2
```

```
 Type                    : SFP+ER (unsupported)
 Connector Type          : LC
 Wavelength              : 3590nm
 Transfer Distance       : 80m (SMF), 0m (OM1), 0m (OM2), 0m (OM3)
 Diagnostic Support      : DOM
 Vendor Name             : INNOLIGHT
 Vendor Part Number      : TR-PX15Z-NHP
 Vendor Part Revision: 1A
 Vendor Serial number: MYxxxxxxx
Status
 Temperature : 28.88C
 Voltage     : 3.30V
 Tx Bias     : 65.53mA
 Rx Power    : 0.00mW, -inf
 Tx Power    : 1.47mW, 1.67dBm
Recent Alarms:
 Rx Power low alarm
 Rx Power low warning
 Recent Errors:
```

Showing transceiver threshold-violations:

```
switch(config)# show interface transceiver threshold-violations
-----------------------------------------------------
Port      Type        Channel  Type(s) of Recent
                               Threshold Violation(s)
-----------------------------------------------------
1/1/1     SFP+SR                Tx bias high warning
                50.52 mA > 40.00 mA
1/1/2     SFP+ER*               ??
1/2/1     QSFP+SR4    1         Tx power low alarm
                -17.00 dBm < -0.50 dBm
              2         Tx bias low warning
                3.12 mA < 4.00 mA
1/2/2     QSFP+ER4*             ??
1/3/1     SFP28DAC3             n/a
* unsupported transceiver
switch(config)# show interface transceiver threshold-violations
----------------------------------------------------------------
Port      Type           Channel#  Recent Threshold Violations
----------------------------------------------------------------
1/1/15    SFP+SR                   none
1/1/16    SFP+SR                   none
switch#
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show interface supported-transceivers`

Syntax

```
show interface {<IFNAME>|<IFRANGE>} supported-transceivers
```

Description

Displays transceivers supported by a specific, or all, interface(s) in the switch. Information varies and is dependent on transceiver types, part number, and short description or interfaces.

Parameter	Description
`IFNAME`	Specifies the name of an interface on the switch. Use the following format: member/slot/port. Example: 1/3/1
`IFRANGE`	Specifies the name of an interface range on the switch. Use the following format: member/slot/port. Example: 1/1/1 - 1/1/2

Example

Showing a full switch interface or chassis switch with multiple line modules:

```
switch# show interface supported-transceivers
--------------------------------------------------------------------------
Type                 Product           Supported
                     Number            Interfaces
--------------------------------------------------------------------------
1G-BT                JL747             1/1/33-1/1/34
1G-BT                J8177             1/1/33-1/1/34
1G-LH                J4860             1/1/33-1/1/34
IG-LX                J4859             1/1/33-1/1/34
8x25G-DAC2           S4B39
1/1/2,1/1/4,1/1/6,1/1/8,1/1/10,1/1/12,1/1/14,

1/1/16,1/1/18,1/1/20,1/1/22,1/1/24,1/1/26,
                                       1/1/28,1/1/30,1/1/32
QSA28                845970            1/1/1-1/1/32
Note: For full product numbers (SKUs) with revision letters,
refer to the transceiver guide.
switch# show interface supported-transceivers
--------------------------------------------------------------------------
Type                 Product           Supported
                     Number            Interfaces
--------------------------------------------------------------------------
100M-FX              J9054             1/10/1-1/10/24,1/12/1-1/12/48
1G-BT                JL747
1/5/49-1/5/52,1/7/25-1/7/28,1/8/25-1/8/28,

1/9/49-1/9/52,1/10/1-1/10/28,1/12/1-1/12/48
10G-BT               JL563
1/5/49-1/5/52,1/6/1-1/6/32,1/7/25-1/7/28,

1/8/25-1/8/28,1/9/49-1/9/52,1/10/1-1/10/28,
                                       1/12/1-1/12/48
10G-BiDi-D           R9X54
1/4/1-1/4/12,1/6/1-1/6/36,1/11/1-1/11/12,
                                       1/12/1-1/12/48
10G-SR               J9150
1/4/1-1/4/12,1/5/49-1/5/52,1/6/1-1/6/36,

1/7/25-1/7/28,1/8/25-1/8/28,1/9/49-1/9/52,

1/10/1-1/10/28,1/11/1-1/11/12,1/12/1-1/12/48
10G-SR               JL782
1/4/1-1/4/12,1/6/33-1/6/36,1/11/1-1/11/12
```

```
25G-AOC15            R0Z21              1/6/1-1/6/32
25G-AOC7             R0M45              1/6/1-1/6/32
25G-BiDi-10-U        S1C98
1/4/1-1/4/12,1/5/49-1/5/52,1/6/1-1/6/36,
8x50G-DAC2           S4B42
1/5/49-1/5/52,1/6/1-1/6/32,1/7/25-1/7/28,

1/8/25-1/8/28,1/9/49-1/9/52,1/10/25-1/10/28,
                                        1/12/1-1/12/48
QSA28                845970
1/4/1-1/4/12,1/6/33-1/6/36,1/11/1-1/11/12
Note: For full product numbers (SKUs) with revision letters,
refer to the transceiver guide.
```

Showing a single interface:

```
switch# show interface 1/1/1 supported-transceivers
-------------------------------------------------------------------------
Type                 Product            Description
                     Number
-------------------------------------------------------------------------
Interface 1/1/1
10G-BiDi-D @         R9X54              10G BX-D 40km
10G-BiDi-U @         R9X55              10G BX-U 40km
100G-LR4             JL310              100G QSFP28 LR4
100G-SR1.2           S4B44              100G SR1.2 Q28 LC
100G-SR2             S1C93              100G MPO SR2
100G-SR4             JL309              100G QSFP28 SR4
QSA28                845970             --
@ This transceiver is only supported when inserted in a QSA
For full product numbers (SKUs) with revision letters, refer to the
Transceiver guide.
```

Showing a range of interfaces:

```
switch# show interface 1/1/1-1/1/2 supported-transceivers
-------------------------------------------------------------------------
Type                 Product            Description
                     Number
-------------------------------------------------------------------------
Interface 1/1/1
10G-BiDi-D @         R9X54              10G BX-D 40km
10G-BiDi-U @         R9X55              10G BX-U 40km
10G-ER @             S2P32              10G SFP+ ER
100G-SR2             S1C93              100G MPO SR2
100G-SR4             JL309              100G QSFP28 SR4
QSA28                845970             --
```

```
Interface 1/1/2
10G-BiDi-D @       R9X54             10G BX-D 40km
10G-BiDi-U @       R9X55             10G BX-U 40km
10G-ER @           S2P32             10G SFP+ ER
25G-BiDi-10-D @    S1C96             25G LC BiDi 10km-D
25G-SR @           S2P33             25G SFP28 SR
40G-eSR4           JH233             QSFP+ eSR4
100G-BiDi          845972            --
100G-SR4           JL309             100G QSFP28 SR4
QSA28              845970            --
@ This transceiver is only supported when inserted in a QSA
For full product numbers (SKUs) with revision letters, refer to the
Transceiver guide.
```

Showing a chassis switch with muliple line modules:

```
switch# show interface supported-transceivers
--------------------------------------------------------------------------
Type             Product     Supported
             Number      Interfaces
--------------------------------------------------------------------------
100M-FX          J9054       1/10/1-1/10/24,1/12/1-1/12/48
1G-BT            JL747       1/5/49-1/5/52,1/7/25-1/7/28,1/8/25-1/8/28,
       1/9/49-1/9/52,1/10/1-1/10/28,1/12/1-1/12/48
10G-BT           JL563       1/5/49-1/5/52,1/6/1-1/6/32,1/7/25-1/7/28,
                             1/8/25-1/8/28,1/9/49-1/9/52,1/10/1-1/10/28,
                             1/12/1-1/12/48
10G-BiDi-D       R9X54       1/4/1-1/4/12,1/6/1-1/6/36,1/11/1-1/11/12,
                             1/12/1-1/12/48
10G-SR           J9150       1/4/1-1/4/12,1/5/49-1/5/52,1/6/1-1/6/36,
                             1/7/25-1/7/28,1/8/25-1/8/28,1/9/49-1/9/52,
                             1/10/1-1/10/28,1/11/1-1/11/12,1/12/1-1/12/48
10G-SR           JL782       1/4/1-1/4/12,1/6/33-1/6/36,1/11/1-1/11/12
25G-AOC15        R0Z21       1/6/1-1/6/32
25G-AOC7         R0M45       1/6/1-1/6/32
25G-BiDi-10-U    S1C98       1/4/1-1/4/12,1/5/49-1/5/52,1/6/1-1/6/36,
8x50G-DAC2       S4B42       1/5/49-1/5/52,1/6/1-1/6/32,1/7/25-1/7/28,
                             1/8/25-1/8/28,1/9/49-1/9/52,1/10/25-1/10/28,
                             1/12/1-1/12/48
QSA28            845970      1/4/1-1/4/12,1/6/33-1/6/36,1/11/1-1/11/12
For full product numbers (SKUs) with revision letters, refer to the
Transceiver guide.
```

Showing a VSF stack:

```
switch# show interface supported-transceivers
 --------------------------------------------------------------------------
```

```
Type            Product          Supported
                Number           Interfaces
 ---------------------------------------------------------------------------
100M-FX         J9054            1/1/1-1/1/24,2/1/1-2/1/48
1G-BT           JL747            1/1/49-1/1/52,3/1/25-3/1/28,4/1/25-4/1/28
10G-BT          JL563            1/1/49-1/1/52,2/1/1-2/1/32,3/1/25-3/1/28,
                                 4/1/25-4/1/28
10G-AOC15       721076           1/1/13,1/1/15-1/1/27,1/1/41-1/1/42
10G-AOC5        Q9S66            1/1/13,1/1/15-1/1/27,1/1/41-1/1/42
25G-AOC15       R0Z21            3/1/1-3/1/32
QSA28           845970           4/1/1-4/1/12,2/1/33-2/1/36,1/1/1-1/1/12
For full product numbers (SKUs) with revision letters, refer to the
Transceiver guide.
```

Showing an empty chassis:

```
switch# show interface supported-transceivers
No available interfaces.
```

Command History

Release	Modification
10.16.1000	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

show interface utilization

Syntax

```
show interface [<IFNNAME>|<IFRANGE>] utilization [non-zero]
```

Description

Displays physical port throughput and utilization.

Parameter	Description
<IFNAME>	Specifies an interface name.
<IFRANGE>	Specifies the port identifier range.
`utilization`	Displays utilization statistics.
`non-zero`	Displays non - zero statistics

Examples

The following example shows port utilization of all interfaces:

```
switch# show interface utilization
-------------------------|------------------------|------------------------|
---------------------------|----------------------
Interval |             RX          |             TX          |    Total
(RX+TX)          |
Interface        seconds |   Mbps  KPkt/s  Util % |   Mbps  KPkt/s  Util %
|    Mbps   KPkt/s   Util % | Description
-------------------------|------------------------|------------------------|
---------------------------|----------------------
1/1/1                 300  9578.02  788.70   95.78    25.70   45.89
0.26   9603.72   834.59    96.04   Aruba-AP
1/1/2                 300    25.71   45.90    0.26  9581.09  788.96
95.81   9606.80   834.86    96.07   Aruba2530-AP-conce...
1/1/3  - lag123       300     0.00    0.00    0.00     0.00    0.00
0.00      0.00     0.00      0.00   ISL: SWRTS-0064-1
1/1/4                 300  9261.79  804.52   92.62  9496.70  823.97   94.97
18758.50  1628.48   187.58   Backup data center...
1/1/5                 300  9496.70  823.97   94.97  9261.79  804.52   92.62
18758.50  1628.48   187.58   --
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show ip interface`

Syntax

```
show ip interface <INTERFACE-ID>|{lag|loopback|vlan|tunnel <ID>}
```

Description

Shows status and configuration information for an IPv4 interface.

Parameter	Description
`<INTERFACE-ID>`	Specifies the name of an interface. Format: **member/slot/port**.
`lag <ID>`	Show LAG interface information, where <ID> is 1 - 256.
`loopback <ID>`	Show loopback interface information, where <ID> is 0 - 255.
`tunnel <ID>`	Show tunnel interface information, where <ID> is 1 - 255.
`vlan <ID>`	Show VLAN interface information, where <ID> is 1 - 4094.

Example

```
switch# show ip interface 1/1/1
Interface 1/1/1 is up
Admin state is up
Hardware: Ethernet, MAC Address: 70:72:cf:fd:e7:b4
IP MTU 1500
IP TCP MSS 1460
IPv4 address 192.168.1.1/24
L3 Counters: Rx Enabled, Tx Enabled
Statistic                          RX                    TX
Total
---------------- -------------------- --------------------
--------------------
L3 Packets                          0
0                    0
L3 Bytes                            0
0                    0
switch# show ip interface 1/1/1
Interface 1/1/1 is up
Admin state is up
Hardware: Ethernet, MAC Address: 70:72:cf:fd:e7:b4
IP MTU 1500
IP TCP MSS 1460
IP Directed Broadcast is Disabled
IPv4 address 2.2.2.2/32 [unnumbered to loopback1]
L3 Counters: Rx Disabled, Tx Disabled
switch# show ip interface loopback 1
Interface loopback1 is up
Admin state is up
Hardware: Loopback
IP Directed Broadcast is Disabled
IPv4 address 192.168.1.1/24
```

Command History

Release	Modification
10.14.1000	Command output can display information for an IP unnumbered interface and IP TCP MSS configuration.
10.07 or earlier	--

show ip source-interface

Syntax

```
show ip source-interface {sflow |  tftp |  radius | tacacs | all} [vrf
<VRF-NAME>]
     [vsx-peer]
```

Description

Shows single source IP address configuration settings.

Parameter	Description
`sflow	tftp	radius	tacacs	all`	Shows single source IP address configuration settings for a spe cific protocol. The **all** option shows the global setting that appli es to all protocols that do not have an address set.
`vrf <VRF-NAME>`	Specifies the name of a VRF.
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output f rom the VSX peer switch is not displayed. This parameter is avai lable on switches that support VSX.

Examples

```
Showing single source IP address configuration settings for sFlow:
switch# show ip source-interface sflow
Source-interface Configuration Information
----------------------------------------
Protocol        Source Interface
--------        ----------------
sflow            10.10.10.1
```

Showing single source IP address configuration settings for all protocols:

```
switch# show ip source-interface all
Source-interface Configuration Information
----------------------------------------
Protocol        Source Interface
--------        ----------------
all             1/1/1
switch# show ip source-interface all
Source-interface Configuration Information
```

```
--------------------------------------------------------------
Protocol        Src-Interface       Src-IP                   VRF
--------------------------------------------------------------
all             vlan2               2.2.2.2             default
switch#
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution righ ts for this command.

`show ipv6 interface`

Syntax

```
show ipv6 interface <INTERFACE-ID>
             [vsx-peer]
```

Description

Shows status and configuration information for an IPv6 interface.

Parameter	Description
`<INTERFACE-ID>`	Specifies an interface ID. Format: **member/slot/port**.
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output f

Parameter	Description
	rom the VSX peer switch is not displayed. This parameter is avai lable on switches that support VSX.

Examples

```
switch# show ipv6 interface 1/1/1
Interface 1/1/1 is up
  Admin state is up
  IPv6 address:
    2001:0db8:85a3:0000:0000:8a2e:0370:7334/24 [VALID]
  IPv6 link-local address: fe80::1e98:ecff:fee3:e800/64 (default)[VALID]
  IPv6 virtual address configured: none
  IPv6 multicast routing: disable
  IPv6 Forwarding feature: enabled
  IPv6 multicast groups locally joined:
    ff02::ff70:7334  ff02::ffe3:e800  ff02::1  ff02::1:ff00:0
    ff02::2
  IPv6 multicast (S,G) entries joined: none
  IPv6 MTU: 1524 (using link MTU)
  IPv6 TCP MSS 1440
  IPv6 unicast reverse path forwarding: none
  IPv6 load sharing: none
  RX
          0 packets, 0 bytes
  TX
          0 packets, 0 bytes
switch# show ipv6 interface vlan2
Interface vlan2 is up
  Admin state is up
    IPv6 address:
      2001::1/64 [VALID]
    IPv6 link-local address: fe80::883a:3080:247:c1c0/64 [VALID]
    IPv6 Forwarding feature: enabled
    IPv6 multicast groups locally joined:
      ff02::1  ff02::1:ff00:1  ff02::1:ff47:c1c0  ff02::1:ff00:0
      ff02::2
    IPv6 MTU 1500
    IPv6 TCP MSS 1440
    IPv6 unicast reverse path forwarding: none
    IPv6 load sharing: none
switch# show ipv6 interface 1/1/14.1
Interface 1/1/14.1 is up
```

```
  Admin state is up
    IPv6 address:
      30::1/64 [VALID]
    IPv6 link-local address: fe80::b86a:97c0:122:2f42/64 [VALID]
    IPv6 virtual address configured: none
    IPv6 multicast routing: disable
    IPv6 Forwarding feature: enabled
    IPv6 multicast groups locally joined:
      ff02::1  ff02::1:ff22:2f42  ff02::1:ff00:1  ff02::1:ff00:0
      ff02::2
    IPv6 multicast (S,G) entries joined: none
    IPv6 MTU 1500
    IPv6 TCP MSS 1440
    IPv6 unicast reverse path forwarding: none
    IPv6 load sharing: none
Encapsulation dot1q ID: 20
switch# show ipv6 interface lag2.1
Interface lag2.1 is up
  Admin state is up
    IPv6 address:
      40::1/64 [VALID]
    IPv6 link-local address: fe80::b86a:97c0:122:2f42/64 [VALID]
    IPv6 virtual address configured: none
    IPv6 multicast routing: disable
    IPv6 Forwarding feature: enabled
    IPv6 multicast groups locally joined:
      ff02::1  ff02::1:ff22:2f42  ff02::1:ff00:1  ff02::1:ff00:0
      ff02::2
    IPv6 multicast (S,G) entries joined: none
    IPv6 MTU 1500
    IPv6 TCP MSS 1440
    IPv6 unicast reverse path forwarding: none
    IPv6 load sharing: none
  Encapsulation dot1q ID: 30
```

Command History

Release	Modification
10.14.1000	Command output can display information for an IP unnumbered interface and I Pv6 TCP MSS configuration.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show ipv6 source-interface`

Syntax

```
show ipv6 source-interface <PROTOCOL> [{all-vrfs | vrf <VRF-NAME>}]
     [vsx-peer]
```

Description

Shows single source IPv6 address configuration settings.

Parameter	Description
`<PROTOCOL>`	Specifies the protocol to configure. • **all**: Selects the source for all protocols covered by this com mand. • **central**: Selects HPE Aruba Networking Central. • **dhcp_relay**: Selects DHCP relay. When you configure a dhc p_relay source interface, you must also enable DHCP relay Option 82 using the **dhcp - relay option 82source - inte rface** command. • **dns**: Selects DNS. • **http**: Selects HTTP. • **ipfix**: Selects ipfix. Configures source interface for IPFIX. • **ntp**: Selects NTP. • **radius**: Selects RADIUS. • **sflow**: Selects sFLow. • **sftp - scp**: Selects SFTP and SCP. • **ssh - client**: Selects SSH Client. • **syslog**: Selects the source for syslog packets. • **tacacs**: Selects the source for TACACS packets.

Parameter	Description
	• **tftp**: Selects TFTP.
`all-vrfs	vrf <VRF-NAME>`	Specifies the name of a VRF.
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output f rom the VSX peer switch is not displayed. This parameter is avai lable on switches that support VSX.

Examples

Showing single source IPv6 address configuration settings for sFlow:

```
switch# show ipv6 source-interface sflow
Source-interface Configuration Information
---------------------------------------
Protocol        Source Interface
--------        ----------------
 sflow            2001:DB8::1
```

Showing single source IPv6 address configuration settings for all protocols:

```
switch# show ipv6 source-interface all
Source-interface Configuration Information
----------------------------------------
Protocol        Source Interface
--------        ----------------
 all             1/1/1
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

show system interface-group

Syntax

```
show system interface-group
    detail
    line-module <slot-id>

    member <member-id>
```

Description

Displays the configuration and status of interface groups.

Parameters

Parameter	Description
detail	Show detailed interface group information
line-module *<slot-id>*	Show information for a line module slot ID (for example, 1/1).
member *<member-id>*	Show information for a single VSF member
parameter	description.

Examples

NOTE

The output in the following commands vary, depending on switch model.

Show basic interface group status

```
switch# show system interface-group
------------------------------------------------
Group  Speed  Member Ports      Mismatched Ports
------------------------------------------------
1      10g    1/1/1-1/1/12
2      25g    1/1/13-1/1/24     1/1/13
3      25g    1/1/25-1/1/36
4      25g    1/1/37-1/1/48
```

Show detailed interface group status, including allowed speeds of each.

```
switch# show system interface-group detail
----------------------------------------------------------------------
```

```
Allowed                Mismatched
Module  Group  Ports      Option    Speeds                 Ports
----------------------------------------------------------------------
1/6     1      1-8          25g @+*   10g,25g
50g      50g
1/6     2      9-16         25g   *   10g,25g
50g @+   50g                    9-16
1/6     3      17-24        25g @+*   10g,25g
50g      50g
1/6     4      25-32        25g   *   10g,25g
50g @+   50g
@ Currently applied setting
+ Configured setting
* Default setting
```
```

| Release | Modification |
|---|---|
| 10.16.1010 | The **map** parameter introduced on 8325 Swith series. |
| 10.12 | Command introduced |

### Command Information

| Platforms | Command context | Authority |
|---|---|---|
| 8400 | `config` | Administrators or local user group members with exe cution rights for this command. |

## `shutdown`

### Syntax

```
shutdown
no shutdown
```

Description

Disables an interface. Interfaces are disabled by default when created.
```

The **no** form of this command enables an interface.

Examples

Disabling an interface:

```
switch(config-if)# shutdown
```

Enabling an interface:

```
switch(config-if)# no shutdown
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`split`

Syntax

```
split [<COUNT>][<SPEED>][confirm]
no split [confirm]
```

Description

Splits a port into multiple interfaces. Only ports capable of supporting breakout cables or SR4/eSR4/eDR4 optics can be split.

Parameter	Description
	Specifies the number of child interfaces to activate upon splittin g the port. Default: 4.

Parameter	Description
<COUNT>	
<SPEED>	Specifies the speed for the child interfaces.
`confirm`	Specifies the confirmation of port splitting.

Usage

- Some switch interfaces support different split counts depending on the installed transceiver. For these interfaces, the number of child interfaces to activate can be specified. If omitted, the default is 4. For transceivers capable of supporting multiple split modes, the closest mode with enough lanes is used.
- Some transceivers also support multiple split modes with different speeds. For example, 2x200G or 2x100G. When a speed is not specified, the highest available speed for the split count is used. To select a different split mode with a lower speed, the desired speed must be specified.

The splittable ports for all models are shown in the table below:

Model	Description	Port info
8400X modules • JL365A • JL366A	8400X 8p 40G QSFP+ Adv Modul e 8400X 6p 40G/100G QSFP28 Ad v Module	1 - 8 (40G) 1 - 6 Only capable of 40G split in to 4 x 10G JL366A modules do not have 25G MACs to support split 100G
8400X modules • JL365A • JL366A	8400X 8p 40G QSFP+ Adv Mod 8400X 6p 40G/100G QSFP28 Ad v Mod	1 - 8 (40G) 1 - 6 Only capable of 40G split in to 4 x 10G JL366A modules do not have 25G MACs to support split 100G

Examples

Splitting an interface:

```
switch(config-if)# interface 1/1/1
switch(config-if)# split
This command will disable the specified port, clear its configuration,
```

```
and split it into multiple interfaces. The split interfaces will not
be available until the next system or line module reboot.
Continue (y/n)? y
switch(config-if)# show interface brief
----------------------------------------------------------------------------
--------
Port     Native Mode    Type          Enabled Status   Reason
Speed    Desc           VLAN
----------------------------------------------------------------------------
--------
1/1/1:1 --      routed QSFP+DA3x4   yes      down     Split reboot pending
--   --
1/1/1:2 --      routed QSFP+DA3x4   yes      down     Split reboot pending
--   --
1/1/1:3 --      routed QSFP+DA3x4   yes      down     Split reboot pending
--   --
1/1/1:4 --      routed QSFP+DA3x4   yes      down     Split reboot pending
--   --
```

Unsplitting a port on a switch that requires a reboot:

```
switch(config)# interface 1/1/1
switch(config-if)# no split
This command will disable the split interfaces for this port and clear
their configuration. The port will not be available until the next
system or line module reboot.
Continue (y/n)? y
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	`config-if`	Administrators or local user group members with execution righ ts for this command.

system interface-group

Syntax

```
system interface-group <GROUP> line-module <SLOT-ID> speed <SPEED>
no system interface-group <GROUP> line-module <SLOT-ID> speed <SPEED>
```

Description

Configures the speed for an interface group. After changing group speed, only transceivers compatible with the new speed will be enabled.

CAUTION

- All speed-mismatched interfaces in the group will be disabled.
- This command can interrupt active network links, user confirmation is required to proceed.

The **no** form of this command resets the specified interface group to its default.

Parameter	Description
<GROUP>	Specifies the interface group to configure.
<SPEED>	Configures transceiver speed (10g or 25g) for a group. Default i s 25g (see the for further detail). On HPE Aruba Networking 8400 Switch Series: • 10g allows 1Gbps or 10Gbps transceivers only. • 25g allows 25Gbps transceivers only.
<SLOT-ID>	Specifies the slot ID of the line module.

Examples

Configuring interface group 1 on line-module 1/1 to allow 10Gbps and slower transceivers:

```
switch(config)# system interface-group 1 line-module 1/1 speed 10g
Changing the group speed will disable all member interfaces that do not
match the new speed.
```

```
Continue (y/n)? y
```

Command History

Release	Modification
10.09.0002	Command introduced on 6400 and 8400 Switch series.

Command Information

Platforms	Command context	Authority
8400	Operator (`>`) or Manager (`#`)	Administrators or local user group members with execution rights for this command.

Source interface selection

The source IP address is determined by the system and is typically the IP address of the outgoing interface in the routing table. However, routing switches may have multiple routing interfaces and outgoing packets can potentially be sent by different paths at different times. This results in different source IP addresses.

AOS-CX provides a configuration model that allows the selection of an IP address to use as the source address for all outgoing traffic. This allows unique identification of the software application on the server site regardless of which local interface has been used to reach the destination server. The source interface selection supports selecting an IP address or interface name.

NOTE

If the source interface and source IP are configured, Source IP will have priority.

VLANs

VLANs are primarily used to provide network segmentation at layer 2. VLANs enable the grouping of users by logical function instead of physical location. They make managing bandwidth usage within networks possible by:

- Allowing grouping of high-bandwidth users on low-traffic segments

- Organizing users from different LAN segments according to their need for common resources and individual protocols
- Improving traffic control at the edge of networks by separating traffic of different protocol types.
- Enhancing network security by creating subnets to control in-band access to specific network resources

VLANs are generally assigned on an organizational basis rather than on a physical basis. For example, a network administrator could assign all workstations and servers used by a particular workgroup to the same VLAN, regardless of their physical locations.

Hosts in the same VLAN can directly communicate with one another. A router or a Layer 3 switch is required for hosts in different VLANs to communicate with one another.

VLANs help reduce bandwidth waste, improve LAN security, and enable network administrators to address issues such as scalability and network management.

> **NOTE**
> Refer to the Layer 2 Bridging Guide for VLAN configuration and commands.

Configuration and firmware management

Subtopics

Upgrade and downgrade scenarios

Upgrade and downgrade scenarios are provided by the Configuration Migration Framework (CMF).

Upgrades

All upgrade scenarios are supported and configuration is migrated.

Downgrades

About this task

The following scenario is supported:

Procedure

1. Create a checkpoint before upgrading.
2. Upgrade.
3. Perform config changes.
4. Set the startup configuration to the checkpoint created in step 1.
5. Downgrade

> **NOTE**
>
> Configuration changes that occur after the checkpoint in step 1 will be lost during the downgrade.

Limitations

The following are the limitations of upgrade and downgrade scenarios provided by CMF.

Table 1. Configuration from and to the same build

From/to	Checkpoint	Running	Startup
Checkpoint	N/A	Supported	Supported
Running	Supported	N/A	Supported
Startup	Supported	Supported	N/A
Config from URL in CLI Format	Supported	Supported	Supported
Config from URL in JSON Format	Supported	Supported	Supported

Table 2. Configuration from an older build to a newer build (upgrade)

From/to	Checkpoint	Running	Startup
Config from URL in CLI Format	Not supported	Not supported	Not supported

From/to	Checkpoint	Running	Startup
Config from URL in JSON Format	Not supported	Not supported	Not supported
Startup	Not supported	Supported	Supported

Table 3. Configuration from a newer build to an older build (downgrade)

From/to	Checkpoint	Running	Startup
Config from URL in CLI Format	Not supported	Not supported	Not supported
Config from URL in JSON Format	Not supported	Not supported	Not supported
Startup	Not supported	Not supported	Not supported

Hot-patch software

Overview

Hot-patching provides users of AOS-CX with a way to update running software without rebooting their system. Hot-patches are distributed as signed .patch files and are applied on top of an official AOS-CX image.

A hot-patch can be downloaded from a remote server onto an AOS-CX switch then applied without rebooting the switch. By default, the software patch remains on the switch after it is applied, and is automatically reapplied when the switch reboots. In order to revert an applied hot-patch update, the patch must be disabled. When the hot-patch is disabled, rebooting the box is not necessary, though the hot-patch will still remain on the system. The hot-patch can be removed from the system after disabling the hot-patch.

Once a downloaded hot-patch is removed, it must be downloaded again before it can be reapplied.

> **NOTE**
> Hot-patches are cumulative, meaning that each subsequent patch is a superset of the previous patch(es) for a given software image. When multiple hot-patches are required, the first should be removed after the second has been applied.

Prerequisites

- AOS-CX hot-patch software can be obtained from HPE Aruba Networking customer support, and is identified with a **.patch** extension. The name of the patch indicates the version of AOS-CX to which it can be applied. For example, the patch **FL.10.12.XXXX-YYYY.patch** indicates that the patch can be applied on top of switch image **FL.10.12.XXXX.swi** and brings software up to date with the **FL.10.12.YYYY** release.

- Hot-patch software can be applied only after it has been downloaded to the switch.

Caveats and limitations

- Hot-patch files cannot be managed when the switch is loaded with a .swi image that does not support the hot-patch feature.
- Only restartable daemons can be updated with a hot-patch file.
- Each hot-patch applies to only one release image and each subsequent hot-patch for that release image is cumulative.This means that applying the most recent hot-patch provides all the same fixes included in previous hot-patches. As such, only one hot-patch can be applied at a time.
- incompatible and un-applied hot-patches are allowed to remain on the system and in the config to support rollback to the previous version of the software, and to automatically apply any hot-patches that have been previously applied.
- A maximum of ten hot-patches are allowed to be present or preconfigured on the system at a time.
- A daemon/service may restart if it has a hard dependency on another service that gets restarted by being hot-patched. Upon rebooting, hot-patches are reapplied after daemons have started.
- The **dpsed** and **hpe-cardd** daemon cannot have a hot-patch applied while a switchover or failover is in progress.

To apply a hot-patch update

1. Request a hot-patch from HPE Aruba Networking customer support, then place the hot-patch on an FTP-SFTP server.
2. Use the **copy** command to copy the hot-patch from the remote server to the standalone switch or VSF.
3. Use the **hot-patch apply** command to apply the patch.

To unapply hot-patch update

This option only unapplies the patch, but does not remove the patch from the device. The patch can be reapplied again since it is still on the device. If an unapplied patch needs to be removed/deleted from the switch, it can be removed with the command **hot-patch remove <patch>.patch**.

```
switch(config)# show hot-patch
Name                        Status
------------------------    ------
FL_10_12_0001-0002.patch    Not applied
switch(config)# hot-patch apply FL_10_12_0001-0002.patch
```

```
switch(config)#
switch(config)# show hot-patch
Name                      Status
------------------------  ------
FL_10_12_0001-0002.patch  Applied
switch(config)# no hot-patch apply FL_10_12_0001-0002.patch
switch(config)# show hot-patch
Name                      Status
------------------------  ------
FL_10_12_0001-0002.patch  Not applied
switch(config)#
switch(config)# hot-patch remove FL_10_12_0001-0002.patch
Do you want to save the current configuration (y/n)? y
The running configuration was saved to the startup configuration.
switch(config)#
switch(config)# show hot-patch
No hot-patch found or configured
switch(config)#
```

To remove/delete a patch:

An applied hot-patch can be both unapplied and removed from the device with the command **hot-patch remove <patch-name>.patch**. The patch cannot be reapplied unless it is downloaded again.

CLI Session Behavior for Standalone ISSU

Overview

During a standalone ISSU upgrade, active CLI sessions are terminated while the upgrade is in progress. User session behavior varies depending on the access method.

> **NOTE**
> It is recommended to avoid running scripts or automated tools that repeatedly create new CLI sessions during ISSU, as these sessions may be terminated and could result in unexpected script failures.

Console Access CLI Session

Active console CLI sessions will be terminated while the ISSU upgrade is in progress, with the message shown below. After session termination, users can log in to the router at any time during the upgrade process.

```
The current session is being terminated due to an ongoing ISSU upgrade.
Please login again to access the router.
```

Users are advised to use their existing console CLI session for running show commands and to avoid starting new sessions using the `vtysh` command from bash while the ISSU is in progress.

Remote Access CLI Session

Active remote CLI sessions will be terminated while the ISSU upgrade is in progress, and the message shown below will be displayed. SSH and Telnet access to the router will be temporarily unavailable during the upgrade.

Console login is recommended for router management until the ISSU upgrade is complete.

```
The current session is being terminated due to an ongoing ISSU upgrade.
Please use the console to log in and access the router.
```

Login During ISSU

If a user logs in while an ISSU upgrade is still in progress, the following warning will be displayed after login:

```
WARNING: ISSU upgrade is in progress. Some CLI output may be incomplete
or unavailable. Please logout and login again after the upgrade completes.
```

Also note that the message will be displayed if users try to configure other features during ISSU.

This warning indicates that certain daemons may still be restarting. As a result, CLI commands that rely on those services may return partial or incomplete output. Users should limit activity to urgent monitoring via show commands and are advised to log out until the upgrade has fully completed.

Caveats and limitations

- IGMP/MLD leave messages may not be processed until Standalone ISSU completes.
- Telemetry data collection is halted during Standalone ISSU.
- Standalone ISSU is intended to succeed when the network is in a **stable** state. Its successful completion is not guaranteed during major network events such as STP or routing re-convergence, significant port flapping, or when a large number of clients are connecting or disconnecting simultaneously.
- Standalone ISSU is not permitted when MACsec is enabled on any system port.
- During Standalone ISSU, LLDP temporarily stops sending the LLDP-MED Network Policy TLV.
- BFD sessions may be interrupted until Standalone ISSU completes.
- While Standalone ISSU is in progress, the following operations are not allowed:
  - Any hardware hot-swap activity
  - Pressing any buttons
  - Link state transitions
- Performing Standalone ISSU simultaneously on two or more devices, especially if OSPF is configured, is not recommended. It could result in possible route flaps, link failure, tunnel flapping and OSPF restarting on these devices.

- During Standalone ISSU, UDLD may cause link flapping if an aggressive UDLD interval is configured. This can result in a brief traffic outage. To minimize this risk, it is recommended to use the default UDLD interval.
- Standalone ISSU will be prevented if the feature readiness check fails. This check ensures that none of the following features are in a failed state:
  - ACLs, object groups, and their associated applications
  - Classifier policies, classes, and their applications
  - Port-access policies, classes, and their applications
  - Port-access group-based policies, classes, and their applications
  - Active PBR actions and their applications

Standalone ISSU Examples

For more information about `show issu` command, refer to: GUID-8ECDD722-3092-4230-8299-E029D1401469.html.

Showing detailed status of an in-progress standalone ISSU on a switch operating in a standalone topology:

```
switch# show issu
ISSU Summary
============
ISSU Status      : In Progress
Current Version : FL.10.18.0000-4093- Upgrade Version : FL.10.18.0000-4093-
Upgrade Image    : secondary           Start Date      : 2025-10-13 09:39:01
Last ISSU Result: --
Rollback timer   : Not configured

ISSU Progress
=============
Upgrade Operation                    Status          Start Date
------------------------------------------------------------------------
Initiate ISSU                        Complete        2025-10-13 09:39:01
Validate System Readiness            Complete        2025-10-13 09:39:01
Upgrade Data Plane Services          In Progress     2025-10-13 09:39:15
Upgrade Control Plane Services       Pending         2025-10-13 09:39:15
Finalize Upgrade                     Pending         --
ISSU Complete                        Pending         --
```

Showing detailed ISSU status when Standalone ISSU is aborted due to topology change from Standalone to other:

```
switch# show issu
ISSU Summary
```

```
============
ISSU Status      : Not Ready
Current Version : FL.10.18.0000-4093- Upgrade Version : FL.10.18.0000-4093-
Upgrade Image    : secondary           Start Date       : 2025-10-13 09:39:01
Last ISSU Result: Aborted (Topology changed from Standalone to other)
Rollback timer   : Not configured

ISSU Progress
=============
Upgrade Operation                    Status          Start Date
------------------------------------------------------------------------
Initiate ISSU                        Complete        2025-10-13 09:39:01
Validate System Readiness            Error           2025-10-13 09:39:01
Upgrade Data Plane Services          Aborted         --
Upgrade Control Plane Services       Aborted         --
Finalize Upgrade                     Aborted         --
ISSU Complete                        Aborted         --
```

NOTE

The setup is considered as standalone when `show vsf` command shows the topology as **Standalone**.

Showing detailed ISSU status when Standalone ISSU is aborted due to insufficient system memory. Note that, if system memory utilization exceeds the 75% threshold, it will be unable to perform ISSU.

Current system memory utilization can be obtained from `show system resource-utilization` command.

```
switch# show issu
ISSU Summary
============
ISSU Status      : Ready
Current Version : FL.10.18.0000-4093- Upgrade Version : FL.10.18.0000-4093-
Upgrade Image    : secondary           Start Date       : 2025-10-13 09:39:01
Last ISSU Result: Aborted (System memory utilization above 75% threshold to
start ISSU)
Rollback timer   : Not configured

ISSU Progress
=============
Upgrade Operation                    Status          Start Date
------------------------------------------------------------------------
Initiate ISSU                        Complete        2025-10-13 09:39:01
Validate System Readiness            Error           2025-10-13 09:39:01
Upgrade Data Plane Services          Aborted         --
```

```
Upgrade Control Plane Services      Aborted        --
Finalize Upgrade                    Aborted        --
ISSU Complete                       Aborted        --

switch#show system resource-utilization

System Resources:
Processes                          : 154
CPU usage(%)                       : 50
CPU usage(% average over 1 minute): 50
CPU usage(% average over 5 minute): 50
Memory usage(%)                    : 80
Open FD's                          : 2189
...
...
```

Showing output from the debug buffer:

```
switch# show debug buffer module issu
----------------------------------------------------------------------------
--------------------------------------------
show debug buffer
----------------------------------------------------------------------------
-------------------------------------------
2026-04-10:17:04:13.002029|issud|LOG_DEBUG|SMM|1/2|ISSU|ISSU|Ignoring db
sync frozen change, to be processed in AMM only
2026-04-10:17:09:18.824108|issud|LOG_DEBUG|AMM|1/2|ISSU|ISSU|Writeable get
name: action is undefined
2026-04-10:17:09:18.824175|issud|LOG_DEBUG|AMM|1/2|ISSU|ISSU|ISSU type set
to undetermined
2026-04-10:17:09:18.824194|issud|LOG_DEBUG|AMM|1/2|ISSU|ISSU|Queued
writable to clear ISSU type
```

Checkpoints

A checkpoint is a snapshot of the running configuration of a switch and its relevant metadata during the time of creation. Checkpoints can be used to apply the switch configuration stored within a checkpoint whenever needed, such as to revert to a previous, clean configuration. Checkpoints can be applied to other switches of the same platform. A switch is able to store multiple checkpoints.

Subtopics

Checkpoint types

The switch supports two types of checkpoints:

- **System generated checkpoints**: The switch automatically generates a system checkpoint whenever a configuration change occurs.
- **User generated checkpoints**: The administrator can manually generate a checkpoint whenever required.

Maximum number of checkpoints

- Maximum checkpoints: 64 (including the startup configuration)
- Maximum user checkpoints: 32
- Maximum system checkpoints: 32

User generated checkpoints

User checkpoints can be created at any time, as long as one configuration difference exists since the last checkpoint was created. Checkpoints can be applied to either the running or startup configurations on the switch.

All user generated checkpoints include a time stamp to identify when a checkpoint was created.

A maximum of 32 user generated checkpoints can be created.

System generated checkpoints

System generated checkpoints are automatically created by default. Whenever a configuration change occurs, the switch starts a timeout counter (300 seconds by default). For each additional configuration change, the timeout counter is restarted. If the timeout expires with no additional configuration changes being made, the switch generates a new checkpoint.

System generated checkpoints are named with the prefix `CPC` followed by a time stamp in the format `<YYYYMMDDHHMMSS>`. For example: `CPC20170630073127`.

System checkpoints can be applied using the checkpoint rollback feature or copy command.

A maximum of 32 system checkpoints can be created. Beyond this limit, the newest system checkpoint replaces the oldest system checkpoint.

Supported remote file formats

You can restore a switch configuration by copying a switch configuration stored on a USB drive or a remote network device through SFTP/TFTP. The remote file formats that the switch supports depends on where you plan to restore the checkpoint.

Restoring a checkpoint to a...	File type supported
Running configuration	• CLI • JSON • Checkpoint
Startup configuration	• JSON • Checkpoint
Specified checkpoint	Specified checkpoint

Rollback

The term rollback is used to refer to when a switch configuration is reverted to a pre-existing checkpoint.

For example, the following command applies the configuration from checkpoint `ckpt1`. All previous configurations are lost after the execution of this command: `checkpoint rollback ckpt1`

You can also specify the rollback of the running configuration or of the startup configuration with a specified checkpoint, as shown with the following command: `copy checkpoint <checkpoint-name> {running-config | startup-config}`

Checkpoint auto mode

Checkpoint auto mode configures the switch with failover support, causing it to automatically revert to a previous configuration if it becomes inoperable or inaccessible due to configuration changes that are being made.

After entering checkpoint auto mode, you have a set amount of time to add, remove, or modify the existing switch configuration. To save your changes, you must execute the `checkpoint auto confirm` command before the auto mode timer expires. If you do not execute the `checkpoint auto confirm` command within the specified time, all configuration changes you made are discarded and the running configuration reverts to the state it was before entering checkpoint auto mode.

Testing a switch configuration in checkpoint auto mode

About this task

Process overview:

Procedure

1. Enable the checkpoint auto mode.
2. To save the configuration, enter the `checkpoint auto confirm` command before the specified time set in step 1.

Checkpoint commands

Select a command from the list in the left navigation menu.

Subtopics

copy <REMOTE-URL> startup-config|running-config
copy {running-config | startup-config} <REMOTE-URL>
copy running-config {startup-config | checkpoint <CHECKPOINT-NAME>}
copy {running-config | startup-config} <STORAGE-URL>
copy startup-config running-config
copy <STORAGE-URL> running-config
erase
show checkpoint <CHECKPOINT-NAME>
show checkpoint <CHECKPOINT-NAME> hash
show checkpoint post-configuration
show checkpoint
show checkpoint date
show running-config hash
show startup-config hash
write memory

checkpoint auto

Syntax

```
checkpoint auto <TIME-LAPSE-INTERVAL>
```

Description

Starts auto checkpoint mode. In auto checkpoint mode, the switch temporarily saves the runtime configuration as a checkpoint only for the specified time lapse interval. Configuration changes must be saved before the interval expires, otherwise the runtime configuration is restored from the temporary checkpoint.

Parameter	Description
<TIME-LAPSE-INTERVAL>	Specifies the time lapse interval in minutes. Range: 1 to 60.

Usage

To save the runtime checkpoint permanently, run the **checkpoint auto confirm** command during the time lapse interval. The filename for the saved checkpoint is named **AUTO <YYYYMMDDHHSS>**. If the **checkpoint auto confirm** command is not entered during the specified time lapse interval, the previous runtime configuration is restored.

Examples

Confirming the auto checkpoint:

```
switch# checkpoint auto 20
Auto checkpoint mode expires in 20 minute(s)
switch# WARNING  Please "checkpoint auto confirm" within 2 minutes
switch# checkpoint auto confirm
checkpoint AUTO20170801011154 created
```

In this example, the runtime checkpoint was saved because the **checkpoint auto confirm** command was entered within the value set by the **time-lapse-interval** parameter, which was 20 minutes.

Not confirming the auto checkpoint:

```
switch# checkpoint auto 20
Auto checkpoint mode expires in 20 minute(s)
switch# WARNING  Please "checkpoint auto confirm" within 2 minutes
WARNING: Restoring configuration. Do NOT add any new configuration.
Restoration successful
```

In this example, the runtime checkpoint was reverted because the **checkpoint auto confirm** command was not entered within the value set by the **time-lapse-interval** parameter, which was 20 minutes.

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution righ ts for this command.

`checkpoint auto confirm`

Syntax

```
checkpoint auto confirm
```

Description

Signals to the switch to save the running configuration used during the auto checkpoint mode. This command also ends the auto checkpoint mode.

Usage

To save the runtime checkpoint permanently, run the **checkpoint auto confirm** command during the time lapse value set by the **checkpoint auto TIME-LAPSE-INTERVAL** command. The generated checkpoint name will be in the format **AUTO <YYYYMMDDHHSS>**. If the **checkpoint auto confirm** command is not entered during the specified time lapse interval, the previous runtime configuration is restored.

Examples

Confirming the auto checkpoint:

```
switch# checkpoint auto confirm
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution righ ts for this command.

checkpoint diff

Syntax

```
checkpoint diff {<CHECKPOINT-NAME1> | running-config | startup-config}
     {<CHECKPOINT-NAME2> | running-config | startup-config | REMOTE_URL |
STORAGE_URL} [vrf <VRF_NAME>] [deep]
```

Description

Shows the difference in configuration between two configurations. Compare checkpoints, the running configuration, or the startup configuration.

Parameter	Description
`{<CHECKPOINT-NAME1> | running-config | startup-config}`	Selects either a checkpoint, the running configuration, or the startup configuration as the baseline.
`{<CHECKPOINT-NAME2> | running-config | startup-config}`	Selects either a checkpoint, the running configuration, or the startup configuration to compare.
`REMOTE_URL | STORAGE-URL`	Retrieves a CLI type configuration from a remote or local file. Configurations must be in CLI format.
`[vrf <VRF_NAME>]`	Specifies the VRF name when using a REMOTE_URL to download a checkpoint.
`[deep]`	Uses a deep mode comparison.

Usage

> **NOTE**
>
> REMOTE_URL or STORAGE_URL configurations must be in CLI format only, checkpoint format is currently not supported.

In order to get accurate diff results, ensure that the imported file has a valid configuration, correct indentation to represent context changes between CLI contexts, and the file uses the Unix end of line format (\n). The imported file should follow the same syntax and format as the output for **show running-config**. The use of 4 spaces is recommended over the use of tab.

When using the default diff mode, the output of a similar Unix **diff-u** is used. The deep option makes the comparison based on unique entries within a given context instead of based on the order of configuration lines.

The output of the **checkpoint diff** command has several symbols:

- The plus sign (+) at the beginning of a line indicates that the line exists in the comparison but not in the baseline.
- The minus sign (-) at the beginning of a line indicates that the line exists in the baseline but not in the comparison.
- An empty space ( ) is used to represent configuration contexts that exist in both configurations (common) but contain added or removed entries.

Examples

In the following example, the configurations of checkpoints **cp1** and **cp2** are displayed before the `checkpoint diff` command, so that you can see the context of the **checkpoint diff** command.

```
switch# show checkpoint cp1
Checkpoint configuration:
!
!Version AOS-CX XL.10.00.0002
!Schema version 0.1.8
module 1/1 product-number jl363a
!
!
!
!
!
!
!
vlan 1,200
interface 1/1/1
    no shutdown
    ip address 1.0.0.1/24
interface 1/1/2
    no shutdown
    ip address 2.0.0.1/24
switch# show checkpoint cp2
Checkpoint configuration:
!
!Version AOS-CX XL.10.00.0002
!Schema version 0.1.8
module 1/1 product-number jl363a
!
!
!
!
!
!
!
vlan 1,200,300
interface 1/1/1
    no shutdown
    ip address 1.0.0.1/24
interface 1/1/2
    no shutdown
    ip address 2.0.0.1/24
```

```
switch# checkpoint diff cp1 cp2
--- /tmp/chkpt11501550258421    2017-08-01 01:17:38.420514016 +0000
+++ /tmp/chkpt21501550258421    2017-08-01 01:17:38.420514016 +0000
@@ -9,7 +9,7 @@
 !
 !
 !
-vlan 1,200
+vlan 1,200,300
 interface 1/1/1
     no shutdown
     ip address 1.0.0.1/24
switch# checkpoint diff chkpt01 chkpt02
--- /tmp/chkpt011607564301327
+++ /tmp/chkpt021607564301353
@@ -1,7 +1,7 @@
 !
 !Version AOS-CX PL.10.06.0100V
 !export-password: default
-hostname Switch
+hostname Switch1
 user admin group administrators password ciphertext
AQBapTyg9tpaiAaTfSVV5eNdFzOORRvZ6CMpglh1P+LQUHQLYgAAAGAhmRqFbkNvrgy2SBVk7H8C
5hvg/Iib8rWYFZLEaSCrobNP9EwMu+hLNM0xmsh45yG8dncP7WkxjwrW4p4Qra6dVfr0EW8xh/
lpQf8F/2Wki20Lc9JLXiYge7ti0H6cVn+G
 radius-server tracking interval 60
 no usb
switch#
```

Showing the difference in configuration between two configurations using the deep option.

```
switch# checkpoint diff running-config checkpoint run deep
-vlan 1, 100
+vlan 1
```

Showing the difference in configuration between two configurations using the deep option when there are warnings related to the file.

```
switch# checkpoint diff running-config sftp://user@172.17.0.1/config2.txt
vrf

               mgmt


             deep(user@172.17.0.1) Password:
sftp> get halon-src/hpe-config/scripts/configdiff/configdiff/config2.txt
userconfig2.txt
```

```
-vlan 1, 100
+vlan 1
WARN: Use of tabs is not recommended. File: dest-config2.txt line: 15
WARN: Use of tabs is not recommended. File: dest-config2.txt line: 16
```

Showing the difference in configuration between two configurations using the deep option when there are errors in the file and no diff output is displayed.

```
switch# checkpoint diff running-config sftp://user@172.17.0.1/config.txt
vrf
                    mgmt

                   deep(user@172.17.0.1) Password:
sftp> get config.txt userconfig.txt
ERROR: Duplicate entry found. File: dest-config.txt (line 25): 'vlan 2'
```

Command History

Release	Modification
10.16	Updated syntax to include REMOTE_URL, STORAGE_URL, and deep.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution righ ts for this command.

checkpoint post-configuration

Syntax

```
checkpoint post-configuration
no checkpoint post-configuration
```

Description

Enables creation of system generated checkpoints when configuration changes occur. This feature is enabled by default.

The **no** form of this command disables system generated checkpoints.

Usage

System generated checkpoints are automatically created by default. Whenever a configuration change occurs, the switch starts a timeout counter (300 seconds by default). For each additional configuration change, the timeout counter is restarted. If the timeout expires with no additional configuration changes being made, the switch generates a new checkpoint.

System generated checkpoints are named with the prefix **CPC** followed by a time stamp in the format **<YYYYMMDDHHMMSS>**. For example: **CPC20170630073127**.

System checkpoints can be applied using the checkpoint rollback feature or copy command.

A maximum of 32 system checkpoints can be created. Beyond this limit, the newest system checkpoint replaces the oldest system checkpoint.

Examples

Enabling system checkpoints:

```
switch(config)# checkpoint post-configuration
```

Disabling system checkpoints:

```
switch(config)# no checkpoint post-configuration
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution righ ts for this command.

`checkpoint post-configuration timeout`

Syntax

```
checkpoint post-configuration timeout <TIMEOUT>
no checkpoint post-configuration timeout <TIMEOUT>
```

Description

Sets the timeout for the creation of system checkpoints. The timeout specifies the amount of time since the latest configuration for the switch to create a system checkpoint.

The **no** form of this command resets the timeout to 300 seconds, regardless of the value of the **<TIMEOUT>** parameter.

Parameter	Description
`timeout <TIMEOUT>`	Specifies the timeout in seconds. Range: 5 to 600. Default: 300.

Examples

Setting the timeout for system checkpoints to 60 seconds:

```
switch(config)# checkpoint post-configuration timeout 60
```

Resetting the timeout for system checkpoints to 300 seconds:

```
switch(config)# no checkpoint post-configuration timeout 1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`checkpoint rename`

Syntax

```
checkpoint rename <OLD-CHECKPOINT-NAME>
           <NEW-CHECKPOINT-NAME>
```

Description

Renames an existing checkpoint.

Parameter	Description
`<OLD-CHECKPOINT-NAME>`	Specifies the name of an existing checkpoint to be renamed.
`<NEW-CHECKPOINT-NAME>`	Specifies the new name for the checkpoint. The checkpoint name can be alphanumeric. It can also contain underscores (_) and dashes (-). **NOTE** Do not start the checkpoint name with **CPC** because it is used for system - generated checkpoints.

Examples

Renaming checkpoint **ckpt1** to **cfg001**:

```
switch# checkpoint rename ckpt1 cfg001
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution righ ts for this command.

checkpoint rollback

Syntax

```
checkpoint rollback {<CHECKPOINT-NAME> | startup-config}
```

Description

Applies the configuration from a pre-existing checkpoint or the startup configuration to the running configuration.

Parameter	Description
<CHECKPOINT-NAME>	Specifies a checkpoint name.
startup-config	Specifies the startup configuration.

Examples

Applying a checkpoint named **ckpt1** to the running configuration:

```
switch# checkpoint rollback ckpt1
Success
```

Applying a startup checkpoint to the running configuration:

```
switch# checkpoint rollback startup-config
Success
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`copy checkpoint <CHECKPOINT-NAME> <REMOTE-URL>`

Syntax

```
copy checkpoint <CHECKPOINT-NAME> <REMOTE-URL> [{cli | json | checkpoint}]
[vrf <VRF-NAME>]
```

Description

Copies a checkpoint configuration to a remote location as a file. The configuration is exported in checkpoint format, which includes switch configuration and relevant metadata.

Parameter	Description
`<CHECKPOINT-NAME>`	Specifies the name of a checkpoint.

`<REMOTE-URL>`	Specifies the remote destination and filename using the syntax:
	TFTP format:
	`tftp://<IP-ADDR>[:<PORT-NUM>] [;blocksize=<Value>]/<FILENAME>`
	SFTP format:
	`sftp://<USERNAME>@<IP-ADDR> [:<PORT-NUM>]/<FILENAME>`

Parameter	Description
	SCP format: `scp://USER@{IP|HOST}[:PORT]/FILE`
`cli | json | checkpoint`	Specifies the transfer format: • Checkpoint - default • CLI • JSON
`vrf <VRF-NAME>`	Specifies a VRF name.

Examples

Copying checkpoint configuration to a remote file through TFTP:

```
switch# copy checkpoint ckpt1 tftp://192.168.1.10/ckptmeta vrf default
#########################################################################
100.0%
Success
```

Copying checkpoint configuration to a remote file through SFTP:

```
switch# copy checkpoint ckpt1 sftp://root@192.168.1.10/ckptmeta vrf default
The authenticity of host '192.168.1.10 (192.168.1.10)' can't be established.
ECDSA key fingerprint is SHA256:FtOm6Uxuxumil7VCwLnhz92H9LkjY+eURbdddOETy50.
Are you sure you want to continue connecting (yes/no)? yes
root@192.168.1.10's password:
sftp> put /tmp/ckptmeta ckptmeta
Uploading /tmp/ckptmeta to /root/ckptmeta
Warning: Permanently added '192.168.1.10' (ECDSA) to the list of known
hosts.
Connected to 192.168.1.10.
Success
```

Command History

Release	Modification
10.18	Optional parameters have been integrated: CLI, JSON, and checkpoint.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`copy checkpoint <CHECKPOINT-NAME> {running-config | startup-config}`

Syntax

```
copy checkpoint <CHECKPOINT-NAME> {running-config | startup-config}
```

Description

Copies an existing checkpoint configuration to the running configuration or to the startup configuration.

Parameter	Description
`<CHECKPOINT-NAME>`	Specifies the name of an existing checkpoint.
{running - config | startup - config}	Selects whether the running configuration or the startup config uration receives the copied checkpoint configuration. If the star tup configuration is already present, the command overwrites t he startup configuration.

Examples

Copying **ckpt1** checkpoint to the running configuration:

```
switch# copy checkpoint ckpt1 running-config
Success
```

Copying **ckpt1** checkpoint to the startup configuration:

```
switch# copy checkpoint ckpt1 startup-config
Success
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution righ ts for this command.

`copy checkpoint <CHECKPOINT-NAME> <STORAGE-URL>`

Syntax

```
copy checkpoint <CHECKPOINT-NAME>
                <STORAGE-URL>
```

Description

Copies an existing checkpoint configuration to a USB drive. The file format is defined when the checkpoint was created.

Parameter	Description
<CHECKPOINT-NAME>	Specifies the name of the checkpoint to copy. The checkpoint n ame can be alphanumeric. It can also contain underscores (_) an d dashes (-).
<STORAGE-URL>>	Specifies the name of the target file on the USB drive using the following syntax: `usb:/<FILE>` The USB drive must be formatted with the FAT file system.

Examples

Copying the **test** checkpoint to the **testCheck** file on the USB drive:

```
switch# copy checkpoint test usb:/testCheck
Success
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution righ ts for this command.

copy <REMOTE-URL> checkpoint <CHECKPOINT-NAME >

Syntax

```
copy <REMOTE-URL> checkpoint <CHECKPOINT-NAME> [vrf <VRF-NAME>]
```

Description

Copies a remote configuration file to a checkpoint. The remote configuration file must be in checkpoint format.

Parameter	Description
<REMOTE-URL>	Specifies a remote file using the following syntax: **TFTP format:** `tftp://<IP-ADDR>[:<PORT-NUM>] [;blocksize=<Value>]/<FILENAME>` **SFTP format:**

Parameter	Description
	sftp://<USERNAME>@<IP-ADDR>[:<PORT-NUM>]/<FILENAME> **SCP format:** scp://USER@{IP|HOST}[:PORT]/FILE
<CHECKPOINT-NAME>	Specifies the name of the target checkpoint. The checkpoint name can be alphanumeric. It can also contain underscores (_) and dashes (-). Required. **NOTE** Do not start the checkpoint name with CPC because it is used for system - generated checkpoints.
vrf <VRF-NAME>	Specifies a VRF name. Default: default.

Examples

Copying a checkpoint format file to checkpoint **ckpt5** on the default VRF:

```
switch# copy tftp://192.168.1.10/ckptmeta checkpoint ckpt5
########################################################################
100.0%
100.0%
Success
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

`copy <REMOTE-URL> startup-config|running-config`

Syntax

```
copy <REMOTE-URL> startup-config|{running-config [overwrite]} [vrf <VRF-
NAME>]
```

Description

Copies a remote file containing a switch configuration to the running configuration or to the startup configuration.

Parameter	Description

`<REMOTE-URL>`	Specifies a remote file with the following syntax:
	TFTP format:
	`tftp://<IP-ADDR>[:<PORT-NUM>] [;blocksize=<Value>]/<FILENAME>`
	SFTP format:
	`sftp://<USERNAME>@<IP-ADDR> [:<PORT-NUM>]/<FILENAME>`
	SCP format:
	`scp://USER@{IP|HOST}[:PORT]/FILE`

`startup-config`	Indicates that the startup configuration receives the copied ch eckpoint configuration. If the startup configuration is already p resent, the command overwrites the startup configuration.
`running-config`	Indicates that the running configuration receives the copied che ckpoint configuration.
`overwrite`	By default, the CLI configuration contained in the coped file is applied on top of the running - config, so the CLI command d oes not clear the running - config before applying the CLI com mands. In the event that a particular CLI command fails then th e failure is logged in event logs and the next line in CLI configur ation is processed. When you include the **overwrite** parameter, the configuration is only applied to the running - config if all the commands succee

Parameter	Description
	ded without errors. If there are any errors during the copy proc ess, the existing running - config will remain intact. You can the n inspect the event logs to check and fix the errors and retry t he operation. If more than 20 errors are present, the operation stops processing any further commands. The event logs will dis play only those errors procesed up to that point.
`vrf <VRF-NAME>`	Specifies the name of a VRF. Default: default.

Usage

The switch copies only certain file types. The format of the file is automatically detected from contents of the file. The **startup-config** option only supports the JSON file format and checkpoints, but the **running-config** option supports the JSON and CLI file formats and checkpoints.

When a file of the CLI format is copied, it overwrites the running configuration. The CLI command does not clear the running configuration before applying the CLI commands. All of the CLI commands in the file are applied line-by-line. If a particular CLI command fails, the switch logs the failure and it continues to the next line in the CLI configuration. The event log (**show events -d hpe-config**) provides information as to which command failed.

Examples

Copying a JSON format file to the running configuration:

```
switch# copy tftp://192.168.1.10/runjson running-config
#########################################################################
100.0%
Configuration may take several minutes to complete according to
configuration file size

--0%----10%----20%----30%----40%----50%----60%----70%----80%----90%----100%-
-
Success
```

Copying a CLI format file to the running configuration with an error in the file:

```
switch# copy tftp://192.168.1.10/runcli running-config
#########################################################################
100.0%
Configuration may take several minutes to complete according to
configuration file size

--0%----10%----20%----30%----40%----50%----60%----70%----80%----90%----100%-
-
Some of the configuration lines from the file were NOT applied. Use 'show
events -d hpe-config' for more info.
```

```
########################################################################
100.0%
Configuration may take several minutes to complete according to
configuration file size

--0%----10%----20%----30%----40%----50%----60%----70%----80%----90%----100%-
-
Some of the configuration lines from the file were NOT applied. Use 'show
events -d hpe-config' for more info.
```

Copying a CLI format file to the startup configuration:

```
switch# copy tftp://192.168.1.10/startjson startup-config
########################################################################
100.0%
100.0%
Success
```

Copying an unsupported file format to the startup configuration:

```
switch# copy tftp://192.168.1.10/startfile startup-config
########################################################################
100.0%
100.0%
unsupported file format
```

Command History

Release	Modification
10.15	The overwrite parameter is introduced.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution righ ts for this command.

copy {running-config | startup-config} *<REMOTE-URL>*

Syntax

```
copy {running-config | startup-config} <REMOTE-URL> {cli | json} [vrf <VRF-
NAME>]
```

Description

Copies the running configuration or the startup configuration to a remote file in either CLI or JSON format.

Parameter	Description
`{running-config	startup-config}`	Selects whether the running configuration or the startup config uration is copied to a remote file.

`<REMOTE-URL>`	Specifies the remote file using the syntax:
	TFTP format:
	`tftp://<IP-ADDR>[:<PORT-NUM>] [;blocksize=<Value>]/<FILENAME>`
	SFTP format:
	`sftp://<USERNAME>@<IP-ADDR> [:<PORT-NUM>]/<FILENAME>`
	SCP format:
	`scp://USER@{IP|HOST}[:PORT]/FILE`

`{cli	json}`	Selects the remote file format: P: CLI or JSON.
`vrf <VRF-NAME>`	Specifies the name of a VRF. Default: default.

Examples

Copying a running configuration to a remote file in CLI format:

```
switch# copy running-config tftp://192.168.1.10/runcli cli
########################################################################
100.0%
Success
```

Copying a running configuration to a remote file in JSON format:

```
switch# copy running-config tftp://192.168.1.10/runjson json
########################################################################
100.0%
Success
```

Copying a startup configuration to a remote file in CLI format:

```
switch# copy startup-config sftp://root@192.168.1.10/startcli cli
root@192.168.1.10's password:
sftp> put /tmp/startcli startcli
Uploading /tmp/startcli to /root/startcli
Connected to 192.168.1.10.
Success
```

Copying a startup configuration to a remote file in JSON format:

```
switch# copy startup-config sftp://root@192.168.1.10/startjson json
root@192.168.1.10's password:
sftp>
root@192.168.1.10's password:
sftp> put /tmp/startjson startjson
Uploading /tmp/startjson to /root/startjson
Connected to 192.168.1.10.
Success
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`copy running-config {startup-config | checkpoint <CHECKPOINT-NAME>}`

Syntax

```
copy running-config {startup-config | checkpoint <CHECKPOINT-NAME>}
```

Description

Copies the running configuration to the startup configuration or to a new checkpoint. If the startup configuration is already present, the command overwrites the existing startup configuration.

Parameter	Description
`startup-config`	Specifies that the startup configuration receives a copy of the r unning configuration.
`checkpoint <CHECKPOINT-NAME>`	Specifies the name of a new checkpoint to receive a copy of th e running configuration. The checkpoint name can be comprise d of alphanumeric character, underscores (_) and dashes (-), a nd must be 32 characters or fewer. **NOTE** Do not start the checkpoint name with **CPC** be cause it is used for system - generated checkpo ints.

Examples

Copying the running configuration to the startup configuration:

```
switch# copy running-config startup-config
Success
```

Copying the running configuration to a new checkpoint named **ckpt1**:

```
switch# copy running-config checkpoint ckpt1
Success
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`copy {running-config | startup-config} <STORAGE-URL>`

Syntax

```
copy {running-config | startup-config} <STORAGE-URL> {cli | json}
```

Description

Copies the running configuration or a startup configuration to a USB drive.

Parameter	Description
`{running-config | startup-config}`	Selects the running configuration or the startup configuration to be copied to the switch USB drive.
`<STORAGE-URL>`	Specifies a remote file with the following syntax: **usb:/** <file>
`{cli | json}`	Selects the format of the remote file: CLI or JSON.

Usage

The switch supports JSON and CLI file formats when copying the running or starting configuration to the USB drive. The USB drive must be formatted with the FAT file system.

The USB drive must be enabled and mounted with the following commands:

```
switch(config)# usb
switch(config)# end
switch# usb mount
```

Examples

Copying a running configuration to a file named **runCLI** on the USB drive:

```
switch# copy running-config usb:/runCLI cli
Success
```

Copying a startup configuration to a file named **startCLI** on the USB drive:

```
switch# copy startup-config usb:/startCLI cli
Success
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`copy startup-config running-config`

Syntax

```
copy startup-config running-config
```

Description

Copies the startup configuration to the running configuration.

Examples

```
switch# copy startup-config running-config
Success
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`copy <STORAGE-URL> running-config`

Syntax

```
copy <STORAGE-URL> {running-config | startup-config | checkpoint
<CHECKPOINT-NAME>}
```

Description

This command copies a specified configuration from the USB drive to the running configuration, to a startup configuration, or to a checkpoint.

Parameter	Description
`<STORAGE-URL>`	Specifies the name of a configuration file on the USB drive with the syntax: **usb:/** <FILE>
`running-config`	Specifies that the configuration file is copied to the running conf iguration. The file must be in CLI, JSON, or checkpoint format o r the copy will fail. the copy will not work.
`startup-config`	Specifies that the configuration file is copied to the startup c onfiguration. The switch stores this configuration between re boots. The startup configuration is used as the operating config uration following a reboot of the switch. The file must be in JSO N or checkpoint format or the copy will fail.

Parameter	Description
`checkpoint <CHECKPOINT-NAME>`	Specifies the name of a new checkpoint file to receive a copy of the configuration. The configuration file on the USB drive must be in checkpoint format. **NOTE** Do not start the checkpoint name with **CPC** be cause it is used for system - generated checkpo ints.

Usage

This command requires that the USB drive is formatted with the FAT file system and that the file be in the appropriate format as follows:

- **running-config**: This option requires the file on the USB drive be in CLI, JSON, or checkpoint format.
- **startup-config**: This option requires the file on the USB drive be in JSON or checkpoint format.
- **checkpoint** <checkpoint-name>: This option requires the file on the USB drive be in checkpoint format.

Examples

Copying the file **runCli** from the USB drive to the running configuration:

```
switch# copy usb:/runCli running-config
Configuration may take several minutes to complete according to
configuration
file size
--0%----10%----20%----30%----40%----50%----60%----70%----80%----90%----100%-
-
Success
```

Copying the file **startUp** from the USB drive to the startup configuration:

```
switch# copy usb:/startUp startup-config
Success
```

Copying the file **testCheck** from the USB drive to the **abc** checkpoint:

```
switch# copy usb:/testCheck checkpoint abc
Success
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`erase`

Syntax

```
erase
   checkpoint <checkpont-name>
   core-dump all|daemon|dsm|kernel
   startup-config
   all
```

Description

Deletes an existing checkpoint, startup configuration, or core-dump.

Parameter	Description
`checkpoint <CHECKPOINT-NAME>`	Specifies the name of a checkpoint.
`core-dump all|daemon <daemon-name> |kernel`	Erase one of the following sets of core - dump files: • all: Erase all core - dump files. • daemon <daemon - name>: Erase daemon core - dump file s. • kerne:l Erase the kernel core - dump. •

Parameter	Description
startup-config	Specifies the startup configuration.
all	Specifies all checkpoints.

Examples

Erasing checkpoint **ckpt1**:

```
switch# erase checkpoint ckpt1
```

Erasing the startup configuration:

```
switch# erase startup-config
```

Erasing all checkpoints:

```
switch# erase checkpoint all
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution righ ts for this command.

`show checkpoint <CHECKPOINT-NAME>`

Syntax

```
show checkpoint <CHECKPOINT-NAME> [json]
```

Description

Shows the configuration of a checkpoint.

Parameter	Description
`<CHECKPOINT-NAME>`	Specifies the name of a checkpoint.
`[json]`	Specifies that the output is displayed in JSON format.

Examples

Showing the configuration of the **ckpt1** checkpoint in CLI format:

```
switch# show checkpoint ckpt1
Checkpoint configuration:
!
!Version AOS-CX PL.10.07.0000K-75-g55e5193
!export-password: default
lacp system-priority 65535
user admin group administrators password ciphertext
AQBapQjwipebv36io0jFfde7ZzrHckncal1D+3n8XFTZKQdmYgAAADEtYOeHSme93xzdD0uz6Vr9
Kl+XBzB+2GB0UBxSF7rvgN2x8KSgkqv7iqXVQ0Te6LkSMnH4BdNaT3Bf25qyvOqmr4YakO1V3rg8
zAOADkPktQD8joTHXflzwomoIzcmv/uX
cli-session
    timeout 0
!
!
!
!
ssh server vrf default
vlan 1
spanning-tree
interface lag 1
    no shutdown
    vlan access 1
interface lag 128
    no shutdown
    vlan access 1
```

```
interface lag 129
    shutdown
    vlan access 1
    lacp mode active
interface 1/1/1
    no shutdown
    lag 128
    lacp port-id 65535
interface 1/1/2
    no shutdown
    vlan access 1
interface 1/1/3
    no shutdown
    vlan access 1
interface 1/1/4
    no shutdown
    vlan access 1
interface 1/1/5
    no shutdown
    vlan access 1
interface 1/1/6
    no shutdown
    vlan access 1
interface 1/1/7
    no shutdown
    vlan access 1
interface 1/1/8
    no shutdown
    vlan access 1
interface 1/1/9
    no shutdown
    vlan access 1
interface 1/1/10
    no shutdown
    vlan access 1
interface 1/1/11
    no shutdown
    vlan access 1
interface 1/1/12
    no shutdown
    vlan access 1
interface 1/1/13
    no shutdown
```

```
    vlan access 1
interface 1/1/14
    no shutdown
    vlan access 1
interface 1/1/15
    no shutdown
    vlan access 1
interface 1/1/16
    no shutdown
    vlan access 1
interface vlan 1
    ip dhcp
snmp-server vrf default
!
!
!
!
!
https-server vrf default
```

Showing the configuration of the **ckpt1** checkpoint in JSON format:

```
switch# show checkpoint ckpt1 json
Checkpoint configuration:
{
    "AAA_Server_Group": {
        "local": {
            "group_name": "local"
        },
        "none": {
            "group_name": "none"
        }
    },
...
...
...
...
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`show checkpoint <CHECKPOINT-NAME> hash`

Syntax

```
show checkpoint <CHECKPOINT-NAME> hash [cli | json]
```

Description

Shows a configuration checkpoint hash calculated with the SHA-256 algorithm. When the output format is not specified, the CLI format is used. This enables you to determine whether there has been a configuration change since a previous hash was calculated.

Parameter	Description
`<CHECKPOINT-NAME>`	Specifies an existing checkpoint name.
`[cli | json]`	Selects either the CLI or JSON format.

Examples

Showing a checkpoint SHA-256 hash in JSON format:

```
switch# show checkpoint ckpt1 hash json
Calculating the hash: [Success]
The SHA-256 hash of the checkpoint in JSON format, created in image
XX.10.08.xxxx:
cc7a57a9bbb4e6600d3b4180296a35f6af9e797ce9c439955dfe5de58b06da9e
This hash is only valid for comparison to a baseline hash if the
configuration has
not been explicitly changed (such as with a CLI command, REST operation,
etc.)
```

```
or implicitly changed (such as by changing a hardware module, upgrading the
SW version, etc.).
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`show checkpoint post-configuration`

Syntax

```
show checkpoint post-configuration
```

Description

Shows the configuration settings for creating system checkpoints.

Examples

```
switch# show checkpoint post-configuration
Checkpoint Post-Configuration feature
-------------------------------------
 Status      : enabled
 Timeout (sec)   : 300
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`show checkpoint`

Syntax

```
show checkpoint
```

Description

Shows a detailed list of all saved checkpoints.

Examples

Showing a detailed list of all saved checkpoints:

```
switch# show checkpoint
NAME                 TYPE         WRITER   DATE(YYYY/MM/DD)       IMAGE VERSION
ckpt1                checkpoint   User     2017-02-23T00:10:02Z   XX.01.01.000X
ckpt2                checkpoint   User     2017-03-08T18:10:01Z   XX.01.01.000X
ckpt3                checkpoint   User     2017-03-09T23:11:02Z   XX.01.01.000X
ckpt4                checkpoint   User     2017-03-11T00:00:03Z   XX.01.01.000X
ckpt5                latest       User     2017-03-14T01:12:27Z   XX.01.01.000X
```

Command History

Release	Modification
10.08	Command syntax `show checkpoint list all` is replaced with `sho w checkpoint`.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`show checkpoint date`

Syntax

```
show checkpoint date <START-DATE>
            <END-DATE>
```

Description

Shows detailed list of all saved checkpoints created within the specified date range.

Parameter	Description
<START-DATE>	Specifies the starting date for the range of saved checkpoints to show. Format: **YYYY - MM - DD**.
<END-DATE>	Specifies the endingdate for the range of saved checkpoints to show. Format: **YYYY - MM - DD**.

Examples

Showing a detailed list of saved checkpoints for a specific date range:

```
switch# show checkpoint date 2017-03-08 2017-03-12
NAME                TYPE        WRITER   DATE(YYYY/MM/DD)      IMAGE VERSION
ckpt2               checkpoint  User     2017-03-08T18:10:01Z  XX.01.01.000X
ckpt3               checkpoint  User     2017-03-09T23:11:02Z  XX.01.01.000X
ckpt4               checkpoint  User     2017-03-11T00:00:03Z  XX.01.01.000X
```

Command History

Release	Modification
10.08	Command syntax `show checkpoint list date <START-DATE> <END-DATE>` is replaced with `show checkpoint date <START-DATE> <END-DATE>`
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`show running-config hash`

Syntax

```
show running-config hash [cli | json]
```

Description

Shows the running-config checkpoint hash, calculated with the SHA-256 algorithm. When the output format is not specified, the CLI format is used. This enables you to determine whether there has been a configuration change since a previous hash was calculated.

Parameter	Description
`[cli	json]`	Selects either the CLI or JSON format.

Examples

Showing the running-config checkpoint SHA-256 hash in CLI format:

```
switch# show running-config hash cli
Calculating the hash: [Success]
SHA-256 hash of the config in CLI format:
8db4e7e10f4b7f1a6ab17ad2b4efe0e72f1849103eaf43da62aa1d715075b89e
```

```
This hash is only valid for comparison to a baseline hash if the
configuration has
not been explicitly changed (such as with a CLI command, REST operation,
etc.)
or implicitly changed (such as by changing a hardware module, upgrading the
SW version, etc.).
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`show startup-config hash`

Syntax

```
show startup-config hash [cli | json]
```

Description

Shows the startup-config checkpoint hash, calculated with the SHA-256 algorithm. When the output format is not specified, the CLI format is used. This enables you to determine whether there has been a configuration change since a previous hash was calculated.

Parameter	Description
`[cli	json]`	Selects either the CLI or JSON format.

Examples

Showing the startup-config checkpoint SHA-256 hash in CLI format:

```
switch# show startup-config hash cli
Calculating the hash: [Success]
SHA-256 hash of the config in CLI format:
8db4e7e10f4b7f1a6ab17ad2b4efe0e72f1849103eaf43da62aa1d715075b89e
This hash is only valid for comparison to a baseline hash if the
configuration has
not been explicitly changed (such as with a CLI command, REST operation,
etc.)
or implicitly changed (such as by changing a hardware module, upgrading the
SW version, etc.).
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`write memory`

Syntax

```
write memory
```

Description

Saves the running configuration to the startup configuration. It is an alias of the command **copy running-config startup-config**. If the startup configuration is already present, this command overwrites the startup configuration.

Examples

```
switch# write memory
Success
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution righ ts for this command.

Boot commands

Select a command from the list in the left navigation menu.

Subtopics

- **boot fabric-module**
- **boot line-module**
- **boot management-module**
- **boot management-module (recovery console)**
- **boot set-default**
- **boot system**
- **show boot-history**

`boot fabric-module`

Syntax

```
boot fabric-module <SLOT-ID>
```

Description

Reboots the specified fabric module.

Parameter	Description
<SLOT-ID>	Specifies the member and slot of the module in the format **me mber/slot**. For example, to specify the module in member 1 slot 3, enter **1/3**.

Usage

The **boot fabric-module** command reboots the specified fabric module. Traffic performance is affected while the module is down.

If the specified module is the only fabric module in an up state, rebooting that module stops traffic switching between line modules and the line modules power down. The line modules power up when one fabric module returns to an up state.

This command is valid for fabric modules only.

Examples

Rebooting the fabric module in slot **1/3** when auto-confirm is not enabled:

```
switch# boot fabric-module 1/3
This command will reboot the specified fabric module.  Traffic performance
may
be affected while the module is down.  Rebooting the last fabric module will
stop traffic switching between line modules.
Do you want to continue (y/n)? y
switch#
```

Rebooting the fabric module in slot **1/1** when auto-confirm is enabled:

```
switch# boot fabric-module 1/3
This command will reboot the specified fabric module.  Traffic performance
may
be affected while the module is down.  Rebooting the last fabric module will
stop traffic switching between line modules.
Do you want to continue (y/n) y (auto-confirm)
switch#
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`boot line-module`

Syntax

```
boot line-module <SLOT-ID>
```

Description

Reboots the specified line module.

Parameter	Description
`<SLOT-ID>`	Specifies the member and slot of the module in the format **me mber/slot**. For example, to specify the module in member 1 slot 3, enter **1/3**.

Usage

Reboots the specified line module. Any traffic for the switch passing through the affected module (SSH, TELNET, and SNMP) is interrupted. It can take up to 2 minutes to reboot the module. During that time, you can monitor progress by viewing the event log.

This command is valid for line modules only.

Examples

Reloading the module in slot 1/1:

```
switch# boot line-module 1/1
This command will reboot the specified line module and interfaces on this
module will not send or receive packets while the module is down. Any
traffic passing through the line module will be interrupted. Management
sessions connected through the line module will be affected. It might take
up to 2 minutes to complete rebooting the module. During that time, you can
```

```
monitor progress by viewing the event log.
Do you want to continue (y/n)? y
switch#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	Manager (#)	Administrators or local user group members with execution righ ts for this command.

boot management-module

Syntax

```
boot management-module {active | standby | <SLOT-ID>}
```

Description

Reboots the specified management module. Choose the management module to reboot by role (active or standby) or by slot number.

Parameter	Description
active	Selects the active management module.
standby	Selects the standby management module.

Parameter	Description
`<SLOT-ID>`	Specifies the member and slot of the management module in th e format **member/slot**. For example, to specify the module in m ember 1 slot 5, enter **1/5**.

Usage

This command reboots a single management module in a chassis. Choose the management module to reboot by role (active or standby) or by slot number.

You can use the **show images** command to show information about the primary and secondary system images.

If you reboot the active management module and the standby management module is available, the active management module reboots and the standby management module becomes the active management module.

If you reboot the active management module and the standby management module is not available, you are warned, you are prompted to save the configuration, and you are prompted to confirm the operation.

If you reboot the standby management module, the standby management module reboots and remains the standby management module.

If you attempt to reboot a management module that is not available, the **boot** command is aborted.

Saving the configuration is not required. However, if you attempt to save the configuration and there is an error during the save operation, the **boot** command is aborted.

> **NOTE**
>
> Hewlett Packard Enterprise recommends that you use the **boot management-module** command instead of pressing the module reset button to reboot a management module because if you are rebooting the only available management module, the **boot management-module** command enables you to save the configuration, cancel the reboot, or both.

Examples

Rebooting the active management module when the standby management module is available:

```
switch# boot management-module active
The management-module in slot 1/5 is going down for reboot now.
```

Rebooting the active management module when the standby management module is not available:

```
switch# boot management-module 1/5
The management module in slot 1/5 is currently active and no
standby management module was found.
This will reboot the entire switch.
Do you want to save the current configuration (y/n)? n
```

```
This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	Manager (#)	Administrators or local user group members with execution righ ts for this command.

boot management-module (recovery console)

Syntax

```
boot management-module {local|remote}
```

Description

Reboots the specified management module by specified location (local or remote).

Parameter	Description
<local>	Reboots the local management module.
<remote>	Reboots the remote management module.

Usage

This command reboots a single management module in a chassis. Choose the management module to reboot by role (active or standby) or by slot number.

You can use the **show images** command to show information about the primary and secondary system images.

If you reboot the active management module and the standby management module is available, the active management module reboots and the standby management module becomes the active management module.

If you reboot the active management module and the standby management module is not available, you are warned, you are prompted to save the configuration, and you are prompted to confirm the operation.

If you reboot the standby management module, the standby management module reboots and remains the standby management module.

If you attempt to reboot a management module that is not available, the **boot** command is aborted.

Saving the configuration is not required. However, if you attempt to save the configuration and there is an error during the save operation, the **boot** command is aborted.

> **NOTE**
>
> Hewlett Packard Enterprise recommends that you use the **boot management-module** command instead of pressing the module reset button to reboot a management module because if you are rebooting the only available management module, the **boot management-module** command enables you to save the configuration, cancel the reboot, or both.

Examples

Booting a remote management module:

```
switch# boot management-module remote
There is no other management module installed.
Aborting.
switch#
```

Command History

Release	Modification
10.12	Command introduced.

Command Information

Platforms	Command context	Authority
8400	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`boot set-default`

Syntax

```
boot set-default {primary | secondary}
```

Description

Sets the default operating system image to use when the system is booted.

Parameter	Description
`primary`	Selects the primary network operating system image.
`secondary`	Selects the secondary network operating system image.

Example

Selecting the primary image as the default boot image:

```
switch# boot set-default primary
Default boot image set to primary.
```

> **NOTE**
>
> For more information on features that use this command, refer to the Fundamentals Guide or the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution rights for this command.

`boot system`

Syntax

```
boot system [primary | secondary | serviceos]
```

Description

Reboots all modules on the switch. By default, the configured default operating system image is used. Optional parameters enable you to specify which system image to use for the reboot operation and for future reboot operations.

Parameter	Description
`primary`	Selects the primary operating system image for this reboot and sets the configured default operating system image to **primary** for future reboots.
`secondary`	Selects the secondary operating system image for this reboot a nd sets the configured default operating system image to **secon dary** for future reboots.
`serviceos`	Selects the service operating system for this reboot. Does not c hange the configured default operating system image. The ser vice operating system acts as a standalone bootloader and rec overy OS for switches running the AOS-CX operating system an d is used in rare cases when troubleshooting a switch.

Usage

This command reboots the entire system. If you do not select one of the optional parameters, the system reboots from the configured default boot image.

You can use the **show images** command to show information about the primary and secondary system images.

Choosing one of the optional parameters affects the setting for the default boot image:

- If you select the **primary** or **secondary** optional parameter, that image becomes the configured default boot image for future system reboots. The command fails if the switch is not able to set the operating system image to the image you selected.
  You can use the **boot set-default** command to change the configured default operating system image.
- If you select **serviceos** as the optional parameter, the configured default boot image remains the same, and the system reboots all management modules with the service operating system.

If the configuration of the switch has changed since the last reboot, when you execute the **boot system** command you are prompted to save the configuration and you are prompted to confirm the reboot operation.

Saving the configuration is not required. However, if you attempt to save the configuration and there is an error during the save operation, the **boot system** command is aborted.

Examples

Rebooting the system from the configured default operating system image:

```
switch# boot system
Do you want to save the current configuration (y/n)? y
The running configuration was saved to the startup configuration.
This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
The system is going down for reboot.
```

Rebooting the system from the secondary operating system image, setting the secondary operating system image as the configured default boot image:

```
switch# boot system secondary
Default boot image set to secondary.
Do you want to save the current configuration (y/n)? n
This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
```

Canceling a system reboot:

```
switch# boot system
Do you want to save the current configuration (y/n)? n
This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? n
Reboot aborted.
switch#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`show boot-history`

Syntax

```
show boot-history [all|{vsf member <1-10>}]
```

Description

Shows boot history information. When no parameters are specified, shows the most recent information about the current boot operation, and the three previous boot operations for the switch. When the **all** parameter is specified, the output of this command shows the boot information for the active management module.

For switches that support line modules (such as 8400 switch series) including the **all** parameter displays information for the active management module and all available line modules.

> **NOTE**
> To view boot-history on a standby, the command must be sent on the conductor console.

Parameter	Description
`all`	Optional. Shows boot information for the active management m odule. For switches that support line modules, including this pa rameter displays information for and all available line modules.
`vsf member <1-10>`	Optional. Display boot history for the specified VSF member

Usage

This command displays the boot-index, boot-ID, and up time in seconds for the current boot. If there is a previous boot, it displays boot-index, boot-ID, reboot time (based on the time zone configured in the system) and reboot reasons. Previous boot information is displayed in reverse chronological order.

The output of this command includes the following information:

Parameter	Description
Index	The position of the boot in the history file. Range: 0 to 3.
Boot ID	A unique ID for the boot . A system - generated 128 - bit strin g.
Current Boot, up for <time>	For the current boot, the **show boot - history** command shows the number of seconds the module has been running on the cur rent software.
<Timestamp>: boot reason	For previous boot operations, the **show boot - history** comma nd shows the time at which the operation occurred and the rea son for the boot. The reason for the boot is one of the following values: • <DAEMON - NAME> **crash**: The daemon identified by <DA EMON - NAME> caused the module to boot. • **Kernel crash**: The operating system software associated wit h the module caused the module to boot. • **Uncontrolled reboot**: The reason for the reboot is not kno wn. • **Reboot requested through database**: The reboot occurre d because of a request made through the CLI or other API.

Table 1. Description of reboots handled through the database

Boot History String	Description
Reboot requested by user	A user requested a switch reboot through the CLI or web UI.
Reset button pressed	The switch detected a short - press of the reset but ton
Backplane fault	A backplane fault occurred.
Configuration change	A configuration change resulted in a reboot.

Boot History String	Description
Configuration version migration	A configuration version migration occurred which re quired a reboot.
Console error	The console failed to start.
Fabric fault	A fabric fault occurred.
All line modules faulted	A zero line card condition occurred.
Redundancy switchover requested	A user requested a redundancy switchover.
Redundant Management communication timeout	The standby management module has taken over fro m an unresponsive active management module.
Redundant Management election timeout	A failure to elect a standby management module in t he allotted time.
Critical service fault (error)	A daemon critical to switch operation has stopped fu nctioning. An extra error string may be present to d escribe the error in detail.
VSF autojoin renumber	Reset triggered by VSF autojoin.
VSF member renumbered	A user requested a renumber of a VSF member.
VSF switchover requested	A user requested a VSF switchover.
VSX software update	Reset triggered by a VSX software update.
Chassis critical temperature	Chassis operating temperature exceeded.
Chassis low critical temperature	Chassis temperature below the minimum operating t hreshold.
Chassis insufficient fans	Insufficient fans to cool the chassis.
Chassis unsupported PSUs/fans	Unsupported or misconfigured PSUs or system fans.
Management module critical temperature	Management module operating temperature exceed ed.
ISSU SMM update	Standby management module reboot triggered by a n In - Service Software Upgrade (ISSU).
ISSU switchover	Redundancy switchover triggered by an In - Service Software Upgrade.

Boot History String	Description
ISSU aborted	Standby management module reset triggered by fail ure during an In - Service Software Upgrade.
Rollback timer expired	Reset triggered by the ISSU rollback timer expiring.

Examples

Showing the boot history of the active management module:

```
switch# show boot-history
Management module
=================
Index : 2
Boot ID : c34a2c2499004a02bbeeff4992e1fdbd
Current Boot, up for 1 days 13 hrs 13 mins 27 secs
Index : 1
Boot ID : bfba9bc486304e57904ac717a0ccbdcd
02 Sep 23 02:55:33 : CPU request reset with 0x20201, Version:
FL.10.14.0000-1619-ga9ec1805bd442~dirty
02 Sep 23 02:55:33 : Switch boot count is 2
Index : 0
Boot ID : a88a71b7ca9a4574af7e3b811ddfdc7e
02 Sep 23 02:49:26 : Reboot requested by user, Version: FL.10.14.0000-1619-
ga9ec1805bd442~dirty
02 Sep 23 02:50:02 : Switch boot count is 1
Index : 3
Boot ID : f00ba10c8c44457f83fee303d014a89a
25 Aug 23 10:27:42 :  Power on reset with 0x1, Version: FL.10.14.0000-1465-
g9df95249d06b0~dirty
25 Aug 23 10:28:18 :  Switch boot count is 3
25 Aug 23 10:29:02 :  Primary overtemperature fault detected with 0x2 in
PSU 1/1
```

Showing the boot history of the active management module and all line modules:

```
switch#
Management module
=================
Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs
Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database
```

```
Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database
Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database
Line module 1/1
=================
Index : 3
10 Aug 17 12:45:46 : dune_agent crashed
...
Management module
=================
Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs
Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database
Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database
Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database
Line module 1/1
=================
Index : 3
10 Aug 17 12:45:46 : dune_agent crashed
...
```

```
switch# show boot-history
Management module
=================
Index : 2
Boot ID : a61ad00d10864c748bc7893a5d4af2e4
15 Dec 23 19:02:02 : Power on reset with 0x1, Version: FL.10.13.1000AF
15 Dec 23 19:02:02 : Switch boot count is 0
15 Dec 23 19:02:17 : PSU 1/1: Fault detected
Index : 1
Boot ID : 30d831bbfdfa425baf50a629ee01b185
15 Dec 23 19:01:58 : Power on reset with 0x1, Version: FL.10.13.1000AF
15 Dec 23 19:01:58 : Switch boot count is 0
```

```
switch# show boot-history vsf member 2
Member-2
=========
Index : 0
Boot ID : df99026c194a44f1944a3e7685fb4d90
Current Boot, up for 3 hrs 31 mins 39 secs
Index : 3
Boot ID : 7bf4104903fe4ad1ba4bce40e8099c76
10 Aug 17 10:02:24 : Reboot requested through database
10 Aug 17 10:02:13 : Switch boot count is 2
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution righ ts for this command.

Firmware management commands

Select a command from the list in the left navigation menu.

Subtopics

- **copy {primary | secondary} <REMOTE-URL>**
- **copy {primary | secondary} <FIRMWARE-FILENAME>**
- **copy primary secondary**
- **copy <REMOTE-URL>**
- **copy secondary primary**
- **copy <STORAGE-URL>**
- **copy hot-patch**
- **hot-patch**
- **show hot-patch**

copy {primary | secondary} <REMOTE-URL>

Syntax

```
copy {primary | secondary} <REMOTE-URL> [vrf <VRF-NAME>]
```

Description

Uploads a firmware image to a TFTP or SFTP server.

Parameter	Description
{primary | secondary}	Selects the primary or secondary image profile to upload. Required
<REMOTE-URL>	Specifies the URL to receive the uploaded firmware using SFTP, TFTP or SCP. **TFTP format:** tftp://*<IP-ADDR>*[:*<PORT-NUM>*] [;blocksize=*<Value>*]/*<FILENAME>* **SFTP format:** sftp://*<USERNAME>*@*<IP-ADDR>* [:*<PORT-NUM>*]/*<FILENAME>* **SCP format:** scp://USER@{IP|HOST}[:PORT]/FILE
vrf *<VRF-NAME>*	Specifies a VRF name. Default: default.

Examples

TFTP upload:

```
switch# copy primary tftp://192.0.2.0/00_10_00_0002.swi
#########################################################################
100.0%
Verifying and writing system firmware...
```

SFTP upload:

```
switch# copy primary sftp://swuser@192.0.2.0/00_10_00_0002.swi
swuser@192.0.2.0's password:
Connected to 192.0.2.0.
sftp> put primary.swi XL_10_00_0002.swi
Uploading primary.swi to /users/swuser/00_10_00_0002.swi
primary.swi                              100%   179MB   35.8MB/s    00:05
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

copy {primary | secondary} <*FIRMWARE-FILENAME*>

Syntax

```
copy {primary | secondary} <FIRMWARE-FILENAME>
```

Description

Copies a firmware image to USB storage.

Parameter	Description
{primary | secondary}	Selects the primary or secondary image from which to copy the firmware. Required
	Specifies the name of the firmware file to create on the USB storage device. Prefix the filename with **usb:/**. For example: **usb:/firmware_v1.2.3.swi**

Parameter	Description
<FIRMWARE-FILENAME>	For information on how to format the path to a firmware file on a USB drive, see USB URL.

Examples

```
switch# copy primary usb:/11.10.00.0002.swi
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution righ ts for this command.

copy primary secondary

Syntax

```
copy primary secondary
```

Description

Copies the firmware image from the primary to the secondary location.

Examples

```
switch# copy primary secondary
The secondary image will be deleted.
Continue (y/n)? y
Verifying and writing system firmware...
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution righ ts for this command.

copy <REMOTE-URL>

Syntax

```
copy <REMOTE-URL> {hot-patch|primary|secondary} [vrf <VRF-NAME>]
```

Description

Downloads a hot-patch or firmware image from a TFTP or SFTP server.

Parameter	Description

<REMOTE-URL>	Specifies the URL from which to download the firmware using S FTP or TFTP.
	TFTP format:
	`tftp://<IP-ADDR>[:<PORT-NUM>] [;blocksize=<Value>]/<FILENAME>`
	SFTP format:
	`sftp://<USERNAME>@<IP-ADDR> [:<PORT-NUM>]/<FILENAME>`
	SCP format:
	`scp://USER@{IP|HOST}[:PORT]/FILE`

Parameter	Description
`{hot-patch|primary|secondary}`	Select a hot - patch or a primary or secondary image profile for receiving the downloaded firmware. Required. **NOTE** For more information about hot - patch, see hot - patch.
`vrf <VRF-NAME>`	Specifies the name of a VRF. Default: default.

TFTP usage

To specify a URL with:

- an IPv4 address: **tftp://** 192.0.2.1/a.txt
- an IPv6 address: **tftp://[** 2000::2]/a.txt
- a hostname: **tftp://** hpe.com/a.txt

To specify TFTP with:

- the port number of the server in the URL: **tftp://** 192.0.2.1:12/a.txt
- the blocksize in the URL: **tftp://** 192.0.2.1 **;blocksize=** 1462 **/** a.txt
  The valid blocksize range is 8 to 65464.
- the port number of the server and blocksize in the URL: **tftp://192.0.2.1** :12 **;blocksize=** 1462 **/** a.txt

To specify a file in a directory of URL: **tftp://** 192.0.2.1 **/** dir **/** a.txt

SFTP usage

To specify:

- A URL with an IPv4 address: **sftp://** user@192.0.2.1/a.txt
- A URL with an IPv6 address: **sftp://** user **@[** 2000::2 **]/** a.txt
- A URL with a hostname: **sftp://** user **@** hpe.com **/** a.txt
- SFTP port number of a server in the URL: **sftp://** user **@** 192.0.2.1:12 **/** a.txt
- A file in a directory of URL: **sftp://** user **@** 192.0.2.1 **/** dir **/** a.txt
- To specify a file with absolute path in the URL: **sftp://** user **@** 192.0.2.1 **//** home/user/a.txt

SCP Usage

To specify:

- A username with an IP address: **scp://user@192.0.2.1:12/a.txt**
- A username with a remote host: **scp://user@hpe.com/a.txt**

Examples

TFTP download for a hot-patch:

```
switch# copy tftp://192.168.1.1/FL.10.12.0001-0002.patch hot-patch vrf vrf1

Fetching /users/swuser/FL.10.10.0001-0002.patch to hotpatch.dnld.uE2YT1
FL.10.12.0001-0002.patch                     100%   62KB  12.4MB/s   00:00
Verifying and writing hot-patch...
```

TFTP download for primary software image:

```
switch#  copy tftp://192.10.12.0/FL_10_12_0001.swi primary
The primary image will be deleted.
Continue (y/n)? y
#########################################################################
100.0%
Verifying and writing system firmware...
```

SFTP download:

```
switch# copy sftp://swuser@192.10.12.0/FL_10_12_0001.swi primary
The primary image will be deleted.
Continue (y/n)? y
The authenticity of host '192.10.12.0 (192.10.12.0)' can't be established.
ECDSA key fingerprint is SHA256:L64khLwlyLgXlARKRMiwcAAK8oRaQ8C0oWP+PkGBXHY.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '192.10.12.0' (ECDSA) to the list of known hosts.
swuser@192.10.12.0's password:
Connected to 192.10.12.0.
Fetching /users/swuser/ss.10.00.0002.swi to ss.10.00.0002.swi.dnld
/users/swuser/ss.10.00.0002.swi                  100%  179MB  25.6MB/s
00:07
Verifying and writing system firmware...
```

Command History

Release	Modification
10.12	The `hot-patch` parameter is supported on all platforms.

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`copy secondary primary`

Syntax

```
copy secondary primary
```

Description

Copies the firmware image from the secondary to the primary location.

Examples

```
switch# copy secondary primary
The primary image will be deleted.
Continue (y/n)? y
Verifying and writing system firmware...
switch# copy sftp://stor@192.22.1.0/im-switch.swi primary vrf mgmt

The primary image will be deleted.
Continue (y/n)? y
The authenticity of host '192.22.1.0 (192.22.1.0)' can't be established.
ECDSA key fingerprint is SHA256:MyI1xbdKnehYut0NLfL69gDpNzCmZqBVvBaRR46m7o8.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '192.22.1.0' (ECDSA) to the list of known hosts.
stor@192.22.1.0's password:
Connected to 192.22.1.0.
sftp> get c8d5b9f-topflite.swi c8d5b9f-topflite.swi.dnld
Fetching /home/dr/im-switch.swi to c8d5b9f-topflite.swi.dnld
/home/dr/im-switch.swi                   100%  226MB  56.6MB/s   00:04
Verifying and writing system firmware...
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution righ ts for this command.

copy <STORAGE-URL>

Syntax

```
copy <STORAGE-URL> {hot-patch|primary|secondary}
```

Description

Copies, verifies, and installs a hot-patch or firmware image from a USB storage device connected to the active management module.

Parameter	Description
<STORAGE-URL>	Specifies the name of the firmware file to copy from the storage device. Required. **USB format:** `usb:/<FILENAME>`
`{hot-patch|primary|secondary}`	Select a hot - patch image or a primary or secondary profile for receiving the copied firmware. **NOTE** For more information about hot - patch, see hot - patch.

USB usage

To specify a file:

- In a USB storage device: **usb:/** a.txt
- In a directory of a USB storage device: **usb:/** dir **/** a.txt

Examples

```
switch# copy usb:/FL.10.12.0001-0002.patch
switch# copy usb:/FL.10.12.0001.swi primary
The primary image will be deleted.
Continue (y/n)? y
Verifying and writing system firmware...
```

Command History

Release	Modification
10.12	The `hot-patch` parameter is supported on all platforms.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`copy hot-patch`

Syntax

```
copy hot-patch <Word>
            {<REMOTE-URL>|<Storage-URL>} [vrf <VRF-NAME>]
```

Description

Copies a hot-patch from a switch to the specified remote URL or storage URL.

Parameter	Description
`<Word>`	Name of the hot - patch software to upload.
`<REMOTE-URL>`	Specifies the URL to receive the uploaded patch using SFTP or TFTP. For information on how to format the remote URL, see URL formatting for copy commands.
`vrf <VRF-NAME>`	[Optional] specify the VRF instance to use for upload.
`<STORAGE-URL>`	Specifies the name of the patch file to create on the USB storage device. Prefix the filename with **usb:/**, for example, **usb:/firmware_FL_10_12_0001 - 0002.patch.**

Examples

```
switch# copy hot-patch FL_10_12_0001-0002.patch tftp:172.21.18.170/
FL_10_12_0001-0002.patch vrf vrf1
```

Related Commands

Command	Description
`copy <REMOTE-URL>`	Downloads a hot - patch image from a TFTP or SFTP server.
`hot-patch apply`	Apply a hot - patch image or remove it from the switch.

Command History

Release	Modification
10.12	Hot - patch is now supported on all platforms.
10.10	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`hot-patch`

Syntax

```
hot-patch apply|remove <name.patch>
no hot-patch apply <name.patch>
```

Description

Apply hot-patch software or remove it from the switch. The **no** form of the **hot-patch apply** command disables the hot-patch image, but does not remove it from the switch. Rebooting the system after disabling or removing the patch is not required.

Profile names	Description
`apply <name.patch>`	Apply the specified hot - patch image to a standalone switch or VSF stack. AOS-CX hot - patch software images can be obtaine d from HPE Aruba Networking customer support, and are identif ied with a **.patch** extension.
`remove <name.patch>`	Disables the hot - patch image and removes the patch from the switch. This removal will also disable the patch. Once removed, a hot - patch must be downloaded again in order to be applied.

Usage

A hot-patch can be downloaded from a remote server onto a switch then applied without rebooting the switch. When the hot-patch is disabled, the hot-patch will still remain on the system. The disabled hot-patch can be removed from the system without the need for a reboot of the system.

If a checkpoint configuration that does not contain a hot-patch is restored to a running configuration that does have a hot-patch, the patch is not deleted, it remains as not applied but is present in the device memory.

Examples

```
switch(config)# hot-patch apply FL_10_12_0001-0002.patch
```

Related Commands

Command	Description
`copy <REMOTE-URL>`	Downloads and installs a hot - patch image from a TFTP or SFTP server.
`copy hot-patch`	Copies a hot - patch software image from a switch to a specified remote URL or storage URL.

Command History

Release	Modification
10.12	Hot - patch is now supported on all platforms.
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`show hot-patch`

Syntax

```
show hot-patch [detail]
```

Parameter	Description
`detail`	Displays the detailed status of all hot - patches present on the system.

Description

the **show hot-patch** command displays the status of all hot-patches present on the system. The **show hot-patch detail** command displays detailed information for all hot patches present on the system.

Examples

```
switch# show hot-patch
Name                         Status
-----------------------      -------
FL_10_12_0001-0002.patch     Applied
switch# show hot-patch detail
Name                   : FL_10_12_0001-0002.patch
Status                 : Applied
Version                : FL_10_12_0001-0002.patch
Compatible Version     : FL.10.12.0001
Issues Fixed           : CR1234, CR2345
Patch Date             : 2022-03-29 20:46:15 UTC
Patch ID               : AOS-CX:FL.10.12.0001-sp1-256-gd457e88d39:20220442009
Patch SHA              : a4038d06a2e5fe7e0d457e868d39e526185b
```

Related Commands

Command	Description
`copy <REMOTE-URL>`	Downloads a hot - patch image from a TFTP or SFTP server.
`hot-patch apply`	Apply a hot - patch image or remove it from the switch.

Command History

Release	Modification
10.12	Command supported on all platforms.
10.10	Command introduced on 6300 Switch series.

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

SNMP

Simple Network Management Protocol (SNMP) is an Internet-standard protocol used for managing and monitoring the devices connected to a network by collecting, organizing and modifying information about managed devices on IP networks.

Subtopics

Configuring SNMP

(The SNMP agent provides read-only access.)

Procedure

1. Enable SNMP on a VRF using the command `snmp-server vrf` .
2. Set the system contact, location, and description for the switch with the following commands:
   - `snmp-server system-contact`
   - `snmp-server system-location`
   - `snmp-server system-description`
3. If required, change the default SNMP port on which the agent listens for requests with the command `snmp-server agent-port`.
4. By default, the agent uses the community string **public** to protect access through SNMPv1/v2c. Set a new community string with the command `snmp-server community`.
5. Configure the trap receivers to which the SNMP agent will send trap notifications with the command `snmp-server host`.

6. Create an SNMPv3 context and associate it with any available SNMPv3 user to perform context specific v3 MIB polling using the command `snmpv3 user`.

7. Create an SNMPv3 context and associate it with an available SNMPv1/v2c community string to perform context specific v1/v2c MIB polling using the command `snmpv3 context`.

8. Review your SNMP configuration settings with the following commands:

   - `show snmp agent-port`
   - `show snmp community`
   - `show snmp system`
   - `show snmpv3 context`
   - `show snmp trap`
   - `show snmp vrf`
   - `show snmpv3 users`
   - `show tech snmp`

Example 1

This example creates the following configuration:

- Enables SNMP on the out-of-band management interface (VRF **mgmt**).
- Sets the contact, location, and description for the switch to: **JaniceM**, **Building2**, **LabSwitch**.
- Sets the community string to **Lab8899X**.

```
switch(config)# snmp-server vrf mgmt

switch(config)# snmp-server system-contact JaniceM
switch(config)# snmp-server system-location Building2
switch(config)# snmp-server system-description LabSwitch
switch(config)# snmp-server community Lab8899X
```

Example 2

This example creates the following configuration:

- Creates an SNMPv3 user named **Admin** using **sha** authentication with the plaintext password **mypassword** and using **des** security with the plaintext password **myprivpass**.
- Associates the SNMPv3 user `Admin` with a context named `newContext`.

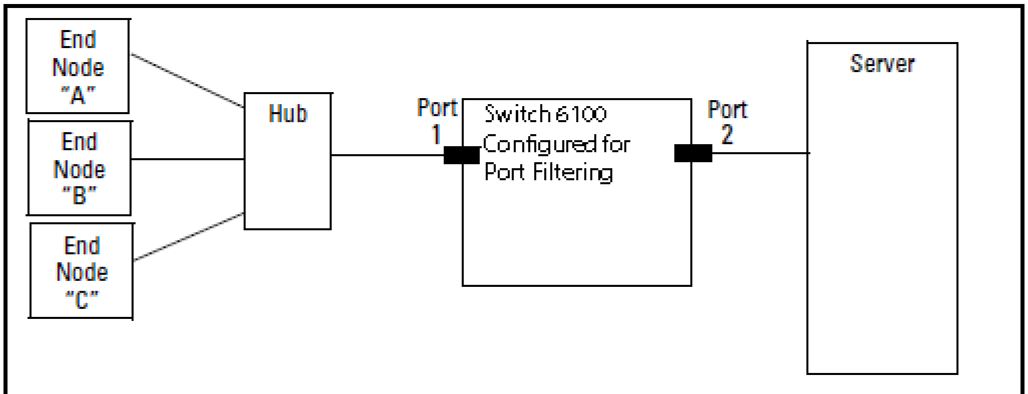
```
switch(config)# snmpv3 user Admin auth sha auth-pass plaintext mypassword
priv des
     priv-pass plaintext myprivpass
switch(config)# snmpv3 user Admin context newContext
```

> **NOTE**
> Refer to the SNMP Guide for SNMP Commands.

Port filtering

Port filtering is a feature in which packets that are ingressed through a source port can be blocked for egressing on a specific set of ports.

Figure 1. Port Filter Application



Configuring a port filter to drop traffic received on port 1/1/1 with an outbound destination of port 1/1/2 means that End Nodes A, B, and C cannot send traffic to the server. To block traffic in the opposite direction, you would also configure a port filter to drop traffic receeived on port 1/1/2 with an outbound destination of port 1/1/1.

Subtopics

Port filtering commands

Select a command from the list in the left navigation menu.

DNS

The Domain Name System (DNS) is the Internet protocol for mapping domain names or hostnames into addresses. DNS allows users to enter more readily memorable and intuitive hostnames, rather than IP addresses, to identify devices connected to a network. It also allows a host to keep the same hostname even if it changes its IP address.

Hostname resolution can be either static or dynamic.

- In static resolution, a local table is defined on the switch that associates hostnames with their IP addresses. Static tables can be used to speed up the resolution of frequently queried hosts.
- Dynamic resolution requires that the switch query a DNS server located elsewhere on the network. Dynamic name resolution takes more time than static name resolution, but requires far less configuration and management.

Configuration

VxLAN IPv4 and IPv6 underlay support the DNS client. To ensure configuration, make sure the following are set up:

- VxLAN overlay
    - Must be configured to establish the VxLAN tunnel.
- VxLAN tunnel
    - Ensure the tunnel is established between the VTEPS via either static or EVPN using the following command: **show interface VxLAN VTEPS**.
-
- **NOTE**

  For more information, refer to the .

Subtopics

[DNS client](#)
[Configuring the DNS client](#)
[DNS client commands](#)

DNS client

The DNS client resolves hostnames to IP addresses for protocols that are running on the switch. When the DNS client receives a request to resolve a hostname, it can do so in one of two ways:

- Forward the request to a DNS name server for resolution.
- Reply to the request without using a DNS name server, by resolving the name using a statically defined table of hostnames and their associated IP addresses.

Configuring the DNS client

Procedure

1. Configure one or more DNS name servers with the command `ip dns server`.
2. To resolve DNS requests by appending a domain name to the requests, either configure a single domain name with the command `ip dns domain-name`, or configure a list of up to six domain names with the command `ip dns domain-list`.
3. To use static name resolution for certain hosts, associate an IP address to a host with the command `ip dns host`.
4. Review your DNS configuration settings with the command `show ip dns`.

Examples

This example creates the following configuration:

- Defines the domain **switch.com** to append to all requests.
- Defines a DNS server with IPv4 address of **1.1.1.1**.
- Defines a static DNS host named **myhost1** with an IPv4 address of **3.3.3.3**.
- DNS client traffic is sent on the default VRF (named **default**).

```
switch(config)# ip dns domain-name switch.com
switch(config)# ip dns server-address 1.1.1.1
switch(config)# ip dns host myhost1 3.3.3.3
switch(config)# exit
switch# show ip dns
VRF Name : vrf_mgmt
Host Name                                                         Address
-----------------------------------------------------------------------------
----
VRF Name : vrf_default
Domain Name : switch.com
DNS Domain list :
Name Server(s) : 1.1.1.1
```

```
Host Name                                                          Address
-----------------------------------------------------------------------------
----
myhost1
```

This example creates the following configuration:

- Defines three domains to append to DNS requests **domain1.com**, **domain2.com**, **domain3.com** with traffic forwarding on VRF **mainvrf**.
- Defines a DNS server with an IPv6 address of **c::13**.
- Defines a DNS host named **myhost** with an IPv4 address of **3.3.3.3**.

```
switch(config)# ip dns domain-list domain1.com vrf mainvrf
switch(config)# ip dns domain-list domain2.com vrf mainvrf
switch(config)# ip dns domain-list domain3.com vrf mainvrf
switch(config)# ip dns server-address c::13
switch(config)# ip dns host myhost 3.3.3.3 vrf mainvrf
switch(config)# quit
switch# show ip dns mainvrf
VRF Name : mainvrf
Domain Name :
DNS Domain list : domain1.com, domain2.com, domain3.com
Name Server(s) : c::13
Host Name                                                          Address
-----------------------------------------------------------------------------
----
myhost                                                             3.3.3.3
```

DNS client commands

Select a command from the list in the left navigation menu.

Subtopics

[Show IP DNS FQDN Resolver Refresh Interval](#)

ip dns domain-list

Syntax

```
ip dns domain-list <DOMAIN-NAME> [vrf
                <VRF-NAME>

             ]
no ip dns domain-list <DOMAIN-NAME> [vrf
                <VRF-NAME>

             ]
```

Description

Configures one or more domain names that are appended to the DNS request. The DNS client appends each name in succession until the DNS server replies. Domains can be either IPv4 or IPv6. By default, requests are forwarded on the default VRF.

The **no** form of this command removes a domain from the list.

Parameter	Description
`list <DOMAIN-NAME>`	Specifies a domain name. Up to six domains can be added to th e list. Length: 1 to 256 characters.

Examples

This example defines a list with two entries: **domain1.com** and **domain2.com**.

```
switch(config)# ip dns domain-list domain1.com
switch(config)# ip dns domain-list domain2.com
```

This example defines a list with two entries, **domain2.com** and **domain5.com**, with requests being sent on **mainvrf**.

```
switch(config)# ip dns domain-list domain2.com vrf mainvrf
switch(config)# ip dns domain-list domain5.com vrf mainvrf
```

This example removes the entry **domain1.com**.

```
switch(config)# no ip dns domain-list domain1.com
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`ip dns domain-name`

Syntax

```
ip dns domain-name <DOMAIN-NAME> [ vrf <VRF-NAME> ]
no ip dns domain-name <DOMAIN-NAME> [ vrf <VRF-NAME> ]
```

Description

Configures a domain name that is appended to the DNS request. The domain can be either IPv4 or IPv6. By default, requests are forwarded on the default VRF. If a domain list is defined with the command `ip dns domain-list`, the domain name defined with this command is ignored.

The **no** form of this command removes the domain name.

Parameter	Description
`<DOMAIN-NAME>`	Specifies the domain name to append to DNS requests. Length: 1 to 256 characters.
`vrf <VRF-NAME>`	Specifies a VRF name. Default: default.

Examples

Setting the default domain name to `domain.com`:

```
switch(config)# ip dns domain-name domain.com
```

Removing the default domain name `domain.com`:

```
switch(config)# no ip dns domain-name domain.com
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`ip dns host`

Syntax

```
ip dns host <HOST-NAME>
            <IP-ADDR> [ vrf <VRF-NAME> ]
no ip dns host <HOST-NAME>
            <IP-ADDR> [ vrf <VRF-NAME> ]
```

Description

Associates a static IP address with a hostname. The DNS client returns this IP address instead of querying a DNS server for an IP address for the hostname. Up to six hosts can be defined. If no VRF is defined, the default VRF is used.

The **no** form of this command removes a static IP address associated with a hostname.

Parameter	Description
`host <HOST-NAME>`	Specifies the name of a host. Length: 1 to 256 characters.

Parameter	Description
`<IP-ADDR>`	Specifies an IP address in IPv4 format (**x.x.x.x**), where **x** is a de cimal number from 0 to 255, or IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where **x** is a hexa decimal number from 0 to F.
`vrf <VRF-NAME>`	Specifies a VRF name. Default: default.

Examples

This example defines an IPv4 address of **3.3.3.3** for **host1**.

```
switch(config)# ip dns host host1 3.3.3.3
```

This example defines an IPv6 address of **b::5** for **host 1**.

```
switch(config)# ip dns host host1 b::5
```

This example defines removes the entry for **host 1** with address **b::5**.

```
switch(config)# no ip dns host host1 b::5
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`ip dns server address`

Syntax

```
ip dns server-address <IP-ADDR> [ vrf <VRF-NAME> ]
no ip dns server-address <IP-ADDR> [ vrf <VRF-NAME> ]
```

Description

Configures the DNS name servers that the DNS client queries to resolve DNS queries. Up to six name servers can be defined. The DNS client queries the servers in the order that they are defined. If no VRF is defined, the default VRF is used.

The **no** form of this command removes a name server from the list.

Parameter	Description
`<IP-ADDR>`	Specifies an IP address in IPv4 format (**x.x.x.x**), where **x** is a decimal number from 0 to 255, or IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where **x** is a hexadecimal number from 0 to F.
`vrf <VRF-NAME>`	Specifies a VRF name. Default: default.

Examples

This example defines a name server at **1.1.1.1**.

```
switch(config)# ip dns server-address 1.1.1.1
```

This example defines a name server at **a::1**.

```
switch(config)# ip dns server-address a::1
```

This example removes a name server at **a::1**.

```
switch(config)# no ip dns server-address a::1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`ip dns fqdn-resolver refresh-interval`

Syntax

```
[no] ip dns fqdn-resolver refresh-interval <3600-86400>
```

Description

Sets the refresh interval for FQDN resolvers. The interval specifies the time in seconds after which the FQDN resolver entries are refreshed. The valid range for the interval is between 3600 seconds (1 hour) to 86400 seconds (1 day). For each FQDN entry refresh timer is applicable from the time entry is installed.

The **no** form of this command removes the refresh interval and sets it back to default value which is 1 day.

Parameter	Description
`<3600-86400>`	Refresh interval in seconds (default: 86400 seconds). Required

Examples

This example sets a custom refresh interval.

```
switch(config)# ip dns fqdn-resolver refresh-interval 6000
switch(config)# ip dns fqdn-resolver refresh-interval 85400
```

Resetting the refresh interval to the default value.

```
switch(config)# no ip dns fqdn-resolver refresh-interval
```

Attempting to set an invalid interval:

```
switch(config)# ip dns fqdn-resolver refresh-interval 20
Invalid interval. Valid range: 3600-86400 seconds.
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`ip dns fqdn-resolver force-refresh`

Syntax

```
ip dns fqdn-resolver force-refresh { fqdn WORD | source port-access | vrf
WORD | status (resolved | stale | unresolved) }
```

Description

This command initiates a force refresh for FQDN resolver entry/entries. It can refresh all FQDN entries, or entries filtered by FQDN, source, VRF or status. If no FQDN is specified, all entries are refreshed.

Parameter	Description
`fqdn`	Refresh entries associated with specific FQDN. Optional
`source`	Refresh entries based on the source of the resolution request. Optional
`vrf`	Refresh entries associated with a specific VRF. Optional

Parameter	Description
`status`	Refresh entries associated with a specific resolution status. Optional

Examples

Refreshing all FQDN entries:

```
switch(config)# ip dns fqdn-resolver force-refresh
```

Refreshing a specific FQDN:

```
switch(config)# ip dns fqdn-resolver force-refresh fqdn www.example.com
```

Refreshing all FQDNs from specific source:

```
switch(config)# ip dns fqdn-resolver force-refresh source port-access
```

Refreshing FQDN in a specific VRF:

```
switch(config)# ip dns fqdn-resolver force-refresh fqdn www.example.com vrf
red
```

Refreshing FQDN for a specific source and vrf:

```
switch(config)# ip dns fqdn-resolver force-refresh source classifier vrf red
```

When a FQDN entry is not found:

```
switch(config)# ip dns fqdn-resolver force-refresh fqdn www.nonexistent.com
No FQDN entries matching the given criteria.
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`show ip dns`

Syntax

```
show ip dns [vrf <VRF-NAME>][vsx-peer]
```

Description

Shows all DNS client configuration settings or the settings for a specific VRF.

Parameter	Description
`vrf <VRF-NAME>`	Specifies the VRF for which to show information. If no VRF is def ined, the default VRF is used.

Usage

DNS client arbitration on the MGMT interface on a MGMT VRF can be updated via three different methods.

1. Using the **domain-name <name>** or **nameservers <servers>** commands in the command-line interface.
2. Using the **ip dns domain-name** <DOMAIN-NAME> **vrf** MGMT or **ip dns server-address <SERVER> vrf MGMT** commands in the command-line interface.
3. Using the **ip dhcp** command in the command-line interface (dynamic enties).

AOS-CX gives the following priority levels to the these three update methods.

- Priority 1 - standalone CLI configuration
- Priority 2 - static ip dns configuration
- Priority 3 - Dynamic config

Examples

These examples define DNS settings and then show how they are displayed with the **show ip dns** command.

```
switch(config)# ip dns domain-name domain.com
switch(config)# ip dns domain-list domain5.com
switch(config)# ip dns domain-list domain8.com
switch(config)# ip dns server-address 4.4.4.4
switch(config)# ip dns server-address 6.6.6.6
switch(config)# ip dns host host3 5.5.5.5
switch(config)# ip dns host host2 2.2.2.2
```

```
switch(config)# ip dns host host3 c::12
switch(config)# ip dns domain-name reddomain.com vrf
                  red
               default
switch(config)# ip dns domain-list reddomain5.com vrf
                  red
               default
switch(config)# ip dns domain-list reddomain8.com vrf
                  red
               default
switch(config)# ip dns server-address 4.4.4.5 vrf
                  red
               default
switch(config)# ip dns server-address 6.6.6.7 vrf
                  red
               default
switch(config)# ip dns host host3 5.5.5.6 vrf
                  red
               default
switch(config)# ip dns host host2 2.2.2.3 vrf
                  red
               default
switch(config)# ip dns host host3 c::13 vrf

               red


switch# show ip dns
VRF Name : default
Domain Name : domain.com
DNS Domain list : domain5.com, domain8.com
Name Server(s) : 4.4.4.4, 6.6.6.6
Host Name    Address
-------------------------------
host2            2.2.2.2
host3            5.5.5.5
host3            c::12

VRF Name : red
Domain Name : reddomain.com
DNS Domain list : reddomain5.com, reddomain8.com
Name Server(s) : 4.4.4.5, 6.6.6.7
Host Name    Address
```

```
-------------------------------
host2            2.2.2.3
host3            5.5.5.6
host3            c::13

switch(config)# ip dns domain-name domain.com vrf red
switch(config)# ip dns domain-list domain5.com vrf red
switch(config)# ip dns domain-list domain8.com vrf red
switch(config)# ip dns server-address 4.4.4.4 vrf red
switch(config)# ip dns server-address 6.6.6.6 vrf red
switch(config)# ip dns host host3 5.5.5.5 vrf red
switch(config)# no ip dns host host2 2.2.2.2 vrf red
switch(config)# ip dns host host3 c::12 vrf red
switch# show ip dns vrf red
VRF Name : red
Domain Name : domain.com
DNS Domain list : domain5.com, domain8.com
Name Server(s) : 4.4.4.4, 6.6.6.6
Host Name    Address
-------------------------------
host3            5.5.5.5
host3            c::12
switch(config)# ip dns domain-name domain.com
switch(config)# ip dns domain-list domain5.com
switch(config)# ip dns domain-list domain8.com
switch(config)# ip dns server-address 4.4.4.4
switch(config)# ip dns server-address 6.6.6.6
switch(config)# ip dns host host3 5.5.5.5
switch(config)# ip dns host host3 c::12
switch# show ip dns
VRF Name : default
Domain Name : domain.com
DNS Domain list : domain5.com, domain8.com
Name Server(s) : 4.4.4.4, 6.6.6.6
Host Name    Address
-------------------------------
host3            5.5.5.5
host3            c::12
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

Show FQDN Resolver

Syntax

```
show ip dns fqdn-resolver
```

Description

This command displays a summary table of all FQDN resolver entries, including the total count of FQDN entries, refresh time, source information, and a tabular display of all entries.

Examples

Show unresolved FQDN entries with ipv4 requested:

```
switch# show ip dns fqdn-resolver
Total FQDN: 1
Refresh Time: 86400 seconds
Source: port-access
VRF         FQDN                            Status        Addr-Family    ip-
address
---------- ------------------------------ ------------- --------------
-------------
default     host1.search.com                unresolved    ipv4           -
```

Show resolved FQDN entries:

```
switch# show ip dns fqdn-resolver
Total FQDN: 2
```

```
Refresh Time: 7200 seconds
Source: port-access
VRF           FQDN                           Status        Addr-Family
ip-address
------------- ------------------------------ ------------- --------------
-------------
default       www.example.com                resolved      ipv4
192.168.1.1, 192.168.1.2
default       api.test.com                   resolved      ipv4
10.0.0.1
                                                           ipv6
2001:db8::1, 2001:db8::2
long_vrf_na   a_very_long_fqdn_name          resolved      ipv4
10.0.0.2
```

When no FQDN entries exist:

```
switch(config)# show ip dns fqdn-resolver
No FQDN resolver entries found.
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution righ ts for this command.

Show IP DNS FQDN Resolver Detail

Syntax

```
show ip dns fqdn-resolver detail {fqdn WORD | source (port-access) | vrf
WORD | requested-address-family (ipv4|ipv6|both)}
```

Description

Displays detailed hierarchical information for FQDN resolver entries, including the FQDN, VRF, requested address family, resolved IPv4 and IPv6 addresses, resolution status, and remaining refresh timer. The command supports filtering by FQDN, source, VRF, and address family. If no filters are provided, all entries are displayed.

Examples

Show detailed view with unresolved entry:

```
switch# show ip dns fqdn-resolver detail
----------------------------------------------------------------------------
----
Source: port-access
----------------------------------------------------------------------------
----
FQDN: host1.search.com
VRF: default
Requested Address Family: ipv4
Status: unresolved
----------------------------------------------------------------------------
----
```

Show detailed view with resolved entries:

```
switch# show ip dns fqdn-resolver detail
Source: port-access
  FQDN: www.example.com
  VRF: default
  Status: resolved
  Requested Address Family: both
       IPv4 Addresses:
      192.168.1.1
      192.168.1.2
    IPv6 Addresses:
    2001:db8::1
      2001:db8::2
----------------------------------------------------------------------------
----
  FQDN: www.hpe.com
  VRF: default
  Status: resolved
  Requested Address Family: ipv4
    IPv4 Addresses:
      10.0.0.1
```

```
--------------------------------------------------------------------------------
----
```

Show detailed view with filter:

```
switch# show ip dns fqdn-resolver detail fqdn www.example.com
Source: port-access
  FQDN: www.example.com
  VRF: default
  Status: resolved
  Requested Address Family: both
  IPv4 Addresses:
    192.168.1.1
    192.168.1.2
   IPv6 Addresses:
    2001:db8::1
    2001:db8::2
--------------------------------------------------------------------------------
----
```

Show detailed view when no entries exist:

```
switch# show ip dns fqdn-resolver detail
No FQDN resolver entries found
```

> **NOTE**
>
> The **source** parameter is not supported in 8320, 8400 or 9300 switch series.

> **NOTE**
>
> For more information on features that use this command, refer to the Security Guide for your switch model.

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution righ ts for this command.

Show IP DNS FQDN Resolver Refresh Interval

Syntax

```
show ip dns fqdn-resolver refresh-interval
```

Description

Displays the current refresh interval for FQDN resolvers.

Examples

Show current refresh interval (default):

```
switch# show ip dns fqdn-resolver refresh-interval
Refresh Interval: 86400 seconds
```

Show refresh interval after configuration change:

```
switch(config)# ip dns fqdn-resolver refresh-interval 7200
switch(config)# exit
switch# show ip dns fqdn-resolver refresh-interval
Refresh Interval: 7200 seconds
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution righ ts for this command.

Device discovery and configuration

The switch provides support for LLDP and CDP to enable automatic discovery and configuration of other devices on the network.

Subtopics

LLDP

The IEEE 802.1AB Link Layer Discovery Protocol (LLDP) provides a standards-based method for network devices to discover each other and exchange information about their capabilities. An LLDP device advertises itself to adjacent (neighbor) devices by transmitting LLDP data packets on all interfaces on which outbound LLDP is enabled, and reading LLDP advertisements from neighbor devices on ports on which inbound LLDP is enabled. Inbound packets from neighbor devices are stored in a special LLDP MIB (management information base). This information can then be queried by other devices through SNMP.

LLDP information is used by network management tools to create accurate physical network topologies by determining which devices are neighbors and through which interfaces they connect. LLDP operates at layer 2 and requires an LLDP agent to be active on each interface that sends and receives LLDP advertisements. LLDP advertisements can contain a variable number of TLV (type, length, value) information elements. Each TLV describes a single attribute of a device such as: system capabilities, management IP address, device ID, port ID.

Packet boundaries

When multiple LLDP devices are directly connected, an outbound LLDP packet travels only to the next LLDP device. An LLDP-capable device does not forward LLDP packets to any other devices, regardless of whether they are LLDP-enabled.

An intervening hub or repeater forwards the LLDP packets it receives in the same manner as any other multicast packets it receives. Therefore, two LLDP switches joined by a hub or repeater handle LLDP traffic in the same way that they would if directly connected.

Any intervening 802.1D device or Layer-3 device that is either LLDP-unaware or has disabled LLDP operation drops the packet.

LLDP-MED

LLDP-MED (ANSI/TIA-1057/D6) extends the LLDP (IEEE 802.1AB) industry standard to support advanced features on the network edge for Voice Over IP (VoIP) endpoint devices with specialized capabilities and LLDP-MED standards-based functionality. LLDP-MED in the switches uses the standard LLDP commands described earlier in this section, with some extensions, and also introduces new commands unique to LLDP-MED operation. The show commands described elsewhere in this section are applicable to both LLDP and LLDP-MED operation. LLDP-MED enables:

- Configure Voice VLAN and advertise it to connected MED endpoint devices.
- Power over Ethernet (PoE) status and troubleshooting support via SNMP.

Subtopics

LLDP agent

When you enable LLDP on the switch, it is automatically enabled on all data plane interfaces. You can customize this behavior by manually enabling/disabling support on each interface.

Supported standards

The LLDP agent supports the following standards: IEEE 802.1AB-2005, Station, and Media Access Control Connectivity Discovery.

Supported interfaces

LLDP is supported on interfaces mapped to a physical port, and the Out-Of-Band Management (OOBM) port. It is not supported on logical interfaces, such as loopback, tunnels, and SVIs.

Operating modes

When LLDP is enabled, the switch periodically transmits an LLDP advertisement (packet) out each active port enabled for outbound LLDP transmissions and receives LLDP advertisements on each active port enabled to receive LLDP traffic.

The LLDP agent can operate in one of the following modes:

- Transmit and receive (TxRx): This is the default setting on all ports. It enables a given port to both transmit and receive LLDP packets and to store the data from received (inbound) LLDP packets in the switch's MIB.
- Transmit only (Tx): Enables a port to transmit LLDP packets that can be read by LLDP neighbors. However, the port drops inbound LLDP packets from LLDP neighbors without reading them. This prevents the switch from learning about LLDP neighbors on that port.
- Receive only (Rx): Enables a port to receive and read LLDP packets from LLDP neighbors and to store the packet data in the switch's MIB. However, the port does not transmit outbound LLDP packets. This prevents LLDP neighbors from learning about the switch through that port.
- Disabled: Disables LLDP packet transmissions and reception on a port. In this state, the switch does not use the port for either learning about LLDP neighbors or informing LLDP neighbors of its presence.

An LLDP agent operating in TxRx mode or Tx mode sends LLDP frames to its directly connected devices both periodically and when the local configuration changes.

Sending LLDP frames

Each time the LLDP operating mode of an LLDP agent changes, its LLDP protocol state machine reinitializes. A configurable reinitialization delay prevents frequent initializations caused by frequent changes to the operating mode. If you configure the reinitialization delay, an LLDP agent must wait the specified amount of time to initialize LLDP after the LLDP operating mode changes.

Receiving LLDP frames

An LLDP agent operating in TxRx mode or Rx mode confirms the validity of TLVs carried in every received LLDP frame. If the TLVs are valid, the LLDP agent saves the information and starts an aging timer. The initial value of the aging timer is equal to the TTL value in the Time To Live TLV carried in the LLDP frame. When the LLDP agent receives a new LLDP frame, the aging timer restarts. When the aging timer decreases to zero, all saved information ages out.

TLV support

By default, the agent sends and receives the following mandatory TLVs on each interface:

- Port ID
- Chassis ID
- TTL

By default, the following ANSI/TIA-1057 TLVs for LLDP Media Endpoint Discovery (MED) are enabled on an agent. Sending them depends on the configuration and reception of any MED TLVs:

- MAC/PHY status. Includes the bit rate and auto negotiation status of the link.
- Power Via MDI: Includes Power Over Ethernet related information for supported interfaces.
- Port description
- System name
- System description
- Management address
- System capabilities
- Port VLAN ID

By default, the agent sends and receives the following ANSI/TIA-1057 TLVs for LLDP Media Endpoint Discovery (MED):

- Capabilities: Indicates MED TLV capability.
- Power Via MDI: Includes Power Over Ethernet related information.
- Network Policy: Includes the VLAN configuration for voice application.
- Location: Location identification information.

- Extended Power Via MDI: Power Over Ethernet related information

TLV advertisements

The LLDP agent transmits the following:

- Chassis-ID: Base MAC address of the switch.
- Port-ID: Port number of the physical port.
- Time-to-Live (TTL): Length of time an LLDP neighbor retains advertised data before discarding it.
- System capabilities: Identifies the primary switch capabilities (bridge, router). Identifies the primary switch functions that are enabled, such as routing.
- System description: Includes switch model name and running software version, and ROM version.
- System name: Name assigned to the switch.
- Management address: Default address selection method unless an optional address is configured.
- Port description: Physical port identifier.
- Port VLAN ID: On an L2 port, contains access or native VLAN ID. On an L3 port, contains a value of 0. Trunk allowed VLANs information are not advertised as part of the Port VLAN ID TLV. (Not supported on the OOBM interface)
- Port VLAN Name: Access or native VLAN name. Enabled by default.
- Port MFS: Maximum Frame Size. Enabled by default.

LLDP MED support

LLDP-MED interoperates with directly connected IP telephony (endpoint) clients and provides the following features:

- Advertisement of the voice VLAN configured on the interface which is used by connected IP telephony (endpoint) clients.
- Advertisement of the configured location on the switch that can be used by the connected endpoint.
- Support for the fast-start capability

> **NOTE**
> LLDP-MED is intended for use with VoIP endpoints and is not designed to support links between network infrastructure devices (such as switch-to-switch or switch-to-router links).

LLDP EEE

Energy Efficient Ethernet (EEE) is a mechanism defined by IEEE 802.3az, which is an extension of 802.3 IEEE standards. EEE defines support for the device to operate in the Low Power Idle (LPI) mode during low link utilization. This allows both sides of a link to disable or turn off respective transmit/receive circuitry to save power. EEE uses LLDP for exchanging optimal set of parameters for both devices.

Guidelines

- An LLDP must not contain more than one EEE TLV.
- LLDP may or may not be enabled on the interfaces supporting EEE. If LLDP is not enabled, it might not be in the optimal operational mode.
- EEE TLV should not be exchanged until it negotiates from L1 and it detects peer EEE capabilities.
- EEE TLV is disabled by default and enabled only when EEE is active.
- EEE and EEE TLV exchange can be enabled or disabled at the interface level.

Configuring the LLDP agent

Procedure

1. By default, the LLDP agent is enabled on all active interfaces. If LLDP was disabled, enable it with the command `lldp`.
2. By default, the LLDP agent transmits and receive on all interfaces. To customize LLDP behavior on a specific interface, use the commands `lldp transmit` and `lldp receive`.
3. By default, the LLDP agent sets the management address in all TLVs in the following order:
   a. LLDP management IP address.
   b. Loopback interface IP.
   c. ROP (L3 ports) or SVI (L2 ports).
   d. OOBM (Management interface IP).
   e. Base MAC.

   On the OOBM port, the following order is used:

   a. LLDP management IP address,

b. IP address of the management interface (OOBM port).

c. IP address of the loopback interface.

d. Base MAC address of the switch.

To specify a different address, use the commands lldp management-ipv4-address and lldp management-ipv6-address

4. By default, all supported TLVs are sent and received. To customize the list, use the command `lldp select-tlv`.

5. By default, support for the LLDP-MED TLV is enabled. To customize settings, use the commands `lldp med` and `lldp med-location`.

6. If required, adjust LLDP timer, holdtime, reinitialization delay, and transmit delay from their default values with the commands `lldp timer`, `lldp holdtime`, `lldp reinit`, and `lldp tx delay`.

Example

This example creates the following configuration:

- Enables LLDP support.
- Disables LLDP transmission on interface **1/1/1**.

```
switch(config)# lldp
switch(config)# interface 1/1/1
switch(config-copp)# no lldp transmit
```

LLDP commands

Select a command from the list in the left navigation menu.

Subtopics

`clear lldp neighbors`

Syntax

```
clear lldp neighbors
```

Description

Clears all LLDP neighbor details.

Examples

Clearing all LLDP neighbor details:

```
switch# clear lldp neighbors
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`clear lldp statistics`

Syntax

```
clear lldp statistics
```

Description

Clears all LLDP neighbor statistics.

Examples

Clearing all LLDP neighbor statistics:

```
switch# clear lldp statistics
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

lldp

Syntax

```
lldp
no lldp
```

Description

Enables LLDP support globally on all active interfaces. By default, LLDP is enabled.

The **no** form of this command disables LLDP support globally on all active interfaces. It does not remove any LLDP configuration settings.

Examples

Enabling LLDP:

```
switch(config)# lldp
```

Disabling LLDP:

```
switch(config)# no lldp
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`lldp dot3`

Syntax

```
lldp dot3 {poe | macphy}
no lldp dot3 {poe | macphy}
```

Description

Sets the 802.3 TLVs to be advertised. By default, advertisement of both POE and MAC/PHY TLVs is enabled. Not supported on the OOBM interface.

The **no** form of this command disables advertisement of 802.3 TLVs.

Parameter	Description
`poe`	Specifies advertisement of power over Ethernet data link classif ication.
`macphy`	Specifies advertisement of media access control and physical l ayer information.

Examples

Enabling advertisement of the POE TLV:

```
switch(config-if)# lldp dot3 poe
```

Disabling advertisement of the POE TLV:

```
switch(config-if)# no lldp dot3 poe
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config-if`	Administrators or local user group members with execution righ ts for this command.

`lldp dot3 mfs`

Syntax

```
lldp dot3 mfs
no lldp dot3 mfs
```

Description

Enables the 802.3 TLV list in LLDP to advertise for maximum frame size (MFS). Enabled by default.

The **no** form of this command disables the advertisement of maximum frame size TLVs.

Examples

Enabling advertisement of maximum frame size TLVs:

```
switch(config)# interface 1/1/1
switch(config-if)# lldp dot3 mfs
```

Disabling advertisement of maximum frame size TLVs:

```
switch(config)# interface 1/1/1
switch(config-if)# no lldp dot3 mfs
```

Command History

Release	Modification
10.11	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`lldp holdtime-multiplier`

Syntax

```
lldp holdtime-multiplier <multiplier>
no lldp holdtime-multiplier
```

Description

Sets the holdtime TTL multiplier value that is used to calculate the LLDP Time-to-Live value. Time-to-Live defines the length of time that neighbors consider LLDP information sent by this agent as valid. When Time-to-Live expires, the information is deleted by the neighbor. Time-to-live is calculated by multiplying holdtime by the value of **lldp timer**.

The **no** form of this command sets the holdtime TTL multiplier to its default value of 4.

Parameter	Description
`<multiplier>`	Specifies the TTL multiplier in the range of 2 to 10. Default: 4.

Formula

TTL = Holdtime-multiplier x lldp timer

where:

TTL = Time-to-Live

Holdtime-multiplier = Multiplying holdtime value

lldp timer = Message transmission interval

Examples

Setting the holdtime to 8 times of the value of lldp timer:

```
switch(config)# lldp holdtime-multiplier 8
```

Setting the holdtime to the default value of 4 times of the value of lldp timer:

```
switch(config)# no lldp holdtime-multiplier
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`lldp management-address vlan`

Syntax

```
lldp management-address vlan <VLAN-ID>
no lldp management-address vlan <VLAN-ID>
```

Description

Sets the VLAN whose IPv4 or IPv6 address is advertised as the LLDP management authority.

The **no** form of this command removes the VLAN whose IPv4 or IPv6 address is advertised as the LLDP management authority.

The following is the precedence for the management IP address TLV in the LLDP packet (in order):

- LLDP management-IP-address and management-ipv6-address, if configured.
- LLDP management VLAN's IPv4 and IPv6 address, if configured.
- Loopback IP address from the smallest configured loopback interface identifier.
- Route-only-port IP address (Layer-3 interface) or IP address of the SVI (Layer-2 interface).
- OOBM IP address.
- Base MAC address of the switch.

Parameter	Description
`<VLAN-ID>`	Specifies the VLAN ID. Range 1 - 4094.

Examples

Setting the management authority for VLAN 10:

```
switch(config)# lldp management-address vlan 10
```

Removing the management authority for VLAN 10:

```
switch(config)# no lldp management-address vlan 10
```

Command History

Release	Modification
10.12	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`lldp management-ip-address`

Syntax

```
lldp management-ip-address <IPV-ADDR>
no lldp management-ip-address
```

Description

Defines the IP management address of the switch which is sent in the management address TLV. One IPv4 and one IPv6 management address can be configured.

If you do not define an LLDP management address, then LLDP uses one of the following (in order):

- IP address of the port
- IP address of the management interface
- Base MAC address of the switch

The **no** form of this command removes the IPv4 management address of the switch.

Parameter	Description
`<IPV4-ADDR>`	Specifies the management address of the switch as an IPv4 for mat (**x.x.x.x**), where **x** is a decimal value from 0 to 255.

Examples

Setting the management address to **10.10.10.2**:

```
switch(config)# lldp management-ip-address 10.10.10.2
```

Removing the management address:

```
switch(config)# no lldp management-ip-address
```

Command History

Release	Modification
10.14	The **management - ipv4 - address** keyword is deprecated and replaced with **management - ip - address**.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`lldp management-ipv6-address`

Syntax

```
lldp management-ipv6-address <IPV6-ADDR>
no lldp management-ipv6-address
```

Description

Defines the IPv6 management address of the switch. The management address is encapsulated in the management address TLV.

If you do not define an LLDP management address, then LLDP uses one of the following (in order):

- IP address of the port
- IP address of the management interface
- Base MAC address of the switch

The **no** form of this command removes the IPv6 management address of the switch.

Parameter	Description
<IPV6-ADDR>	Specifies an IP address in IPv6 format (**xxxx:xxxx:xxxx:xxxx:x xxx:xxxx:xxxx:xxxx**), where **x** is a hexadecimal number from 0 to F.

Examples

Setting the management address to 2001:db8:85a3::8a2e:370:7334:

```
switch(config)# lldp management-ipv6-address 2001:0db8:85a3::8a2e:0370:7334
```

Removing the management address:

```
switch(config)# no lldp management-ipv6-address
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`lldp med`

Syntax

```
lldp med [poe [priority-override] | capability | network-policy]
no med [poe [priority-override] | capability | network-policy]
```

Description

Configures support for the LLDP-MED TLV. LLDP-MED (media endpoint devices) is an extension to LLDP developed by TIA to support interoperability between VoIP endpoint devices and other networking end-devices. The switch only sends the LLDP MED TLV after receiving a MED TLV from and connected endpoint device.

Not supported on the OOBM interface.

The **no** form of this command disables support for the LLDP MED TLV.

Parameter	Description
`poe [priority-override]`	Specifies advertisement of power over Ethernet data link classif ication. The **priority - override** option overrides user - config ured port priority for Power over Ethernet. When both **lldp dot3 poe** and **lldp med poe** are enabled, the **lldp dot3 poe3** setting takes precedence. Default: enabled.

Parameter	Description
`capability`	Specifies advertisement of supported LLDP MED TLVs. The cap ability TLV is always sent with other MED TLVs, therefore it can not be disabled when other MED TLVs are enabled. Default: ena bled.
`network-policy`	Network policy discovery lets endpoints and network devices a dvertise their VLAN IDs, and IEEE 802.1p (PCP and DSCP) valu es for voice applications. This TLV is only sent when a voice VL AN policy is present. Default: enabled.

Examples

Enabling advertisement of the network policy TLV:

```
switch(config-if)# lldp med network-policy
```

Disabling advertisement of the network policy TLV:

```
switch(config-if)# no lldp med network-policy
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config-if`	Administrators or local user group members with execution righ ts for this command.

`lldp med location`

Syntax

```
lldp med location {civic-addr   elin-addr }
no med location {civic-addr  elin-addr }
```

Description

Configures support for the LLDP-MED TLV. Supports only civic address and emergency location information number (ELIN). Coordinate-based location is not supported.

The **no** form of this command disables support for the LLDP MED TLV.

Parameter	Description
`civic-addr`	Configures the LLDP MED civic location TLV.
`elin-addr`	Configures support for the LLDP MED emergency location TLV. This feature is intended for use in ECS applications to support c lass 3 LLDP - MED VoIP telephones connected to a switch in a n MLTS infrastructure. An ELIN is a valid NANP format telephon e number assigned to MLTS operators in North America by the appropriate authority. The ELIN is used to route emergency (E9 11) calls to a PSAP. (Range: 1 - 15 numeric characters)

The **lldp med location civic-addr** command requires a minimum of one type/value pair, but typically includes multiple type/value pairs as needed to configure a complete set of data describing a given location.

- CA-TYPE: This is the first entry in a type/value pair and is a number defining the type of data contained in the second entry in the type/value pair (CA-VALUE.) Some examples of CA-TYPE specifiers include: 3=city 6=street (name) 25=building name (Range: 0 - 255)
- CA-VALUE: This is the second entry in a type/value pair and is an alphanumeric string containing the location information corresponding to the immediately preceding CA-TYPE entry. Strings are delimited by either blank spaces, single quotes (' ... '), or double quotes ("... ".) Each string should represent a specific data type in a set of unique type/value pairs comprising the description of a location, and each string must be preceded by a CA-TYPE number identifying the type of data in the string.

The following LLDP-MED TLV values are supported. For details on these value types, refer to **RFC 4776**

- 1: national subdivisions (state, canton, region, province, prefecture)
- 2: county, parish, gun (JP), district (IN)
- 3: city, township, shi (JP)
- 4: city division, borough, city district, ward, chou (JP)
- 5: neighborhood, block
- 6: group of streets below the neighborhood level
- 16: leading street direction N

- 17: trailing street suffix SW
- 18: street suffix or type
- 19: house number
- 20: house number suffix
- 21: landmark or vanity
- 22: location
- 23: name
- 24: postal/zip code
- 25: building (structure)
- 26: unit (apartment, suite)
- 27: floor
- 28: room
- 29: type of place
- 30: postal community name
- 31: post office box
- 32: additional code 13203000003
- 33: seat (desk, cubicle workstation)
- 34: primary road name 35 road section
- 36: branch road name
- 37: sub-branch road name
- 38: street name pre-modifier
- 39: street name post-modifier

Examples

Enabling support for the LLDP MED emergency location TLV:

```
switch(config-if)# lldp med location elin-addr 408-555-1212
```

Disabling support for the LLDP MED emergency location TLV:

```
switch(config-if)# no lldp med location elin-addr 408-555-1212
```

Enabling support for the LLDP MED civic address TLV:

```
switch(config-if)# lldp med location civic-addr US 1 19 123 6 Fake 18 Street
```

Disabling support for the LLDP MED civic address TLV:

```
switch(config-if)# no lldp med location civic-addr US 1 19 123 6 Fake 18
Street
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config-if`	Administrators or local user group members with execution righ ts for this command.

`lldp receive`

Syntax

```
lldp receive
no lldp receive
```

Description

Enables reception of LLDP information on an interface. By default, LLDP reception is enabled on all active interfaces, including the OOBM interface.

The **no** form of this command disables reception of LLDP information on an interface.

Examples

Enabling LLDP reception on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# lldp receive
```

Disabling LLDP reception on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no lldp receive
```

Enabling LLDP reception on the OOBM interface:

```
switch(config)# interface mgmt
switch(config-if)# lldp receive
```

Disabling LLDP reception on the OOBM interface:

```
switch(config)# interface mgmt
switch(config-if)# no lldp receive
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config-if`	Administrators or local user group members with execution righ ts for this command.

`lldp reinit`

Syntax

```
lldp reinit <TIME>
no lldp reinit
```

Description

Sets the amount of time (in seconds) to wait before performing LLDP initialization on an interface.

The **no** form of this command sets the reinitialization time to its default value of 2 seconds.

Parameter	Description
<TIME>	Specifies the reinitialization time in seconds. Range: 1 to 10. De fault: 2 seconds.

Examples

Setting the reinitialization time to 5 seconds:

```
switch(config)# lldp reinit 5
```

Setting the reinitialization time to the default value of 2 seconds:

```
switch(config)# no lldp reinit
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`lldp select-tlv`

Syntax

```
lldp select-tlv <TLV-NAME>
no lldp select-tlv <TLV-NAME>
```

Description

Selects a TLV that the LLDP agent will send and receive. By default, all supported TLVs are sent and received.

The **no** form of this command stops the LLDP agent from sending and receiving a specific TLV.

> **NOTE**
>
> LLDP supports Organization Unique Identifiers (OUIs) with the following Organization-specific TLVs:
>
> - IEEE 802.1 (DOT1) (oui:0x00, 0x80, 0xc2)
> - IEEE 802.3 (DOT3) (oui:0x00, 0x12, 0x0f)
> - HPE Aruba Networking (oui:0x88, 0x3a, 0x30)

Parameter	Description
`select-tlv <TLV-NAME>`	Specifies the TLV name to send. The following TLV names are s upported: • **management - address:** Selection is based on priority in t he following list (for example if first TLV name isn't selected , the next will be, progressing through this list until a selecti on is made): 1. IPv4 or IPV6 management address. 2. IP address of the lowest configured loopback interface. 3. If layer 3, then the route - only port IP address. If layer 2, the IP address of the SVI. 4. OOBM interface IP address. 5. Base MAC address of the switch. • **port - description:** Select port - description TLV. • **port - vlan - id:** Select port - vlan - id TLV. • **port - vlan - name:** Select port - vlan - name TLV. • **system - capabilities:** Select system - capabilities TLV. • **system - description:** Select system - description TLV. • **system - name:** Select system - name TLV.

Examples

Stopping the LLDP agent from sending the **port-description** TLV:

```
switch(config)# no lldp select-tlv port-description
```

Enabling the LLDP agent to send the **port-description** TLV:

```
switch(config)# lldp select-tlv oui
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`lldp timer`

Syntax

```
lldp timer <TIME>
no lldp timer
```

Description

Sets the interval (in seconds) at which local LLDP information is updated and TLVs are sent to neighboring network devices by the LLDP agent. The minimum setting for this timer must be four times the value of **lldp txdelay**.

For example, this is a valid configuration:

- **lldp timer** = 16
- **lldp txdelay** = 4

And, this is an invalid configuration:

- **lldp timer** = 5

- **lldp txdelay** = 2

> **NOTE**
>
> When copying a saved configuration to the running configuration, the value for **lldp timer** is applied before the value of **lldp txdelay**. This can result in a configuration error if the saved configuration has a value of **lldp timer** that is not four times the value of **lldp txdelay** in the running configuration.
>
> For example, if the saved configuration has the settings:
>
> - **lldp timer** = 16
> - **lldp txdelay** = 4
>
> And the running configuration has the settings:
>
> - **lldp timer** = 30
> - **lldp txdelay** = 7
>
> Then you will see an error indicating that certain configuration settings could not be applied, and you will have to manually adjust the value of **lldp txdelay** in the running configuration.

The **no** form of this command sets the update interval to its default value of 30 seconds.

Parameter	Description
<TIME>	Specifies the update interval (in seconds). Range: 5 to 32768. Default: 30.

Examples

Setting the update interval to 7 seconds:

```
switch(config)# lldp timer 7
```

Setting the update interval to the default value of 30 seconds:

```
switch(config)# no lldp timer
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

`lldp tlv-enable`

Syntax

```
lldp tlv-enable basic-tlv management-address-tlv {ip <X.X.X.X>}|{ipv6
<X:X::X:X>}
no lldp select-tlv <TLV-NAME>
```

Description

A **TLV** is an information element that contains the type, length, and value fields. This command configures the IPv4/IPv6 address to be advertised in the management address TLV. This configured value will take the highest precedence in the selection logic the switch uses to identify the IP address to be advertised in the management address TLV.

Parameter	Description
ip <X.X.X.X>	Specify an IP address in IPv4 format **(X.X.X.X)**, where **X** is a decimal number from 0 to 255.
ipv6 <X:X::X:X>	Specify an IP address in IPv6 format (`X:X::X:X`), where **X** is a hexadecimal number from **0** to **F**.

Examples

Configuring an IP address in IPv4 format:

```
switch(config)# interface 1/3/1
switch(config-if)# lldp tlv-enable basic-tlv management-address-tlv ip
10.0.0.1
```

Configuring an IP address in IPv6 format:

```
switch(config)# interface 1/3/1
switch(config-if)# lldp tlv-enable basic-tlv management-address-tlv ipv6
2001:db8:85a3::8a2e:370:7334
```

Command History

Release	Modification
10.14.1000	Command introduced.

Command Information

Platforms	Command context	Authority
8400	`config-if`	Administrators or local user group members with execution righ ts for this command.

`lldp transmit`

Syntax

```
lldp transmit
no lldp transmit
```

Description

Enables transmission of LLDP information on specific interface. By default, LLDP transmission is enabled on all active interfaces, including the OOBM interface.

The **no** form of this command disables transmission of LLDP information on an interface.

Examples

Enabling LLDP transmission on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# lldp transmit
```

Disabling LLDP transmission on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no lldp transmit
```

Enabling LLDP transmission on the OOBM interface:

```
switch(config)# interface mgmt
switch(config-if)# lldp transmit
```

Disabling LLDP transmission on the OOBM interface:

```
switch(config)# interface mgmt
switch(config-if)# no lldp transmit
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config-if`	Administrators or local user group members with execution righ ts for this command.

`lldp txdelay`

Syntax

```
lldp txdelay <TIME>
no lldp txdelay
```

Description

Sets the amount of time (in seconds) to wait before sending LLDP information from any interface. The maximum value for **txdelay** is 25% of the value of **lldp tx timer**.

The **no** form of this command sets the delay time to its default value of 2 seconds.

Parameter	Description
<TIME>	Specifies the delay time in seconds. Range: 0 to 10. Default: 2.

Examples

Setting the delay time to 8 seconds:

```
switch(config)# lldp txdelay 8
```

Setting the delay time to the default value of 2 seconds:

```
switch(config)# no lldp txdelay
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`lldp trap enable`

Syntax

```
lldp trap enable
no lldp trap enable
```

Description

Enables sending SNMP traps for LLDP related events from a particular interface. LLDP trap generation is enabled by default on all the interfaces and has to be disabled for interfaces on which traps are not required to be generated.

The **no** form of this command disables the LLDP trap generation.

> **NOTE**
> LLDP trap generation is disabled by default at the global level and must be enabled before any LLDP traps are sent.

Examples

Enabling LLDP trap generation on global level:

```
switch(config)# lldp trap enable
```

Enabling LLDP trap generation on interface level:

```
switch(config-if)# lldp trap enable
```

Disabling LLDP trap generation on global level:

```
switch(config)# no lldp trap enable
```

Disabling LLDP trap generation on interface level:

```
switch(config-if)# no lldp trap enable
```

Displaying LLDP global configuration:

```
switch# show lldp configuration
LLDP Global Configuration
=========================
LLDP Enabled                 : No
LLDP Transmit Interval       : 30
LLDP Hold Time Multiplier    : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Timer Interval   : 2
LLDP Trap Enabled            : No
TLVs Advertised
===============
```

```
Management Address
Port Description
Port VLAN-ID
System Description
System Name
LLDP Port Configuration
=======================
PORT           TX-ENABLED          RX-ENABLED          INTF-TRAP-ENABLED
---------------------------------------------------------------------------
1/1/1          Yes                 Yes                 Yes
1/1/2          Yes                 Yes                 Yes
1/1/3          Yes                 Yes                 Yes
1/1/4          Yes                 Yes                 Yes
1/1/5          Yes                 Yes                 Yes
1/1/6          Yes                 Yes                 Yes
...........
...........
mgmt           Yes                 Yes                 Yes
```

Displaying LLDP Configuration for the interface:

```
switch# show lldp configuration 1/1/1
LLDP Global Configuration
=========================
LLDP Enabled                  : Yes
LLDP Transmit Interval        : 30
LLDP Hold Time Multiplier     : 4
LLDP Transmit Delay Interval  : 2
LLDP Reinit Timer Interval    : 2
LLDP Trap Enabled             : No
LLDP Port Configuration
=======================
PORT           TX-ENABLED          RX-ENABLED          INTF-TRAP-ENABLED
---------------------------------------------------------------------------
1/1/1          Yes                 Yes                 Yes
```

Displaying LLDP Configuration for the management interface:

```
switch# show lldp configuration mgmt
LLDP Global Configuration
=========================
LLDP Enabled                  : Yes
LLDP Transmit Interval        : 30
LLDP Hold Time Multiplier     : 4
LLDP Transmit Delay Interval  : 2
LLDP Reinit Timer Interval    : 2
LLDP Trap Enabled             : Yes
```

```
LLDP Port Configuration
=======================
PORT            TX-ENABLED          RX-ENABLED          INTF-TRAP-ENABLED
--------------------------------------------------------------------------
mgmt            Yes                 Yes                 Yes
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config` and `config-if`	Administrators or local user group members with execution rights for this command.

`show lldp configuration`

Syntax

```
show lldp configuration [<INTERFACE-ID>][vsx-peer]
```

Description

Shows LLDP configuration settings for all interfaces or a specific interface.

Parameter	Description
`<INTERFACE-ID>`	Specifies an interface. Format: **member/slot/port**.
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Example

Showing configuration settings for all interfaces:

```
switch# show lldp configuration
LLDP Global Configuration
=========================
LLDP Enabled                 : No
LLDP Transmit Interval       : 30
LLDP Hold Time Multiplier    : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Timer Interval   : 2
LLDP Trap Enabled            : No
TLVs Advertised
===============
Management Address
Port Description
Port VLAN-ID
System Description
System Name
LLDP Port Configuration
=======================
PORT          TX-ENABLED          RX-ENABLED          INTF-TRAP-ENABLED
--------------------------------------------------------------------------
1/1/1         Yes                 Yes                 Yes
1/1/2         Yes                 Yes                 Yes
1/1/3         Yes                 Yes                 Yes
1/1/4         Yes                 Yes                 Yes
1/1/5         Yes                 Yes                 Yes
1/1/6         Yes                 Yes                 Yes
...........
...........
mgmt          Yes                 Yes                 Yes
```

This example shows configuration settings for interface **1/1/1**.

```
switch# show lldp configuration 1/1/1
LLDP Global Configuration
=========================
LLDP Enabled                 : Yes
LLDP Transmit Interval       : 30
LLDP Hold Time Multiplier    : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Timer Interval   : 2
LLDP Trap Enabled            : No
LLDP Port Configuration
```

```
========================
Auto Flush On Link Down        : Yes
Management IPv4 Address        : 10.1.1.1
Management IPv6 Address        : 3001:1::1:1
```

Command History

Release	Modification
10.14.1000	When displaying the LLDP configuration for a specific interface, the output of th is command displays any configured management IP addresses.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Operators or Administrators or l ocal user group members with exe cution rights for this command. O perators can execute this comman d from the operator context (>) on ly.

`show lldp configuration mgmt`

Syntax

```
show lldp configuration mgmt
```

Description

Shows LLDP configuration settings for the OOBM interface.

Example

Showing configuration settings for all interfaces:

```
switch# show lldp configuration mgmt
LLDP Global Configuration
=========================
LLDP Enabled                   : Yes
```

```
LLDP Transmit Interval        : 30
LLDP Hold Time Multiplier     : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Timer Interval    : 2
LLDP Trap Enabled             : Yes
LLDP Port Configuration
=======================
PORT           TX-ENABLED          RX-ENABLED          INTF-TRAP-ENABLED
---------------------------------------------------------------------------
mgmt           Yes                 Yes                 Yes
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

show lldp local-device

Syntax

```
show lldp local-device[vsx-peer]
```

Description

Shows global LLDP information advertised by the switch, as well as port-based data. If VLANs are configured on any active interfaces, the VLAN ID is only shown for trunk native or untagged VLAN IDs on access interfaces.

Parameter	Description
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output f rom the VSX peer switch is not displayed. This parameter is avai lable on switches that support VSX.

Example

Showing global LLDP information only (all ports including OOBM port are administratively down):

```
switch# show lldp local-device
Global Data
===========
Chassis-ID              : 1c:98:ec:e3:45:00
System Name             : switch
System Description      : HPE ANW JL375A 8400X XL.01.01.0001
Management Address      : 192.168.10.1
Capabilities Available : Bridge, Router
Capabilities Enabled    : Bridge, Router
TTL                     : 120
switch# show lldp local-device
Global Data
===========
Chassis-ID              : 88:3a:30:47:c1:c0
System Name             : 6100
System Description      : HPE ANW JL679A  PL.10.06.0001-346-g56a12a8f4cf15
Management Address      : 88:3a:30:47:c1:c0
Capabilities Available : Bridge, Router
Capabilities Enabled    : Bridge, Router
TTL                     : 120
```

Showing all ports except **1/1/11** and OOBM as administratively down:

```
switch# show lldp local-device
Global Data
===========
Chassis-ID              : 1c:98:ec:e3:45:00
System Name             : switch
System Description      : HPE ANW
Management Address      : 192.168.10.1
Capabilities Available : Bridge, Router
Capabilities Enabled    : Bridge, Router
TTL                     : 120
Port Based Data
```

```
===============
Port-ID             : 1/1/11
Port-Desc           : "1/1/11"
Port Mgmt-Address : 164.254.21.220
Port VLAN ID        : 1
Port-ID             : mgmt
Port-Desc           : "mgmt"
Port Mgmt-Address : 164.254.21.220
switch# show lldp local-device
Global Data
===========
Chassis-ID                : 88:3a:30:47:c1:c0
System Name               : 6100
System Description        : HPE ANW JL679A  PL.10.06.0001-346-g56a12a8f4cf15
Management Address        : 11.11.11.11
Capabilities Available : Bridge, Router
Capabilities Enabled    : Bridge, Router
TTL                       : 120
Port Based Data
===============
Port-ID             : 1/1/11
Port-Desc           : "1/1/11"
Port Mgmt-Address : 11.11.11.11
Port VLAN ID        : 1
Parent Interface   : interface 1/1/11
switch#
switch# show lldp local-device
Global Data
===========
Chassis-ID                : 88:3a:30:47:c1:c0
System Name               : 4100i
System Description        : HPE ANW JL817A  RL.10.08.0001
Management Address        : 11.11.11.11
Capabilities Available : Bridge, Router
Capabilities Enabled    : Bridge, Router
TTL                       : 120
Port Based Data
===============
Port-ID             : 1/1/11
Port-Desc           : "1/1/11"
Port Mgmt-Address : 11.11.11.11
Port VLAN ID        : 1
```

```
Parent Interface  : interface 1/1/11
switch#
```

In this example, all the ports except **1/1/11** are administratively down, and VLAN ID 100 is configured on this access interface.

```
switch# show lldp local-device
Global Data
===========
Chassis-ID                : 1c:98:ec:e3:45:00
System Name               : switch
System Description        : HPE ANW
Management Address        : 192.168.10.1
Capabilities Available : Bridge, Router
Capabilities Enabled    : Bridge, Router
TTL                       : 120
Port Based Data
===============
Port-ID            : 1/1/11
Port-Desc          : "1/1/11"
Port VLAN ID       : 100
Parent Interface  : interface 1/1/11
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or l ocal user group members with exe cution rights for this command. O perators can execute this comman d from the operator context (>) on ly.

show lldp neighbor-info

Syntax

```
show lldp neighbor-info [<INTERFACE-NAME>][vsx-peer]
```

Description

Displays information about neighboring devices for all interfaces or for a specific interface. The information displayed varies depending on the type of neighbor connected and the type of TLVs sent by the neighbor.

Parameter	Description
`<INTERFACE-NAME>`	Specifies the interface for which to show information for neigh boring devices. Use the format **member/slot/port** (for example, **1/3/1**).
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output f rom the VSX peer switch is not displayed. This parameter is avai lable on switches that support VSX.

Examples

Showing LLDP information for all interfaces:

```
switch# show lldp neighbor-info
LLDP Neighbor Information
=========================
Total Neighbor Entries          : 3
Total Neighbor Entries Deleted  : 0
Total Neighbor Entries Dropped  : 0
Total Neighbor Entries Aged-Out : 0
LOCAL-PORT  CHASSIS-ID         PORT-ID        PORT-DESC      TTL      SYS-NAME
-----------------------------------------------------------------------------
----
1/1/1       70:72:cf:a4:7d:50  1/1/1          1/1/1          32       switch
1/1/2       48:0f:cf:af:73:80  1/1/2          1/1/2          120      switch
1/1/46      48:0f:cf:af:73:80  1/1/46         1/1/46         120      switch
mgmt        48:0f:cf:af:73:80  mgmt           mgmt           120      switch
switch# show lldp neighbor-info
LLDP Neighbor Information
=========================
Total Neighbor Entries          : 1
```

```
Total Neighbor Entries Deleted  : 5
Total Neighbor Entries Dropped  : 0
Total Neighbor Entries Aged-Out : 3
LOCAL-PORT  CHASSIS-ID          PORT-ID        PORT-DESC      TTL      SYS-NAME
----------------------------------------------------------------------------
----
1/1/2       38:21:c7:5c:df:40  1/1/2          1/1/2          120      switch
1/1/3       f8:60:f0:c9:e0:a0  1/1/3          1/1/3          120      switch
```

Showing information for interface **1/3/1** when it has only one switch connected as a neighbor:

```
switch# show lldp neighbor-info 1/3/1
Port                            : 1/1/1
Neighbor Entries                : 1
Neighbor Entries Deleted        : 0
Neighbor Entries Dropped        : 0
Neighbor Entries Aged-Out       : 0
Neighbor Chassis-Name           : HP-3800-24G-PoEP-2XG
Neighbor Chassis-Description    : HP J9587A 3800-24G-PoE+-2XG Switch,
revision...
Neighbor Chassis-ID             : 10:60:4b:39:3e:80
Neighbor Management-Address     : 192.168.1.1
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled    : Bridge
Neighbor Port-ID                : 1/1/1
Neighbor Port-Desc              : 1/1/1
Neighbor Port VLAN ID           : 1
Neighbor Port VLAN Name         : DEFAULT_VLAN_1
Neighbor Port MFS               : 1500
TTL                             : 120
```

```
Switch2# show lldp neighbor-info 1/1/3
Port                            : 1/1/3
Neighbor Entries                : 1
Neighbor Entries Deleted        : 1
Neighbor Entries Dropped        : 0
Neighbor Entries Aged-Out       : 1
Neighbor Chassis-Name           : 6100
Neighbor Chassis-Description    : HPE ANW JL679A PL.10.15.0001
Neighbor Chassis-ID             : 88:3a:30:47:d1:c0
Neighbor Management-Address     : 88:3a:30:47:d1:c0
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled    : Bridge, Router
Neighbor Port-ID                : 1/1/3
Neighbor Port-Desc              : 1/1/3
Neighbor Port VLAN ID           : 1
```

```
Neighbor Port VLAN Name          : DEFAULT_VLAN_1
Neighbor Port MFS                : 1500
TTL                              : 120
Neighbor Mac-Phy details
Neighbor Auto-neg Supported      : True
Neighbor Auto-neg Enabled        : True
Neighbor Auto-ned Advertised     : 1000 BASE_TFD, 100 BASE_T4, 10 BASET_FD
Neighnor MAU Type                : 1000 BASETFD
```

Showing information for interface **1/3/10** when the neighbor sends a DOT3 power TLV:

```
switch# show lldp neighbor-info 1/3/10
Port                             : 1/3/10
Neighbor Entries                 : 1
Neighbor Entries Deleted         : 0
Neighbor Entries Dropped         : 0
Neighbor Entries Aged-Out        : 0
Neighbor Chassis-Name            : 84:d4:7e:ce:5d:68
Neighbor Chassis-Description     : ArubaOS (MODEL: 325), Version HPE_ANW IAP
Neighbor Chassis-ID              : 84:d4:7e:ce:5d:68
Neighbor Management-Address      : 169.254.41.250
Chassis Capabilities Available : Bridge, WLAN
Chassis Capabilities Enabled     : WLAN
Neighbor Port-ID                 : 84:d4:7e:ce:5d:68
Neighbor Port-Desc               : eth0
TTL                              : 120
Neighbor Port VLAN ID            : 1
Neighbor Port VLAN Name          : DEFAULT_VLAN_1
Neighbor Port MFS                : 1500
Neighbor PoE information         : DOT3
Neighbor Power Type              : TYPE2 PD
Neighbor Power Priority          : Unkown
Neighbor Power Source            : Primary
PD Requested Power Value         : 25.0 W
PSE Allocated Power Value: 25.0 W
Neighbor Power Supported         : Yes
Neighbor Power Enabled           : Yes
Neighbor Power Class             : 5
Neighbor Power Paircontrol       : No
PSE Power Pairs                  : Signal
```

Showing information for interface **1/1/1** when it has multiple neighbors (displays a maximum of four):

```
switch# show lldp neighbor-info 1/1/1
Port                             : 1/1/1
Neighbor Entries                 : 4
Neighbor Entries Deleted         : 0
```

```
Neighbor Entries Dropped        : 0
Neighbor Entries Aged-Out       : 0
Neighbor Chassis-Name           : switch
Neighbor Chassis-Description    : hpe_anw JL375A 8400X XL.01.01.0001
Neighbor Chassis-ID             : 1c:98:ec:fe:25:00
Neighbor Management-Address     : 10.1.1.2
Chassis Capabilities Available  : Bridge, Router
Chassis Capabilities Enabled    : Bridge, Router
Neighbor Port-ID                : 1/1/1
Neighbor Port-Desc              : 1/1/1
Neighbor Port VLAN ID           : 1
Neighbor Port VLAN Name         : DEFAULT_VLAN_1
Neighbor Port MFS               : 1500
TTL                             : 120
Neighbor Chassis-Name           : switch
Neighbor Chassis-Description    : hpe_anw JL375A 8400X XL.01.01.0001
Neighbor Chassis-ID             : 1c:98:ec:fe:25:01
Neighbor Management-Address     : 10.1.1.3
Chassis Capabilities Available  : Bridge, Router
Chassis Capabilities Enabled    : Bridge, Router
Neighbor Port-ID                : 1/1/1
Neighbor Port-Desc              : 1/1/1
Neighbor Port VLAN ID           : 1
Neighbor Port VLAN Name         : DEFAULT_VLAN_1
Neighbor Port MFS               : 1500
TTL                             : 120
Neighbor Chassis-Name           : switch
Neighbor Chassis-Description    : HPE ANW JL375A 8400X XL.01.01.0001
Neighbor Chassis-ID             : 1c:98:ec:fe:25:02
Neighbor Management-Address     : 10.1.1.4
Chassis Capabilities Available  : Bridge, Router
Chassis Capabilities Enabled    : Bridge, Router
Neighbor Port-ID                : 1/1/1
Neighbor Port-Desc              : 1/1/1
Neighbor Port VLAN ID           : 50
Neighbor Port VLAN Name         : VLAN_50
Neighbor Port MFS               : 1500
TTL                             : 120
Neighbor Chassis-Name           : switch
Neighbor Chassis-Description    : hpe_anw JL375A 8400X XL.01.01.0001
Neighbor Chassis-ID             : 1c:98:ec:fe:25:03
Neighbor Management-Address     : 10.1.1.5
Chassis Capabilities Available  : Bridge, Router
```

```
Chassis Capabilities Enabled   : Bridge, Router
Neighbor Port-ID               : 1/1/1
Neighbor Port-Desc             : 1/1/1
Neighbor Port VLAN ID          : 100
Neighbor Port VLAN Name        : VLAN_100
Neighbor Port MFS              : 1500
TTL                            : 120
```

```
switch# show lldp neighbor-info 1/3/2
Port                           : 1/3/2
Neighbor Entries               : 1
Neighbor Entries Deleted       : 1
Neighbor Entries Dropped       : 0
Neighbor Entries Aged-Out      : 1
Neighbor Chassis-Name          : BLDG01-F1-6300
Neighbor Chassis-Description   : hpe_anw JL668A  FL.10.15.0001
Neighbor Chassis-ID            : 88:3a:30:92:a5:c0
Neighbor Management-Address    : 10.6.9.15
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled   : Bridge, Router
Neighbor Port-ID               : 1/1/1
Neighbor Port-Desc             : 1/1/1
Neighbor Port VLAN ID          : 1
Neighbor Port VLAN Name        : DEFAULT_VLAN_1
Neighbor Port MFS              : 1500
TTL                            : 120
Neighbor Mac-Phy details
Neighbor Auto-neg Supported    : true
Neighbor Auto-Neg Enabled      : true
Neighbor Auto-Neg Advertised   : 1000 BASE_TFD, 100 BASE_T4, 10 BASET_FD
Neighbor MAU type              : 1000 BASETFD
Neighbor EEE information       : DOT3
Neighbor TX Wake time          : 17 us
Neighbor RX Wake time          : 17 us
Neighbor Fallback time         : 17 us
Neighbor TX Echo time          : 17 us
Neighbor RX Echo time          : 17 us
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show lldp neighbor-info detail`

Syntax

```
show lldp neighbor-info detail [vsx-peer]
```

Description

Shows detailed LLDP neighbor information for all LLDP neighbor connected interfaces.

Parameter	Description
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output f rom the VSX peer switch is not displayed. This parameter is avai lable on switches that support VSX.

Examples

Showing detailed LLDP information for all interfaces:

```
switch# show lldp neighbor-info detail
LLDP Neighbor Information
=========================
Total Neighbor Entries           : 6
Total Neighbor Entries Deleted   : 2
Total Neighbor Entries Dropped   : 0
Total Neighbor Entries Aged-Out : 2
----------------------------------------------------------------------------
----
Port                            : 1/1/1
Neighbor Entries                : 1
Neighbor Entries Deleted        : 0
```

```
Neighbor Entries Dropped       : 0
Neighbor Entries Aged-Out      : 0
Neighbor Chassis-Name          : 6300
Neighbor Chassis-Description   : hpe_anw ...
Neighbor Chassis-ID            : 38:11:17:1a:d5:00
Neighbor Management-Address    : 38:11:17:1a:d5:00
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled   : Bridge, Router
Neighbor Port-ID               : 1/1/4
Neighbor Port-Desc             : 1/1/4
Neighbor Port VLAN ID          : 1
Neighbor Port VLAN Name        : DEFAULT_VLAN_1
Neighbor Port MFS              : 1500
TTL                            : 120
Neighbor Mac-Phy details
Neighbor Auto-neg Supported    : true
Neighbor Auto-Neg Enabled      : true
Neighbor Auto-Neg Advertised   : 1000 BASE_TFD, 100 BASE_T4, 10 BASET_FD
Neighbor MAU type              : 1000 BASETFD
----------------------------------------------------------------------------
----
Port                           : 1/1/2
Neighbor Entries               : 1
Neighbor Entries Deleted       : 0
Neighbor Entries Dropped       : 0
Neighbor Entries Aged-Out      : 0
Neighbor Chassis-Name          : 6300
Neighbor Chassis-Description   : hpe_anw...
Neighbor Chassis-ID            : 38:11:17:1a:d5:00
Neighbor Management-Address    : 38:11:17:1a:d5:00
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled   : Bridge, Router
Neighbor Port-ID               : 1/1/5
Neighbor Port-Desc             : 1/1/5
Neighbor Port VLAN ID          : 1
Neighbor Port VLAN Name        : DEFAULT_VLAN_1
Neighbor Port MFS              : 1500
TTL                            : 120
Neighbor Mac-Phy details
Neighbor Auto-neg Supported    : true
Neighbor Auto-Neg Enabled      : true
Neighbor Auto-Neg Advertised   : 1000 BASE_TFD, 100 BASE_T4, 10 BASET_FD
Neighbor MAU type              : 1000 BASETFD
```

```
--------------------------------------------------------------------------------
----
Port                           : 1/1/3
Neighbor Entries               : 1
Neighbor Entries Deleted       : 0
Neighbor Entries Dropped       : 0
Neighbor Entries Aged-Out      : 0
Neighbor Chassis-Name          : 6300
Neighbor Chassis-Description   : hpe_anw ...
Neighbor Chassis-ID            : 38:11:17:1a:d5:00
Neighbor Management-Address    : 38:11:17:1a:d5:00
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled   : Bridge, Router
Neighbor Port-ID               : 1/1/6
Neighbor Port-Desc             : 1/1/6
Neighbor Port VLAN ID          : 1
Neighbor Port VLAN Name        : DEFAULT_VLAN_1
Neighbor Port MFS              : 1500
TTL                            : 120
Neighbor Mac-Phy details
Neighbor Auto-neg Supported    : true
Neighbor Auto-Neg Enabled      : true
Neighbor Auto-Neg Advertised   : 1000 BASE_TFD, 100 BASE_T4, 10 BASET_FD
Neighbor MAU type              : 1000 BASETFD
--------------------------------------------------------------------------------
----
Port                           : 1/1/46
Neighbor Entries               : 1
Neighbor Entries Deleted       : 0
Neighbor Entries Dropped       : 0
Neighbor Entries Aged-Out      : 0
Neighbor Chassis-Name          : 6300
Neighbor Chassis-Description   : hpe_anw ...
Neighbor Chassis-ID            : 38:11:17:1a:d5:00
Neighbor Management-Address    : 38:11:17:1a:d5:00
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled   : Bridge, Router
Neighbor Port-ID               : 1/1/19
Neighbor Port-Desc             : 1/1/19
Neighbor Port VLAN ID          : 1
Neighbor Port VLAN Name        : DEFAULT_VLAN_1
Neighbor Port MFS              : 1500
TTL                            : 120
```

```
Neighbor Mac-Phy details
Neighbor Auto-neg Supported     : true
Neighbor Auto-Neg Enabled       : true
Neighbor Auto-Neg Advertised    : 1000 BASE_TFD, 100 BASE_T4, 10 BASET_FD
Neighbor MAU type               : 1000 BASETFD
----------------------------------------------------------------------------
----
Port                            : 1/1/47
Neighbor Entries                : 1
Neighbor Entries Deleted        : 0
Neighbor Entries Dropped        : 0
Neighbor Entries Aged-Out       : 0
Neighbor Chassis-Name           : 6300
Neighbor Chassis-Description    : hpe_anw ...
Neighbor Chassis-ID             : 38:11:17:1a:d5:00
Neighbor Management-Address     : 38:11:17:1a:d5:00
Chassis Cap
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

show lldp neighbor-info mgmt

Syntax

```
show lldp neighbor-info mgmt
```

Description

Displays information about neighboring devices connected to the OOBM interface.

Examples

Showing LLDP information for the OOBM interface:

```
switch# show lldp neighbor-info mgmt
Port                              : mgmt
Neighbor Entries                  : 1
Neighbor Entries Deleted          : 0
Neighbor Entries Dropped          : 0
Neighbor Entries Aged-Out         : 0
Neighbor Chassis-Name             : HP-3800-24G-PoEP-2XG
Neighbor Chassis-Description      : HP J9587A 3800-24G-PoE+-2XG Switch,
revision...
Neighbor Chassis-ID               : 10:60:4b:39:3e:80
Neighbor Management-Address       : 192.168.1.1
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled      : Bridge
Neighbor Port-ID                  : mgmt
Neighbor Port-Desc                : mgmt
TTL                               : 120
```

Showing LLDP information for the OOBM interface when there are four neighbors:

```
switch# show lldp neighbor-info mgmt
Port                              : mgmt
Neighbor Entries                  : 4
Neighbor Entries Deleted          : 0
Neighbor Entries Dropped          : 0
Neighbor Entries Aged-Out         : 0
Neighbor Chassis-Name             : switch
Neighbor Chassis-Description      : HPE ANW JL375A 8400X XL.01.01.0001
Neighbor Chassis-ID               : 1c:98:ec:fe:25:00
Neighbor Management-Address       : 10.1.1.2
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled      : Bridge, Router
Neighbor Port-ID                  : 1/1/1
Neighbor Port-Desc                : 1/1/1
TTL                               : 120
Neighbor Chassis-Name             : switch
Neighbor Chassis-Description      : HPE ANW JL375A 8400X XL.01.01.0001
Neighbor Chassis-ID               : 1c:98:ec:fe:25:01
Neighbor Management-Address       : 10.1.1.3
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled      : Bridge, Router
Neighbor Port-ID                  : 1/1/1
Neighbor Port-Desc                : 1/1/1
```

```
TTL                               : 120
Neighbor Chassis-Name             : switch
Neighbor Chassis-Description      : HPE ANW JL375A 8400X XL.01.01.0001
Neighbor Chassis-ID               : 1c:98:ec:fe:25:02
Neighbor Management-Address       : 10.1.1.4
Chassis Capabilities Available    : Bridge, Router
Chassis Capabilities Enabled      : Bridge, Router
Neighbor Port-ID                  : 1/1/1
Neighbor Port-Desc                : 1/1/1
TTL                               : 120
Neighbor Chassis-Name             : switch
Neighbor Chassis-Description      : HPE ANW JL375A 8400X XL.01.01.0001
Neighbor Chassis-ID               : 1c:98:ec:fe:25:03
Neighbor Management-Address       : 10.1.1.5
Chassis Capabilities Available    : Bridge, Router
Chassis Capabilities Enabled      : Bridge, Router
Neighbor Port-ID                  : 1/1/1
Neighbor Port-Desc                : 1/1/1
TTL                               : 120
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show lldp neighbor-info json`

Syntax

```
show lldp neighbor-info json
```

Description

Displays LLDP neighbor information in structured JSON format. This command provides the same information as the standard `show lldp neighbor-info` command, but it prints the output in JSON format.

Example

```
switch# show lldp neighbor-info json
{
    "total neighbor entries": 3,
    "total neighbor entries deleted": 0,
    "total neighbor entries dropped": 0,
    "total neighbor entries aged out": 0,
    "neighbors": [
        {
            "local port": "1/1/25",
            "chassis id": "90:20:c2:1c:b5:40",
            "port id": "1/1/26",
            "port description": "1/1/26",
            "ttl": "120",
            "sys name": "switch"
        },
        {
            "local port": "1/1/26",
            "chassis id": "90:20:c2:1c:b5:40",
            "port id": "1/1/25",
            "port description": "1/1/25",
            "ttl": "120",
            "sys name": "switch"
        },
        {
            "local port": "mgmt",
            "chassis id": "d4:e0:53:99:78:80",
            "port id": "25",
            "port description": "25",
            "ttl": "120",
            "sys name": "Uplinks-213"
        }
    ]
}
```

Command History

Release	Modification
10.18 or later	- -

`show lldp statistics`

Syntax

```
show lldp statistics [<INTERFACE-ID>][vsx-peer]
```

Description

Shows global LLDP statistics or statistics for a specific interface.

Parameter	Description
`<INTERFACE-ID>`	Specifies an interface. Format: **member/slot/port**.
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output f rom the VSX peer switch is not displayed. This parameter is avai lable on switches that support VSX.

Example

Showing global statistics for all interfaces:

```
switch# show lldp statistics
LLDP Global Statistics
======================
Total Packets Transmitted            : 19
Total Packets Received               : 19
Total Packets Received And Discarded : 0
Total TLVs Unrecognized              : 0
LLDP Port Statistics
====================
```

```
PORT-ID          TX-PACKETS      RX-PACKETS      RX-DISCARDED   TLVS-UNKNOWN
---------------------------------------------------------------------------
1/1/1            7               7               0              0
1/1/2            7               7               0              0
1/1/3            0               0               0              0
1/1/4            0               0               0              0
1/1/5            0               0               0              0
...
mgmt             5               5               0              0
```

```
switch# show copp-policy statistics
switch# show lldp statistics
LLDP Global Statistics
============================================================================
Total Packets Transmitted            : 25
Total Packets Received               : 20
Total Packets Received And Discarded : 0
Total TLVs Unrecognized              : 0
LLDP Port Statistics
============================================================================
PORT-ID          TX-PACKETS      RX-PACKETS      RX-DISCARDED   TLVS-UNKNOWN
----------------------------------------------------------------------------
1/1/1            25              20              0              0
1/1/2            0               0               0              0
1/1/3            0               0               0              0
1/1/4            0               0               0              0
1/1/5            0               0               0              0
1/1/6            0               0               0              0
1/1/7            0               0               0              0
1/1/8            0               0               0              0
1/1/9            0               0               0              0
1/1/10           0               0               0              0
1/1/11           0               0               0              0
1/1/12           0               0               0              0
1/1/13           0               0               0              0
1/1/14           0               0               0              0
1/1/15           0               0               0              0
1/1/16           0               0               0              0
1/1/17           0               0               0              0
1/1/18           0               0               0              0
1/1/19           0               0               0              0
1/1/20           0               0               0              0
1/1/21           0               0               0              0
```

```
1/1/22          0               0               0               0
1/1/23          0               0               0               0
1/1/24          0               0               0               0
1/1/25          0               0               0               0
1/1/26          0               0               0               0
1/1/27          0               0               0               0
1/1/28          0               0               0               0
```

Showing statistics for interface **1/1/1**:

```
switch# show lldp statistics 1/1/1
LLDP Statistics
===============
Port Name                          : 1/1/1
Packets Transmitted                : 159
Packets Received                   : 163
Packets Received And Discarded     : 0
Packets Received And Unrecognized : 0
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show lldp statistics mgmt`

Syntax

```
show lldp statistics mgmt
```

Description

Shows LLDP statistics for the OOBM interface.

Example

Showing LLDP statistics for the OOBM interface:

```
switch# show lldp statistics mgmt
LLDP Statistics
===============
Port Name                          : mgmt
Packets Transmitted                : 20
Packets Received                   : 23
Packets Received And Discarded     : 0
Packets Received And Unrecognized : 0
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

show lldp tlv

Syntax

```
show lldp tlv[vsx-peer]
```

Description

Shows the LLDP TLVs that are configured for send and receive.

Parameter	Description
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output f rom the VSX peer switch is not displayed. This parameter is avai lable on switches that support VSX.

Example

```
switch# show lldp tlv
TLVs Advertised
===============
Management Address
Port Description
Port VLAN-ID
System Capabilities
System Description
System Name
VLAN Name
MFS
OUI
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

Cisco Discovery Protocol (CDP)

Cisco Discovery Protocol (CDP) is a proprietary layer 2 protocol supported by most Cisco devices. It is used to exchange information, such as software version, device capabilities, and voice VLAN information, between directly connected devices, such as a VoIP phone and a switch.

Subtopics

CDP support
CDP commands

CDP support

By default, CDP is enabled on each active switch port. This is a read-only capability, which means the switch can receive and store information about adjacent CDP devices, but does not generate CDP packets (except when communicating with Cisco IP phones.)

The switch supports CDPv2 only and does not support SNMP MIB traps.

When a CDP-enabled port receives a CDP packet from another CDP device, it enters data for that device into the CDP Neighbors table, along with the port number on which the data was received. It does not forward the packet. The switch also periodically purges the table of any entries that have expired. (The holdtime for any data entry in the switch CDP Neighbors table is configured in the device transmitting the CDP packet and cannot be controlled in the switch receiving the packet.) A switch reviews the list of CDP neighbor entries every three seconds and purges any expired entries.

Support for legacy Cisco IP phones

Autoconfiguration of legacy Cisco IP phones for tagged voice VLAN support requires CDPv2.

On initial boot-up, and sometimes periodically, a Cisco phone queries the switch and advertises information about itself using CDPv2. When the switch receives the VoIP VLAN Query TLV (type 0x0f) from the phone, the switch immediately responds with the voice VLAN ID in a reply packet using the VoIP VLAN Reply TLV (type 0x0e). This enables the Cisco phone to boot properly and send traffic on the advertised voice VLAN ID.

The switch CDP packet includes these TLVs:

- CDP Version: 2
- CDP TTL: 180 seconds
- Checksum
- Capabilities (type 0x04): 0x0008 (is a switch)
- Native VLAN: The PVID of the port
- VoIP VLAN Reply (type 0xe): voice VLAN ID (same as advertised by LLDP-MED)
- Trust Bitmap (type 0x12): 0x00

- Untrusted port CoS (type 0x13): 0x00

CDP commands

Select a command from the list in the left navigation menu.

Subtopics

`cdp`

Syntax

```
cdp
```

Description

Configures CDP support globally on all active interfaces or on a specific interface. By default, CDP is enabled on all active interfaces.

When CDP is enabled, the switch adds entries to its CDP Neighbors table for any CDP packets it receives from neighboring CDP devices.

When CDP is disabled, the CDP Neighbors table is cleared and the switch drops all inbound CDP packets without entering the data in the CDP Neighbors table.

The **no** form of this command disables CDP support globally on all active interfaces or on a specific interface.

Examples

Enabling CDP globally:

```
switch(config)# cdp
```

Disabling CDP globally:

```
switch(config)# no cdp
```

Enabling CDP on interface **1/1/1**:

```
switch(config)# interface 1/1/1s
switch(config-if)# cdp
```

Disabling CDP on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no cdp
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config` `config-if`	Administrators or local user group members with execution righ ts for this command.

`clear cdp counters`

Syntax

```
clear cdp counters
```

Description

Clears CDP counters.

Examples

Clearing CDP counters:

```
switch(config) clear cdp counters
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`clear cdp neighbor-info`

Syntax

```
clear cdp neighbor-info
```

Description

Clears CDP neighbor information.

Examples

Clearing CDP neighbor information:

```
switch(config) clear neighbor-info
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`show cdp`

Syntax

```
show cdp
```

Description

Shows CDP information for all interfaces.

Examples

Showing CDP information:

```
switch(config)# show cdp
CDP Global Information
======================
CDP           : Enabled
CDP Mode      : Rx only
CDP Hold Time : 180 seconds
Port         CDP
--------  --------------
1/1/1          Enabled
1/1/2          Enabled
1/1/3          Enabled
1/1/4          Enabled
1/1/5          Enabled
1/1/6          Enabled
1/1/7          Enabled
1/1/8          Enabled
1/1/9          Enabled
1/1/10         Enabled
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show cdp neighbor-info`

Syntax

```
show cdp neighbor-info <INTERFACE-ID>
```

Description

Shows CDP information for all neighbors or for CDP information on a specific interface.

Parameter	Description
`<INTERFACE-ID>`	Specifies an interface. Format: **member/slot/port**.

Examples

Showing all CDP neighbor information:

```
switch(config)# show cdp neighbor-info
Total Neighbor Entries  : 1
Port        Device ID                  Platform                 Capability
---------------------------------------------------------------------------
1/1/1       HPE_ANW-3810M-24G-1-slot... HPE ANW Sw                S
```

Showing CDP information for interface **1/1/1**:

```
switch(config)# show cdp neighbor-info 1/1/1
Local Port          : 1/1/1
MAC                 : 70:10:6f:86:78:7f
Device ID           : HPE_ANW-3810M-24G-1-slot(70106f-867800)
Address             : 127.0.0.1
Platform            : HPE ANW Sw
Duplex              : half
Version             : Revision KB.16.07.0002, ROM KB.16.01....
Capability          : switch
Native VLAN         : 1
Voice VLAN Support : No
Neighbor Port-ID    : 1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show cdp traffic`

Syntax

```
show cdp traffic
```

Description

Shows CDP statistics for each interface.

Examples

Showing CDP traffic statistics:

```
switch(config)# show cdp traffic
CDP Statistics
====================
Port         Transmitted Frames        Received Frames          Discarded
Frames
----------------------------------------------------------------------------
----
1/1/1            0                         4                        0
1/1/2            0                         0                        0
1/1/3            0                         2                        0
1/1/4            0                         0                        0
1/1/5            0                         0                        0
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show cdp voice-vlan mode`

Syntax

```
show cdp voice-vlan mode
```

Description

Shows CDP voice-vlan and mode.

Examples

Showing CDP voice-vlan and mode:

```
switch(config)# show cdp voice-vlan mode
CDP voice VLAN mode
====================
Port        Voice VLAN        Mode
--------    -----------       ----------
1/1/1       N/A               Rx only
1/1/2       N/A               Rx only
1/1/3       N/A               Rx only
1/1/4       N/A               Rx only
1/1/5       N/A               Rx only
1/1/6       N/A               Rx only
1/1/7       N/A               Rx only
1/1/8       N/A               Rx only
1/1/9       N/A               Rx only
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

Zero Touch Provisioning

Zero Touch Provisioning (ZTP) enables the auto-configuration of factory default switches without a network administrator onsite.

When a switch is booted from its factory default configuration, ZTP autoprovisions the switch by automatically downloading and installing a firmware file, a configuration file, or both. With ZTP, even a nontechnical user (for example: a store manager in a retail chain or a teacher in a school) can deploy devices at a site.

ZTP connection priorities

An HPE Aruba Networking CX switch supports ZTP provisioning on an Out-of-Band Management (OOBM) port, a data port (vlan 1), or an image and config file provided via USB.

Priority is given to ZTP options received on the OOBM port. If the switch receives ZTP configuration options on a data port first or from a DHCPv6 server connected to an OOBM port, the switch waits for an additional 30 seconds to see if higher-priority ZTP options will be received on an OOBM port also.

Priority is given to ZTPv4 options over ZTPv6 options, so connection priorities are as follows, from highest to lowest priority:

1. ZTP options received via USB.
2. ZTP options received from a DHCPv4 server on an OOBM port.
3. ZTP options received from a DHCPv4 server on a data port.
4. ZTP options received from a DHCPv6 server on an OOBM port.

Subtopics

ZTP support

The switch supports standards-based Zero Touch Provisioning (ZTP) operations as follows:

- ZTP operations are supported over IPv4 and IPv6 connections.
- The switch must be running the factory default configuration.
- The switch can connect to the DHCP server from the OOBM management port.
- The configuration file is a text file or JSON file that becomes the startup and running configuration on the switch after the ZTP operation is complete. The configuration can be in CLI or in JSON format.
- When the switch is started using the factory default configuration, the ZTP operation is started automatically and is active until any running configuration of the switch is modified. There is no CLI command required to start the operation.

- You must configure the DHCP server to provide a standards-based ZTP server solution. Options and features that are specific to Network Management Solution (NMS) tools, such as AirWave, are not supported.
  - Aruba Central on-premise can manage AOS-CX switches on supported models through DHCP ZTP using two approaches:
    - On the DHCP server, configure DHCP option-60 as "ArubaInstantAP" 90 and provide the value in option-43 in the format <group-details>, <aruba-central-on-prem-ip-or-fqdn>, <shared-token>.
    - On the DHCP server, configure DHCP option-60 as HPE vendor VCI and provide the value in option-43 in the tag-length-value (TLV) format. Next, enter the Aruba Central on-premise fully qualified domain name (FQDN) or IPv4 address as the value for sub-option code 146 using the format: <group-details>, <aruba-central-on-prem-ip-or-fqdn>, <shared-token>.

  Supported ZTP options from a DHCPv4 server are:

DHCP option	Description
43	Vendor Specific Information
43 suboption 144	Name of the configuration file
43 suboption 145	Name of the firmware image file
43 suboption 146	FQDN or IPv4 address of the Aruba Central on - p remise server
43 suboption 148	FQDN or IPv4 or IPv6 address of the HTTP Proxy
60	Vendor Class Identifier (VCI)
66	IPv4 or IPv6 address of the TFTP or HTTP server (Specifying a host name instead of an IP address is not supported.)
67	Name of the configuration file (Option 43 suboptio n 144 takes precedence over this option.)

Unique ZTP IPv6 options can be sent from the DHCPv6 Server based on MAC address of the switch or based on the DHCP Vendor class Identifier of the switch. Supported ZTP options from a DHCPv6 server are:

DHCP option	Description
59	IPv6 address of the TFTP/HTTP server. The image file name also can be given in this URL or in the VCA option - 17. **NOTE** the boot file URL will support TFTPv6 Server or HTTP Server option with IPv6 address or FQDN name.
17 suboption 145	Name of the image file.
17 suboption 144	Name of the configuration file.

The switch supports the following standards:

- **RFC 2131**, Dynamic Host Configuration Protocol.
- **RFC 2132**, DHCP Options and BOOTP Vendor Extensions. Support is limited to the options listed in the table "Supported DHCP options for ZTP on AOS-CX."
- **RFC 5970**, DHCPv6 Options for Network Boot. Support is limited to the options listed in the table "Supported DHCP options for ZTP on AOS-CX."

Hewlett Packard Enterprise recommends that you implement ZTP in a secure and private environment. Any public access can compromise the security of the switch, as follows:

- ZTP is enabled only in the factory default configuration of the switch, DHCP snooping is not enabled. The Rogue DHCP server must be manually managed.
- The DHCP offer is in plain data without encryption.

Setting up ZTP on a trusted network

The following procedure is an overview of setting up a Zero Touch Provisioning (ZTP) environment to provision newly installed switches automatically. The procedure is intended for network administrators who are familiar with automatically provisioning switches in a network, and does not provide detailed information about configuring or managing switches.

Procedure

1. For each switch model to be provisioned using ZTP, do the following:

a. Obtain the switch firmware image file.

b. Prepare the switch configuration file. The configuration file becomes the running configuration and the startup configuration on the switch.

2. Set up a TFTP or HTTP server and record its IPv4 or IPv6 address. The address is required when you set up the DHCP server. The switch must be able to reach the TFTP or HTTP server and DHCP server, either on the same subnet, or on a remote subnet via DHCP relay.
For switches that do not support ZTP connections through a data port, use the management port and management network.

3. Publish the configuration files and image files to the TFTP or HTTP server. You need to know the locations of the files and the IPv4/IPv6 address of theTFTP or HTTP server when you set up the vendor class options on the DHCP server.

4. On the DHCP server, set up vendor classes for each switch model you plan to provision. To do this you need the following information:

   - The IPv4/IPv6 address of the TFTP or HTTP server. Using a host name is not supported.

   - The path to the switch configuration and firmware image files on the TFTP or HTTP server.

   - The vendor class identifier (VCI) for each switch model.
     You can obtain the VCI by entering the **show dhcp client vendor-class-identifier** command from a switch CLI command prompt in the manager context. The VCI is the text string in the response that starts with **Aruba**.

     For example:

     ```
     switch# show dhcp client vendor-class-identifier
     Vendor Class Identifier: Aruba xxxxx xxxx
     ```
     Where x indicates the switch model number.

5. At the installation site, provide the switch installer with a Cat6 network cable connected to the network that includes the DHCP and TFTP or HTTP servers, and information about the switch port to use. The switch installer plugs the cable into the data port you specify.
The ZTP operation begins when power is applied to the switch after the network cable is installed.

6. Assuming the downloaded configuration includes a way to access the CLI of the switch, you can enter the following command to show the options offered by the DHCP server and the status of the ZTP operation:
show ztp information

ZTP using USB

About this task

AOS-CX supports EXT4, FAT-16, and FAT-32 USB file systems for ZTP. To perform zero-touch provisioning via USB, the image and config files should be in one of the following folders on the USB device.

> **NOTE**
>
> The file with extension **.swi** is recognized as an image file and the file with extension **.cfg** is recognized as a configuration file. These file extensions are case sensitive.

A serial-number specific folder:

- AOSCX/CNXXXXXXXX/ZTP/<image.swi>
- AOSCX/CNXXXXXXXX/ZTP/<config.cfg>

A vendor-class specific folder:

- AOSCX/PRODUCTNAME_PARTNUM/ZTP/<image.swi>
- AOSCX/PRODUCTNAME_PARTNUM/ZTP/<config.cfg> (for example: AOSCX/8320_JL479A/ZTP)

A generic configuration folder:

- AOSCX/ZTP/<image.swi>
- AOSCX/ZTP/<config.cfg>

If multiple paths are present, then the priority is given in the following order:

Procedure

1. The serial number specific folder with an image file or configuration file.
2. The vendor class specific folder with an image file or configuration file.
3. The generic configuration folder with an image file or configuration file.

Results

> **NOTE**
>
> You can identify the switch product name and part number you should use to create a ZTP folder by issuing the command **show dhcp client vendor-class-identifier**. For example:
>
> ```
> switch# show dhcp client vendor-class-identifier
> Vendor Class Identifier:  Aruba <xxxx> <yyyy>
> ```
>
> Where **<xxxx>** is the part number and **<yyyy>** is the product name in the output of this command.

If multiple image or configuration files are present in these ZTP folders, the first file that shows up in the output of the **ls -1 <usb_ztp_path**> command will be used for ZTP provisioning

If the switch boots up with a factory default configuration and a USB device is connected to the switch, the switch looks for the presence of ZTP-related folders. The switch then will be provisioned with the image and configuration files provided via USB. If either an image or a configuration file is present in the ZTP related folders on USB, then the switch will be provisioned with the respective file provided from the USB device.

ZTP using USB will not be triggered and the switch will continue with ZTP using DHCP in the following scenarios:

- There is no USB device connected.
- The USB device does not have the ZTP-specific folder.
- The ZTP folder has no image or configuration files in the first 127 files listed in the folder.
- The USB drive containing ZTP specific files is inserted into the device after it has booted up. (A device reboot is required to initiate the ZTP process using the USB.)

To confirm that the ZTP process has been triggered, issue the command **show ztp information**.

Performing an upgrade from a USB device without ZTP

To perform a manual upgrade using an image and configuration file from a USB device, issue the following copy commands to download the image and config file from the USB device directly on to the switch.

```
copy usb:/file <running-config|startup-config|primary|secondary
```

ZTP process during switch boot

Procedure

1. The switch boots up with the factory default configuration. If the ZTP operation detects that the switch configuration is different from the factory default configuration, the ZTP operation ends. The switch must be configured at the installation site.
2. The switch sends out a DHCP discovery from the management port .

   The switch waits to receive DHCP options indefinitely or until the running configuration is modified. If a DHCP IP address is received but no DHCP options are received, the switch waits an additional minute before ending the ZTP operation.

   After the ZTP operation ends, there is no automatic retry. You can either attempt to boot the switch with the factory default configuration again, configure the switch at the installation site, or use the ZTP force-provision CLI to trigger the ZTP process, ignoring the present running configuration of the switch.

   - Once force-provision is enabled, new DHCP requests are sent from the switch. Disabling force-provision does not stop the DHCP already in progress, but only changes the switch configuration status of force-provision.
   - If ZTP fails while force-provision is enabled, there is no automatic retry. To retry, **ztp force-provision** should be disabled and re-enabled to clear the current ZTP state and send a new DHCP request. When **ztp force-provision** is already enabled on the switch, re-enabling it results in no operation.
   - If the DHCP server is configured to provide both ZTP image and configuration options and there is a non-default startup configuration present on the switch, clearing the non-default startup configuration before triggering **ztp force-provision** is recommended. If an image is downloaded via ZTP, the switch reboots once the image download is complete and ZTP force-provision configuration is lost, causing ZTP to enter into a failed state. ZTP force-provision will need to be enabled again to continue the process.
3. The DHCP server responds with an offer containing the following:
   - The IPv4 or IPv6 address of the TFTP or HTTP server
   - One or both of the following:
     - The name of the firmware image file
     - The name of the configuration file
   - Aruba Central Location (optional) including the shared token value for the on-premise server
   - HTTP Proxy Location (optional)
4. If a firmware image file is offered, the ZTP operation downloads the image file from the TFTP or HTTP server to the switch. If the current switch image and downloaded firmware image version do not match, then the switch boots with the downloaded image:

- If the image upgrade fails, the switch retains its original firmware image and the ZTP operation ends with a failed status.
- If the image upgrade succeeds, the ZTP operation is started again after the switch reboots. Because the downloaded image file matches the image file installed on the switch, the ZTP operation continues, and checks if a configuration file is offered.

5. If a configuration file is offered, the ZTP operation downloads the configuration file copies the file to the running-config and then to the startup-config of the switch:
   - If the startup configuration update fails, the switch retains its factory-default running configuration and the ZTP operation ends with a failed status.
     - If the copy operation fails, the ZTP operation ends with a failed status.
     - If the copy operation succeeds, the ZTP operation ends successfully.

ZTP VSF switchover support

ZTP status is not synced in the VSF stack. When the VSF stack is formed, configuration changes are applied on the conductor switch, which is then synced to standby switch. When the switchover is performed on the VSF stack, the standby becomes the new conductor switch.

As part of the switchover process, the ZTP daemon starts on the new conductor. The status of the ZTP is failed because there are configuration changes present.

ZTP commands

Select a command from the list in the left navigation menu.

Subtopics

show ztp information
ztp force provision

`show ztp information`

Syntax

```
show ztp information
```

Description

Shows information about Zero Touch Provisioning (ZTP) operations performed on the switch.

Usage

When a switch configured to use ZTP is booted from a factory default configuration, the switch contacts a DHCP server, which offers options for obtaining files used to provision the switch:

- The IPv4/IPv6 address of TFTP Server or the IPv6 address of HTTP server
- The name of the image file
- The name of the configuration file
- The HPE Aruba Networking Central FDQN or IPv4/IPv6 address
- The HTTP proxy FDQN or IPv4/IPv6 address

The **show ztp information** command shows the options offered by the DHCP server and the status of the ZTP operation.

The status of the ZTP operation is one of the following:

Success	The ZTP operation succeeded. One of the following is true: • Both the running configuration and the startup c onfiguration were updated. • The IP address of the TFTP/HTTP server was re ceived, but the offer did not include a configurati on file or a firmware image file. • Any combination of vendor encapsulated DHCP options are received as configured, along with th e firmware image and switch configuration file. • Only vendor encapsulated DHCP options are conf igured and are received accordingly.
Failed - Custom startup configuration detected	The switch was booted from a configuration that is n ot the factory default configuration. For example, the administrator password has been set.
Failed - Timed out while waiting to receive ZTP options	Either the switch received the DHCP IPv4 or IPv6 ad dress but no ZTP options were received within 1 min ute or ZTP force - provision is triggered and no ZTP options are received within 3 minutes.

Failed - Detected change in running configuratio n	The running configuration was modified by a user w hile the ZTP operation was in progress.
Failed - TFTP server unreachable	The TFTP server is not reachable at the specified IP address.
Failed - TFTP server information unavailable	The image file name or config file name is provided without the TFTP server location to fetch the files f rom and ZTP enters failed state.
Failed - Invalid configuration file received	Either the file transfer of the configuration file failed , or the configuration file is invalid (an error occurred while attempting to apply the configuration).
Failed - Invalid image file received	Either the file transfer of the firmware image file faile d, or the firmware image file is invalid (an error occur red while verifying the image).
Failed - HTTP server unreachable	The HTTP server is not reachable at the specified IP address
Failed - Invalid configuration file from USB	The configuration file is invalid or it is not in the pro per format.
Failed - Invalid image file from USB	The image file is invalid or it is not in the proper for mat.
Failed - Error while copying the configuration f rom USB	ZTP failed to copy the configuration file from the US B device due to a file permissions, file access or syste m error (for example, if the switch is out of memory), if the user interrupts the ZTP process, or if the fille is deleted from the USB device. This message may also display in the event of an unknown error.
Failed - Error while copying the image from USB	ZTP failed to copy the image file from the USB devi ce due to a file permissions, file access or system er ror (for example, if the switch is out of memory), if th e user interrupts the ZTP process, or if the fille is de

	leted from the USB device. This message may also di splay in the event of an unknown error.

NOTE

In the case of reconnection, connect with a main or alternative location to the COP instance as a user. The current connection is shown in the **Central location** field.

Scenario 1: If the location the device is currently connected on is updated, the system reconnects in order to connect with the new location.

Scenario 2: If the location in which the device is not currently connected on is updated, the DUT does not go through the reconnection process.

Examples

Showing switch image and config download from an IPv4 TFTP server in progress after receiving ZTP options:

```
switch# show ztp information
TFTP Server                                      : 10.0.0.2
Image File                                       : TL_10_02_0001.swi
Configuration File                               : ztp.cfg
ZTP Status                                       : In-progress - Image download and
verification
HPE ANW Central Location                                 : NA
Alternative HPE ANW Central Location                         : NA
HPE ANW Central Shared Token                                 : NA
Force-Provision                                  : Enabled
HTTP Proxy Location                              : NA
```

Showing switch image download failure from an IPv4 TFTP server because the server was unreachable

```
switch# show ztp information
TFTP Server                                      : 10.0.0.2
Image File                                       : TL_10_02_0001.swi
Configuration File                               : ztpv6.cfg
ZTP Status                                       : Failed - TFTP server unreachable
HPE ANW Central Location                                 : NA
Alternative HPE ANW Central Location                         : NA
HPE ANW Central Shared Token                                 : NA
Force-Provision                                  : Enabled
HTTP Proxy Location                              : NA
```

Showing a successful switch configuration download from an IPv6 HTTP server

```
switch# show ztp information
HTTP Server                                       : http://[2001::50]
```

```
Image File                                   : TL_10_15_0001.swi
Configuration File                           : config_file
ZTP Status                                   : Success
HPE ANW Central Location                             : NA
Alternative HPE ANW Central Location                     : NA
HPE ANW Central Shared Token                             : NA
Force-Provision                              : Enabled
HTTP Proxy Location                          : NA
```

The switch received image and configuration files from a serial-number specific folder on a USB device, and ZTP was successful.

```
switch# show ztp information
TFTP Server                                : NA
Image File                                 : usb:/AOSCX/TW88KHD005/
GL_10_16_00001.swi
Configuration File                         : usb:/AOSCX/TW88KHD005/ztp.cfg
Status                                     : Success
HPE ANW Central Location                   : NA
Alternative HPE ANW Central Location : NA
HPE ANW Central Shared Token               : NA
Force-Provision                            : Disabled
HTTP Proxy Location                        : NA
```

The switch received image and configuration files from a generic folder on a USB device, and ZTP was successful.

```
switch# show ztp information
TFTP Server                                : NA
Image File                                 : usb:/AOSCX/ztp/ML_10_15_1000F.swi
Configuration File                         : usb:/AOSCX/ztp/config.cfg
Status                                     : Success
HPE ANW Central Location                   : NA
Alternative HPE ANW Central Location : NA
HPE ANW Central Shared Token               : NA
Force-Provision                            : Disabled
HTTP Proxy Location                        : NA
```

The switch received an image file from a vendor-class specific folder on a USB device, and ZTP was successful.

```
switch# show ztp information
TFTP Server                                : NA
Image File                                 : usb:/AOSCX/8325_JL636A/
GL_10_16_1000A.swi
Configuration File                         : NA
Status                                     : Success
HPE ANW Central Location                   : NA
Alternative HPE ANW Central Location : NA
```

```
HPE ANW Central Shared Token          : NA
Force-Provision                       : Disabled
HTTP Proxy Location                   : NA
```

Command History

Release	Modification
10.16	The output of this command can show the status of ZTP over USB.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`ztp force provision`

Syntax

```
ztp force-provision
no ztp force-provision
```

Description

Starts on-demand ZTP via DHCP.

Usage

DHCP options received are processed independent of the current state of configuration on the switch. Previous ZTP options are cleared, such as the ZTP TFTP server, image file, configuration file, HPE Aruba Networking Central location, and HTTP proxy location, and the switch sends a DHCP request. ZTP using USB is not triggered by this command.

Examples

In the following example, force-provision is enabled.

```
switch# configure terminal
switch(config)# ztp force-provision
```

In the following example, force-provision status is checked while enabled.

```
switch# show ztp information
TFTP Server                      : 10.0.0.2
Image File                       : TL_10_02_0001.swi
Configuration File               : ztp.cfg
Status                           : Success
HPE ANW Central Location                     : NA
HPE ANW Central Shared Token                    : NA
Force-Provision                  : Enabled
HTTP Proxy Location              : NA
```

In the following example, force-provision is disabled.

```
switch# configure terminal
switch(config)# no ztp force-provision
```

In the following example, force-provision status is checked while disabled.

```
switch# show ztp information
TFTP Server                      : 10.0.0.2
Image File                       : TL_10_02_0001.swi
Configuration File               : ztp.cfg
Status                           : Success
HPE ANW Central Location                     : NA
HPE ANW Central Shared Token                    : NA
Force-Provision                  : Disabled
HTTP Proxy Location              : NA
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manage r (#)	Administrators or local user group members with execution righ ts for this command.

Thermal Management

Thermal management consists of the **fand** and **tempd** daemons, that manage the fans and temperature sensors for the system respectively. They are responsible for monitoring temperatures, adjusting fan speeds to keep the system at nominal temperatures, initiating shutdown sequences for chassis, subsystems, or transceivers during over-temperature conditions, and reporting temperatures, sensor states, fan speeds, and fan states.

Subtopics

Thermal management support

	Description	Platforms
Transceiver Thermal Management	When the transceiver thermal management capability is enabled, a temperature sensor will be available for each transceiver that supports thermal monitoring.	
Thermal Shutdown Behavior	Thermal shutdown is a critical safety mechanism designed to prevent hardware damage caused by overheating. When critical temperature thresholds are exceeded, thermal daemons log an event and initiate a shutdown sequence to power down the affected components gracefully.	All platforms
Chassis (Switch) Thermal Shutdown	If a temperature sensor exceeds its critical threshold and the associated subsystem does not support being powered off, the entire chassis will shut down for a five minute cool down period. If the critical temperature is detected again after reboot, the shutdown sequence will repeat.	All platforms

	Description	Platforms
Subsystem (Line Card, Fabric Card) Thermal Shutdown	If a temperature sensor exceeds i ts critical threshold and the associ ated subsystem supports being po wered off, that subsystem will shu t down for a five minute cool down period. If the critical temperature i s detected again after reboot, the s hutdown sequence will repeat.	5420 6400 8400
Transceiver Thermal Shutdown	If a transceiver's temperature sens or exceeds its critical threshold an d the transceiver is configured to s hut down upon exceeding the criti cal threshold, it will power down fo r a five - minute cool down period . If the critical temperature is dete cted again after reboot, the shutdo wn sequence will repeat.	8325 8325H 8325P 9300 - 32C8D 9300 - 32D

Transceiver Thermal Management

Transceiver temperature sensors are visible on REST API calls, SNMP and CLI (show environment temperature). Events logs also are triggered upon thermal status changes, shutdown and recovery. If a transceiver exceeds the temperature threshold and it is hot swapped, the cool time timer is canceled and the transceiver is re-enabled. There is no limit to the number of times a transceiver is thermally disabled and re-enabled if it's continuously above the critical temperature threshold.

This is an example of the temperature information on the normal conditions.

```
switch# show environment temperature
Temperature information
---------------------------------------------------------------------------
--
                                                     Current
Mbr/Slot-Sensor                 Module Type        temperature   Status
---------------------------------------------------------------------------
--
1/1-xcv-5                line-card-module     66.45 C    normal
...
```

If the temperature raises to 70 degrees Celsius, the status changes to **overtemp**.

```
switch# show environment temperature
Temperature information
```

```
--------------------------------------------------------------------------
--
                                                    Current
Mbr/Slot-Sensor                 Module Type        temperature  Status
--------------------------------------------------------------------------
--
1/1-xcv-5                 line-card-module     70.00 C    overtemp
...
```

The status change will generate this event log: **Temperature sensor 1/1-xcv-5 status is overtemp at 70.000000 degC**

If the temperature raises to 82 degrees Celsius, for example, the status changes to **emergency**. The transceiver is disabled and the cool own period begins. The status change and the disabling of the interface reflect in the event logs.

```
Temperature sensor 1/1-xcv-5 status is overtemp at 70.000000 degC
Temperature sensor 1/1-xcv-5 status is critical at 82.000000 degC
Disabling interface 1/1/5 transceiver for 5 minutes - temperature sensor
1/1-xcv-5 status is emergency at 82.000000 degC
```

This is an example of the **show interface** command when the interface is **disabled:**

```
switch# show int 1/1/5
Interface 1/1/5 is down
Admin state is down
State information: Transceiver disabled - Module disabled by system thermal
management
Link state: down for 1 hour (since Wed May 14:22:17:42 UTC 2025)
Link transitions: 0
Description:
Hardware: Ethernet, MAC Address: 70:72:cf:fd:e7:b4
MTU 1500
Type 40G-eSR4 / QSPF+ eSR4
Full-duplex
qos trust none
Speed 0 Mb/s
Auto-negotiation is off
Flow-control: off
Error-control: off
L3 Counters: Rx Desabled, Tx Disabled
Rate collection interval: 300 seconds
Rates                          RX                    TX        Total (RX+TX)
------------- -------------------- -------------------- --------------------
Mbits / sec                  0.00                 0.00                 0.00
KPkts / sec                  0.00                 0.00                 0.00
  Unicast                    0.00                 0.00                 0.00
  Multicast                  0.00                 0.00                 0.00
```

```
  Broadcast                     0.00                 0.00                 0.00
Utilization %                   0.00                 0.00                 0.00
Statistics                        RX                   TX                Total
------------- -------------------- -------------------- --------------------
Packets                            0                    0                    0
  Unicast                          0                    0                    0
  Multicast                        0                    0                    0
  Broadcast                        0                    0                    0
Bytes                              0                    0                    0
Jumbos                             0                    0                    0
Dropped                            0                    0                    0
Filtered                           0                    0                    0
Pause Frames                       0                    0                    0
L3 Packets                         0                    0                    0
L3 Bytes                           0                    0                    0
Errors                             0                    0                    0
  CRC/FCS                          0                  n/a                    0
  Collision                      n/a                    0                    0
  Runts                            0                  n/a                    0
  Giants                           0                  n/a                    0
Other                              0                    0                    0
```

Once the cool down period is over, the transceiver is enabled. Therefore, the status changes to normal and the corresponding event log is triggered:

```
Temperature sensor 1/1-xcv-5 status is overtemp at 70.000000 degC
Temperature sensor 1/1-xcv-5 status is critical at 82.000000 degC
Disabling interface 1/1/5 transceiver for 5 minutes - temperature sensor
1/1-xcv-5 status is emergency at 82.000000 degC
Enabling interface 1/1/5 transceiver - thermal disablement period has
expired
```

If the transceiver is hot swapped, the interface is re-enabled and the event logs are:

```
Temperature sensor 1/1-xcv-5 status is overtemp at 70.000000 degC
Temperature sensor 1/1-xcv-5 status is critical at 82.000000 degC
Disabling interface 1/1/5 transceiver for 5 minutes - temperature sensor
1/1-xcv-5 status is emergency at 82.000000 degC
Enabling interface 1/1/5 transceiver - thermal disablement cancelled due to
hot-swap of transceiver
```

NOTE

The examples above apply to the 9300-32D Switch Series. The output of these commands may vary in other platforms. The temperature thresholds shown here are for illustration purposes only.

Switch system and hardware commands

Subtopics

- Switch system and hardware commands

Switch system and hardware commands

Select a command from the list in the left navigation menu..

Subtopics

- allow-non-failsafe-updates
- bluetooth disable
- bluetooth enable
- clear events
- clear ip errors
- console baud-rate
- copy support-files top-cpu-processes
- domain-name
- fabric admin-state
- hostname
- led locator
- module admin-state
- module product-number
- mtrace
- power consumption-average-period
- Reboot reasons
- show bluetooth
- show boot-history
- show capacities
- show capacities-status
- show console
- show core-dump
- show deprecated commands
- show domain-name
- show environment fan
- show environment led
- show environment power-consumption
- show environment power-consumption
- show environment power-supply
- show environment rear-display-module
- show environment temperature

- **show events**
- **show fabric**
- **show hostname**
- **show images**
- **show ip errors**
- **show module**
- **show process monitor statistics**
- **show process top cpu**
- **show process top-memory**
- **show running-config**
- **show running-config current-context**
- **show startup-config**
- **show system**
- **show system error-counter-monitor**
- **show system resource-utilization**
- **show tech**
- **show usb**
- **show usb file-system**
- **show version**
- **system error-counter-monitor poll-interval**
- **system error-counter-monitor**
- **system resource-utilization poll-interval**
- **top cpu**
- **top memory**
- **usb**
- **usb mount | unmount**

`allow-non-failsafe-updates`

Syntax

```
allow-non-failsafe-updates <num_minutes>
```

Description

Allow non-failsafe updates of programmable devices for the specified number of minutes, or **enter** 0 to disallow unsafe updates. The maximum permitted value is 120 minutes.

Example

The following example will enable non-failsafe updates of programmable devices for the next 30 minutes.

```
switch(config)#  allow-non-failsafe-updates 30
This command will enable non-failsafe updates of programmable devices for
```

```
the next 30 minutes.  You will first need to wait for all line and fabric
modules to reach the ready state, and then reboot the switch to begin
applying any needed
updates.  Ensure that the switch will not lose power, be rebooted again, or
have any
modules removed until all updates have finished and all line and fabric
modules have
returned to the ready state
WARNING: Interrupting these updates may make the product unusable!
Continue [y/n] ? y
Unsafe updates      : allowed (less than 30 minute(s) remaining)
```

Command History

Release	Modification
10.15.1010	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`bluetooth disable`

Syntax

```
bluetooth disable
no bluetooth disable
```

Description

Disables the Bluetooth feature on the switch. The Bluetooth feature includes both Bluetooth Classic and Bluetooth Low Energy (BLE). Bluetooth is enabled by default.

The **no** form of this command enables the Bluetooth feature on the switch.

Example

Disabling Bluetooth on the switch. <XXXX> is the switch platform and <NNNNNNNNNN> is the device identifier.

```
switch(config)# bluetooth disable
switch# show bluetooth
Enabled              : No
Device name          : <XXXX>-<NNNNNNNNNN>
switch(config)# show running-config
...
bluetooth disabled
...
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	`config`	Administrators or local user group members with execution righ ts for this command.

`bluetooth enable`

Syntax

```
bluetooth enable
no bluetooth enable
```

Description

This command enables the Bluetooth feature on the switch. The Bluetooth feature includes both Bluetooth Classic and Bluetooth Low Energy (BLE).

Default: Bluetooth is enabled by default.

The **no** form of this command disables the Bluetooth feature on the switch.

Usage

The default configuration of the Bluetooth feature is `enabled`. The output of the `show running-config` command includes Bluetooth information only if the Bluetooth feature is disabled.

The Bluetooth feature includes both Bluetooth Classic and Bluetooth Low Energy (BLE).

The Bluetooth feature requires the USB feature to be enabled. If the USB feature has been disabled, you must enable the USB feature before you can enable the Bluetooth feature.

Examples

```
switch(config)# bluetooth enable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	`config`	Administrators or local user group members with execution rights for this command.

`clear events`

Syntax

```
clear events
```

Description

Clears up event logs. Using the `show events` command will only display the logs generated after the `clear events` command.

Examples

Clearing all generated event logs:

```
switch# show events
---------------------------------------------------
show event logs
---------------------------------------------------
2018-10-14:06:57:53.534384|hpe-sysmond|6301|LOG_INFO|MSTR|1|System resource
utilization poll interval is changed to 27
2018-10-14:06:58:30.805504|lldpd|103|LOG_INFO|MSTR|1|Configured LLDP tx-
timer to 36
2018-10-14:07:01:01.577564|hpe-sysmond|6301|LOG_INFO|MSTR|1|System resource
utilization poll interval is changed to 49
switch# clear events
switch# show events
---------------------------------------------------
show event logs
---------------------------------------------------
2018-10-14:07:03:05.637544|hpe-sysmond|6301|LOG_INFO|MSTR|1|System resource
utilization poll interval is changed to 34
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution righ ts for this command.

clear ip errors

Syntax

```
clear ip errors
```

Description

Clears all IP error statistics.

Example

Clearing and showing ip errors:

```
switch# clear ip errors
switch# show ip errors
----------------------------------
Drop reason                Packets
----------------------------------
Malformed packets                0
IP address errors                0
...
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	Manager (#)	Administrators or local user group members with execution righ ts for this command.

console baud-rate

Syntax

```
console baud-rate <SPEED>
no console baud-rate <SPEED>
```

Description

Sets the console serial port speed.

The no form of this command resets the console port speed to its default of 115200 bps.

Parameter	Description
<SPEED>	Selects the console port speed in bps, either `9600` or `115200`.

Usage

The speed change occurs immediately for the active console session. The console will be inaccessible until the client terminal settings are updated to match the console port speed that you set. After the command is executed you will be prompted to log in again.

Examples

Setting the console port speed to 9600 bps:

```
switch(config)# console baud-rate 9600
This command will configure the baud rate immediately for the active serial
console session. After the command is executed the user will be prompted to
re-login. The serial console will be inaccessible until the terminal client
settings are updated to match the baud rate of the switch.
Continue (y/n)? y
```

Resetting the console port to its default speed 115200 bps:

```
switch(config)# no console baud-rate
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

`copy support-files top-cpu-processes`

Syntax

```
copy support-files top-cpu <REMOTE_URL> {vrf <VRF_NAME>} | <STORAGE_URL> |
<local-file>
```

Description

Copies CPU high-utilization diagnostic bundles from /tmp/system_utilization_snapshots (files matching cpu_high_*.tgz). These archives are created automatically when the 1-minute average CPU exceeds the upper threshold and are maintained as a rolling set (maximum 5 files).

Table 1.

Parameter	Description

`REMOTE_URL`	URL to copy support-files. Use one of the following syntax:
	`sftp://USER@{IP|HOST}[:PORT]/{DIRECTORY/}`
	`tftp://{IP|HOST}[:PORT][;blocksize=VAL]/`
	`scp://USER@{IP|HOST}[:PORT]/{DIRECTORY/}`
	Required

`STORAGE_URL`	USB or local SFTP server storage to copy support-files top-cpu. Syntax: `{usb|local}:/{[DIRECTORY/]}` Optional
`VRF_NAME`	Specifies the VRF name. Optional Default: default VRF

Examples

Copying CPU high-utilization diagnostic bundles to a remote URL:

```
switch# copy support-files top-cpu sftp://user@10.0.0.12/cpu_diags/ vrf mgmt
Attempting to copy CPU diag archives...
Done
```

Copying CPU high-utilization diagnostic bundles to a storage URL:

```
switch# copy support-files top-cpu usb:/cpu_diags/
Copying CPU diag archives to /mnt/usb/cpu_diags/
Done
```

Copying CPU high-utilization diagnostic bundles to a local file:

```
switch# copy support-files top-cpu local-file
Preserving CPU diag archives into local-files...
support-files are preserved successfully in the local-files.
```

NOTE

- Archives copied are located under **/tmp/system_utilization_snapshots** and named **cpu_high_*.tgz**.
- Up to 5 archives are retained (oldest overwritten).
- Remote copies support SFTP, SCP, TFTP with optional VRF; storage copies support USB/local SFTP directories.

Command History

Release	Modification
10.18	Command introduced.

Command Information

Platforms	Command context	Authority
8400	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`domain-name`

Syntax

```
domain-name <NAME>
no domain-name [<NAME>]
```

Description

Specifies the domain name of the switch.

The **no** form of this command sets the domain name to the default, which is no domain name.

Parameter	Description
<NAME>	Specifies the domain name to be assigned to the switch. The first character of the name must be a letter or a number. Length: 1 to 32 characters.

Examples

Setting and showing the domain name:

```
switch# show domain-name
switch# config
switch(config)# domain-name example.com
switch(config)# show domain-name
example.com
switch(config)#
```

Setting the domain name to the default value:

```
switch(config)# no domain-name
switch(config)# show domain-name
switch(config)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution rights for this command.

`fabric admin-state`

Syntax

```
fabric <SLOT-ID> admin-state
   diagnostic
   down
   up
```

Description

Sets the administrative state of the specified fabric module.

Parameter	Description
`<SLOT-ID>`	Specifies the member and slot of the module. For example, to s pecify the module in member 1, slot 2, enter the following: `1/2`
`diagnostic`	Selects the **diagnostic** administrative state. Network traffic d oes not pass through the module.
`down`	Selects the **down** administrative state. Network traffic does not pass through the module.
`up`	Selects the **up** administrative state. The module is fully operatio nal. The **up** state is the default administrative state.

Usage

This command is valid for fabric modules only.

Examples

Setting the administrative state of the fabric module 2 to **down**:

```
switch(config)# fabric 1/2 admin-state down
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	`config`	Administrators or local user group members with execution righ ts for this command.

`hostname`

Syntax

```
hostname <HOSTNAME>
no hostname [<HOSTNAME>]
```

Description

Sets the host name of the switch.

The **no** form of this command sets the host name to the default value, which is `switch`.

Parameter	Description
<HOSTNAME>	Specifies the host name. The first character of the host name m ust be a letter or a number. Length: 1 to 32 characters. Default: `switch`

Examples

Setting and showing the host name:

```
switch# show hostname
switch
switch# config
switch(config)# hostname myswitch
myswitch(config)# show hostname
myswitch
```

Setting the host name to the default value:

```
myswitch(config)# no hostname
switch(config)#
```

Error messages:

```
switch(config)# hostname
ThisIsAVeryLongHostnameThatExceedsTheMaximumLengthLimitOf64Characters
The hostname cannot exceed 63 characters in length.
switch(config)#
switch(config)# hostname -abd-123-
The hostname must begin and end with a letter or number.
switch(config)#
switch(config)# hostname abc.123@12
To be compliant with RFC 1123, the hostname must contain only letters,
numbers and hyphens, and must not start or end with a hyphen.
switch(config)#
switch(config)# hostname 123_123
Hostname contains one or more non-RFC 1123 compliant characters. Some
features may not work as intended.
Do you want to continue (y/n)? y
123_123(config)#
switch(config)# hostname test.name
Hostname contains one or more non-RFC 1123 compliant characters. Some
features may not work as intended.
Do you want to continue (y/n)? y
test.name(config)#
```

Command History

Release	Modification
10.17	Enhanced **hostname** to support dot (.) and underscore (_) characters.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`led locator`

Syntax

```
led locator
   on
   off
   flashing

            slow_blink


            fast_blink
            half_bright
   no...
```

Description

Sets the state of the locator LED.

Parameter	Description
`on`	Turns on the LED.
`off`	Turns off the LED, which is the default value.
`flashing`	Sets the LED to blink on and off repeatedly.
`slow_blink`	Sets the LED to slow blink on and off.
`fast_blink`	Sets the LED to fast blink on and off.
`half_bright`	Sets the LED intensity to half bright.
`no`	Negates any configured parameter

Example

Setting the state of the locator LED:

```
switch# led locator flashing
```

Command History

Release	Modification
10.13	Starting with AOS - CX 10.13, this command will not take effect when it is sent to the switch from HPE Aruba Networking Central.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`module admin-state`

Syntax

```
module <SLOT-ID> admin-state {diagnostic | down | up}
no module <SLOT-ID> [admin-state [diagnostic | down | up]]
```

Description

Sets the administrative state of the specified line module.

The **no** form of the command configures administrative state to the default **up**.

Parameter	Description
<SLOT-ID>	Specifies the member and slot of the module. For example, to s pecify the module in member 1, slot 3, enter the following: `1/3`
`diagnostic`	Selects the **diagnostic** administrative state. Network traffic d oes not pass through the module.
`down`	Selects the **down** administrative state. Network traffic does not pass through the module.

Parameter	Description
`up`	Selects the **up** administrative state. The line module is fully ope rational. The **up** state is the default administrative state.

Example

Setting the administrative state of the module in slot **1/3** to **down**:

```
switch(config)# module 1/3 admin-state down
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	`config`	Administrators or local user group members with execution righ ts for this command.

`module product-number`

Syntax

```
module <SLOT-ID> product-number [<PRODUCT-NUM>]
no module <SLOT-ID> [product-number [<PRODUCT-NUM>]]
```

Description

Changes the configuration of the switch to indicate that the specified member and slot number contains, or will contain, a line module.

The **no** form of this command removes the line module and its interfaces from the configuration. If there is a line module installed in the slot, the line module is powered off and then powered on.

Parameter	Description
<SLOT-ID>	Specifies the member and slot in the form **m/s**, where **m** is the member number, and **s** is the slot number.
<PRODUCT-NUM>	Specifies the product number of the line module. For example: `JL363A` If there is a line module installed in the slot when you execute t his command, `<PRODUCT-NUM>` is optional. The switch re ads the product number information from the module that is in stalled in the slot. If there is no line module installed in the slot when you execute this command, `<PRODUCT-NUM>` is required.

Usage

The default configuration associated with a line module slot is:

- There is no module product number or interface configuration information associated with the slot. The slot is available for the installation with any supported line module.
- The Admin State is Up (which is the default value for Admin State).

To add a line module to the configuration, you must use the **module** command either before or after you install the physical module.

If you execute the **module** command after you install a line module in an empty slot, you can omit the `<PRODUCT-NUM>` variable. The switch reads the product information from the installed module.

If the module is not installed in the slot when you execute the module command, you must specify a value for the `<PRODUCT-NUM>` variable:

- The switch validates the product number of the module against the slot number you specify to ensure that the right type of module is configured for the specified slot.
  For example, the switch returns an error if you specify the product number of a line module for a slot reserved for management modules.
- You can configure the line module interfaces before the line module is installed.

When you install the physical line module in a preconfigured slot, the following actions occur:

- If a product number was specified in the command and it matches the product number of the installed module, the switch initializes the module.
- If a product number was specified in the command and the product number of the module does not match what was specified, the module device initialization fails.

The **no** form of the command removes the line module and its interfaces from the configuration and restores the line module slot to the default configuration.

If there is a line module installed in the slot when you execute the **no** form of the command, the command also powers off and then powers on the module. Traffic passing through the line module is stopped. Management sessions connected through the line module are also affected.

If the slot associated with the line module is in the default configuration, you can remove the module from the chassis without disrupting the operation of the switch.

Examples

Configuring slot 1/1 for future installation of a line module:

```
switch(config)# module 1/1 product-number jl363a
```

Configuring a line module that is already installed in slot 1/1:

```
switch(config)# module 1/1 product-number
```

Attempting to configure slot 1/1 for the future installation of a line module without specifying the product number (returned error shown):

```
switch(config)# module 1/1 product-number
Line module '1/4' is not physically available.  Please provide the product
number to preconfigure the line module.
```

Configuring a JL363A line module in a slot that has already been used to configure a different module:

```
switch(config)# no module 1/4
switch(config)# module 1/4 product-number jl363a
```

Removing a module from the configuration:

```
switch(config)# no module 1/1
This command will power cycle the specified line module and restore its
default
configuration. Any traffic passing through the line module will be
interrupted.
Management sessions connected through the line module will be affected. It
might take a few minutes to complete this operation.
Do you want to continue (y/n)? y
switch(config)#
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	`config`	Administrators or local user group members with execution righ ts for this command.

`mtrace`

Syntax

```
mtrace <IPV4-SRC-ADDR>
             <IPV4-GROUP-ADDR> [lhr <IPV4-LHR-ADDR>] [ttl <HOPS>]
   [vrf <VRF-NAME>]
```

Description

Traces the specified IPv4 source and group addresses.

Parameter	Description
IPV4-SRC-ADDR	Specifies the source IPv4 address to trace.
IPV4-GROUP-ADDR	Specifies the group IPv4 address to trace.
`lhr <IPV4-LHR-ADDR>`	Specifies the last hop router address from which to start the tra ce.
`ttl <HOPS>`	Specifies the Time - To - Live duration in hops. Range: 1 to 25 5 hops. Default: 8 hops.
`vrf <VRF-NAME>`	Specifies the name of the VRF. If a name is not specified the de fault VRF will be used.

Examples

Tracing with source, group, and LHR addresses and TTL:

```
(switch)# mtrace 20.0.0.1 239.1.1.1 lhr 10.1.1.1 ttl 10
Type escape sequence to abort.
Mtrace from 10.0.0.1 for Source 20.0.0.1 via Group 239.1.1.1
From destination(?) to source (?)...
Querying ful reverse path...
0  10.0.0.1
-1  30.0.0.1 PIM  0 ms
-2  40.0.0.1 PIM  2 ms
-3  50.0.0.1 PIM  100 ms
-4  60.0.0.1 PIM  156 ms
-5  20.0.0.1 PIM  123 ms
```

Tracing with source and group addresses:

```
(switch)# mtrace 200.0.0.1 239.1.1.1
Type escape sequence to abort.
Mtrace from self for Source 200.0.0.1 via Group 239.1.1.1
From destination(?) to source (?)...
Querying ful reverse path...
0  10.0.0.1
-1  30.0.0.1 PIM  0 ms
-2  40.0.0.1 PIM  2 ms
-3  50.0.0.1 PIM  100 ms
-4  60.0.0.1 PIM  156 ms
-5  200.0.0.1 PIM  123 ms
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	Manager (#)	Administrators or local user group members with execution righ ts for this command.

power consumption-average-period

Syntax

```
power consumption-average-period <PERIOD-IN-SECONDS>
```

Description

Configures a time period for average power consumption in seconds.

Parameter	Description
`<PERIOD-IN-SECONDS>`	Specifies the period in seconds for average power consumed. Range: 60 - 3600. Default: 600

Example

Configuring a time period of 60 seconds for average power consumption:

```
switch(config)# power consumption-average-period 60
```

Command History

Release	Modification
10.13	Command introduced.

Command Information

Platforms	Command context	Authority
8400	`config`	Administrators or local user group members with execution righ ts for this command.

Reboot reasons

The **show boot-history** command displays the following reboot reasons for the management module and line module:

Reboot reasons for management module

Parameter	Description
Reboot requested by user	A user requested a switch reboot through the CLI or web UI.
Reset button pressed	The switch detected a short - press of the reset button.
Backplane fault	A backplane fault occured.
Configuration change	A configuration change resulted in a reboot.
Console error	Console failed to start.
Fabric fault	A fabric fault occurred.
All line modules faulted	A zero line card condition occurred.
Redundacy switchover requested	A user requested a redundancy switchover.
Redundant Management communication timeout	The standby management module has taken over from an un responsive active management module.
Redundant Management election timeout	A failure to elect a standby management module in the alloted t ime.
Critical service fault (error)	A daemon critical to switch operation has stopped functioning. An extra error string may be present to describe the error in d etail.
VSX software update	Reset triggered by a VSX software update.
Chassis critical temperature	Chassis operating temperature exceeded.
Chassis insufficient fans	Insufficient fans to cool the chassis.
Chassis unsupported PSUs/fans	Unsupported or misconfigured PSUs or system fans.
Management module critical temperature	Management module operating temperature exceeded.

Uncontrolled reboots

- ops-switchd crashed
- ovsdb-server crashed
- Reset

- Software thermal reset
- Power on reset
- Watchdog reset
- CPU request reset
- cold reset
- Long press reset
- Jumper reset

- switchd_agent crashed

Reboot reasons for line module

- Line module reboot
- Line module crashed

`show bluetooth`

Syntax

```
show bluetooth
```

Description

Shows general status information about the Bluetooth wireless management feature on the switch.

Usage

This command shows status information about the following:

- The USB Bluetooth adapter
- Clients connected using Bluetooth
- The switch Bluetooth feature.

The output of the **show running-config** command includes Bluetooth information only if the Bluetooth feature is disabled.

The device name given to the switch includes the switch serial number to uniquely identify the switch while pairing with a mobile device.

The management IP address is a private network address created for managing the switch through a Bluetooth connection.

Examples

Example output when Bluetooth is enabled but no Bluetooth adapter is connected. <XXXX> is the switch platform and <NNNNNNNNNN> is the device identifier.

```
switch# show bluetooth
Enabled                : Yes
Device name            : <XXXX>-<NNNNNNNNNN>
Adapter State          : Absent
```

Example output when Bluetooth is enabled and there is a Bluetooth adapter connected:

```
switch# show bluetooth
Enabled                : Yes
Device name            : <XXXX>-
Adapter State          : Ready
Adapter IP address     : 192.168.99.1
Adapter MAC address    : 480fcf-af153a
Connected Clients
-----------------
Name              MAC Address       IP Address     Connected Since
--------------    --------------    ------------   -------------------------
Mark's iPhone     089734-b12000     192.168.99.10   2018-07-09 08:47:22 PDT
```

Example output when Bluetooth is disabled:

```
switch# show bluetooth
Enabled                : No
Device name            : <XXXX>-<NNNNNNNNNN>
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	Operator (>) or Manage r (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

show boot-history

Syntax

```
show boot-history [all|{vsf member <1-10>}]
```

Description

Shows boot history information. When no parameters are specified, shows the most recent information about the current boot operation, and the three previous boot operations for the switch. When the **all** parameter is specified, the output of this command shows the boot information for the active management module.

For switches that support line modules (such as 8400 switch series) including the **all** parameter displays information for the active management module and all available line modules.

> **NOTE**
>
> To view boot-history on a standby, the command must be sent on the conductor console.

Parameter	Description
`all`	Optional. Shows boot information for the active management m odule. For switches that support line modules, including this pa rameter displays information for and all available line modules.
`vsf member <1-10>`	Optional. Display boot history for the specified VSF member

Usage

This command displays the boot-index, boot-ID, and up time in seconds for the current boot. If there is a previous boot, it displays boot-index, boot-ID, reboot time (based on the time zone configured in the system) and reboot reasons. Previous boot information is displayed in reverse chronological order.

The output of this command includes the following information:

Parameter	Description
`Index`	The position of the boot in the history file. Range: 0 to 3.
`Boot ID`	A unique ID for the boot . A system - generated 128 - bit strin g.
`Current Boot, up for <time>`	For the current boot, the **show boot - history** command shows the number of seconds the module has been running on the cur rent software.

Parameter	Description
`<Timestamp>: boot reason`	For previous boot operations, the **show boot - history** command shows the time at which the operation occurred and the reason for the boot. The reason for the boot is one of the following values: • <DAEMON - NAME> **crash**: The daemon identified by <DAEMON - NAME> caused the module to boot. • **Kernel crash**: The operating system software associated with the module caused the module to boot. • **Uncontrolled reboot**: The reason for the reboot is not known. • **Reboot requested through database**: The reboot occurred because of a request made through the CLI or other API.

Table 1. Description of reboots handled through the database

Boot History String	Description
Reboot requested by user	A user requested a switch reboot through the CLI or web UI.
Reset button pressed	The switch detected a short - press of the reset button
Backplane fault	A backplane fault occurred.
Configuration change	A configuration change resulted in a reboot.
Configuration version migration	A configuration version migration occurred which required a reboot.
Console error	The console failed to start.
Fabric fault	A fabric fault occurred.
All line modules faulted	A zero line card condition occurred.
Redundancy switchover requested	A user requested a redundancy switchover.
Redundant Management communication timeout	The standby management module has taken over from an unresponsive active management module.
Redundant Management election timeout	A failure to elect a standby management module in the allotted time.

Boot History String	Description
Critical service fault (error)	A daemon critical to switch operation has stopped fu nctioning. An extra error string may be present to d escribe the error in detail.
VSF autojoin renumber	Reset triggered by VSF autojoin.
VSF member renumbered	A user requested a renumber of a VSF member.
VSF switchover requested	A user requested a VSF switchover.
VSX software update	Reset triggered by a VSX software update.
Chassis critical temperature	Chassis operating temperature exceeded.
Chassis low critical temperature	Chassis temperature below the minimum operating t hreshold.
Chassis insufficient fans	Insufficient fans to cool the chassis.
Chassis unsupported PSUs/fans	Unsupported or misconfigured PSUs or system fans.
Management module critical temperature	Management module operating temperature exceed ed.
ISSU SMM update	Standby management module reboot triggered by a n In - Service Software Upgrade (ISSU).
ISSU switchover	Redundancy switchover triggered by an In - Service Software Upgrade.
ISSU aborted	Standby management module reset triggered by fail ure during an In - Service Software Upgrade.
Rollback timer expired	Reset triggered by the ISSU rollback timer expiring.

Examples

Showing the boot history of the active management module:

```
switch# show boot-history
Management module
=================
Index : 2
Boot ID : c34a2c2499004a02bbeeff4992e1fdbd
Current Boot, up for 1 days 13 hrs 13 mins 27 secs
Index : 1
```

```
Boot ID : bfba9bc486304e57904ac717a0ccbdcd
02 Sep 23 02:55:33 : CPU request reset with 0x20201, Version:
FL.10.14.0000-1619-ga9ec1805bd442~dirty
02 Sep 23 02:55:33 : Switch boot count is 2
Index : 0
Boot ID : a88a71b7ca9a4574af7e3b811ddfdc7e
02 Sep 23 02:49:26 : Reboot requested by user, Version: FL.10.14.0000-1619-
ga9ec1805bd442~dirty
02 Sep 23 02:50:02 : Switch boot count is 1
Index : 3
Boot ID : f00ba10c8c44457f83fee303d014a89a
25 Aug 23 10:27:42 :  Power on reset with 0x1, Version: FL.10.14.0000-1465-
g9df95249d06b0~dirty
25 Aug 23 10:28:18 :  Switch boot count is 3
25 Aug 23 10:29:02 :  Primary overtemperature fault detected with 0x2 in
PSU 1/1
```

Showing the boot history of the active management module and all line modules:

```
switch#
Management module
=================
Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs
Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database
Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database
Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database
Line module 1/1
=================
Index : 3
10 Aug 17 12:45:46 : dune_agent crashed
...
Management module
=================
Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs
Index : 2
```

```
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database
Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database
Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database
Line module 1/1
=================
Index : 3
10 Aug 17 12:45:46 : dune_agent crashed
...
```

```
switch# show boot-history
Management module
=================
Index : 2
Boot ID : a61ad00d10864c748bc7893a5d4af2e4
15 Dec 23 19:02:02 : Power on reset with 0x1, Version: FL.10.13.1000AF
15 Dec 23 19:02:02 : Switch boot count is 0
15 Dec 23 19:02:17 : PSU 1/1: Fault detected
Index : 1
Boot ID : 30d831bbfdfa425baf50a629ee01b185
15 Dec 23 19:01:58 : Power on reset with 0x1, Version: FL.10.13.1000AF
15 Dec 23 19:01:58 : Switch boot count is 0
```

```
switch# show boot-history vsf member 2
Member-2
=========
Index : 0
Boot ID : df99026c194a44f1944a3e7685fb4d90
Current Boot, up for 3 hrs 31 mins 39 secs
Index : 3
Boot ID : 7bf4104903fe4ad1ba4bce40e8099c76
10 Aug 17 10:02:24 : Reboot requested through database
10 Aug 17 10:02:13 : Switch boot count is 2
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`show capacities`

Syntax

```
show capacities <FEATURE> [vsx-peer]
```

Description

Shows system capacities and their values for all features or a specific feature.

Parameter	Description
`<FEATURE>`	Specifies a feature. For example, **aaa** or **vrrp** .
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output f rom the VSX peer switch is not displayed. This parameter is avai lable on switches that support VSX.

Usage

Capacities are expressed in user-understandable terms. Thus they may not map to a specific hardware or software resource or component. They are not intended to define a feature exhaustively.

Examples

Showing all available capacities for BGP:

```
switch# show capacities bgp
System Capacities: Filter BGP
Capacities Name
Value
----------------------------------------------------------------------------
-------
Maximum number of AS numbers in as-path
```

```
attribute                                         32
...
```

```
switch# show capacities port-access
System Capacities: Filter Port-Access
Capacities Name
Value
----------------------------------------------------------------------------
--------
Maximum number of Clients that can be authenticated on a
port                                                    256
Maximum number of Device Profiles allowed to be created on the
system                                             16
Maximum number of unique FQDN entries per port access
policy                                                     128
Maximum number of MAC Address that can be authorized on a
port                                                    32
Maximum number of Port Access Role VLAN IDs allowed to be created on the
system                        256
Maximum number of Port Access Role VLAN names allowed to be created on the
system                       50
Maximum number of Port Access Roles allowed to be created on the
system                                 512
```

```
switch# show capacities
System Capacities:
Capacities
Name                                                                   Value
----------------------------------------------------------------------------
--------------
Maximum number of Access Control Entries configurable in a
system                   4096
Maximum number of Access Control Lists configurable in a
system                      512
Maximum number of class entries configurable in a
system                            4096
Maximum number of classes configurable in a
system                                   512
Maximum number of entries in an Access Control
List                                    1024
Maximum number of entries in a
class                                                   1024
Maximum number of entries in a
policy                                                    64
Maximum number of classifier policies configurable in a
```

```
system                          512
Maximum number of policy entries configurable in a
system                              4096
Maximum number of clients supported for tracking the IP address in the
system        128
Maximum number of dynamic VLANs that can be allowed using
MVRP                          256
Maximum number of nexthops per IP ECMP
group                                                 1
Maximum number of IP neighbors (IPv4+IPv6) supported in the
system                  1024
Maximum number of IPv4 neighbors(# of ARP entries) supported in the
system          1024
Maximum number of IPv6 neighbors(# of ND entries) supported in the
system            512
Maximum number of L2 MAC addresses supported in the
system                         8192
Maximum number of L3 Groups for IP Tunnels and ECMP
Groups                                 1
Maximum number of L3 Destinations for Routes, Nexthops in Tunnels and ECMP
groups   1024
Maximum number of configurable LAG
ports                                                     8
Maximum number of members supported by a LAG
port                                        8
Maximum number of VLANs across ports allowed in loop-
protect                        3328
Maximum number of IGMP/MLD groups
supported                                            512
Maximum number of IGMP/MLD snooping groups
supported                                   512
Maximum number of Mirror Sessions configurable in a
system                                 4
Maximum number of enabled Mirror Sessions in a
system                                      4
Maximum number of mstp instances configurable in a
system                                 16
Maximum number of Clients that can be authenticated on a
port                            32
Maximum number of Device Profiles allowed to be created on the
system                   8
Maximum number of Port Access Roles allowed to be created on the
system                32
```

```
Maximum number of MAC Address that can be authorized on a
port                        32
Maximum number of Port Access Role VLAN IDs allowed to be created on the
system       50
Maximum number of Port Access Role VLAN names allowed to be created on the
system     50
Maximum number of RBAC rules per user
group                                              1024
Maximum number of RPVST VLANs configurable on the
system                                   16
Maximum number of RPVST VPORTs supported in a
system                                      512
Maximum number of SVIs supported in the
system                                             16
Maximum number of routes (IPv4+IPv6) on the
system                                        512
Maximum number of IPv4 routes on the
system                                               512
Maximum number of IPv6 routes on the
system                                               512
Maximum number of VLANs supported in the
system                                           512
System Capacities:
Capacities
Name                                                                       Value
----------------------------------------------------------------------------
--------------
Maximum number of Access Control Entries configurable in a
system                     4096
Maximum number of Access Control Lists configurable in a
system                        512
Maximum number of class entries configurable in a
system                              4096
Maximum number of classes configurable in a
system                                     512
Maximum number of entries in an Access Control
List                                 1024
Maximum number of entries in a
class                                                     1024
Maximum number of entries in a
policy                                                      64
Maximum number of classifier policies configurable in a
system                         512
```

```
Maximum number of policy entries configurable in a
system                          4096
Maximum number of clients supported for tracking the IP address in the
system        128
Maximum number of dynamic VLANs that can be allowed using
MVRP                        256
Maximum number of nexthops per IP ECMP
group                                                1
Maximum number of IP neighbors (IPv4+IPv6) supported in the
system                  1024
Maximum number of IPv4 neighbors(# of ARP entries) supported in the
system          1024
Maximum number of IPv6 neighbors(# of ND entries) supported in the
system            512
Maximum number of L2 MAC addresses supported in the
system                         8192
Maximum number of L3 Groups for IP Tunnels and ECMP
Groups                            1
Maximum number of L3 Destinations for Routes, Nexthops in Tunnels and ECMP
groups   1024
Maximum number of configurable LAG
ports                                                    8
Maximum number of members supported by a LAG
port                                          8
Maximum number of VLANs across ports allowed in loop-
protect                       3328
Maximum number of IGMP/MLD groups
supported                                           512
Maximum number of IGMP/MLD snooping groups
supported                                  512
Maximum number of Mirror Sessions configurable in a
system                            4
Maximum number of enabled Mirror Sessions in a
system                                 4
Maximum number of mstp instances configurable in a
system                            16
Maximum number of Clients that can be authenticated on a
port                         32
Maximum number of Device Profiles allowed to be created on the
system                  8
Maximum number of Port Access Roles allowed to be created on the
system               32
Maximum number of MAC Address that can be authorized on a
```

```
port                        32
Maximum number of Port Access Role VLAN IDs allowed to be created on the
system       50
Maximum number of Port Access Role VLAN names allowed to be created on the
system     50
Maximum number of RBAC rules per user
group                                              1024
Maximum number of RPVST VLANs configurable on the
system                              16
Maximum number of RPVST VPORTs supported in a
system                                 512
Maximum number of SVIs supported in the
system                                             16
Maximum number of routes (IPv4+IPv6) on the
system                                   512
Maximum number of IPv4 routes on the
system                                               512
Maximum number of IPv6 routes on the
system                                               512
Maximum number of VLANs supported in the
system                                          512
switch# show capacities
System Capacities:
Capacities
Name                                                                       Value
----------------------------------------------------------------------------
--------------
Maximum number of Access Control Entries configurable in a
system                   4096
Maximum number of Access Control Lists configurable in a
system                      512
Maximum number of class entries configurable in a
system                            4096
Maximum number of classes configurable in a
system                                   512
Maximum number of entries in an Access Control
List                                 1024
Maximum number of entries in a
class                                                    1024
Maximum number of entries in a
policy                                                     64
Maximum number of classifier policies configurable in a
system                       512
```

```
Maximum number of policy entries configurable in a
system                           4096
Maximum number of clients supported for tracking the IP address in the
system        128
Maximum number of dynamic VLANs that can be allowed using
MVRP                           256
Maximum number of nexthops per IP ECMP
group                                                  1
Maximum number of IP neighbors (IPv4+IPv6) supported in the
system                  1024
Maximum number of IPv4 neighbors(# of ARP entries) supported in the
system          1024
Maximum number of IPv6 neighbors(# of ND entries) supported in the
system            512
Maximum number of L2 MAC addresses supported in the
system                          8192
Maximum number of L3 Groups for IP Tunnels and ECMP
Groups                            1
Maximum number of L3 Destinations for Routes, Nexthops in Tunnels and ECMP
groups   1024
Maximum number of configurable LAG
ports                                                      8
Maximum number of members supported by a LAG
port                                        8
Maximum number of VLANs across ports allowed in loop-
protect                        3328
Maximum number of IGMP/MLD groups
supported                                          512
Maximum number of IGMP/MLD snooping groups
supported                                  512
Maximum number of Mirror Sessions configurable in a
system                            4
Maximum number of enabled Mirror Sessions in a
system                                 4
Maximum number of mstp instances configurable in a
system                            16
Maximum number of Clients that can be authenticated on a
port                           32
Maximum number of Device Profiles allowed to be created on the
system                  8
Maximum number of Port Access Roles allowed to be created on the
system               32
Maximum number of MAC Address that can be authorized on a
```

```
port                            32
Maximum number of Port Access Role VLAN IDs allowed to be created on the
system       50
Maximum number of Port Access Role VLAN names allowed to be created on the
system     50
Maximum number of RBAC rules per user
group                                              1024
Maximum number of RPVST VLANs configurable on the
system                                  16
Maximum number of RPVST VPORTs supported in a
system                                 512
Maximum number of SVIs supported in the
system                                            16
Maximum number of routes (IPv4+IPv6) on the
system                                   512
Maximum number of IPv4 routes on the
system                                            512
Maximum number of IPv6 routes on the
system                                            512
Maximum number of VLANs supported in the
system                                        512
System Capacities:
Capacities
Name                                                                          Value
----------------------------------------------------------------------------
--------------
Maximum number of Access Control Entries configurable in a
system                   4096
Maximum number of Access Control Lists configurable in a
system                      512
Maximum number of class entries configurable in a
system                            4096
Maximum number of classes configurable in a
system                                   512
Maximum number of entries in an Access Control
List                                 1024
Maximum number of entries in a
class                                                     1024
Maximum number of entries in a
policy                                                      64
Maximum number of classifier policies configurable in a
system                       512
Maximum number of policy entries configurable in a
```

```
system                              4096
Maximum number of clients supported for tracking the IP address in the
system        128
Maximum number of dynamic VLANs that can be allowed using
MVRP                          256
Maximum number of nexthops per IP ECMP
group                                              1
Maximum number of IP neighbors (IPv4+IPv6) supported in the
system                 1024
Maximum number of IPv4 neighbors(# of ARP entries) supported in the
system         1024
Maximum number of IPv6 neighbors(# of ND entries) supported in the
system           512
Maximum number of L2 MAC addresses supported in the
system                             8192
Maximum number of L3 Groups for IP Tunnels and ECMP
Groups                               1
Maximum number of L3 Destinations for Routes, Nexthops in Tunnels and ECMP
groups   1024
Maximum number of configurable LAG
ports                                                  8
Maximum number of members supported by a LAG
port                                        8
Maximum number of VLANs across ports allowed in loop-
protect                        3328
Maximum number of IGMP/MLD groups
supported                                         512
Maximum number of IGMP/MLD snooping groups
supported                                512
Maximum number of Mirror Sessions configurable in a
system                               4
Maximum number of enabled Mirror Sessions in a
system                                    4
Maximum number of mstp instances configurable in a
system                               16
Maximum number of Clients that can be authenticated on a
port                         32
Maximum number of Device Profiles allowed to be created on the
system                   8
Maximum number of Port Access Roles allowed to be created on the
system                32
Maximum number of MAC Address that can be authorized on a
port                        32
```

```
Maximum number of Port Access Role VLAN IDs allowed to be created on the
system       50
Maximum number of Port Access Role VLAN names allowed to be created on the
system     50
Maximum number of RBAC rules per user
group                                         1024
Maximum number of RPVST VLANs configurable on the
system                              16
Maximum number of RPVST VPORTs supported in a
system                                 512
Maximum number of SVIs supported in the
system                                        16
Maximum number of routes (IPv4+IPv6) on the
system                                   512
Maximum number of IPv4 routes on the
system                                          512
Maximum number of IPv6 routes on the
system                                          512
Maximum number of VLANs supported in the
system                                      512
```

Showing all available capacities for mirroring:

```
switch# show capacities mirroring
System Capacities: Filter Mirroring
Capacities Name
Value
----------------------------------------------------------------------------
-------
Maximum number of Mirror Sessions configurable in a
system                        4
Maximum number of enabled Mirror Sessions in a
system                             4
```

Showing all available capacities for MSTP:

```
switch# show capacities mstp
System Capacities: Filter MSTP
Capacities Name
Value
----------------------------------------------------------------------------
-------
Maximum number of mstp instances configurable in a
system                        64
```

Showing all available capacities for VLAN count:

```
switch# show capacities vlan-count
System Capacities: Filter VLAN Count
Capacities Name
Value
----------------------------------------------------------------------------
-------
Maximum number of VLANs supported in the
system                                      4094
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution righ ts for this command.

show capacities-status

Syntax

```
show capacities-status <FEATURE>
            [vsx-peer]
```

Description

Shows system capacities status and their values for all features or a specific feature.

Parameter	Description
<FEATURE>	Specifies the feature, for example **aaa** or **vrrp** for which to displ ay capacities, values, and status. Required.

Parameter	Description
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output f rom the VSX peer switch is not displayed. This parameter is avai lable on switches that support VSX.

Examples

Showing the system capacities status for all features:

```
switch# show capacities-status
System Capacities Status
Capacities Status Name                                              Value
Maximum
----------------------------------------------------------------------------
--
Number of active gateway mac addresses in a system                 0      16
Number of aspath-lists configured                                  0      64
Number of community-lists configured                               0      64
...
```

```
switch# show capacities-status
System Capacities Status
Capacities Status Name
Value Maximum
----------------------------------------------------------------------------
----------
Number of Access Control Entries currently configured
0    4096
Number of Access Control Lists currently configured
0     512
Number of class entries currently configured
0    4096
Number of classes currently configured
0     512
Number of policies currently configured
0     512
Number of policy entries currently configured
0    4096
Number of dynamic VLANs currently learnt using MVRP
0     256
Number of IP neighbor (IPv4+IPv6) entries
1    1024
Number of IPv4 neighbor(ARP) entries
1    1024
```

```
Number of IPv6 neighbor(ND) entries
0     512
Number of L3 Groups for IP Tunnels and ECMP Groups currently configured
0       1
Number of L3 Destinations for Routes, Nexthops in ECMP groups and
   Tunnels currently configured
0    1024
Number of Mirror Sessions currently configured
0       4
Number of Mirror Sessions currently enabled
0       4
Number of mstp instances currently configured
0      16
Number of RPVST VLANs currently configured
0      16
Number of routes (IPv4+IPv6) currently configured
1     512
Number of IPv4 routes currently configured
1     512
Number of IPv6 routes currently configured
0     512
Number of VLANs currently configured
1     512
```

Showing the system capacities status for BGP:

```
switch# show capacities-status bgp
System Capacities Status: Filter BGP
Capacities Status Name                                             Value
Maximum
----------------------------------------------------------------------------
---
Number of aspath-lists configured                                  0      64
Number of community-lists configured                               0      64
Number of neighbors configured across all VRFs                     0      50
Number of peer groups configured across all VRFs                   0      25
Number of prefix-lists configured                                  0      64
Number of route-maps configured                                    0      64
Number of routes in BGP RIB                                        0  256000
Number of route reflector clients configured across all VRFs       0      16
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`show console`

Syntax

```
show console
```

Description

Shows the serial console port current speed.

Examples

Showing the console port current speed:

```
switch# show console
Baud Rate: 9600
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show core-dump`

Syntax

```
show core-dump [all | <SLOT-ID>]
```

Description

Shows core dump information about the specified module. When no parameters are specified, shows only the core dumps generated in the current boot of the management module. When the **all** parameter is specified, shows all available core dumps.

Parameter	Description
`all`	Shows all available core dumps.
<SLOT-ID>	Shows the core dumps for the management module or line mod ule in **<SLOT - ID>. <SLOT - ID>** specifies a physical location o n the switch. Use the format **member/slot/port** (for example, **1/ 3/1**) for line modules. Use the format **member/slot** for manage ment modules. You must specify the slot ID for either the active management module, or the line module.

Usage

When no parameters are specified, the **show core-dump** command shows only the core dumps generated in the current boot of the management module. You can use this command to determine when any crashes are occurring in the current boot.

If no core dumps have occurred, the following message is displayed: **No core dumps are present**

To show core dump information for the standby management module, you must use the **standby** command to switch to the standby management module and then execute the **show core-dump** command.

In the output, the meaning of the information is the following:

`Daemon Name`

Identifies name of the daemon for which there is dump information.

`Instance ID`

Identifies the specific instance of the daemon shown in the **Daemon Name** column.

`Present`

Indicates the status of the core dump:

`Yes`

The core dump has completed and available for copying.

`In Progress`

Core dump generation is in progress. Do not attempt to copy this core dump.

`Timestamp`

Indicates the time the daemon crash occurred. The time is the local time using the time zone configured on the switch.

`Build ID`

Identifies additional information about the software image associated with the daemon.

Examples

Showing core dump information for the current boot of the active management module only:

```
switch# show core-dump
===========================================================================
======
Daemon Name     | Instance ID | Present    | Timestamp            | Build
ID
===========================================================================
======
hpe-fand          1399          Yes         2017-08-04 19:05:34    1246d2a
hpe-sysmond       957           Yes         2017-08-04 19:05:29    1246d2a
===========================================================================
======
Total number of core dumps : 2
===========================================================================
======
===========================================================================
======
Daemon Name     | Instance ID | Present    | Timestamp            | Build
ID
===========================================================================
```

```
======
hpe-fand           1399          Yes          2017-08-04 19:05:34      1246d2a
hpe-sysmond        957           Yes          2017-08-04 19:05:29      1246d2a
=============================================================================
======
Total number of core dumps : 2
=============================================================================
======
```

Showing all core dumps:

```
switch# show core-dump all
=============================================================================
=
Management Module core-dumps
=============================================================================
=
Daemon Name      | Instance ID | Present     | Timestamp           | Build ID
=============================================================================
=
hpe-sysmond        513           Yes          2017-07-31 13:58:05    e70f101
hpe-tempd          1048          Yes          2017-08-13 13:31:53    e70f101
hpe-tempd          1052          Yes          2017-08-13 13:41:44    e70f101
Line Module core-dumps
=============================================================================
=
Line Module : 1/1
=============================================================================
=
dune_agent_0       18958         Yes          2017-08-12 11:50:17    e70f101
dune_agent_0       18842         Yes          2017-08-12 11:50:09    e70f101
=============================================================================
=
Total number of core dumps : 5
=============================================================================
=
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`show deprecated commands`

Syntax

```
show deprecated-commands [<feature>]
```

Description

Shows the list of CLI commands that will be deprecated in a future release along with the new form of the same command which is recommended for use.

Both the command options will be supported until a certain release, after which only the newer replacement command will be supported.

Parameter	Description
`feature`	Optional. Specify feature name. The list of features for which you can view deprecated comman ds are:

Examples

Check the deprecated CLI commands for a specific feature:

```
switch# show deprecated-commands
-----------------------------------------------------------------------------
-----------------
The following commands with ipv4 keyword will be replaced with ip
-----------------------------------------------------------------------------
-----------------
Deprecated:  vrrp <1-255> address-family (ipv4 | ipv6)
Replacement: vrrp <1-255> address-family (ip | ipv6)
Deprecated:  show bgp ipv4 unicast
Replacement: show bgp ip unicast
```

```
...
switch# show deprecated-commands vrrp
---------------------------------------------------------------------------
-----------------
The following commands with ipv4 keyword will be replaced with ip
---------------------------------------------------------------------------
-----------------
Deprecated:  vrrp <1-255> address-family (ipv4 | ipv6)
Replacement: vrrp <1-255> address-family (ip | ipv6)
Deprecated:  show vrrp (ipv4 | ipv6 | brief | detail)(<1-255>)
Replacement: show vrrp (ip | ipv6 | brief | detail)(<1-255>)
...
switch# show deprecated-commands vsf
Feature vsf has no deprecated commands.
```

Command History

Release	Modification
10.14	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show domain-name`

Syntax

```
show domain-name [vsx-peer]
```

Description

Shows the current domain name.

Parameter	Description
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output f rom the VSX peer switch is not displayed. This parameter is avai lable on switches that support VSX.

Usage

If there is no domain name configured, the CLI displays a blank line.

Example

Setting and showing the domain name:

```
switch# show domain-name
switch# config
switch(config)# domain-name example.com
switch(config)# show domain-name
example.com
switch(config)#
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`show environment fan`

Syntax

```
show environment fan [vsx-peer]
```

Description

Shows the status information for all fans and fan trays (if present) in the system.

Parameter	Description
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output f rom the VSX peer switch is not displayed. This parameter is avai lable on switches that support VSX.

Usage

For fan trays, **Status** is one of the following values:

- ready:The fan tray is operating normally.
- fault:The fan tray is in a fault event. The status of the fan tray does not indicate the status of fans.
- empty:The fan tray is not installed in the system.

For fans:

```
Speed :Indicates the relative speed of the fan based on the nominal speed
range of the fan. Values are:
   Slow:The fan is running at less than 25% of its maximum speed.
   Normal:The fan is running at 25-49% of its maximum speed.
   Medium:The fan is running at 50-74% of its maximum speed.
   Fast:The fan is running at 75-99% of its maximum speed.
   Max:The fan is running at 100% of its maximum speed.
   N/A:The fan is not installed.
   front-to-back:Air flows from the front of the system to the back of the
system.
   N/A:The fan is not installed.
Status: Fan status. Values are:
   uninitialized:The fan has not completed initialization.
   ok: The fan is operating normally.
   fault: The fan is in a fault state.
   empty: The fan is not installed.
```

Examples

Showing output for systems with fan trays for HPE Aruba Networking 8400 Switch series:

```
switch# show environment fan
Fan tray information
-----------------------------------------------------------------------------
--
```

```
Mbr/Tray  Description                                    Status   Serial Number
Fans
----------------------------------------------------------------------------
--
1/1       JL369A 8400 Fan tray                           ready    SGXXXXXXXXXX   6
1/2       JL369A 8400 Fan tray                           ready    SGXXXXXXXXXX   6
1/3       N/A                                            empty    N/A            0
Fan information
------------------------------------------------------------------------
Mbr/Tray/Fan  Serial Number  Speed   Direction      Status        RPM
------------------------------------------------------------------------
1/1/1         SGXXXXXXXXXX   slow    front-to-back  ok            6000
1/1/2         SGXXXXXXXXXX   normal  front-to-back  ok            8000
1/1/3         SGXXXXXXXXXX   medium  front-to-back  ok            11000
1/1/4         SGXXXXXXXXXX   fast    front-to-back  ok            14000
1/1/5         SGXXXXXXXXXX   max     front-to-back  fault         16500
1/1/6         N/A            N/A     N/A            empty         0
1/2/1         SGXXXXXXXXX    slow    front-to-back  ok            6000
...
Fan tray information
----------------------------------------------------------------------------
--
Mbr/Tray  Description                                    Status   Serial Number
Fans
----------------------------------------------------------------------------
--
1/1       JL369A 8400 Fan tray                           ready    SGXXXXXXXXXX   6
1/2       JL369A 8400 Fan tray                           ready    SGXXXXXXXXXX   6
1/3       N/A                                            empty    N/A            0
Fan information
------------------------------------------------------------------------
Mbr/Tray/Fan  Serial Number  Speed   Direction      Status        RPM
------------------------------------------------------------------------
1/1/1         SGXXXXXXXXXX   slow    front-to-back  ok            6000
1/1/2         SGXXXXXXXXXX   normal  front-to-back  ok            8000
1/1/3         SGXXXXXXXXXX   medium  front-to-back  ok            11000
1/1/4         SGXXXXXXXXXX   fast    front-to-back  ok            14000
1/1/5         SGXXXXXXXXXX   max     front-to-back  fault         16500
1/1/6         N/A            N/A     N/A            empty         0
1/2/1         SGXXXXXXXXX    slow    front-to-back  ok            6000
...
```

```
switch# show environment fan
Fan tray informationFan tray information
```

```
---------------------------------------------------------------------------
--
Name  Description                     Status       Serial Number  Fans
---------------------------------------------------------------------------
--
1/1   JL669A  X751 FB Fan Tray        ready        CN04KN9913     2
1/2   JL669A  X751 FB Fan Tray        ready        CN04KN97XJ     2
Fan information
---------------------------------------------------------------------------
--
Location-      Product  Serial Number  Speed   Direction      Status  RPM
Mbr/Slot/Fan   Name
---------------------------------------------------------------------------
--
Tray-1/1/1     N/A      N/A            slow    front-to-back  ok      4024
Tray-1/1/2     N/A      N/A            slow    front-to-back  ok      4041
Tray-1/2/1     N/A      N/A            slow    front-to-back  ok      4030
Tray-1/2/2     N/A      N/A            slow    front-to-back  ok      4028
switch# show environment fan
Fan information
--------------------------------------------------------------------------
Mbr/Fan       Product  Serial Number  Speed   Direction      Status  RPM
              Name
--------------------------------------------------------------------------
1/1           N/A      N/A            N/A     left-to-right  ok      N/A
```

Showing output for a system without a fan tray:

```
switch# show environment fan
Fan information
---------------------------------------------------------------
Fan   Serial Number  Speed   Direction      Status        RPM
---------------------------------------------------------------
1     SGXXXXXXXXXX   slow    front-to-back  ok            6000
2     SGXXXXXXXXXX   normal  front-to-back  ok            8000
3     SGXXXXXXXXXX   medium  front-to-back  ok            11000
4     SGXXXXXXXXXX   fast    front-to-back  ok            14000
5     SGXXXXXXXXXX   max     front-to-back  fault         16500
6     N/A            N/A     N/A            empty
...
```

switch# **show environment fan**

```
switch# show environment fan
Fan tray information
---------------------------------------------------------------------------
--
```

```
Name  Description                          Status        Serial Number
Fans
---------------------------------------------------------------------------
--
1/1   JL630A HPE ANW F2B Fan               ready         N/A             2
1/2   JL630A HPE ANW F2B Fan               ready         N/A             2
1/3   JL630A HPE ANW F2B Fan               ready         N/A             2
1/4   JL630A HPE ANW F2B Fan               ready         N/A             2
1/5   JL630A HPE ANW F2B Fan               ready         N/A             2
1/6   JL630A HPE ANW F2B Fan               ready         N/A             2
Fan information
---------------------------------------------------------------------------
--
Location-      Product  Serial Number  Speed   Direction      Status  RPM
Mbr/Slot/Fan   Name
---------------------------------------------------------------------------
--
Tray-1/1/1     JL630A   N/A            slow    front-to-back  ok      6400
Tray-1/1/2     JL630A   N/A            slow    front-to-back  ok      5400
Tray-1/2/1     JL630A   N/A            slow    front-to-back  ok      6300
Tray-1/2/2     JL630A   N/A            slow    front-to-back  ok      5400
Tray-1/3/1     JL630A   N/A            slow    front-to-back  ok      6500
Tray-1/3/2     JL630A   N/A            slow    front-to-back  ok      5500
Tray-1/4/1     JL630A   N/A            slow    front-to-back  ok      6300
Tray-1/4/2     JL630A   N/A            slow    front-to-back  ok      5400
Tray-1/5/1     JL630A   N/A            slow    front-to-back  ok      6000
Tray-1/5/2     JL630A   N/A            slow    front-to-back  ok      5200
Tray-1/6/1     JL630A   N/A            slow    front-to-back  ok      6200
Tray-1/6/2     JL630A   N/A            slow    front-to-back  ok      5400
```

Command History

Release	Modification
10.18	Introduced support for 8325 and 8325P Switch series.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show environment led`

Syntax

```
show environment led
             [vsx-peer]
```

Description

Shows state and status information for all the configurable LEDs in the system.

Parameter	Description
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output f rom the VSX peer switch is not displayed. This parameter is avai lable on switches that support VSX.

Example

Showing state and status for LED:

```
switch# show environment led
Name               State      Status
-----------------------------------
locator            flashing   ok
switch# show environment led
Mbr/Name           State      Status
-------------------------------
1/locator          off        ok
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show environment power-consumption`

Syntax

```
show environment power-consumption [detail]
```

Description

Displays power consumption information.

Parameter	Description
`[detail}`	Displays detailed power consumption information.
`<MEMBER-ID>`	For VSF supported platforms only. Displays the power consum ption information for the specified VSF member. Range: 1 - 10.

Usage

Power consumed values are updated every 5 seconds. The total power consumed is the total power used in a chassis. The power consumed average is calculated from the total power consumed as a running average over a period of time. The average period has a default of 10 minutes. The period can be configured using **power-consumption-average-period.**

On chassis where power input is not supported, power consumed is displayed as Power Usage and PSU Output Power is displayed as N/A.

The following information is provided for a summary of power consumption:

- line module: power used by line module
- management module: power used by management module
- fabric module: power used by fabric module
- chassis module: power used by chassis module
- fan module: power used by fan module
- power total: total power consumption
- average power: average for total power consumption that is calculated over a given period
- average period: time to calculate power average

For more information, see the video **AOS-CX 10.14 Release Update: Sustainability Update**.

Example

Showing the power consumption for a 1RU switch:

```
switch# show environment power-consumption
Power Consumption Averaging Period : 600 seconds
                                      Power Usage (W)      PSU Output (W)
Name Description                     Instant   Average   Instant   Average
----------------------------------------------------------------------------
1    8360-32Y4C v2 Switch             120.00    111.16    111.00    102.62
```

Showing the power consumption for a VSF stack:

```
switch# show environment power-consumption
Power Consumption Averaging Period : 600 seconds
                                      Power Usage (W)      PSU Output (W)
Name Description                     Instant   Average   Instant   Average
----------------------------------------------------------------------------
1    6200M 24G 4SFP+ Sw               350.00    349.62    300.00    311.50
2    6200M 24G 4SFP+ Sw               300.00    299.50    280.00    275.50
```

Showing the power consumption for a switch where input reading is not supported:

```
switch# show environment power-consumption
Power Consumption Averaging Period : 600 seconds
                                      Power Usage (W)      PSU Output (W)
Name Description                     Instant   Average   Instant   Average
----------------------------------------------------------------------------
1    6200M 24G 4SFP+ Sw               321.00    330.30       N/A       N/A
```

Showing the power consumption for a switch with multiple line cards, in brief:

```
switch# show environment power-consumption
Power Consumption Averaging Period : 600 seconds
```

```
                                            Power Usage (W)      PSU Output (W)
Name Description                           Instant    Average    Instant    Average
---------------------------------------------------------------------------------
1    6410 v2 Chassis                        900.00     905.62     798.00     803.23
```

Showing the power consumption for a switch with multiple line cards, in detail:

```
switch# show environment power-consumption detail
Power Consumption Averaging Period : 600 seconds
                        Power Usage (W)      PSU Output (W)
Name Module Type       Instant    Average    Instant    Average
--------------------------------------------------------------
1    chassis total     1495.00    1499.99    1307.00    1300.67
1/3   line               97.00
1/4   line               95.00
1/5   line               90.00
1/6   line               58.00
1/7   line               90.00
1/8   line               93.00
1/9   line               96.00
1/10  line               92.00
1/11  line              100.00
1/12  line               99.00
1/1   management         14.00
1/2   management         13.00
      other             370.00
```

Showing the power consumption for 4 member stack, in detail:

```
switch# show environment power-consumption detail
Power Consumption Averaging Period : 600 seconds
                        Power Usage (W)      PSU Output (W)
Name Module Type       Instant    Average    Instant    Average
--------------------------------------------------------------
1    chassis total      930.00     905.17     830.00     803.40
2    chassis total      900.00     920.00     875.00     880.25
3    chassis total      850.00     857.25     802.00     805.00
4    chassis total     1200.00    1100.56     921.00     950.00
Total Power Consumption      3428.00
```

Showing the power consumption for VSF member 2:

```
switch# show environment power-consumption member 2
Power Consumption Averaging Period : 600 seconds
                                            Power Usage (W)      PSU Output (W)
Name Description                           Instant    Average    Instant    Average
---------------------------------------------------------------------------------
2    6200M 24G 4SFP+ Sw                     300.00     299.50     280.00     275.50
```

Showing the power consumption for VSF invalid member:

```
switch# show environment power-consumption member 5
Member 5 is not present
```

Command History

Release	Modification
10.14.1000	Introduced enhancement to power consumption and power consumption averag e.
10.13	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Operator (`>`) or Manage r (`#`)	Administrators or local user group members with execution righ ts for this command.

`show environment power-consumption`

Syntax

```
show environment power-consumption [vsx-peer]
```

Description

Shows the power being consumed by each management module, line card, and fabric card subsystem, and shows power consumption for the entire chassis.

Parameter	Description
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output f rom the VSX peer switch is not displayed. This parameter is avai lable on switches that support VSX.

Usage

This command is only applicable to systems that support power consumption readings.

The power consumption values are updated once every minute.

The output of this command includes the following information:

Parameter	Description
`Name`	Shows the member number and slot number of the manageme nt module, line module, or fabric card module.
`Type`	Shows the type of module installed at the location specified by Name.
`Description`	Shows the product name and brief description of the module.
`Usage`	Shows the instantaneous power consumption of the module. Po wer consumption is shown in Watts.
`Module Total Power Usage`	Shows the total power consumption of all the modules listed. Po wer consumption is shown in Watts.
`Chassis Total Power Usage`	Shows the total instantaneous power consumed by the entire c hassis, including modules and components that do not support individual power reporting. Power consumption is shown in Wat ts.
`Chassis Total Power Available`	Shows the total amount of power, in Watts, that can be supplied to the chassis.
`Chassis Total Power Allocated`	Shows total power, in Watts, that is allocated to powering the ch assis and its installed modules.
`Chassis Total Power Unallocated`	Shows the total amount of power, in Watts, that has not been all ocated to powering the chassis or its installed modules. This po wer can be used for additional hardware you install in the chassi s.

For more information, see the video **AOS-CX 10.14 Release Update: Sustainability Update**.

Example

Showing power consumption usage for an HPE Aruba Networking 8400 switch

```
switch> show environment power-consumption
Name     Type                 Description
Usage
----------------------------------------------------------------------------
--
1/5      management-module   JL368A 8400X Mgmt Mod                       97 W
```

```
1/6      management-module    JL368A 8400X Mgmt Mod                         49 W
1/1      line-card-module     JL363A 8400X 32P 10G SFP/SFP+ Msec Mod        139
W
1/2      line-card-module     JL365A 8400X 8P 40G QSFP+ Adv Mod             158
W
1/3      line-card-module     JL366A 8400X 6P 40G/100G QSFP28 Adv Mod       127
W
1/4      line-card-module     JL363A 8400X 32P 10G SFP/SFP+ Msec Adv Mod   148
W
1/7      line-card-module     JL363A 8400X 32P 10G SFP/SFP+ Msec Adv Mod   152
W
1/8      line-card-module     JL363A 8400X 32P 10G SFP/SFP+ Msec Adv Mod   125
W
1/9      line-card-module     JL363A 8400X 32P 10G SFP/SFP+ Msec Adv Mod   132
W
1/10     line-card-module     JL363A 8400X 32P 10G SFP/SFP+ Msec Adv Mod   143
W
1/1      fabric-card-module   JL367A 8400X 7.2Tbps Fab Mod                  107
W
1/2      fabric-card-module   JL367A 8400X 7.2Tbps Fab Mod                  93 W
1/3      fabric-card-module   JL367A 8400X 7.2Tbps Fab Mod                  87 W
Module Total Power Usage
1557 W
Chassis Total Power Usage
1807 W
Chassis Total Power Available
9990 W
Chassis Total Power Allocated (total of all max wattages)
4130 W
Chassis Total Power Unallocated
5860 W
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show environment power-supply`

Syntax

```
show environment power-supply
   input-voltage [events] [vsf member <MEMBER-ID>}
   vsx-peer
```

Description

Shows status information about all power supplies in the switch.

Parameter	Description
`input-voltage`	Display the statistic and status of the Power Supply Unit (PSU) input voltage. This parameter is supported on 5420, 6200M, 63 00M, 6400, 8320, 8325/8325H/8325P, 8100, 8360, 8400, 93 00/9300S and 10000 Switch series.
`events`	Include the optional **events** parameter to display events for the power supply's input voltage. the output of the command displ ays information for the power supplies of the entire chassis or system (including the PSUs within a VSF member.)
`vsx-peer`	Shows the output from the VSX peer switch. This parameter is only available on switches with profiles that support VSX. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed.

Usage

The following information is provided for each power supply:

Parameter	Description
`Mbr/PSU`	Shows the member and slot number of the power supply.
`Product Number`	Shows the product number of the power supply.
`Serial Number`	Shows the serial number of the power supply, which uniquely id entifies the power supply.
`PSU Status`	The status of the power supply. Values are: Here is your list with the numbers removed: • **OK**: power supply is operating normally • **OK***: power supply is operating normally, although there is i nsufficient power to fully power the chassis. • **Empty**: no power supply is installed • **Absent**: no power supply in installed when one is expected • **Input Fault**: there is a fault condition on the input of the po wer supply • **Output Fault**: there is a fault condition on the output of the power supply • **PoE Fault**: there is a fault condition on the PoE output of th e power supply • **Airflow Fault**: airflow direction does not match the expecte d chassis airflow direction • **Redundancy Fault**: type of capacity does not match the ex pected redundancy type and capacity • **Mixed Input Fault**: the input type of the inserted power s upplies does not match. AC and DC power supplies cannot be mixed in the same switch. • **Warning**: there is an internal fault condition but the power supply is still operational • **Non - recoverable Fault**: there is an internal fault conditio n that the power supply cannot recover from • **Unsupported**: power supply is not of a known supported ty pe • **Unknown**: power supply has experienced an unknown fault condition • **Alert**: power supply has experienced an abnormal event bu t is still operational • **Undervoltage**: Power supply's input voltage dips below the minimum operating voltage

Parameter	Description
input type	Power supply's input type Identification. It can be **AC** or **DC**.
Voltage Range	Valid operating voltage range for the power supply to support t he current wattage maximum.
Wattage Maximum	Maximum amount of wattage that the power supply can provid e

If you include the optional **input-voltage** parameter, the output includes the following information:

Output Column	Description
Mbr/PSU	Shows the member and slot number of the power supply.
Product Number	Shows the product number of the power supply.
Input Type	Power supply's input type Identification. It can be **AC** or **DC**.
Normal Input (V)	Normal operating voltage range for the power supply to support current wattag e maximum - according to regulatory requirements.
Present Input (V)	Instantaneous input voltage reading acquired from power supply
Average Input	The average input voltage over the last 15 minutes (or since power was applied if less than 15 minutes ago).
Input voltage Violations: Count	Number of occurrences for which PSU's input voltage crosses outside of the nor mal input.
Input voltage Violations: Most Recent Event	Timestamp of the most recent violation where the PSU input voltage crosses outside of the normal input.
Recovery Count	Number of occurrences when the PSU's input voltage recovers to within the nor mal input range.
Recovery Time	Timestamp of the most recent input voltage's recovery.

Examples

Showing example output for an 8400 Switch series with two power supplies in the chassis:

```
switch# show environment power-supply
-----------------------------------------------------------------------------
----
      Product   Serial             PSU                 Input   Voltage    Wattage
```

```
Mbr/PSU  Number   Number             Status             Type   Range
Maximum
---------------------------------------------------------------------------
----
1/1      JL372A   M031R002JAFC       OK*                AC     200V-240V  2700
1/2      JL372A   M031R003JAFC       Output Fault       AC     200V-240V  2430
1/3      N/A      N/A                Absent             --     --         --
1/4      N/A      N/A             Absent             --     --         --
```

```
switch# show environment power-supply
---------------------------------------------------------------------------
----
         Product  Serial                PSU            Input  Voltage
Wattage
Mbr/PSU  Number   Number                Status         Type   Range
Maximum
---------------------------------------------------------------------------
-------
1/1      R0X36A   CN91KMM2H3            OK             AC     200V-240V
3000
1/2      N/A      N/A                   Absent         --     --      0
1/3      N/A      N/A                   Absent         --     --      0
1/4      N/A      N/A                   Absent         --     --          0
```

```
switch# show environment power-supply
```

The output of the following command shows input voltage data for a power supply. In the following example, the output of the **show environment power-supply input-voltage** command has been broken into two sections to better fit in this document. In the product command-line interface, the output appears as a single, wide table.

```
switch# show environment power-supply input-voltage
Input Voltage Averaging Period : 15 minutes
---------------------------------------------------------------------------
--
Mbr/  Product  Input  Normal     Present    Average    Input Voltage
Violations
PSU   Number   Type   Input (V)  Input (V)  Input (V)  Count
---------------------------------------------------------------------------
----
1/1   JL670A   AC     180-264        231.7      231.2      1
1/2   JL670A   AC     180-264           --         --      0
---------------------------------------------------------------------------
----
                     Recovery  Recovery
Most Recent Event            Count   Time
```

```
------------------------------------------------------------
Mon Jun 23 02:45:28 UTC 2025    1   Mon Jun 23 02:45:41 UTC 2025
N/A                             0   N/A
```

```
switch#  show environment power-supply input-voltage vsf member 2
Input Voltage Averaging Period : 15 minutes
Mbr    Product Input       Normal
/PSU   Number  Type        Input (V)
--------------------------------------------------------
2/1 JL086A  AC      90-264
2/2 JL758A  DC      40-60
Present     AveragE
Input (V)      Input (V)
--------------------------------------------
123.5       122.4
42.5        48.1
Input Voltage Violations:       Recovery  Recovery
Count   Most Recent Event          Count  Time
----------------------------------------------------------------------------
--
   0    N/A                     0   N/A
   1    Mon Nov 18 06:37:53 UTC 2024          1   Mon Nov 18 09:01:44 UTC
2024
```

```
switch# show env power-supply input-voltage events vsf member 2
Mbr/PSU           : 2/1
Product Number    : JL670A
Serial Number     : CN1AL6H326
Status            : OK
Input Type        : AC
Normal Input (V) : 180-264
Present Input     : 201.8 V
Average Input     : 200.2 V
Averaging Period : 15 minutes
Max Input         : 236.0 V
Min Input         : 176.1 V
ID    Type        Voltage   Timestamp
--------------------------------------------------------
0     violation     177.8   Tue Jun 11 13:20:23 UTC 2025
1     recovered     180.1   Tue Jun 17 06:51:45 UTC 2025
Mbr/PSU           : 2/2
951 Product Number    : N/A
Serial Number     : N/A
Status            : Absent
Input Type        : --
```

```
Normal Input (V) : --
Present Input    : --
Average Input    : --
Averaging Period : 15 minutes
Max Input        : --
Min Input        : --
ID    Type         Voltage   Timestamp
---------------------------------------------------------
No Input Voltage Violation/Recovery events
```

NOTE

Showing individual input voltage events for a switch power supply:

```
switch# show environment power-supply input-voltage events
Mbr/PSU          : 1/2
Product Number   : JL758A
Serial Number    : CN1AL6H326
Status           : OK
Input Type       : DC
Normal Input (V) : 40-60
Present Input    : 44.8 V
Average Input    : 43.1 V
Averaging Period : 15 minutes
Max Input        : 48.9 V
Min Input        : 36.9 V
ID    Type         Voltage   Timestamp
---------------------------------------------------------
0     violation       37.8   Mon Jul 21 06:37:53 UTC 2025
1     recovered       40.1   Mon Jul 21 09:01:44 UTC 2025
Mbr/PSU          : 1/2
Product Number   : JL670A
Serial Number    : CN1AL6H326
Status           : OK
Input Type       : AC
Normal Input (V) : 99-140
Present Input    : 100.8 V
Average Input    : 104.2 V
Averaging Period : 15 minutes
Max Input        : 119.8 V
Min Input        : 95.5 V
ID    Type         Voltage   Timestamp
---------------------------------------------------------
```

```
0      violation       96.5   Wed Jul 02 08:51:07 UTC 2025
1      recovered       99.1   Wed Jul 02 08:54:55 UTC 2025
2      violation       96.6   Wed Jul 02 09:11:14 UTC 2025
3      recovered       99.3   Wed Jul 02 10:24:19 UTC 2025
```
```

**Command History**

| Release | Modification |
|---|---|
| 10.16.1000 | The **input - voltage** parameter is introduced. |
| 10.07 or earlier | - - |

**Command Information**

| Platforms | Command context | Authority |
|---|---|---|
| All platforms | Operator ( `>` ) or Manager ( `#` ) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## `show environment rear-display-module`

**Syntax**

```
show environment rear-display-module [vsx-peer]
```

Description

Shows information about the display module on the back of the switch.

Parameter	Description
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.
```

Example

Showing the rear display module information on the back of the switch:

```
switch> show environment rear-display-module
Rear display module is ready
Description: 8400 Rear Display Mod
Full Description: 8400 Rear Display Module
Serial number: SG00000000
Part number: 5300_0272
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show environment temperature`

Syntax

```
show environment temperature [detail]

            [vsx-peer]
```

Description

Shows the temperature information from sensors in the switch that affect fan control.

Parameter	Description
`detail`	Shows detailed information from each temperature sensor.

Parameter	Description
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output f rom the VSX peer switch is not displayed. This parameter is avai lable on switches that support VSX.

Usage

Temperatures are shown in Celsius.

Valid values for status are the following: \

Parameter	Description
`normal`	Sensor is within nominal temperature range.
`max`	Highest temperature from this sensor.
`low_critical`	Lowest threshold temperature for this sensor.
`critical`	Highest threshold temperature for this sensor.
`fault`	Fault event for this sensor.
`emergency`	Over temperature event for Over temperature event for this sen sor.

Examples

Showing current temperature information for an 8400 switch:

```
switch# show environment temperature
Temperature information
----------------------------------------------------------------------------
--
                                                    Current
Mbr/Slot-Sensor                 Module Type        temperature  Status
----------------------------------------------------------------------------
--
1/1-PCIE-Switch                 line-card-module     95.82 C     normal
1/1-Processor                   line-card-module     0.00 C      fault
1/1-Switch-ASIC                 line-card-module     116.36 C    emergency
1/1-Switch-ASIC-Internal        line-card-module     108.25 C    critical
1/2-PCIE-Switch                 line-card-module     95.82 C     normal
...
```

```
switch# show environment temperature
Temperature information
---------------------------------------------------------------------------
--
                                                    Current
Mbr/Slot-Sensor                 Module Type        temperature  Status
---------------------------------------------------------------------------
--
1/1-PHY-01-04                   line-card-module     45.00 C    normal
1/1-PHY-05-08                   line-card-module     45.00 C    normal
1/1-PHY-09-12                   line-card-module     46.00 C    normal
1/1-PHY-13-16                   line-card-module     47.00 C    normal
1/1-PHY-17-20                   line-card-module     47.00 C    normal
1/1-PHY-21-24                   line-card-module     50.00 C    normal
1/1-PHY-25-28                   line-card-module     45.00 C    normal
1/1-PHY-29-32                   line-card-module     47.00 C    normal
1/1-PHY-33-36                   line-card-module     48.00 C    normal
1/1-PHY-37-40                   line-card-module     47.00 C    normal
1/1-PHY-41-44                   line-card-module     48.00 C    normal
1/1-PHY-45-48                   line-card-module     49.00 C    normal
1/1-Switch-ASIC-Internal        line-card-module     56.25 C    normal
1/1-CPU-Zone-0                  management-module    50.00 C    normal
1/1-CPU-Zone-1                  management-module    50.00 C    normal
1/1-CPU-Zone-2                  management-module    50.00 C    normal
1/1-CPU-Zone-3                  management-module    51.00 C    normal
1/1-CPU-Zone-4                  management-module    51.00 C    normal
1/1-CPU-diode                   management-module    53.12 C    normal
1/1-DDR                         management-module    45.25 C    normal
1/1-Inlet-Air                   management-module    24.88 C    normal
1/1-MB-IBC                      management-module    45.62 C    normal
1/1-Switch-ASIC-diode           management-module    58.06 C    normal
```

```
switch# show environment temperature
Temperature information
---------------------------------------------------------------------------
--
                                                    Current
Mbr/Slot-Sensor                 Module Type        temperature  Status
---------------------------------------------------------------------------
--
1/1-COMe-Daughter-Boar          line-card-module     66.45 C    normal
1/1-PCIE-Switch                 line-card-module     95.82 C    normal
1/1-Processor                   line-card-module     00.00 C    fault
1/1-Switch-ASIC                 line-card-module     116.36 C   emergency
```

```
1/1-Switch-ASIC-Internal        line-card-module     108.25 C    critical
1/2-COMe-Daughter-Boar          line-card-module     67.29 C     normal
1/2-PCIE-Switch                 line-card-module     95.82 C     normal
1/2-Processor-1                 line-card-module     72.92 C     normal
1/2-Processor-2                 line-card-module     73.05 C     normal
1/2-Switch-ASIC                 line-card-module     97.41 C     normal
1/2-Switch-ASIC-Internal        line-card-module     97.62 C     normal
...
```

Showing detailed temperature information for an 8400 switch:

```
switch# show environment temperature detail
Detailed temperature information
---------------------------------------------------
Mbr/Slot-Sensor       : 1/1-PCIE-Switch
Module Type           : line-card-module
Module Description    : JL363A 8400X 32P 10G SFP/SFP+ Msec Mod
Status                : normal
Fan-state             : normal
Current temperature   : 95.82 C
Minimum temperature   : 93.52 C
Maximum temperature   : 96.15 C
...
Detailed temperature information
---------------------------------------------------
Mbr/Slot-Sensor       : 1/1-PCIE-Switch
Module Type           : line-card-module
Module Description    : JL363A 8400X 32P 10G SFP/SFP+ Msec Mod
Status                : normal
Fan-state             : normal
Current temperature   : 95.82 C
Minimum temperature   : 93.52 C
Maximum temperature   : 96.15 C
...
```

```
switch# show environment temperature
Temperature information
-----------------------------------------------------------------------------
--
                                                     Current
Mbr/Slot-Sensor                 Module Type        temperature  Status
-----------------------------------------------------------------------------
--
1/1-PHY-01-08                   line-card-module     51.00 C     normal
1/1-PHY-09-16                   line-card-module     48.00 C     normal
1/1-PHY-17-24                   line-card-module     50.00 C     normal
```

```
1/1-PHY-25-32                  line-card-module      52.00 C     normal
1/1-PHY-33-40                  line-card-module      53.00 C     normal
1/1-PHY-41-48                  line-card-module      54.00 C     normal
1/1-Switch-ASIC-Internal       line-card-module      63.25 C     normal
1/1-CPU                        management-module     47.56 C     normal
1/1-CPU-Zone-0                 management-module     46.00 C     normal
1/1-CPU-Zone-1                 management-module     46.00 C     normal
1/1-CPU-Zone-2                 management-module     46.00 C     normal
1/1-CPU-Zone-3                 management-module     47.00 C     normal
1/1-CPU-Zone-4                 management-module     47.00 C     normal
1/1-DDR                        management-module     37.75 C     normal
1/1-DDR-Inlet                  management-module     30.69 C     normal
1/1-Inlet-Air                  management-module     22.00 C     normal
1/1-Switch-ASIC-Remote         management-module     64.06 C     normal
```

```
switch# show environment temperature detail
Detailed temperature information
----------------------------------------------------------------
Mbr/Slot-Sensor      : 1/1-PHY-01-04
Module Type          : line-card-module
Module Description   : JL659A 6300M 48SR5 CL6 PoE 4SFP56 Swch
Status               : normal
Fan-state            : normal
Current temperature  : 45.00 C
Minimum temperature  : 41.00 C
Maximum temperature  : 50.00 C
Mbr/Slot-Sensor      : 1/1-PHY-05-08
Module Type          : line-card-module
Module Description   : JL659A 6300M 48SR5 CL6 PoE 4SFP56 Swch
Status               : normal
Fan-state            : normal
Current temperature  : 45.00 C
Minimum temperature  : 41.00 C
Maximum temperature  : 50.00 C
...
Detailed temperature information
----------------------------------------------------------------
Mbr/Slot-Sensor      : 1/1-PHY-01-04
Module Type          : line-card-module
Module Description   : JL659A 6300M 48SR5 CL6 PoE 4SFP56 Swch
Status               : normal
Fan-state            : normal
Current temperature  : 45.00 C
Minimum temperature  : 41.00 C
```

```
Maximum temperature  : 50.00 C
Mbr/Slot-Sensor      : 1/1-PHY-05-08
Module Type          : line-card-module
Module Description   : JL659A 6300M 48SR5 CL6 PoE 4SFP56 Swch
Status               : normal
Fan-state            : normal
Current temperature  : 45.00 C
Minimum temperature  : 41.00 C
Maximum temperature  : 50.00 C
...
```

Command History

Release	Modification
10.18	Adds support for 8325 Switch series.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show events

Syntax

```
show events [ -e <EVENT-ID> |
    -s {emergency | alert | critical | error | warning | notice | info |
debug} |
    -r |
    -a |
    -n <COUNT> |
    -i  <MEMBER-SLOT> |
    -m {active | standby} |
```

```
    -c {lldp | ospf | ...} |
    -d {lldpd | bgpd | fand | ...}]
```

Description

Shows event logs generated by the switch modules since the last reboot.

Parameter	Description
`-e <EVENT-ID>`	Shows the event logs for the specified event ID. Event ID range: 101 through 99999.
`-s {emergency | alert | critical | error | warning | notice | info | debug}`	Shows the event logs for the specified severity. Select the severi ty from the following list: • **emergency**: Displays event logs with severity emergency o nly. • **alert**: Displays event logs with severity alert and above. • **critical**: Displays event logs with severity critical and above. • **error**: Displays event logs with severity error and above. • **warning**: Displays event logs with severity warning and a bove. • **notice**: Displays event logs with severity notice and above. • **info**: Displays event logs with severity info and above. • **debug**: Displays event logs with all severities.
`-r`	Shows the most recent event logs first.
`-a`	Shows all event logs, including those events from previous boots.
`-n <COUNT>`	Displays the specified number of event logs.
`-i <MEMBER-SLOT>`	Shows the event logs for the specified slot ID.
`-m {active | standby}`	Shows the event logs for the specified management card role. Selecting **active** displays the event log for the AMM manageme nt card role and **standby** displays event logs for the SMM mana gement card role.
`-c {lldp | ospf | ...}`	Shows the event logs for the specified event category. Enter **sh ow event -c** for a full listing of supported categories with desc riptions.

Parameter	Description
`-d {lldpd | bgpd | fand | ...}`	Shows the event logs for the specified process. Enter `show event -d` for a full listing of supported daemons with descriptions.

Examples

Showing event logs:

```
switch# show events
---------------------------------------------------
show event logs
---------------------------------------------------
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to
   70:72:cf:51:50:7c
2016-12-01:12:37:31.734541|intfd|4001|INFO|AMM|1|Interface port_admin set to
   up for bridge_normal interface
2016-12-01:12:37:32.583256|switchd|24002|ERR|AMM|1|Failed to create VLAN 1
   in Hardware
```

Showing the most recent event logs first:

```
switch# show events -r
---------------------------------------------------
show event logs
---------------------------------------------------
2016-12-01:12:37:32.583256|switchd|24002|ERR|AMM|1|Failed to create VLAN 1
   in Hardware
2016-12-01:12:37:31.734541|intfd|4001|INFO|AMM|1|Interface port_admin set to
   up for bridge_normal interface
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to
   70:72:cf:51:50:7c
```

Showing all event logs:

```
switch# show events -a
---------------------------------------------------
show event logs
---------------------------------------------------
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to
   70:72:cf:51:50:7c
2016-12-01:12:37:31.734541|intfd|4001|INFO|AMM|1|Interface port_admin set to
   up for bridge_normal interface
2016-12-01:12:37:32.583256|switchd|24002|ERR|AMM|1|Failed to create VLAN 1
   in Hardware
```

Showing event logs related to LACP:

```
switch# show events -c lacp

---------------------------------------------------
show event logs
---------------------------------------------------
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to
   70:72:cf:51:50:7c
```

Showing event logs as per the specified management card role:

```
switch# show events -m active
---------------------------------------------------
show event logs
---------------------------------------------------
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to
   70:72:cf:51:50:7c
2016-12-01:12:37:31.734541|intfd|4001|INFO|AMM|1|Interface port_admin set to
   up for bridge_normal interface
2016-12-01:12:37:32.583256|switchd|24002|ERR|AMM|1|Failed to create VLAN 1
   in Hardware
```

Showing event logs as per the specified member/slot ID:

```
switch# show events -i 1/1

---------------------------------------------------
show event logs
---------------------------------------------------
2017-08-17:22:32:25.743991|hpe-sysmond|6301|LOG_INFO|LC|1/1|System resource
   utilization poll interval is changed to 313
2017-08-17:22:33:01.692860|hpe-sysmond|6301|LOG_INFO|LC|1/1|System resource
   utilization poll interval is changed to 23
2017-08-17:22:33:06.181436|hpe-sysmond|6301|LOG_INFO|LC|1/1|System resource
   utilization poll interval is changed to 512
2017-08-17:22:33:06.181436|systemd-coredump|1201|LOG_CRIT|LC|1/1|hpe-sysmond
   crashed due to signal:11
```

Showing event logs as per the specified process:

```
switch# show events -d lacpd

---------------------------------------------------
show event logs
---------------------------------------------------
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to
   70:72:cf:51:50:7c
```

Displaying the specified number of event logs:

```
switch# show events -n 5
---------------------------------------------------
show event logs
---------------------------------------------------
2018-03-21:06:12:15.500603|arpmgrd|6101|LOG_INFO|AMM|-|ARPMGRD daemon has
started
2018-03-21:06:12:17.734405|lldpd|109|LOG_INFO|AMM|-|Configured LLDP tx-
delay to 2
2018-03-21:06:12:17.740517|lacpd|1307|LOG_INFO|AMM|-|LACP system ID set to
   70:72:cf:d4:34:42
2018-03-21:06:12:17.743491|vrfmgrd|5401|LOG_INFO|AMM|-|Created a vrf entity
   42cc3df7-1113-412f-b5cb-e8227b8c22f2
2018-03-21:06:12:17.904008|vrfmgrd|5401|LOG_INFO|AMM|-|Created a vrf entity
   4409133e-2071-4ab8-adfe-f9662c06b889
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this co mmand from the auditor context (auditor>) only.

show fabric

Syntax

```
show fabric [<SLOT-ID>] [vsx-peer]
```

Description

Shows information about the installed fabrics.

Parameter	Description
`<SLOT-ID>`	Specifies the member and slot of the fabric to show. For exampl e, to show the module in member 1, slot 2, enter the following: `1/2`
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output f rom the VSX peer switch is not displayed. This parameter is avai lable on switches that support VSX.

Examples

Showing all fabrics:

```
switch# show fabric
Fabric Modules
==============
     Product                                    Serial
Slot Number  Description                        Number     Status
---- ------- ---------------------------------- ---------- ------
1/1  JL367A  8400X 7.2Tbps Fab Mod              SG00000000 Ready
1/2  JL367A  8400X 7.2Tbps Fab Mod              SG00000000 Initializing
1/3  JL367A  8400X 7.2Tbps Fab Mod              SG00000000 Initializing
```

Showing a single fabric:

```
switch# show fabric 1/1
Fabric module 1/1 is ready
Admin state: Up
Description: 8400X 7.2Tbps Fab Mod
Full Description: HPE_ANW 8400X 7.2Tbps Fabric Module
Serial number: SG00000000
Product number: JL367A
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show hostname`

Syntax

```
show hostname [vsx-peer]
```

Description

Shows the current host name.

Parameter	Description
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Example

Setting and showing the host name:

```
switch# show hostname
switch
switch# config
switch(config)# hostname myswitch
myswitch(config)# show hostname
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`show images`

Syntax

```
show images [vsx-peer]
```

Description

Shows information about the software in the primary and secondary images.

Parameter	Description
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output f rom the VSX peer switch is not displayed. This parameter is avai lable on switches that support VSX.

Example

Showing the primary and secondary images:

```
switch# show images
---------------------------------------------------------------------------
 AOS-CX Primary Image

---------------------------------------------------------------------------
 Version : XL.xx.xx.xxxx
```

```
 Size    : 141 MB
 Date    : 2017-06-30 14:02:34 PDT
 SHA-256 : 2cf24dba5fb0a30e26e83b2ac5b9e29e1b161e5c1fa7425e73043362938b9824

-----------------------------------------------------------------------------
 AOS-CX Secondary Image

-----------------------------------------------------------------------------
 Version : XL.xx.xx.xxxx
 Size    : 143 MB
 Date    : 2017-06-30 14:02:34 PDT
 SHA-256 : 2cf24dba5fb0a30e26e83b2ac5b9e29e1b161e5c1fa7425e73043362938b9824
 Default Image : secondary
 ------------------------------------------------------
 Management Module 1/5 (Active)
 ------------------------------------------------------
 Active Image       : primary
 Service OS Version : GT.01.01.0001
 BIOS Version       : GT-01-0013
 ------------------------------------------------------
 Management Module 1/6 (Standby)
 ------------------------------------------------------
 Active Image       : secondary
 Service OS Version : GT.01.01.0001
 BIOS Version       : GT-01-0013
```

```
switch(config)# show images
---------------------------------------------------------------------------
AOS-CX Primary Image
---------------------------------------------------------------------------
Version : FL.xx.xx.xxxx
Size    : 722 MB
Date    : 2019-10-22 17:00:46 PDT
SHA-256 : 4c84e49c0961fc56b5c7eab064750a333f1050212b7ce2fab587d13469d24cfa
---------------------------------------------------------------------------
 Primary Image
---------------------------------------------------------------------------
Version : FL.xx.xx.xxxx
Size    : 722 MB
Date    : 2019-10-22 17:00:46 PDT
SHA-256 : 4c84e49c0961fc56b5c7eab064750a333f1050212b7ce2fab587d13469d24cfa
---------------------------------------------------------------------------
AOS-CX Secondary Image
---------------------------------------------------------------------------
```

```
Version : FL.xx.xx.xxxx
Size    : 722 MB
Date    : 2019-10-22 17:00:46 PDT
SHA-256 : 4c84e49c0961fc56b5c7eab064750a333f1050212b7ce2fab587d13469d24cfa
Default Image : secondary
------------------------------------------------------
Management Module 1/1 (Active)
------------------------------------------------------
Active Image       : secondary
Service OS Version : FL.01.05.0001-internal
BIOS Version       : FL.01.0001
```

```
switch(config)# show images
---------------------------------------------------------------------------
AOS-CX Primary Image
---------------------------------------------------------------------------
Version : FL.xx.xx.xxxxQ-2710-gd4ac39f30c9
Size    : 766 MB
Date    : 2019-10-30 17:22:01 PDT
SHA-256 : e560ca9141f425d19024d122573c5ff730df2a9a726488212263b45ea00382cf
---------------------------------------------------------------------------
AOS-CX Secondary Image
---------------------------------------------------------------------------
Version : FL.xx.xx.xxxx
Size    : 722 MB
Date    : 2019-10-21 19:36:26 PDT
SHA-256 : 657e28adc1b512217ce780e3523c37c94db3d3420231deac1ab9aaa8324dc6b9
Default Image : secondary
------------------------------------------------------
Management Module 1/1 (Active)
------------------------------------------------------
Active Image       : secondary
Service OS Version : FL.01.05.0001-internal
BIOS Version       : FL.01.0001
```

```
switch# show images
---------------------------------------------------------------------------
AOS-CX Primary Image
---------------------------------------------------------------------------
Version : TL.10.05.0001I
Size    : 405 MB
Date    : 2020-04-23 02:49:04 PDT
SHA-256 :
7efe86a445e87e40f47de156add25720b7277cae1a8db2f9c4ea5f49e74f2a5a
---------------------------------------------------------------------------
```

```
AOS-CX Secondary Image
---------------------------------------------------------------------------
Version : TL.10.05.0001I
Size    : 405 MB
Date    : 2020-04-23 02:49:04 PDT
SHA-256 :
7efe86a445e87e40f47de156add25720b7277cae1a8db2f9c4ea5f49e74f2a5a
Default Image : primary
------------------------------------------------------
Management Module 1/1 (Active)
------------------------------------------------------
Active Image       : primary
Service OS Version : TL.01.05.0002-internal
BIOS Version       : TL-01-0013
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show ip errors`

Syntax

```
show ip errors [{vrf <vrf-name>}|all-vrfs]
```

Description

Shows harware and software IP error statistics for packets received by the switch since the switch was last booted. If none of the optoinal VRF parameters are included, the **show ip errors** command displays the current system-wide hardware IP error counters and the default VRF's software IP error counters on the switch

Parameter	Description
`vrf <vrf-name>`	Optional. Display hardware and software IP error counters for a specific VRF.
`all-vrfs`	Optional. Display hardware and software IP error counters for al l VRFs

Usage

IP error info about received packets is collected from each active line card on the switch and is preserved during failover events. Error counts are cleared when the switch is rebooted.

Drop reasons may include the following:

- **Malformed packet**

The packet does not conform to TCP/IP protocol standards such as packet length or internet header length.

A large number of malformed packets can indicate that there are hardware malfunctions such as loose cables, network card malfunctions, or that a DOS (denial of service) attack is occurring.

- **IP address error**
  The packet has an error in the destination or source IP address. Examples of IP address errors include the following:
  - The source IP address and destination IP address are the same.
  - There is no destination IP address.
  - The source IP address is a multicast IP address.
  - The forwarding header of an IPv6 address is empty.
  - There is no source IP address for an IPv6 packet.
- **Invalid TTLs**
  The TTL (time to live) value of the packet reached zero. The packet was discarded because it traversed the maximum number of hops permitted by the TTL value.

  TTLs are used to prevent packets from being circulated on the network endlessly.

Example

Showing the current system-wide hardware IP error counters and the default VRF's software IP error counters on the switch:

```
switch# show ip errors
System-wide (hardware) IP errors
--------------------------------------------------
```

```
Drop reason                                       Packets
--------------------------------------------------
Malformed packets                                       1
IP address errors                                      10
Per-VRF (software) IP errors
--------------------------------------------------
VRF
Drop reason                                   Packets
--------------------------------------------------
default
IPv4 maximum transmission unit                    3
IPv4 time to live                                 0
IPv6 maximum transmission unit                    0
IPv6 hop limit                                    6
------------------------------
Malformed packets                 1
...
```

Shoiwng the current system-wide hardware IP error counter and the software IP error counter for all VRFs on the switch.

```
switch# show ip errors all-vrfs
System-wide (hardware) IP errors
--------------------------------------------------
Drop reason                                       Packets
--------------------------------------------------
Malformed packets                                       1
IP address errors                                      10
Per-VRF (software) IP errors
--------------------------------------------------
VRF
Drop reason                                   Packets
--------------------------------------------------
default
IPv4 maximum transmission unit                    3
IPv4 time to live                                 0
IPv6 maximum transmission unit                    0
IPv6 hop limit                                    6
red
IPv4 maximum transmission unit                    3
IPv4 time to live                                 7
IPv6 maximum transmission unit                    0
IPv6 hop limit                                    6
blue
IPv4 maximum transmission unit                    0
```

```
IPv4 time to live                           1
IPv6 maximum transmission unit              0
IPv6 hop limit                              6
...
```

Command History

Release	Modification
10.14.1000	Command updated to display both hardware and software IP errors in the comm and output.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show module`

Syntax

```
show module [<SLOT-ID>]
            [vsx-peer]
```

Description

Shows information about installed line modules and management modules.

Parameter	Description
`<SLOT-ID>`	Specifies the member and slot numbers in format `member/s lot`. For example, to show the module in member 1, slot 3, en ter `1/3`.

Parameter	Description
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output f rom the VSX peer switch is not displayed. This parameter is avai lable on switches that support VSX.

Usage

Identifies and shows status information about the line modules and management modules that are installed in the switch.

If you use the `<SLOT-ID>` parameter to specify a slot that does not have a line module installed, a message similar to the following example is displayed:

```
Module 1/4 is not physically present.
```

To show the configuration information—if any—associated with that line module slot, use the **show running-configuration** command.

Status is one of the following values:

`Active`
This module is the active management module.

`Standby`
This module is the standby management module.

`Deinitializing`
The module is being deinitialized.

`Diagnostic`
The module is in a state used for troubleshooting.

`Down`
The module is physically present but is powered down.

`Empty`
The module hardware is not installed in the chassis.

`Failed`
The module has experienced an error and failed.

`Failover`
This module is a fabric module or a line module, and it is in the process of connecting to the new active management module during a management module failover event.

`Initializing`
The module is being initialized.

`Present`
The module hardware is installed in the chassis.

`Ready`

The module is available for use.

`Updating`

A firmware update is being applied to the module .

Examples

Showing all installed modules on an 8400 switch:

```
switch# show module
Management Modules
==================
     Product                                    Serial
Slot Number  Description                        Number     Status
---- ------- ---------------------------------- ---------- ------
1/5  JL368A  8400 Mgmt Mod                      SG00000000 Active
1/6  JL368A  8400 Mgmt Mod                      SG00000000 Standby
Line Modules
=================
     Product                                    Serial
Slot Number  Description                        Number     Status
---- ------- ---------------------------------- ---------- ------
1/1  JL363A  8400X 32P 10G SFP/SFP+ Msec Mod    SG00000000 Down
1/3  JL365A  8400X 8P 40G QSFP+ Adv Mod         SG00000000 Ready
1/4  JL366A  8400X 6P 40G/100G QSFP28 Adv Mod   SG00000000 Initializing
1/8  JL365A  8400X 8P 40G QSFP+ Adv Mod         SG00000000 Ready
1/10 JL365A  8400X 8P 40G QSFP+ Adv Mod         SG00000000 Ready
...
```

NOTE

The output of this show command may seem to indicate that a JL363A Line card supports MACsec. This line card is hardware-ready to support MACsec, but this feature is not currently support for this line card in AOS-CX software.

```
switch(config)# show module
Management Modules
==================
     Product                                        Serial
Name Number  Description                            Number     Status
---- ------- -------------------------------------- ----------
----------------
1/1  JL581A  8320 Mgmt Mod                          TW87KCW00X Ready
Line Modules
============
     Product                                        Serial
```

```
Name Number  Description                             Number     Status
---- ------- --------------------------------------- ----------
----------------
1/1  JL581A  8320                                    TW87KCW00X Ready
```

```
switch(config)# show module
Management Modules
==================
     Product                                         Serial
Name Number  Description                             Number     Status
---- ------- --------------------------------------- ----------
----------------
1/1  JL659A  6300M 48SR5 CL6 PoE 4SFP56 Swch        ID9ZKHN090 Active
(local)
Line Modules
============
     Product                                         Serial
Name Number  Description                             Number     Status
---- ------- --------------------------------------- ----------
----------------
1/1  JL659A  6300M 48SR5 CL6 PoE 4SFP56 Swch        ID9ZKHN090 Ready
Management Modules
==================
     Product                                         Serial
Name Number  Description                             Number     Status
---- ------- --------------------------------------- ----------
----------------
1/1  JL659A  6300M 48SR5 CL6 PoE 4SFP56 Swch        ID9ZKHN090 Active
(local)
Line Modules
============
     Product                                         Serial
Name Number  Description                             Number     Status
---- ------- --------------------------------------- ----------
----------------
1/1  JL659A  6300M 48SR5 CL6 PoE 4SFP56 Swch        ID9ZKHN090 Ready
```

```
switch# show module
Management Modules
==================
     Product                                         Serial
Name Number  Description                             Number     Status
---- ------- --------------------------------------- ----------
----------------
1/1  JL678A  6100F 24G 4SFP+ Swch                    SG0ZKW600P Active
```

```
(local)
Line Modules
============
     Product                                         Serial
Name Number  Description                             Number     Status
---- ------- --------------------------------------- ----------
----------------
1/1  JL678A  6100F 24G 4SFP+ Swch                    SG0ZKW600P Ready
Management Modules
==================
     Product                                         Serial
Name Number  Description                             Number     Status
---- ------- --------------------------------------- ----------
----------------
1/1  JL678A  6100F 24G 4SFP+ Swch                    SG0ZKW600P Active
(local)
Line Modules
============
     Product                                         Serial
Name Number  Description                             Number     Status
---- ------- --------------------------------------- ----------
----------------
1/1  JL678A  6100F 24G 4SFP+ Swch                    SG0ZKW600P Ready
```

Showing a management module on an 8400 switch:

```
switch# show module 1/5
Management module 1/5 is active
Admin state: Up
Description: 8400 Mgmt Mod
Full Description: 8400 Management Module
Serial number: SG00000000
Product number: JL368A
```

```
switch# show module 1/3
Line module 1/3 is ready
 Admin state: Up
 Description: 6400 24p 10GT 4SFP56 Mod
 Full Description: 6400 24-port 10GBASE-T and 4-port SFP56 Module
 Serial number: SG9ZKMS045
 Product number: R0X42A
 Power priority: 128
```

Showing a slot that does not contain a line module:

```
switch(config)# show module 1/3
Module 1/3 is not physically present
```

Showing a line module after removing it from the configuration with **no module** command, but the hardware remains installed in the slot:

```
switch(config)# show module 1/1
Line module 1/1 is ready
Admin state: Up
Description: 8400X 32P 10G SFP/SFP+ Msec Mod
Full Description: 8400X 32-port 10GbE SFP/SFP+ with MACsec Advanced Module
Serial number: SG00000000
Product number: JL363A
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show process monitor statistics`

Syntax

```
show process monitor statistics [standby | module <SLOT-ID>]
```

Description

Shows the current status of all processes running on the switch. The status of the process information is available in the daemon table, the CLI reads each process status from the daemon table.

Parameter	Description
`standby`	Shows only the resource utilization data for the standby manag ement module.

Parameter	Description
`module <SLOT-ID>`	Shows only the resource utilization data for the line module identified by `<SLOT-ID>`.

Usage

For more details on the type of information displayed in the output of this command, refer to the video on the **HPE Aruba Networking Airheads Broadcasting Channel.**

Examples

Showing process monitor statistics:

```
switch# show process monitor statistics
Process Monitor Statistics
--------------------------
Name                          State             Restarts Allowed
Restarts    Timeout Value    Time Started
----                          -----             ----------------
--------    -------------    -------------
irq/102-uio_vcb               PROCESS_RUNNING   0
0           --               Tue May 14 12:51:46 2024
tempd                         PROCESS_RUNNING   0
0           --               Tue May 14 12:51:48 2024
irq/21-mmc0                   PROCESS_RUNNING   0
0           30               Tue May 14 12:51:51 2024
acctd                         PROCESS_RUNNING   0
0           60               Tue May 14 12:52:01 2024
khungtaskd                    PROCESS_RUNNING   0
0           --               Tue May 14 12:52:02 2024
dbus-daemon                   PROCESS_RUNNING   3
3           --               Tue May 14 12:52:02 2024
irq/105-uio_vcb               PROCESS_RUNNING   3
2           --               Tue May 14 12:52:02 2024
kworker/0:1-events            PROCESS_RUNNING   3
0           240              Tue May 14 12:52:02 2024
dnsmasq                       PROCESS_RUNNING   3
0           240              Tue May 14 12:52:02 2024
```

Showing process monitor statistics with the standby parameter:

```
switch# show process monitor statistics standby
Process Monitor Statistics
--------------------------
Name                          State             Restarts Allowed
Restarts    Timeout Value    Time Started
```

```
----                           -----              ----------------
--------   -------------     -------------
irq/102-uio_vcb                PROCESS_RUNNING    0
0          --                Tue May 14 12:51:46 2024
tempd                          PROCESS_RUNNING    0
0          --                Tue May 14 12:51:48 2024
irq/21-mmc0                    PROCESS_RUNNING    0
0          30                Tue May 14 12:51:51 2024
acctd                          PROCESS_RUNNING    0
0          60                Tue May 14 12:52:01 2024
khungtaskd                     PROCESS_RUNNING    0
0          --                Tue May 14 12:52:02 2024
dbus-daemon                    PROCESS_RUNNING    3
3          --                Tue May 14 12:52:02 2024
irq/105-uio_vcb                PROCESS_RUNNING    3
2          --                Tue May 14 12:52:02 2024
kworker/0:1-events             PROCESS_RUNNING    3
0          240               Tue May 14 12:52:02 2024
dnsmasq                        PROCESS_RUNNING    3
0          240               Tue May 14 12:52:02 2024
```

```
switch# show process monitor statistics member 3
Process Monitor Statistics
--------------------------
Name                           State              Restarts Allowed
Restarts   Timeout Value     Time Started
----                           -----              ----------------
--------   -------------     -------------
irq/102-uio_vcb                PROCESS_RUNNING    0
0          --                Tue May 14 12:51:46 2024
tempd                          PROCESS_RUNNING    0
0          --                Tue May 14 12:51:48 2024
irq/21-mmc0                    PROCESS_RUNNING    0
0          30                Tue May 14 12:51:51 2024
acctd                          PROCESS_RUNNING    0
0          60                Tue May 14 12:52:01 2024
khungtaskd                     PROCESS_RUNNING    0
0          --                Tue May 14 12:52:02 2024
dbus-daemon                    PROCESS_RUNNING    3
3          --                Tue May 14 12:52:02 2024
irq/105-uio_vcb                PROCESS_RUNNING    3
2          --                Tue May 14 12:52:02 2024
kworker/0:1-events             PROCESS_RUNNING    3
0          240               Tue May 14 12:52:02 2024
```

```
dnsmasq                               PROCESS_RUNNING    3
0            240                     Tue May 14 12:52:02 2024
```

Showing process monitor statistics with the module parameter:

```
switch# show process monitor statistics module 1/4
Process Monitor Statistics
--------------------------
Name                                  State             Restarts Allowed
Restarts    Timeout Value            Time Started
----                                  -----             ----------------
--------    -------------            -------------
irq/102-uio_vcb                       PROCESS_RUNNING    0
0            --                      Tue May 14 12:51:46 2024
tempd                                 PROCESS_RUNNING    0
0            --                      Tue May 14 12:51:48 2024
irq/21-mmc0                           PROCESS_RUNNING    0
0            30                      Tue May 14 12:51:51 2024
acctd                                 PROCESS_RUNNING    0
0            60                      Tue May 14 12:52:01 2024
khungtaskd                            PROCESS_RUNNING    0
0            --                      Tue May 14 12:52:02 2024
dbus-daemon                           PROCESS_RUNNING    3
3            --                      Tue May 14 12:52:02 2024
irq/105-uio_vcb                       PROCESS_RUNNING    3
2            --                      Tue May 14 12:52:02 2024
kworker/0:1-events                    PROCESS_RUNNING    3
0            240                     Tue May 14 12:52:02 2024
dnsmasq                               PROCESS_RUNNING    3
0            240                     Tue May 14 12:52:02 2024
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution righ ts for this command.

`show process top cpu`

Syntax

```
show process top-cpu {standby | module <SLOT-NUMBER>}
```

Description

Displays the current top CPU consuming processes for the selected subsystem (management module, standby, VSF member, or line card module).

> **NOTE**
>
> Only the current top 10 CPU consuming processes are captured in the output. CPU process data is only refreshed when the respective utilization thresholds are crossed or updated by the daemon. If the daemon is down, the lists may be empty.

Parameters

Parameter	Description
`<SLOT-NUMBER>`	Specifies the slot where the module is present.

Examples

Displaying the current top CPU consuming processes for a specific module:

```
switch# show process top-cpu module 1/1
Top CPU Processes
-----------------
Last Updated: Wed Dec 03 11:30:41 2025

PID      Name                         CPU(%)
-------- ---------------------------- -------
40825    stress-ng-cpu                60.00
```

Command History

Table 1.

Release	Modification
10.18	Command introduced.

Command Information

Platforms	Command context	Authority
8400	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`show process top-memory`

Syntax

```
show process top-memory {standby | module <SLOT-NUMBER>}
```

Description

Displays the current top memory consuming processes. Columns include percentage within captured set, user memory (KB), and kernel memory (KB) if reported.

> **NOTE**
>
> Memory process data is only refreshed when the respective utilization thresholds are crossed or updated by the daemon. If the daemon is down, the lists may be empty.

Parameters

Table 1.

Parameter	Description
`<SLOT-NUMBER>`	Specifies the slot where the module is present.

Examples

Displaying the current top memory consuming processes:

```
switch# show process top-memory
Top Memory Processes
--------------------
Last Updated: Wed Dec 03 11:30:41 2025

PID      Name                            Mem(%)  UserMem(KB)   KernelMem(KB)
-------- ---------------------------- ------- ------------  -------------
```

```
2791      dhcp-server-ada                 72.50    120000         0
3296      hpe-config                      60.00    95000          0
```

Displaying the current top memory consuming processes for a specific module:

```
switch# show process top-memory module 1/1
Top Memory Processes
--------------------
Last Updated: Wed Dec 03 11:30:41 2025

PID  Name               Mem(%) UserMem(KB) KernelMem(KB)
---- ------------------ ------ ----------- -------------
3576 prometheus         65.19   238988       0
...
```

`show running-config`

Syntax

```
show running-config [<FEATURE>] [all] [vsx-peer]
```

Description

Shows the current nondefault configuration running on the switch. No user information is displayed.

Parameter	Description
<FEATURE>	Specifies the name of a feature. For a list of feature names, ente r the `show running-config` command, followed by a s pace, followed by a question mark (?). When the json parameter is used, the `vsx-peer` parameter is not applicable.
all	Shows all default values for the current running configuration.
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output f rom the VSX peer switch is not displayed. This parameter is avai lable on switches that support VSX.

Examples

Showing the current running configuration:

```
switch> show running-config
Current configuration:
!
!Version AOS-CX 10.0X.XXXX
!
lldp enable
linecard-module LC1 part-number JL363A
                vrf green

!
!
!
!
!
!
aaa authentication login default local
aaa authorization commands default none
!
!
!
!
interface vlan20
    no shutdown
    vrf attach green
    ip address 20.0.0.44/24
    ip ospf 1 area 0.0.0.0
    ip pim-sparse enable
interface vlan30
    no shutdown
    vrf attach green
    ip address 30.0.0.44/24
    ip ospf 1 area 0.0.0.0
    ip pim-sparse enable
    ip pim-sparse hello-interval 100
```

Showing the current running configuration in json format:

```
switch> show running-config json
Running-configuration in JSON:
{
    "Monitoring_Policy_Script": {
        "system_resource_monitor_mm1.1.0": {
```

```
            "Monitoring_Policy_Instance": {
                "system_resource_monitor_mm1.1.0/
system_resource_monitor_mm1.1.0.default": {
                    "name": "system_resource_monitor_mm1.1.0.default",
                    "origin": "system",
                    "parameters_values": {
                        "long_term_high_threshold": "70",
                        "long_term_normal_threshold": "60",
                        "long_term_time_period": "480",
                        "medium_term_high_threshold": "80",
                        "medium_term_normal_threshold": "60",
                        "medium_term_time_period": "120",
                        "short_term_high_threshold": "90",
                        "short_term_normal_threshold": "80",
                        "short_term_time_period": "5"
                    }
                }
            },
...
...
...
...
```

Show the current running configuration without default values:

```
switch(config)# show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.04.0000-6523-gbb15c03~dirty
led locator on
!
!
!
!
!
!
!
!
!
vlan 1
switch(config)# show running-config all
Current configuration:
!
!Version AOS-CX Virtual.10.04.0000-6523-gbb15c03~dirty
led locator on
```

```
!
!
!
!
!
!
!
!
!
vlan 1
switch(config)#
switch(config)# show running-config
Current configuration:
!
!Version AOS-CX PL.10.06.0001-346-g56a12a8f4cf15
!export-password: default
!
!
!
!
ssh server vrf default
vlan 1
spanning-tree
interface 1/1/1
    no shutdown
    vlan access 1
interface 1/1/2
    no shutdown
    vlan access 1
interface 1/1/3
    no shutdown
    vlan access 1
interface 1/1/4
    no shutdown
    vlan access 1
interface 1/1/5
    no shutdown
    vlan access 1
interface 1/1/6
    no shutdown
    vlan access 1
interface 1/1/7
    no shutdown
```

```
    vlan access 1
interface 1/1/8
    no shutdown
    vlan access 1
interface 1/1/9
    no shutdown
    vlan access 1
interface 1/1/10
    no shutdown
    vlan access 1
interface 1/1/11
    no shutdown
    vlan access 1
interface 1/1/12
    no shutdown
    vlan access 1
interface 1/1/13
    no shutdown
    vlan access 1
interface 1/1/14
    no shutdown
    vlan access 1
interface 1/1/15
    no shutdown
    vlan access 1
interface 1/1/16
    no shutdown
    vlan access 1
interface vlan 1
    ip dhcp
!
!
!
!
!
https-server vrf default
switch#
switch#                                                 PL.10.06.0001-346-
g56a12a8f4cf15
!export-password: default
!
!
!
```

```
!
ssh server vrf default
vlan 1
spanning-tree
interface 1/1/1
    no shutdown
    vlan access 1
interface 1/1/2
    no shutdown
    vlan access 1
interface 1/1/3
    no shutdown
    vlan access 1
interface 1/1/4
    no shutdown
    vlan access 1
interface 1/1/5
    no shutdown
    vlan access 1
interface 1/1/6
    no shutdown
    vlan access 1
interface 1/1/7
    no shutdown
    vlan access 1
interface 1/1/8
    no shutdown
    vlan access 1
interface 1/1/9
    no shutdown
    vlan access 1
interface 1/1/10
    no shutdown
    vlan access 1
interface 1/1/11
    no shutdown
    vlan access 1
interface 1/1/12
    no shutdown
    vlan access 1
interface 1/1/13
    no shutdown
    vlan access 1
```

```
interface 1/1/14
    no shutdown
    vlan access 1
interface 1/1/15
    no shutdown
    vlan access 1
interface 1/1/16
    no shutdown
    vlan access 1
interface vlan 1
    ip dhcp
!
!
!
!
!
https-server vrf default
switch#
switch#
```

Show the current running configuration with default values:

```
switch(config)# snmp-server vrf mgmt
switch(config)# show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.04.0000-6523-gbb15c03~dirty
led locator on
!
!
!
!
snmp-server vrf mgmt
!
!
!
!
!
vlan 1
switch(config)#
switch(config)#
switch(config)# show running-config all
Current configuration:
!
!Version AOS-CX Virtual.10.04.0000-6523-gbb15c03~dirty
```

```
led locator on
!
!
!
!
snmp-server vrf mgmt
snmp-server agent-port 161
snmp-server community public
!
!
!
!
!
vlan 1
switch(config)#
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`show running-config current-context`

Syntax

```
show running-config current-context
```

Description

Shows the current non-default configuration running on the switch in the current command context.

Usage

You can enter this command from the following configuration contexts:

- Any child of the global configuration (**config**) context. If the child context has instances—such as interfaces—you can enter the command in the context of a specific instance. Support for this command is provided for one level below the **config** context. For example, entering this command for a child of a child of the `config` context not supported. If you enter the command on a child of the **config** context, the current configuration of that context and the children of that context are displayed.
- The global configuration (**config**) context. If you enter this command in the global configuration (**config**) context, it shows the running configuration of the entire switch. Use the **show running-configuration** command instead.

Examples

Showing the running configuration for the current interface:

```
switch(config-if)# show running-config current-context
interface 1/1/1
vsx-sync qos vlans
    no shutdown
    description Example interface
    no routing
vlan access 1
    exit
```

Showing the current running configuration for the management interface:

```
switch(config-if-mgmt)# show running-config current-context
interface mgmt
    no shutdown
    ip static 10.0.0.1/24
    default-gateway 10.0.0.8
    nameserver 10.0.0.1
switch(config)# interface vlan 1
switch(config-if-vlan)#description IN-BAND Management Interface
switch(config-if-vlan)#ip dhcp
switch(config-if-vlan)#no shutdown
switch(config-if-vlan)#end
switch#
switch(config)# interface vlan 1
switch(config-if-vlan)#description IN-BAND Management Interface
switch(config-if-vlan)#no ip dhcp
switch(config-if-vlan)#ip address 192.168.10.1/24
switch(config-if-vlan)#no shutdown
```

```
switch(config-if-vlan)#end
switch#
```

Showing the running configuration for the external storage share named **nasfiles**:

```
switch(config-external-storage-nasfiles)# show running-config current-
context
external-storage nasfiles
   address 192.168.0.1
   vrf default
   username nasuser
   password ciphertext
AQBapalKj+XMsZumHEwIc9OR6YcOw5Z6Bh9rV+9ZtKDKzvbaBAAAAB1CTrM=
   type scp
   directory /home/nas
   enable
switch(config-external-storage-nasfiles)#
```

Showing the running configuration for a context that does not have instances:

```
switch(config-vsx)# show run current-context
vsx
    inter-switch-link 1/1/1
    role secondary
    vsx-sync sflow time
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config` or a child of `config`. See Usage.	Administrators or local user group members with execution righ ts for this command.

`show startup-config`

Syntax

```
show startup-config [json]
```

Description

Shows the contents of the startup configuration.

> **NOTE**
>
> Switches in the `factory-default` configuration do not have a startup configuration to display.

Parameter	Description
json	Display output in JSON format.

Examples

Showing the startup-configuration in non-JSON format:

```
switch# show startup-config
Startup configuration:
!
!Version AOS-CX XL.xx.xx.xxxx
!export-password: default
hostname BLDG03-F1
user admin group administrators password ciphertext AQBapfYzQNiCPJ/
JNSu7YNfaCzlWEnWFlwLfkARd8OG6yPpcYgAAABXRe9joTRtF1S1b4b09teMfMN3POKdQk+
br6SjSXGG40BB3ilZ8ym9qqSRkr84FEdq6w5uR2IGciVC5tBnZMrWCim0KR20XcQ5rFj/
TMBvTkHq8bJpSYG8nMh
module 1/3 product-number jl365a
module 1/1 product-number jl363a
module 1/2 product-number jl363a
cli-session
    timeout 0
!
!
!
ssh server vrf mgmt
ssh certified-algorithms-only
!
```

```
!
!
!
!
vlan 1,195,197-200,4000
interface mgmt
    no shutdown
    ip static 10.6.9.24/24
    default-gateway 10.6.9.1
interface 1/1/1
    no shutdown
    no routing
    vlan trunk native 1
    vlan trunk allowed 1,197-200
interface vlan200
    ip address 10.3.200.3/24
ip route 0.0.0.0/0 10.3.200.1
https-server rest access-mode read-write
https-server vrf mgmt
```

```
Leaf2(config)# show startup-config
Startup configuration:
!
!Version AOS-CX TL.xx.xx.xxxx
hostname Leaf2
user admin group administrators password ciphertext
AQBapaGi+KZp4g8gw63UqK+zCtvO5zigFLv2DFBEH+lztqjdYgAAABwrJ+5GayUWArgv9tVFo9Az
MY6gmI7x/
KBEkGBJDXjpFson2qM83CXBUI673qWHDQ0pEIZXeuig0XogCVuId4oZiQVZlOe2MfxnqZL+E9hXa
MNVowBwbD0
cli-session
    timeout 0
!
!
!
ssh server vrf mgmt
```

```
switch(config)# show startup-config
Startup configuration:
!
!Version AOS-CX FL.xx.xx.xxxx
!export-password: default
hostname BLDG01-F1
user admin group administrators password ciphertext
AQBapWl8I2ZunZ43NE/8KlbQ7zYC4gTT6uSFYi6n6wyY9PdBYgAAACONCR/
```

```
3+AcNvzRBch0DoG7W9z84LpJA+6C9SKfNwCqi5/
nUPk/ZOvN91/
EQXvPNkHtBtQWyYZqfkebbEH78VWRHfWZjApv4II9qmQfxpA79wEvzshdzZmuAKrm
user ateam group administrators password ciphertext
AQBapcPqMXoF+H10NKrqAedXLvlSRwf4wUEL22hXGD6ZBhicYgAAAGsbh70DKg1u+Ze1wxgmDXjk
GO3bseYiR3LKQg66vrfrqR/
M3oLlliPdZDnq9XMMvCL+7jBbYhYes8+uDxuSTh8kdkd/
qj3lo5FUuC5fENgCjU0YI1l7qtU+YEnsj
!
!
!
!
radius-server host 10.10.10.15
!
radius dyn-authorization enable
ssh server vrf default
ssh server vrf mgmt
!
!
!
!
!
router ospf 1
    router-id 1.63.63.1
    area 0.0.0.0
vlan 1
vlan 66
    name vlan66
vlan 67
    name vlan67
vlan 999
    name vlan999
vlan 4000
spanning-tree
interface mgmt
    no shutdown
    ip static 10.6.9.15/24
    default-gateway 10.6.9.1
```

```
switch# show startup-config
Startup configuration:
!
!Version AOS-CX ML.10.05.0001L
!export-password: default
```

```
no lldp
user admin group administrators password ciphertext
AQBapeGVVYk45m4sYxZDE6ufzB2CWmH2wDncy5Can9iEFZjmYgAAAMl31OWIxExNwi3xahHktOL6
81amYg/yg2ezRrlbMUtlU7fVlASiVbmIq8gVUj01Q4STp9/
su3pnopopuWPxwk765zqofKyhL0E0Gj9yhoxCeZfyNeNpNbm6upKjC5LrfZt9
cli-session
    timeout 0
!
!
```

Showing the startup-configuration in JSON format:

```
switch# show startup-config json
Startup configuration:
{
    "AAA_Server_Group": {
        "local": {
            "group_name": "local"
        },
        "none": {
            "group_name": "none"
        }
    },
...
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution righ ts for this command.

show system

Syntax

```
show system [serviceos password-prompt] [vsx-peer]
```

Description

Shows general status information about the system.

Parameter	Description
`serviceos password-prompt`	Shows the Service OS password prompt status.
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output f rom the VSX peer switch is not displayed. This parameter is avai lable on switches that support VSX.

Usage

CPU utilization represents the average utilization across all the CPU cores.

System Description, System Contact, and System Location can be set with the `snmp-server` command.

When `vsx-peer` is specified, the `Up Time` value is not shown because it is not synchronized between VSX peers.

Examples

Showing system information:

```
switch# show system
Hostname                   : switch
System Description         : switch description
System Contact             : contact
System Location            : location
Vendor                     : Aruba
Product Name               : Xxxxxx ...
Chassis Serial Nbr         : XXXXXXXXXX
Base MAC Address           : xxxxxx-xxxxxx
AOS-CX Version             : XX.99.99.9999
Time Zone                  : UTC
Up Time                    : 1 week, 5 hours, 28 minutes
CPU Util (%)               : 5
CPU Util (% avg 1 min)     : 11
CPU Util (% avg 5 min)     : 10
Memory Usage (%)           : 35
```

Showing the Service OS password prompt status:

```
switch# show system serviceos password-prompt
password-prompt: disabled
```

Showing system information for a VSX primary and secondary (peer) switch:

```
switch# show system
Hostname                : vsx-primary
System Description      : switch description
System Contact          : contact
System Location         : location
Vendor                  : Aruba
Product Name            : Xxxxxx ...
Chassis Serial Nbr      : XXXXXXXXXX
Base MAC Address        : xxxxxx-xxxxxx
Hostname                : vsx-primary
System Description      : switch description
System Contact          : contact
System Location         : location
Vendor                  : Aruba
Product Name            : Xxxxxx ...
Chassis Serial Nbr      : XXXXXXXXXX
Base MAC Address        : xxxxxx-xxxxxx
AOS-CX Version          : XX.99.99.9999
Time Zone               : UTC
Up Time                 : 1 week, 2 hours, 15 minutes
CPU Util (%)            : 15
CPU Util (% avg 1 min)  : 12
CPU Util (% avg 5 min)  : 8
Memory Usage (%)        : 37
switch# show system vsx-peer
Hostname                : vsx-secondary
System Description      : switch description
System Contact          : contact
System Location         : location
Vendor                  : Aruba
Product Name            : Xxxxxx ...
Chassis Serial Nbr      : XXXXXXXXXX
Base MAC Address        : xxxxxx-xxxxxx
AOS-CX Version          : XX.99.99.9999
Time Zone               : UTC
CPU Util (%)            : 7
CPU Util (% avg 1 min)  : 13
CPU Util (% avg 5 min)  : 9
Memory Usage (%)        : 32
```

Command History

Release	Modification
10.12	Added `CPU Util (% avg 1 min)` and `CPU Util (% avg 5 min)`.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show system error-counter-monitor`

Syntax

```
show system error-counter-monitor {basic <PORT-NUM> | extended} [vsx-peer]
```

Description

Shows error counter statistics.

Parameter	Description
`basic <PORT-NUM>`	Specifies a physical port on the switch. Use the format `member/slot/port` (for example, `1/3/1`).
`extended`	Shows statistics for all interfaces.

Examples

Showing error counter statistics for interface 1/1/1:

```
switch# show system error-counter-monitor basic 1/1/1
Interface error counter statistics for 1/1/1
Error Counter                          Value
```

```
-----------------------------------------
EtherStatsOversizePkts              983
EtherStatsUndersizePkts             1024
EtherStatsJabbers                   10
Dot3StatsAlignmentErrors            462
Dot3StatsFCSErrors                  321
Dot3StatsLateCollisions             2024
EtherStatsFragments                 121
Dot3StatsExcessiveCollisions        1025
IfInBroadcastPkts                   2001
```

Showing error counter statistics for all interfaces:

```
switch# show system error-counter-monitor extended
Interface error counter statistics for 1/1/1
Error Counter                       Value
-----------------------------------------
EtherStatsOversizePkts              983
EtherStatsUndersizePkts             1024
EtherStatsJabbers                   10
Dot3StatsAlignmentErrors            462
Dot3StatsFCSErrors                  321
Dot3StatsLateCollisions             2024
EtherStatsFragments                 121
Dot3StatsExcessiveCollisions        1025
IfInBroadcastPkts                   2001
...
...
Interface error counter statistics for 1/8/32
Error Counter                      Value
-----------------------------------------
EtherStatsOversizePkts              0
...
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	Manager (`#`)	Administrators or local user group members with execution rights for this command.

`show system resource-utilization`

Syntax

```
show system resource-utilization [all | daemon <DAEMON-NAME>] |
    standby | module <SLOT-ID>] [vsx-peer]
```

Description

Shows the system resource utilization data.

Parameter	Description
`all`	Shows the resource utilization data for the entire switch.
`daemon <DAEMON-NAME>`	Shows only the resource utilization data for the process identified by `<DAEMON-NAME>`.
`standby`	Shows only the resource utilization data for the standby management module.
`module <SLOT-ID>`	Shows only the resource utilization data for the line module identified by `<SLOT-ID>`.
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Usage

For a list of daemons that log events, enter `show events -d ?` from a switch prompt in the manager (`#`) context.

Examples

Showing system resource utilization data:

```
switch# show system resource-utilization
System Resources:
Processes                          :  144
CPU usage(%)                       :   10
CPU usage(% average over 1 minute) :   11
CPU usage(% average over 5 minute) :   15
Memory usage(%)                    :   22
Open FD's                          : 1358
Storage 1: Endurance utilization = 10-20% (mmc-type-a), 0-10% (mmc-type-b),
   Health = normal
Data written to various partitions since boot
Nos      :   5 MB
Log      :   1 MB
Coredump :  23 MB
Security :   2 MB
Selftest : 405 KB
Swap     :  14 MB
Storage partition usage(%)
Nos      :   5
Log      :  60
Coredump :  23
Security :   2
Selftest :   1
Swap     :   0
Process                 CPU Usage(%)    Memory Usage(%)   Open FD's
--------------------------------------------------------------------------
hpe-sysmond                 1                2                 11
hpe-mgmdd                   0                1                 5
...
```

```
switch# show system resource-utilization
System Resources:
Processes                          :  144
CPU usage(%)                       :   10
CPU usage(% average over 1 minute) :   11
CPU usage(% average over 5 minute) :   15
Memory usage(%)                    :   22
Open FD's                          : 1358
Storage 1: Endurance utilization = 10-20% (mmc-type-a), 0-10% (mmc-type-b),
   Health = normal
Data written to various partitions since boot
Nos      :   5 MB
```

```
Log      :   1 MB
Coredump :  23 MB
Security :   2 MB
Selftest : 405 KB
Swap     :  14 MB
Storage partition usage(%)
Nos      :   5
Log      :  60
Coredump :  23
Security :   2
Selftest :   1
Swap     :   0
```

Attempting to show resource utilization data when system resource utilization polling is disabled:

```
switch# show system resource-utilization
System resource utilization data poll is currently disabled
```

Showing the resource utilization data for a particular process:

```
switch# show system resource-utilization daemon hpe-sysmond
Process                  CPU Usage(%)    Memory Usage(%)   Open FD's
---------------------------------------------------------------------------
hpe-sysmond                  1                2                 11
```

```
switch# show system resource-utilization all
---------------------------------------------------------------------------
Resource utilization data for Management Module
---------------------------------------------------------------------------
System Resources:
Processes                            :  244
CPU usage(%)                         :   10
CPU usage(% average over 1 minute)   :   11
CPU usage(% average over 5 minute)   :   15
Memory usage(%)                      :   11
Open FD's                            : 1854
Storage 1: Endurance utilization = 0-10% (mmc-type-a), 0-10% (mmc-type-b),
   Health = normal
Data written to various partitions since boot
Nos      :  15 MB
Log      :   1 MB
Coredump :  23 MB
Security :   2 MB
Selftest : 405 KB
Swap     :   0 KB
Storage partition usage(%)
Nos      :   5
Log      :  60
```

```
Coredump :  23
Security :   2
Selftest :   1
Swap     :   0
Process                 CPU Usage(%)    Memory Usage(%)   Open FD's
--------------------------------------------------------------------------
(sd-pam)                0               0                7
aaautilspamcfg          0               1                10
...
```

```
switch# show system resource-utilization all
--------------------------------------------------------------------------
Resource utilization data for Management Module
--------------------------------------------------------------------------
System Resources:
Processes                           :  244
CPU usage(%)                        :   10
CPU usage(% average over 1 minute) :   11
CPU usage(% average over 5 minute) :   15
Memory usage(%)                     :   11
Open FD's                           : 1854
Storage 1: Endurance utilization = 0-10% (mmc-type-a), 0-10% (mmc-type-b),
   Health = normal
Data written to various partitions since boot
Nos      :  15 MB
Log      :   1 MB
Coredump :  23 MB
Security :   2 MB
Selftest : 405 KB
Swap     :   0 KB
Storage partition usage(%)
Nos      :   5
Log      :  60
Coredump :  23
Security :   2
Selftest :   1
Swap     :   0
```

```
aaa
```

```
switch# show system resource-utilization standby
System Resources:
Processes                           :  244
CPU usage(%)                        :   10
CPU usage(% average over 1 minute) :   11
CPU usage(% average over 5 minute) :   15
```

```
Memory usage(%)                    :   11
Open FD's                          : 1854
Storage 1: Endurance utilization = 10-20% (mmc-type-a), 0-10% (mmc-type-b)
   Health = normal
Data written to various partitions since boot
Nos       :  15 MB
Log       :   1 MB
Coredump :  23 MB
Security :   2 MB
Selftest : 405 KB
Swap      :  14 MB
Storage partition usage(%)
Nos       :   5
Log       :  60
Coredump :  23
Security :   2
Selftest :   1
Swap      :   0
Process                  CPU Usage(%)    Memory Usage(%)   Open FD's
--------------------------------------------------------------------------
hpe-sysmond                  1                2                11
hpe-mgmdd                    0                1                5
...
```

```
switch# show system resource-utilization module 1/5
--------------------------------------------------------------------------
System Resource utilization for line card module: 1/5
--------------------------------------------------------------------------
CPU usage(%)                       :   10
CPU usage(% average over 1 minute) :   11
CPU usage(% average over 5 minute) :   15
Memory usage(%)                    :   11
Open FD's                          :  754
```

```
switch# show system resource-utilization all
--------------------------------------------------------------------------
Resource utilization data for Management Module
--------------------------------------------------------------------------
System Resources:
Processes                          :  244
CPU usage(%)                       :   10
CPU usage(% average over 1 minute) :   11
CPU usage(% average over 5 minute) :   15
Memory usage(%)                    :   11
Open FD's                          : 1854
```

```
Storage 1: Endurance utilization = 10-20% (mmc-type-a), 0-10% (mmc-type-b),
   Health = normal
Data written to various partitions since boot
Nos      :  15 MB
Log      :   1 MB
Coredump :  23 MB
Security :   2 MB
Selftest : 405 KB
Swap     :  14 MB
Storage partition usage(%)
Nos      :   5
Log      :  60
Coredump :  23
Security :   2
Selftest :   1
Swap     :   0
Process                 CPU Usage(%)    Memory Usage(%)   Open FD's
---------------------------------------------------------------------------
(sd-pam)                0               0               7
aaa
utilspamcfg          0               1               10
---------------------------------------------------------------------------
Resource utilization data for Standby Management Module
---------------------------------------------------------------------------
System Resources:
Processes                          :  244
CPU usage(%)                       :   10
CPU usage(% average over 1 minute) :   11
CPU usage(% average over 5 minute) :   15
Memory usage(%)                    :   11
Open FD's                          : 1854
Storage 1: Endurance utilization = 10-20% (mmc-type-a), 0-10% (mmc-type-b),
   Health = normal
Data written to various partitions since boot
Nos      :  15 MB
Log      :   1 MB
Coredump :  23 MB
Security :   2 MB
Selftest : 405 KB
Swap     :   0 KB
Storage partition usage(%)
Nos      :   5
Log      :  60
```

```
Coredump :  23
Security :   2
Selftest :   1
Swap     :   0
---------------------------------------------------------------------------
System Resource utilization for line card module: 1/7
---------------------------------------------------------------------------
CPU usage(%)                          :   10
CPU usage(% average over 1 minute)    :   11
CPU usage(% average over 5 minute)    :   15
Memory usage(%)                       :   11
Open FD's                             :  854
---------------------------------------------------------------------------
System Resource utilization for line card module: 1/8
---------------------------------------------------------------------------
CPU usage(%)                          :   10
CPU usage(% average over 1 minute)    :   11
CPU usage(% average over 5 minute)    :   15
Memory usage(%)                       :   11
Open FD's                             :  980
```

Showing resource utilization data for the standby management module:

```
switch# show system resource-utilization standby
System Resources:
Processes                             :  244
CPU usage(%)                          :   10
CPU usage(% average over 1 minute)    :   11
CPU usage(% average over 5 minute)    :   15
Memory usage(%)                       :   11
Open FD's                             : 1854
Storage 1: Endurance utilization = 18% (ssd), Health = normal
Data written to various partitions since boot
Nos      :  15 MB
Log      :   1 MB
Coredump :  23 MB
Security :   2 MB
Selftest : 405 KB
Swap     :  14 MB
Storage partition usage(%)
Nos      :   5
Log      :  60
Coredump :  23
Security :   2
Selftest :   1
```

```
Swap     :   0
Process                 CPU Usage(%)    Memory Usage(%)   Open FD's
--------------------------------------------------------------------------
hpe-sysmond                 1                2                11
hpe-mgmdd                   0                1                5
...
```

Showing resource utilization data for a line module:

```
switch# show system resource-utilization module 1/1
--------------------------------------------------------------------------
System Resource utilization for line card module: 1/1
--------------------------------------------------------------------------
CPU usage(%)                         :   10
CPU usage(% average over 1 minute) :   11
CPU usage(% average over 5 minute) :   15
Memory usage(%)                      :   11
Open FD's                            :  754
Data written to various partitions since boot
Coredump :  23 MB
Storage partition usage(%)
Coredump :  45
```

Showing resource utilization data for all modules:

```
switch# show system resource-utilization all
--------------------------------------------------------------------------
Resource utilization data for Management Module
--------------------------------------------------------------------------
System Resources:
Processes                            :  244
CPU usage(%)                         :   10
CPU usage(% average over 1 minute) :   11
CPU usage(% average over 5 minute) :   15
Memory usage(%)                      :   11
Open FD's                            : 1854
Storage 1: Endurance utilization = 17% (ssd), Health = normal
Data written to various partitions since boot
Nos      :  15 MB
Log      :   1 MB
Coredump :  23 MB
Security :   2 MB
Selftest : 405 KB
Swap     :   0 KB
Storage partition usage(%)
Nos      :   5
Log      :  60
```

```
Coredump :  23
Security :   2
Selftest :   1
Swap     :   0
Process                   CPU Usage(%)    Memory Usage(%)   Open FD's
--------------------------------------------------------------------------
(sd-pam)                  0               0                7
aaa
utilspamcfg            0               1                10
--------------------------------------------------------------------------
Resource utilization data for Standby Management Module
--------------------------------------------------------------------------
System Resources:
Processes                              :  244
CPU usage(%)                           :   10
CPU usage(% average over 1 minute) :   11
CPU usage(% average over 5 minute) :   15
Memory usage(%)                        :   11
Open FD's                              : 1854
Storage 1: Endurance utilization = 17% (ssd), Health = normal
Data written to various partitions since boot
Nos      :  15 MB
Log      :   1 MB
Coredump :  23 MB
Security :   2 MB
Selftest : 405 KB
Swap     :   0 KB
Storage partition usage(%)
Nos      :   5
Log      :  60
Coredump :  23
Security :   2
Selftest :   1
Swap     :   0
--------------------------------------------------------------------------
System Resource utilization for line card module: 1/7
--------------------------------------------------------------------------
CPU usage(%)                           :   10
CPU usage(% average over 1 minute) :   11
CPU usage(% average over 5 minute) :   15
Memory usage(%)                        :   11
Open FD's                              :  480
Data written to various partitions since boot
```

```
Coredump :  23 MB
Storage partition usage(%)
Coredump :  23
--------------------------------------------------------------------------
System Resource utilization for line card module: 1/8
--------------------------------------------------------------------------
CPU usage(%)                           :   10
CPU usage(% average over 1 minute) :   11
CPU usage(% average over 5 minute) :   15
Memory usage(%)                        :   11
Open FD's                              :  485
Data written to various partitions since boot
Coredump :  23 MB
Storage partition usage(%)
Coredump :  47
```

Showing memory usage data for all switches:

```
Memory usage in kilobytes:
Physical Memory   :  7274948 KB
Reserved Memory   :  1580224 KB
Used Memory       :  1614416 KB
Free Memory       :  5660532 KB
```

Command History

Release	Modification
10.17.1000	Added the following new fields: physical memory, reserved memory, used memory, and free memory.
10.12	The output of this command includes **CPU usage(% average over 1 minute)** and **CPU usage(% average over 5 minute)**.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

`show tech`

Syntax

```
show tech [basic | <FEATURE>] [local-file]
```

Description

Shows detailed information about switch features by automatically running the `show` commands associated with the feature. If no parameters are specified, the `show tech` command shows information about all switch features. Technical support personnel use the output from this command for troubleshooting.

Parameter	Description
`basic`	Specifies showing a basic set of information.
<FEATURE>	Specifies the name of a feature. For a list of feature names, en ter the `show tech` command, followed by a space, followed by a question mark (?).
`local-file`	Shows the output of the `show tech` command to a local text file.

Usage

To terminate the output of the `show tech` command, enter **Ctrl+C**.

If the command was not terminated with **Ctrl+C**, at the end of the output, the `show tech` command shows the following:

- The time consumed to execute the command.
- The list of failed `show` commands, if any.

To get a copy of the local text file content created with the show tech command that is used with the local-file parameter, use the `copy show-tech local-file` command.

Example

Showing the basic set of system information:

```
switch# show tech basic
==============================================================
Show Tech executed on Wed Sep  6 16:50:37 2017
==============================================================
==============================================================
[Begin] Feature basic
```

```
==============================================================
******************************
Command : show core-dump all
******************************
no core dumps are present
...
==============================================================
[End] Feature basic
==============================================================
==============================================================
1 show tech command failed
==============================================================
Failed command:
  1. show boot-history
==============================================================
Show tech took 3.000000 seconds for execution
```

Directing the output of the **show tech basic** command to the local text file:

```
switch# show tech basic local-file
Show Tech output stored in local-file. Please use 'copy show-tech local-
file'
to copy-out this file.
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution righ ts for this command.

show usb

Syntax

```
show usb [vsx-peer]
```

Description

Shows the USB port configuration and mount settings.

Parameter	Description
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Examples

If USB has not been enabled:

```
switch> show usb
Enabled: No
Mounted: No
```

If USB has been enabled, but no device has been mounted:

```
switch> show usb
Enabled: Yes
Mounted: No
```

If USB has been enabled and a device mounted:

```
switch> show usb
Enabled: Yes
Mounted: Yes
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show usb file-system`

Syntax

`show usb file-system [<PATH>]`

Description

Shows directory listings for a mounted USB device. When entered without the <PATH> parameter the top level directory tree is shown.

Parameter	Description
<PATH>	Specifies the file path to show. A leading "/" in the path is optional.

Usage

Adding a leading "/" as the first character of the <PATH> parameter is optional.

Attempting to enter '..' as any part of the <PATH> will generate an invalid path argument error. Only fully-qualified path names are supported.

Examples

Showing the top level directory tree:

```
switch# show usb file-system
/mnt/usb:
'System Volume Information'  dir1'
/mnt/usb/System Volume Information':
IndexerVolumeGuid  WPSettings.dat
/mnt/usb/dir1:
dir2  test1
```

```
/mnt/usb/dir1/dir2:
test2
```

Showing available path options from the top level:

```
switch# show usb file-system /
total 64
drwxrwxrwx 2 32768 Jan 22 16:27 'System Volume Information'
drwxrwxrwx 3 32768 Mar  5 15:26 dir1
```

Showing the contents of a specific folder:

```
switch# show usb file-system /dir1
total 32
drwxrwxrwx 2 32768 Mar  5 15:26 dir2
-rwxrwxrwx 1     0 Feb  5 18:08 test1
switch# show usb file-system dir1/dir2
total 0
-rwxrwxrwx 1 0 Feb  6 05:35 test2
```

Attempting to enter an invalid character in the path:

```
switch# show usb file-system dir1/../..
Invalid path argument
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show version`

Syntax

```
show version [vsx-peer]
```

Description

Shows version information about the network operating system software, service operating system software, and BIOS.

Parameter	Description
`vsx-peer`	Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output f rom the VSX peer switch is not displayed. This parameter is avai lable on switches that support VSX.

Example

```
DocTeam-4100i# show version
----------------------------------------------------------------------------
-
ArubaOS-CX
(c) Copyright 2017-2023 Hewlett Packard Enterprise Development LP
----------------------------------------------------------------------------
-
Version      : RL.10.13.1000S
Build Date   : 2023-11-15 15:29:10 UTC
Build ID     : ArubaOS-CX:RL.10.13.1000S:ac4c3941a38c:202311151508
Build SHA    : ac4c3941a38c5d66f8f0212706af8f35fa01adfd
Hot Patches  :
Active Image : secondary
Service OS Version : RL.01.09.0003
BIOS Version       : RL.01.0003
```

Showing version information for an 8400 switch::

```
switch> show version
----------------------------------------------------------------------------
-
AOS-CX
(c) Copyright 2022 Hewlett Packard Enterprise Development LP
----------------------------------------------------------------------------
-
Version      : XL.xx.xx.xxxx
Build Date   : 2022-05-27 17:00:46
PDT
```

```
Build ID     : AOS-CX:XL.xx.xx.xxxx:2b4032a4282b:201707241803
Build SHA    :
2b4032a4282b47ab94ea92a1436e1194f34bb5ba
Active Image : secondary
Service OS Version : GT.01.01.0010
BIOS Version       : GT-01-0020
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`system error-counter-monitor poll-interval`

Syntax

```
system error-counter-monitor poll-interval <INTERVAL>
```

Description

Sets the polling interval used for the collection and recording of error counter data.

Parameter	Description
`<INTERVAL>`	Specifies the poll interval in seconds. Range: 10 to 3600. Default: 10.

Example

Setting the system error counter monitor poll interval:

```
switch(config)# system error-counter-monitor poll-interval 20
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
8400	`config`	Administrators or local user group members with execution righ ts for this command.

system error-counter-monitor

Syntax

```
system error-counter-monitor
no system error-counter-monitor
```

Description

Enables the system error counter monitoring feature, which records error counter data every 10 seconds. Default: Disabled.

The **no** form of the command disables error counter monitoring and stops the recording of error counter data.

Example

Enabling the system error counter monitor:

```
switch(config)# system error-counter-monitor
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8400	`config`	Administrators or local user group members with execution rights for this command.

`system resource-utilization poll-interval`

Syntax

```
system resource-utilization poll-interval <SECONDS>
```

Description

Configures the polling interval for system resource information collection and recording such as CPU and memory usage.

Parameter	Description
<SECONDS>	Specifies the poll interval in seconds. Range: 10 - 3600. Default: 10.

Example

Configuring the system resource utilization poll interval:

```
switch(config)# system resource-utilization poll-interval 20
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`top cpu`

Syntax

```
top cpu
```

Description

Shows CPU utilization information.

Example

Showing top CPU information:

```
switch# top cpu
top - 09:42:55 up 3 min, 3 users, load average: 3.44, 3.78, 1.70
Tasks:  76 total, 2 running, 74 sleeping,  0 stopped,  0 zombie
%Cpu(s): 31.4 us, 32.7 sy, 0.5 ni, 34.4 id, 04. wa, 0.0 hi, 0.6 si, 0.0 st
KiB Mem : 4046496 total,  2487508 free,  897040 used,   661948 buff/cache
KiB Swap:       0 total,        0 free,        0 used,  2859196 avail Mem
  PID USER      PRI  NI   VIRT    RES    SHR S  %CPU %MEM       TIME+ COMMAND
...
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`top memory`

Syntax

```
top memory
```

Description

Shows memory utilization information.

Example

Showing top memory:

```
switch> top memory
top - 09:42:55 up 3 min, 3 users, load average: 3.44, 3.78, 1.70
Tasks:  76 total, 2 running, 74 sleeping,  0 stopped,  0 zombie
%Cpu(s): 31.4 us, 32.7 sy, 0.5 ni, 34.4 id, 04. wa, 0.0 hi, 0.6 si, 0.0 st
KiB Mem : 4046496 total,  2487508 free,  897040 used,   661948 buff/cache
KiB Swap:       0 total,        0 free,       0 used,  2859196 avail Mem
  PID USER      PRI  NI    VIRT    RES    SHR S  %CPU %MEM      TIME+ COMMAND
...
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

usb

Syntax

```
usb
no usb
```

Description

Enables the USB ports on the switch. This setting is persistent across switch reboots and management module failovers. Both active and standby management modules are affected by this setting.

The **no** form of this command disables the USB ports.

Example

Enabling USB ports:

```
switch(config)# usb
```

Disabling USB ports when a USB drive is mounted:

```
switch(config)# no usb
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`usb mount | unmount`

Syntax

```
usb {mount | unmount}
```

Description

Enables or disables the inserted USB drive.

Parameter	Description
`mount`	Enables the inserted USB drive.
`unmount`	Disables the inserted USB drive in preparation for removal.

Usage

If USB has been enabled in the configuration, the USB port on the active management module is available for mounting a USB drive. The USB port on the standby management module is not available.

An inserted USB drive must be mounted each time the switch boots or fails over to a different management module.

A USB drive must be unmounted before removal.

The supported USB file systems are FAT16 and FAT32.

Examples

Mounting a USB drive in the USB port:

```
switch# usb mount
```

Unmounting a USB drive:

```
switch# usb unmount
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution righ ts for this command.

Support and Other Resources

Access HPE Aruba Networking support and updates, and view warranty and regulatory information

Subtopics

Accessing HPE Aruba Networking Support

HPE Aruba Networking Support Services	**https://www.hpe.com/us/en/networking/hpe-aruba-networking-support-services.html**
AOS-CX Switch Software Documentation Portal	**https://arubanetworking.hpe.com/techdocs/ArubaDocPortal/content/new-portal/aoscx.html**
HPE Aruba Networking Support Portal	**https://networkingsupport.hpe.com/home**

North America telephone	1 - 800 - 943 - 4526 (US & Canada Toll - Free Number) +1 - 650 - 750 - 0350 (Backup—Toll Number)
International telephone	**https://www.hpe.com/psnow/doc/a50011948enw**

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

HPE Aruba Networking Developer Hub	**https://developer.arubanetworks.com/hpe - aruba - networking - aoscx/docs/about**
Airheads social forums and Knowledge Base	**https://community.arubanetworks.com/**
AOS - CX Software Technical Update channel on You Tube.	Videos on new features introduced in this release : **https://www.youtube.com/playlist?list=PLsYGHuNuBZcbWPEjjHuVMqP - Q_UL3CskS**
HPE Aruba Networking Hardware Documentation and Translations Portal	
HPE Aruba Networking software	**https://networkingsupport.hpe.com/downloads https://networkingsupport.hpe.com/downloads**
Software licensing and Feature Packs	**https://licensemanagement.hpe.com/**

End - of - Life information	**https://networkingsupport.hpe.com/end - of - life**

Accessing Updates

You can access updates from the HPE Aruba Networking Support Portal at **https://networkingsupport.hpe.com**.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

https://networkingsupport.hpe./notifications/subscriptions (requires an active HPE Aruba Networking Support Portal account to manage subscriptions). Security notices are viewable without an HPE Aruba Networking Support Portal account.

Warranty Information

To view warranty information for your product, go to **https://www.arubanetworks.com/support-services/product-warranties/**.

Regulatory Information

To view the regulatory information for your product, view the Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products, available at **https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts**

Additional regulatory information

HPE Aruba Networking is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see **https://www.arubanetworks.com/company/about-us/environmental-citizenship/**.

Documentation Feedback

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